2008 RESTRAINTS

Supplemental Restraint System - Edge & MKX

SPECIFICATIONS

TORQUE SPECIFICATIONS

Description	Nm	lb-ft	lb-in
Front impact severity sensor bolt	11	8	-
Inner floor console support bolts	12	9	-
Manual seat track front cross brace screws	5	-	44
Manual seat track rear cross brace screws	5	-	44
Occupant classification sensor (OCS) weight sensor bolt nuts	33	24	-
Passenger air bag module bolts (Edge)	8	-	71
Passenger air bag module nuts (MKX)	7	-	62
Passenger air bag module support bracket-to-passenger air bag module bolts (MKX)	8	-	71
Passenger air bag module support bracket-to-cross vehicle beam bolts (MKX)	8	-	71
Power seat track front cross brace nuts	20	15	-
Restraints control module (RCM) bolts	12	9	-
Safety belt buckle pretensioner nut	40	30	-
Safety canopy module bolts (all)	9	-	80
Side air bag module bolts	23	17	-
Side impact sensor (B-pillar) bolt	11	8	-
Side impact sensor (C-pillar) bolt	11	8	-

DESCRIPTION AND OPERATION

AIR BAG AND SAFETY BELT PRETENSIONER SUPPLEMENTAL RESTRAINT SYSTEM (SRS)

The air bag supplemental restraint system (SRS) is designed to provide increased collision protection for front seat and second row outboard occupants in addition to that provided by the 3-point safety belt system. Safety belt use is necessary to obtain the best occupant protection and to receive the full advantage of the SRS.

This vehicle contains dual-stage deployment (advanced restraint system) driver and front passenger air bag modules. These vehicles are also equipped with driver and front passenger safety belt retractor pretensioners as well as safety canopies. These canopies deploy from the A-pillar to the C-pillar during a side impact or if a rollover condition is detected. Vehicles with safety canopies are also equipped with seat side air bag modules in the front seats. Seat side air bag modules deploy from the outboard front seat backrest during a side impact. A unique restraints control module (RCM) that detects a potential rollover condition is another standard feature.

In addition, there are 2 front impact severity sensors mounted to the hood latch bracket, 2 first row side impact sensors mounted near the base of the B-pillar and 2 second row impact sensors mounted near the base of the C-pillar. A seat position sensor mounted to the driver seat and a usage detection switch is added to the front driver and passenger outboard buckles.

Vehicles are also equipped with an occupant classification sensor (OCS) system mounted to the front passenger seat track assembly. The weight of any occupant or object on the front passenger seat is electronically communicated to the occupant classification system module (OCSM). The RCM uses this information in determining if the passenger air bag module or passenger seat side air bag module is to be deployed in the event of a deployable collision.

The OCS system includes 4 OCS weight sensor bolts, OCSM and a passenger air bag deactivation (PAD) indicator lamp.

Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS) Components

This vehicle is equipped with following SRS components:

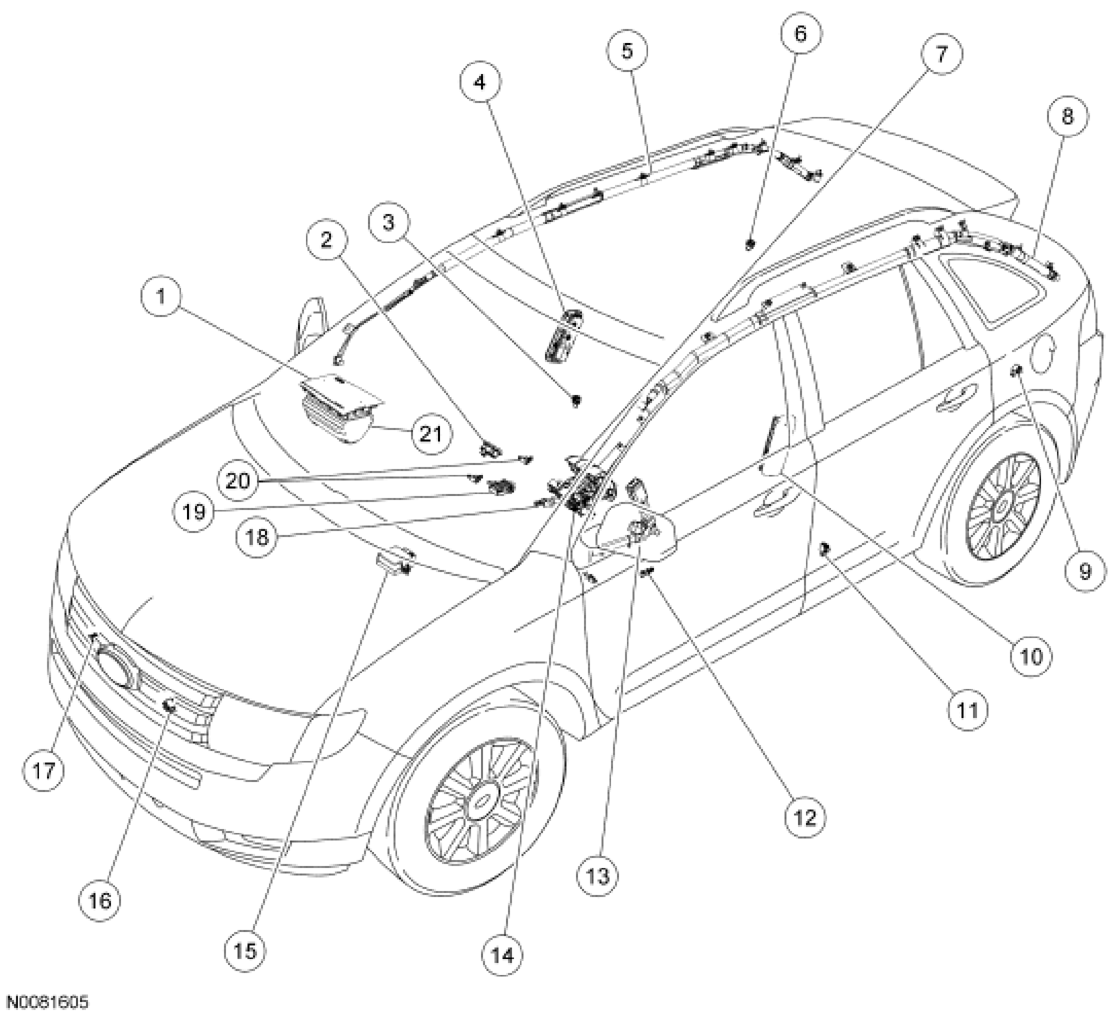


Fig. 1: Identifying Supplemental Restraint System (SRS) Components
Courtesy of FORD MOTOR CO.

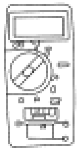
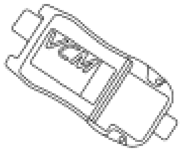
Item	Part Number	Description
1	04338	Passenger air bag cover
2	14B418	Passenger air bag deactivation (PAD) indicator
3	14B345	First row, passenger, side impact sensor
4	611D10	Passenger side air bag module
5	042D94	Passenger side safety canopy module
6	14B345	Second row, passenger, C-pillar side impact sensor
7	043B13	Driver air bag module
8	042D95	Driver side safety canopy module
9	14B345	Second row, driver, C-pillar side impact sensor
10	611D10	Driver side air bag module
11	14B345	First row, driver, side impact sensor
12	14B416	Driver seat track position sensor
13	61203	Driver safety belt buckle (includes safety belt buckle switch)
14	14A664	Clockspring
15	14B321	Restraints control module (RCM)
16	14B006	Driver/center front impact severity sensor
17	14B006	Passenger front impact severity sensor
18	61202	Passenger safety belt buckle (includes safety belt buckle switch)
19	14B422	Occupant classification system module (OCSM)
20	-	Occupant classification weight sensor bolts (2 shown in illustration) (4 required)
21	044A74	Passenger air bag module assembly

DIAGNOSTIC TESTS

AIR BAG AND SAFETY BELT PRETENSIONER SUPPLEMENTAL RESTRAINT SYSTEM (SRS)

Special Tools

Illustration	Tool Name	Tool Number
	Flex Probe Kit	105-R025C or equivalent

		
	FLUKE 73III Automotive Meter	105-R0057 or equivalent
	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS)	software with appropriate hardware, or equivalent scan tool

Principles of Operation

The supplemental restraint system (SRS) consists of a driver and passenger air bag module, safety belt retractor pretensioners and safety canopies (containing an inflator/squib or gas generator and an air bag), impact sensors, a restraints control module (RCM), a clockspring, an air bag warning indicator, occupant classification sensor (OCS) system and a passenger air bag deactivation (PAD) indicator lamp. These components are all interconnected by a wiring harness and powered by the vehicle's battery. The RCM includes a backup power supply. This feature provides sufficient backup power to deploy the SRS in the event that the ignition circuit is lost or damaged during impact. The backup power supply will deplete its stored energy approximately one minute after the battery ground cable is disconnected. If a SRS fault exists, the air bag warning indicator will illuminate and remain illuminated for the rest of the key cycle. In addition to the self-test at start up, the RCM continuously monitors all of its external and internal circuitry for faults.

In a frontal collision, the impact sensors located in the front of the vehicle detect the sudden deceleration and send an electrical signal to the RCM. The RCM uses the information from the impact sensors and the OCS system in the deployment determination. If the RCM determines that deployment is required, the RCM sends voltage and current to the squib(s) causing the solid chemical propellant to undergo a rapid chemical reaction. This controlled reaction produces harmless nitrogen gas that fills the air bag(s)/safety canopies and/or activates the safety belt pretensioners to remove slack from the safety belt(s).

The RCM communicates through the data link connector (DLC) the current and historical DTCs on the high-speed controller area network (HS-CAN). The RCM also communicates over the HS-CAN to the instrument cluster (IC) module, PCM and the occupant classification system module (OCSM).

Air Bag Warning Indicator

The air bag warning indicator:

- is located in the instrument cluster (IC) module.

- lamp and circuitry prove out is a function of the IC module. The IC module will prove out the air bag warning indicator by lighting the air bag warning indicator for 6 seconds and then turn off.
- will flash and/or illuminate based on the message the IC module receives from the RCM.
- will illuminate if the IC module does not receive a message from the RCM.

Air Bag Module Second Stage Deployment Check

Because the driver and passenger front air bags each have 2 deployment stages, it is possible that Stage 1 has deployed and Stage 2 has not.

If a front air bag module has deployed, it is **mandatory** that the front air bag module be remotely deployed using the appropriate air bag disposal procedure.

- For information on driver air bag module and/or passenger air bag module remote deployment, refer to **Pyrotechnic Device Disposal**.

Clockspring

The clockspring:

- is mounted on the steering column, behind the steering wheel.
- allows for continuous electrical connections between the driver air bag module and the restraints control module (RCM) when the steering wheel is turned.

Driver Air Bag Module

The driver air bag module:

- is installed as an assembly.
- is mounted in the center of the steering wheel.
- cannot be interchanged between Edge and MKX vehicles.

High-Speed Controller Area Network (HS-CAN)

This vehicle utilizes a communication system called a high-speed controller area network (HS-CAN). The HS-CAN consists of a twisted pair of wires connected to the following:

- ABS module
- Instrument cluster (IC) module
- PCM
- Transmission control module (TCM)
- Restraints control module (RCM)
- Occupant classification system module (OCSM)
- Air suspension (vehicle dynamics module [VDM]) (if equipped)

- Data link connector (DLC)

The HS-CAN circuits use a bias voltage of approximately 2.5 volts, one is a positive 2.5 volts while the other is a negative 2.5 volts. The HS-CAN also uses 2 terminating resistors, one contained within the PCM, the other in the IC module. The terminating resistors are not serviced separately. The terminating resistors have a value of 120 ohms each, for a normal operating system total of 60 ohms. The HS-CAN may operate with only one terminating resistor and may communicate some messages to some of the control modules with only one circuit functioning. Refer to **MODULE COMMUNICATIONS NETWORK** article.

Impact Sensors

WARNING: **If a vehicle has been in a crash, inspect the restraints control module (RCM) and the impact sensor (if equipped) mounting areas for deformation. If damaged, restore the mounting areas to the original production configuration. A new RCM and sensors must be installed whether or not the air bags have deployed. Failure to follow these instructions may result in serious personal injury or death in a crash.**

For this vehicle line, the supplemental restraint system (SRS) uses 6 satellite sensors in addition to the restraints control module (RCM). The RCM is mounted to the center tunnel beneath the console. All vehicles have 2 front impact severity sensors located in the front-center area of the vehicle, behind the grille mounted on the hood latch bracket. The first row impact sensors are mounted behind the trim panel near the floor on the B-pillar, the second row sensors are located on each C-pillar. Mounting orientation is critical for correct operation of all impact sensors.

Loops/Squibs

All deployable devices contain an initiating device called a squib. The squib is part of the deployment loop. Air bag/safety canopy modules can contain more than one squib, some vehicles may have up to 4 squibs in one air bag module. Squibs are often referred to as loops during the diagnostic process.

Occupant Classification Sensor (OCS) System

The occupant classification sensor (OCS) system is found only on the front passenger seat. The front passenger OCS system is comprised of the following: 4 OCS weight sensor bolts that are mounted to each corner of the seat track and an occupant classification system module (OCSM) which is mounted to the electrical bracket underneath the seat cushion pan. All the components that make up the OCS system are all serviced separately. The weight of any occupant or object on the front passenger seat is electronically communicated to the OCSM. The RCM uses this information in determining if the passenger air bag module or passenger seat side air bag module is to be deployed in the event of a deployable collision.

The OCSM will inform the restraints control module (RCM), via a high-speed controller area network (HS-CAN), of the weight of any occupant or object on the front passenger seat.

The OCSM monitors the OCS system for faults and communicates on-demand and continuous DTCs via the data link connector (DLC) with the use of a scan tool.

The OCS system is also used for operation of the passenger Belt-Minder®. For information on the passenger Belt-Minder® feature, refer to **SAFETY BELT SYSTEM** article. To deactivate or reactivate the passenger Belt-Minder® feature, refer to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES** article or the Owner's Literature.

When the front passenger seat is removed for service, the Zero Seat Weight Test must be carried out after the installation of the front passenger seat. When an OCS system component is installed new, the System Reset must be carried out after the installation of the front passenger seat. The Zero Seat Weight Test and/or System Reset must be carried out only as instructed to do so in the appropriate service information. For information on the Zero Seat Weight Test and/or System Reset, refer to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test** or **Occupant Classification Sensor (OCS) System Reset**.

In the event of a deployable crash, a new passenger seat track with OCS weight sensor bolts must be installed. **DTC B1231 must be cleared before carrying out Occupant Classification Sensor (OCS) System Reset.** Refer to **Occupant Classification Sensor - Manual Seat Track** or **Occupant Classification Sensor - Power Seat Track**.

Passenger Air Bag Deactivation (PAD) Indicator

The passenger air bag deactivation (PAD) indicator is a visual indicator used to inform the front seat occupants of the passenger air bag deactivation state. The PAD indicator is a stand-alone lamp installed into the vehicle instrument panel in a position visible to each front seat occupant.

The restraints control module (RCM) controls the state of the PAD indicator through a direct hardwire connection, based on information provided by the occupant classification sensor (OCS) system. The PAD indicator illuminates to indicate the passenger air bag module is disabled. An exemption to this is when the front passenger seat is determined to be empty and the passenger safety belt is unbuckled, therefore indication of a deactivated passenger air bag module is not necessary. In all other cases, the PAD indicator is unlit when the passenger air bag module is enabled.

When the ignition switch is in the ON position, the PAD indicator prove-out period is initiated by the RCM. The RCM briefly activates the PAD indicator to prove-out the indicator function and verify to the front occupants correct functional operation of the PAD indicator.

When the OCSM detects a fault that causes DTC B2290 and/or B1013 to set on-demand in the OCS system, the OCSM sends a message to the RCM. Upon receiving the fault message from the OCSM, the RCM disables the passenger air bag module, illuminates the PAD indicator indicating the passenger air bag module is disabled and sends a message to the instrument cluster (IC) module to illuminate the air bag warning indicator. The passenger air bag module will remain disabled until the cause for the on-demand DTC has been corrected.

The following table indicates the passenger air bag status and the PAD indicator status based the size of the front outboard passenger occupant.

PASSENGER AIR BAG AND PAD INDICATOR STATUS

Occupant Size	Passenger Safety Belt Buckle Status	Pass. Air Bag Status	PAD Ind. Status
None	Unbuckled	Disabled	Unlit

None	Buckled	Disabled	Lit
Infant or Small Child	Buckled/Unbuckled	Disabled	Lit
Adult Sized	Buckled/Unbuckled	Enabled	Unlit

Passenger Air Bag Module

The passenger air bag module:

- is installed as an assembly.
- is mounted in the passenger side of the instrument panel.
- cannot be interchanged between Edge and MKX vehicles.

Restraints Control Module (RCM)

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **If a vehicle has been in a crash, inspect the restraints control module (RCM) and the impact sensor (if equipped) mounting areas for deformation. If damaged, restore the mounting areas to the original production configuration. A new RCM and sensors must be installed whether or not the air bags have deployed. Failure to follow these instructions may result in serious personal injury or death in a crash.**

NOTE: **Prior to removal of the restraints control module (RCM) module, it is necessary to upload module configuration information to the scan tool. This information needs to be downloaded into the new RCM module once installed. Refer to MODULE CONFIGURATION article. Failure to follow these instructions may result in incorrect operation of the supplemental restraint system (SRS) and may cause system failure.**

NOTE: **When installing a new restraints control module (RCM), always make sure the correct RCM is being installed. If an incorrect RCM is installed, erroneous DTCs will result.**

The RCM carries out the following functions:

- deploys the air bag(s)/safety canopies in the event of a deployable collision.
- activates the safety belt retractor pretensioners to remove slack from the safety belt in the event of a deployable collision.
- monitors the supplemental restraint system (SRS) for faults.

- sends a message to the instrument cluster (IC) module to illuminate the air bag warning indicator if a fault is detected.
- communicates through the data link connector (DLC) the current or historical DTCs.

The RCM monitors the SRS for possible faults. If a fault is detected, the RCM will request illumination of the air bag warning indicator.

When the ignition is cycled (turned OFF and then ON), the IC module will prove out the air bag warning indicator by lighting for 6 seconds and then turn off. If a current SRS fault exists, the RCM requests illumination of the air bag warning indicator and will remain illuminated for the rest of the key cycle. During that same key cycle, some faults that recover (Loss of Communication with the PCM), the RCM will send a request to the IC module to turn the air bag warning indicator off **IF** no other fault is present. The RCM will also communicate the on-demand (current) and continuous (historical) DTCs through the DLC, to the scan tool. If the RCM requests illumination of the air bag warning indicator and the air bag warning indicator does not function, the IC module will automatically activate an audible chime. The chime is a series of 5 sets of 5 tone bursts. If the chime is heard, the SRS and the air bag warning indicator require repair.

The RCM includes a backup power supply. This feature provides sufficient backup power to deploy the air bags in the event that the ignition circuit is lost or damaged during impact. The backup power supply will deplete its stored energy approximately one minute after power and/or ground has been removed from the RCM.

Safety Belt Buckle Switches

As part of the supplemental restraint system (SRS), the driver and front passenger safety belt buckles are equipped with safety belt buckle switches. The safety belt buckle switches indicate to the restraints control module (RCM) whether the safety belts are connected or disconnected. The RCM uses this information in determining the deployment of the dual-stage driver and passenger air bag modules.

Safety Belt Pretensioners

As part of the supplemental restraint system (SRS), the driver and front passenger safety belt retractors are equipped with pretensioners. The safety belt retractor pretensioners remove excess slack from the safety belt webbing. The pretensioners are activated by the restraints control module (RCM) when the module detects a collision exceeding a programmed limit.

Safety Canopy Module

Vehicles are equipped with safety canopies for protection during side impacts or rollovers.

Safety canopies require a specific headliner. The word AIRBAG will appear on the headliner where it meets each B-pillar trim panel.

The safety canopy module:

- is installed as an assembly.
- is mounted above the headliner.

- attaches from the A-pillar frame to the C-pillar frame.
- is standard equipment.
- cannot be interchanged from side-to-side.

Seat Track Position Sensor

The seat track position sensor is mounted to a bracket attached to the inboard side of the driver seat track. The seat track position sensor informs the restraints control module (RCM) of the driver seat position. The RCM uses this information in determining the deployment of the dual-stage driver air bag module.

Side Air Bag Module

NOTE: **References to side air bag modules refer to the seat side air bag module, not to the steering wheel or instrument panel mounted air bag components of the supplemental restraint system (SRS).**

A side air bag module provides protection of the thorax area (between the neck and abdomen) of the body, working in conjunction with the head protection provided by a safety canopy module.

The side air bag module:

- will deploy upon receiving a flow of current from the restraints control module (RCM), initiated by the side impact sensor and internal RCM circuitry.
- is installed as an assembly.
- is mounted in the driver or passenger seat backrest.
- is used in conjunction with a safety canopy module.

Secondary Air Bag Warning (Chime)

The secondary air bag warning, is an audible tone controlled by the instrument cluster (IC) module. If a fault is detected with the air bag warning indicator a DTC will be stored in memory of the IC module. Upon receiving the message from the restraints control module (RCM) that a supplemental restraint system (SRS) fault has been detected, the IC module will sound the secondary air bag warning chime in a pattern of 5 sets of 5 beeps.

Diagnostic Instructions

The symptom chart can be used to help locate supplemental restraint system (SRS) concerns if no DTCs are retrieved and the listed symptoms are observed. Whether or not the listed symptoms are observed, always carry out the following:

1. Run the Self Test to determine what on-demand and continuous memory DTCs are being sensed by the restraints control module (RCM) and occupant classification system module (OCSM).
2. Retrieve all SRS DTCs and fault PIDs stored in the RCM and OCSM memory.
3. If on-demand DTCs are different than continuous memory DTCs, always diagnose the on-demand DTCs first.

A DTC can indicate several concerns. The DTCs are to assist in system diagnosis and are not to be considered definitive. Always refer to the pinpoint test corresponding to the DTC to determine where the concern lies and to repair the concern correctly.

Diagnostic Test Options

Scan tool options:

- Self Test/On-Demand
- Self Test/Continuous Memory and Clear DTCs

Refer to the manufacturer's literature for the scan tool being used for correct scan tool test options.

Self Test/On-Demand

The on-demand Self Test option is used to verify that no electrical concerns exist with the air bag supplemental restraint system (SRS). Upon entering the self-test, the restraints control module (RCM) and the occupant classification system module (OCSM) will make an electrical check of each electrical component in the system. If a concern is detected, a DTC is displayed on the scan tool with a brief description of the DTC. The self-test should always be carried out after any repair to verify that the repair was successful.

To run the on-demand Self Test, follow these steps:

1. Connect the scan tool to the data link connector (DLC).
2. Turn the ignition switch to the ON position.
3. Follow the manufacturer's instructions for the scan tool being used.
4. Select Self-Test.
5. Select RCM or OCSM.
6. The module will run the self-test and display on-demand (reflecting hard system concerns) and continuous memory (historic) DTCs on the scan tool.

Self Test/Continuous Memory and Clear DTCs

During vehicle operation, the restraints control module (RCM) and the occupant classification system module (OCSM) will detect and store on-demand (reflecting hard system concerns) and continuous memory (historic) DTCs. The DTC strategy employed by the RCM incorporates a time-out scheme for determining when a concern exists in the system. This requires a concern may exist for up to one minute in the system before the RCM will detect it. For the RCM to determine that a concern no longer exists, the concern must be absent for up to one minute. The actual detection time-outs vary with each DTC. DTCs can be retrieved with a scan tool using the Self Test option. All DTCs stored in the RCM and OCSM will be displayed on the scan tool along with a brief description of the DTC. If no DTCs are present, the scan tool will display a SYSTEM PASSED message. This option can also be used to clear DTCs from the RCM and OCSM memory, as long as the concern no longer exists.

Once 75 hours of operation have been recorded by the RCM/OCSM since the concern was last detected, all continuous memory DTCs will automatically be removed from memory.

To retrieve or clear DTCs, follow these steps:

1. Connect the scan tool to the data link connector (DLC).
2. Turn the ignition switch to the ON position.
3. Follow the manufacturer's instructions for the scan tool being used.
4. Select Self Test.
5. Select RCM or OCSM (selecting Restraints retrieves DTCs from both modules).

NOTE: Before proceeding with the clearing operation, make note of the DTCs displayed. Once cleared, continuous DTCs cannot be retrieved.

6. All DTCs will be displayed on the screen.
7. Clear the DTCs. After clearing the DTCs, cycle the ignition OFF, then ON.
    - Continuous memory DTCs that have been cleared will **not** reoccur as "continuous memory" in the same key cycle. Only new DTCs which were **not** present before clearing can occur as "continuous memory" after clearing.

Occupant Classification Sensor (OCS) System Zero Seat Weight Test

The Zero Seat Weight Test verifies the occupant classification sensor (OCS) measures zero weight for an empty seat. It is necessary to carry out the Zero Seat Weight Test any time the front passenger seat is serviced or removed from the vehicle or as directed in the service information procedures. Refer to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.

Occupant Classification Sensor (OCS) System Reset

The System Reset active command resets the zero point of the occupant classification sensor (OCS) system.

The System Reset active command should only be carried out if a new OCS component is installed or as directed by service information .

DataLogger PID/Data

The DataLogger PID/Data option allows the scan tool operator to read the state of several PIDs to aid in diagnosing the system. PIDs are real time measurements of parameters, such as voltages and resistances, calculated by the restraints control module (RCM) or the occupant classification system module (OCSM) and sent to the scan tool for display. Many of the PIDs supported by the modules are calculated periodically and are, therefore, not true real time readings.

To retrieve PIDs, follow these steps:

1. Connect the scan tool to the data link connector (DLC).
2. Turn the ignition switch to the ON position.

3. Follow the manufacturer's instructions for the scan tool being used.
4. Select DataLogger.
5. Select RCM or OCSM.

Active Commands

System Reset

System Reset active command sets the zero set point or rezeros the occupant classification sensor (OCS). Refer to **Occupant Classification Sensor (OCS) System Reset**.

Zero Seat Weight Test

The Zero Seat Weight Test verifies the OCS measures zero weight for an empty seat. Refer to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.

Instrument Cluster (IC) Module

These commands allow the technician to verify the operation of instrument cluster (IC) module components and subsystems.

Lamp Fault Codes (LFCs)

This vehicle supports lamp fault codes (LFCs) **only** when the restraints control module (RCM) is in plant mode. A new RCM installed to a vehicle will be in plant mode until:

- programmable module installation (PMI) has been carried out.
- the RCM sees a fault-free supplemental restraint system (SRS) (no DTCs present).

If a LFC is present after completing PMI, a fault is present in the SRS and on-demand DTCs must be retrieved and diagnosed.

Diagnosing Customer Concerns With On-Demand DTCs

NOTE: **Most supplemental restraint system (SRS) diagnostic procedures will require depowering and repowering of the SRS. Depowering and repowering requires disconnecting of the battery and removal of the restraints control module (RCM) fuse. This reduces the risk of accidental deployment of SRS components while diagnostic procedures are being carried out.**

If the air bag warning indicator is reported ON by the customer when the vehicle comes in for service, connect the scan tool and follow the Diagnostic Instruction procedures in this service information to identify the concern.

Once the DTC is known, read the Normal Operation of the pinpoint test for the DTC involved.

Using the scan tool with the use of PIDs and active command(s) may be of assistance in diagnosing the concern.

- Follow the depowering procedure as directed in **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Determine the location of components involved in creating the DTC.
- Carry out a thorough visual inspection of:
  - components.
  - connectors.
  - splices and wiring harnesses.
  - insulation on conductors.

Diagnosing Customer Concerns With Continuous Memory DTCs

If the air bag warning indicator was reported ON by the customer but is not present when the vehicle comes in for service, follow the Diagnostic Instruction procedures in this service information to identify the intermittent DTC.

Once the DTC is known, read the Normal Operation of the pinpoint test for the DTC involved.

- Follow the depowering procedure as directed in **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Determine the location of components involved in creating the DTC.
- Carry out a thorough visual inspection of:
  - components.
  - connectors.
  - splices and wiring harnesses.
  - insulation on conductors.

Refer to the Possible Causes section of the pinpoint test for the DTC involved, which lists the common concerns that relate to the DTC. Concerns are listed according to priority.

Fault PIDs

There are 2 types of fault PIDs that can be reported by the restraints control module (RCM). The first type, considered conventional, has only one level of fault reporting and identifies a specific concern for a given component and points to a particular diagnostic path, example: DTC B1317 (Battery Voltage High).

The second type uses a process within the software of the controller that maps the byte and bit to name a specific device and fault condition. This process is called Bit-mapping and referred to as fault PIDs in the diagnosis of the vehicle. This type does not identify the specific concern or component on the first level of fault reporting, Example: DTC B2293 (Restraint System - Airbag Fault). DTC B2293 can have up to 28 specific on-demand fault PIDs (areas of concern) associated with this DTC.

Those associated fault PIDs are an extension of the information provided by the DTC and are identified by the same DTC number. A scan tool must be used to view DTCs and their fault PIDs. Once a scan tool has retrieved a DTC, use the scan tool to view the fault PIDs. In the diagnostic path, other types of PIDs are sometimes used to determine the root cause (example: resistance or voltage PIDs).

When viewing of fault PIDs has been carried out, the scan tool can display the PIDs associated with that DTC, including the status or state that exists (on-demand [active] DTC) or existed (continuous memory [historic]) DTC. Refer to the manufacturer instructions for the scan tool being used on how to view fault PIDs.

Prove Out Procedure

Turn the ignition switch from the OFF to the ON position and visually monitor the air bag warning indicator with all supplemental restraint system (SRS) components connected. The air bag warning indicator will light continuously for approximately 6 seconds and then turn off. If an SRS fault is present, the air bag warning indicator will:

- fail to light.
- remain lit continuously.
- flash.

The air bag warning indicator may not illuminate until approximately 30 seconds after the ignition switch has been turned from the OFF to the ON position. This is the time required for the restraints control module (RCM) to complete the testing of the SRS. If the air bag warning indicator is inoperative and an SRS fault exists, a chime will sound in a pattern of 5 sets of 5 beeps. If this occurs, the air bag warning indicator will need to be repaired before diagnosis can continue.

Supplemental Restraint System (SRS) Reconnect Checklist

The checklist below should be completed following diagnosis or repair of any air bag system concern:

- All in-seat harness connectors connected?
- All air bag modules connected?
- Occupant classification system module (OCSM) connected?
- All OCS weight sensor bolts connected?
- Safety canopy modules connected?
- Safety belt pretensioner connectors connected?
- All sensors (front and side impact sensors) connected?
- Restraints control module (RCM) connected?
- RCM fuse installed?
- Battery connected?

Air Bag Module Second Stage Deployment Check

Because the driver and passenger front air bags each have 2 deployment stages, it is possible that Stage 1 has deployed and Stage 2 has not.

If a front air bag module has deployed, it is **mandatory** that the front air bag module be remotely deployed using the appropriate air bag disposal procedure to make sure the second stage has been deployed.

- For information on driver or passenger air bag module remote deployment, refer to **Pyrotechnic Device Disposal**.

Glossary

Disconnect the Component

Disconnect the component means to disconnect the component vehicle harness connector, not to remove the component. Do not reconnect a disconnected component unless instructed to do so.

Deactivate the Supplemental Restraint System (SRS)

Deactivate the SRS means to carry out a deactivation procedure. Refer to **Supplemental Restraint System (SRS) Deactivation and Reactivation**.

Depower the SRS

Depower the SRS means to disconnect the battery and remove the restraints control module (RCM) fuse. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

Deployment Loop

The deployment loop is made up of the RCM, deployable device, air bag(s), safety belt pretensioners (safety canopies, deployable steering columns, load limiting retractor, adaptive tether, adaptive vent, if equipped) and associated circuits.

Install a New Component

Install a new component means to remove the existing component and install a new authorized part obtained from Ford Customer Service Division.

Prove Out the SRS

Prove out the SRS means to turn the ignition switch from the OFF to the ON position and visually monitor the air bag warning indicator with all SRS components connected. The air bag warning indicator will light continuously for approximately 6 seconds and then turn off. If an SRS fault is present, the air bag warning indicator will either fail to light or remain lit continuously. The air bag warning indicator may not illuminate until approximately 30 seconds after the ignition switch has been turned from the OFF to the ON position. This is the time required for the RCM to complete the testing of the SRS. If the air bag warning indicator is inoperative and an SRS fault exists, a chime will sound in a pattern of 5 sets of 5 beeps. If this occurs, the air bag warning indicator will need to be repaired before diagnosis can continue.

Reactivate the SRS

Reactivate the SRS means to carry out the reactivation procedure. Refer to **Supplemental Restraint System (SRS) Deactivation and Reactivation**.

Reconnect the SRS

Reconnect the SRS means to reconnect all system components. Refer to **Supplemental Restraint System (SRS) Reconnect Checklist**.

Repower the SRS

Repower the SRS means, turn the ignition ON, install the RCM fuse and connect the battery ground cable. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

Secondary Air Bag Warning

The secondary air bag warning is an audible fault format that consists of 5 sets of 5 tone bursts, with each set of 5 tone bursts separated by a 5-second quiet period. One tone burst cycle will consist of one-second ON and one-second OFF. The tone pattern will repeat periodically until the problem and warning light are repaired.

Squib

A squib (igniter) is a device designed to convert electrical energy to the heat energy necessary to deploy a pyrotechnic restraints system device.

Verify the System

Verify the system means to prove out the SRS with all restraint system components in place.

Inspection and Verification

1. Verify the customer concern by checking the air bag warning indicator in the instrument cluster. Prove Out the System. Refer to step 4 under **Repowering Procedure**.
2. Visually inspect for obvious signs of mechanical or electrical damage. Do not disconnect electrical connectors until directed to do so within the pinpoint test.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Damaged restraints control module (RCM) or loose mounting • Damaged front impact severity sensor(s) or loose mounting • Damaged side impact sensor(s) or loose mounting	• Open smart junction box (SJB) fuse 46 (7.5A) • Damaged wiring harness

• Damaged or disconnected passenger air bag deactivation (PAD) indicator • Damaged front passenger seat track assembly • Damaged occupant classification sensor (OCS) weight sensor bolt(s) or loose mounting	• Loose, damaged or corroded connectors • Circuitry open/shorted • Damaged shorting bars • Loose, damaged or pinched passenger seat wire harness

NOTE: The smart junction box (SJB) may also be identified as the generic electronic module (GEM).

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.

5. If the scan tool does not communicate with the VCM:
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM.
   - refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
   - verify the ignition key is in the ON position.
   - verify the scan tool operation with a known good vehicle.
   - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
   - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
   - If the network test passes, retrieve and record on-demand and continuous memory DTCs from the RCM and OCSM.
8. If the DTCs retrieved are related to the concern, go to Supplemental Restraint System (SRS) DTC Chart.

For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.

9. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

The DTCs can be retrieved from the restraints control module (RCM) and the occupant classification system module (OCSM) with a scan tool via the data link connector (DLC).

RESTRAINTS CONTROL MODULE (RCM) DTC CHART

DTC[a]	Description	Retrieved From	Action To Take
Continuous lamp	The Air Bag Warning Indicator is Illuminated Continuously	Restraints control module (RCM)	Go to **Pinpoint Test A** .
B1231	Event Threshold Exceeded	RCM	INSTALL a new RCM and impact sensors. REFER to **Inspection and Repair After a Supplemental Restraint System (SRS) Deployment**.
B1317	Battery Voltage High	RCM	CHECK battery voltage; to be below 18 volts. REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article.
B1318	Battery Voltage Low	RCM	CHECK battery voltage; to be above 8 volts. REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article.
B1342	ECU (RCM) Is Faulted	RCM	INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**.
B1868	Lamp Air Bag Warning Indicator Circuit Failure	RCM	Go to **Pinpoint Test C** .
B1884	PAD Warning Lamp Circuit Failure	RCM	Go to **Pinpoint Test D** .
B1890	PAD Warning Lamp Circuit Short to Battery	RCM	Go to **Pinpoint Test E** .
B2290	Occupant Classification System Fault	RCM	**NOTE:** **If no on-demand or continuous DTC is present in the OCSM and DTC B2290 is only present in the RCM as a continuous DTC, CLEAR all RCM DTCs and PROVE OUT the system.** **NOTE:** **On-Demand DTC B2290 is reported by the RCM indicating the presence of a fault in the OCS system.**

			Go to **Pinpoint Test F** .
B2292	Restraint System - Seatbelt Pretensioner Fault	RCM	Go to **Pinpoint Test G** .
B2293	Restraint System - Air Bag Fault	RCM	Go to **Pinpoint Test H** .
B2294	Restraint System - Curtain Fault	RCM	Go to **Pinpoint Test I** .
B2295	Restraint System - Side Air Bag Fault	RCM	Go to **Pinpoint Test J** .
B2296	Restraint System - Impact Sensor Fault	RCM	Go to **Pinpoint Test K** .
B2432	Drivers Seat Belt Buckle Switch Circuit Open	RCM	Go to **Pinpoint Test L** .
B2433	Drivers Seat Belt Buckle Switch Circuit Short to Battery	RCM	Go to **Pinpoint Test M** .
B2434	Drivers Seat Belt Buckle Switch Circuit Short to Ground	RCM	Go to **Pinpoint Test N** .
B2435	Drivers Seat Belt Buckle Switch Resistance Out of Range	RCM	Go to **Pinpoint Test O** .
B2436	Passengers Seat Belt Buckle Switch Circuit Open	RCM	Go to **Pinpoint Test P** .
B2437	Passengers Seat Belt Buckle Switch Circuit Short to Battery	RCM	Go to **Pinpoint Test Q** .
B2438	Passengers Seat Belt Buckle Switch Short to Ground	RCM	Go to **Pinpoint Test R** .
B2439	Passengers Seat Belt Buckle Switch Resistance Out of Range	RCM	Go to **Pinpoint Test S** .
B2477	Module Configuration Failure	RCM	**NOTE:** **When installing a new RCM, always make sure the correct RCM is being installed. If an incorrect RCM is installed, erroneous DTCs will result.** INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**.
B2792	Cross Link Between Firing Loops	RCM	Go to **Pinpoint Test T** .
	Incorrect Module Design		INSTALL a new RCM. REFER to

C1414	Level	RCM	**Restraints Control Module (RCM)**.
C1946	Front Driver's Seat Track Position Switch Circuit Open	RCM	Go to **Pinpoint Test V** .
C1947	Front Driver's Seat Track Position Switch Circuit Short to Ground	RCM	Go to **Pinpoint Test W** .
C1948	Front Driver's Seat Track Position Switch Circuit Resistance Out of Range	RCM	Go to **Pinpoint Test X** .
C1982	Front Driver's Seat Track Position Switch Circuit Short to Battery	RCM	Go to **Pinpoint Test Y** .
U0100	Lost Communication With ECM/PCM	RCM	REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the communication concern.
U0154	Lost Communication With Restraints Occupant Classification System Module	RCM	REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the communication concern.
U0401	Invalid Data received From ECM/PCM A	RCM	RETRIEVE PCM DTCs. If DTCs are present in the PCM, REFER to **Introduction - Gasoline Engines** article to diagnose the concern. If DTCs are not present in the PCM and DTC U0401 is present in the RCM as a on-demand DTC, INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**.
U0455	Invalid Data Received From Restraints Occupant Classification System Module	RCM	REPAIR all on-demand OCS DTCs, CLEAR all RCM DTCs and PROVE OUT the system. If no on-demand or continuous DTCs are present in the OCSM and DTC U0455 is only present in the RCM continuous, CLEAR all RCM DTCs and PROVE OUT the system. If no on-demand or continuous DTCs are present in the OCSM and DTC U0455 is present on-demand in the RCM, INSTALL a new OCSM. REFER to **Occupant Classification System Module (OCSM) - Manual Seat Track** or **Occupant Classification System Module**

			(OCSM) - Power Seat Track.
U2050	No Application Present	RCM	INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**.
U2051	One or More Calibration Files Missing/Corrupt	RCM	INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**.

[a] DTC: Retrieved using scan tool.

OCCUPANT CLASSIFICATION SENSOR (OCS) SYSTEM DTC CHART

DTC[a]	Description	Retrieved From	Action To Take
B229B	Occupant Classification System Obstruction	Occupant classification system module (OCSM)	**NOTE:** **If no on-demand or continuous DTC is present in the RCM and DTC B229B is only present in the OCSM as continuous, RESOLVE and CLEAR all OCSM DTCs and PROVE OUT the SRS.** Go to **Pinpoint Test U** .
B1013	Occupant Classification System Calibration Fault	OCSM	Go to **Pinpoint Test Z** .
B1231	Event Threshold Exceeded	OCSM	INSTALL a new passenger seat track assembly with OCS weight sensor bolts. REFER to **SEATING** article and **Inspection and Repair After a Supplemental Restraint System (SRS) Deployment**.
B1317	Battery Voltage High	OCSM	**NOTE:** **If no on-demand or continuous DTC is present in the RCM and DTC B1317 is only present in the OCSM continuous, CLEAR all OCSM DTCs and PROVE OUT the system.** CHECK battery voltage; to be below 18 volts. REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article.
B1318	Battery Voltage Low	OCSM	**NOTE:** **If no on-demand or continuous DTC is present in the RCM and DTC B1318 is only present in the OCSM continuous, CLEAR all OCSM DTCs and PROVE OUT the system.** CHECK battery voltage; to be above 8 volts. REFER to **CHARGING**

			SYSTEM - GENERAL INFORMATION article.
B1342	ECU (OCSM) is Faulted	OCSM	Go to **Pinpoint Test B** .
B2290	Occupant Classification System Fault	OCSM	Go to **Pinpoint Test F** .
C1941	Zero Seat Weight Test Failure	OCSM	REFER to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.
U0100	Lost Communication With ECM/PCM	OCSM	REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the communication concern.
U0151	Lost Communication With Restraints Control Module	OCSM	REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the communication concern.
U2023	Fault Received From External Node	OCSM	DTC is set when a module receives invalid network data from another module with a faulted input. RETRIEVE and REPAIR all non-network DTCs in the other modules on the network. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article for a list of all DTCs.
U2050	No Application Present	OCSM	INSTALL a new OCSM.

[a] DTC: Retrieved using scan tool.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
• Air bag warning indicator is illuminated continuously	• Fuse • DTC • Wiring, terminals or connectors • Faulty data link connector (DLC) • Restraint control module (RCM)	• Go to **Pinpoint Test A** .
• Air bag warning indicator	• RCM not	• CARRY OUT programmable module installation (PMI). REFER to **MODULE CONFIGURATION** article. If the air bag warning indicator

flashing at 5 Hz rate	configured	is flashing or has a continuous light after carrying out PMI, REFER to **DTC Charts** to diagnose the concern.
• Audible tone - DTCs retrieved	• SRS fault and air bag warning indicator fault	• REFER to **DTC Charts** .
• No communication with the restraints control module (RCM)	• Fuse • Circuitry • Scan tool • RCM	• REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the no communication concern.
• No communication with the occupant classification system module (OCSM)	• Fuse • Circuitry • Scan tool • OCSM	• REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the no communication concern.

Pinpoint Test - Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)

Pinpoint Test A: The Air Bag Warning Indicator is Illuminated Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

During normal operation, the instrument cluster (IC) module will illuminate the air bag indicator continuously for 6 seconds after the ignition switch is placed to the RUN or ON position. If the supplemental restraint system (SRS) is fault free, the air bag warning indicator will turn off and remain off. If a fault is detected in the SRS, the air bag warning indicator will illuminate and remain illuminated for the rest of the key cycle. The restraints control module (RCM) will communicate to the IC module via the high-speed controller area network (HS-CAN). The IC module will illuminate the air bag indicator based on messaging from the RCM or if there is no communication between the RCM and IC module.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- IC module
- RCM

PINPOINT TEST A: THE AIR BAG WARNING INDICATOR IS ILLUMINATED CONTINUOUSLY

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in**

the accidental deployment of the safety canopy and cause serious personal injury or death.

WARNING: **Always tighten the fasteners of the restraints control module (RCM) and impact sensor (if equipped) to the specified torque. Failure to do so may result in incorrect restraint system operation, which increases the risk of personal injury or death in a crash.**

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: SRS components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

A1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Were any continuous memory or on-demand DTCs retrieved?**

  YES : If on-demand or continuous memory DTCs were retrieved, go to the **DTC Charts** for diagnostic direction.

  Do not clear any DTCs until all DTCs have been resolved.

  NO : If the RCM **does** communicate with the scan tool, REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES** article to diagnose the IC module concern.

  If the RCM **does not** communicate with the scan tool, REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the no communication concern.

  Do not clear any DTCs until all DTCs have been resolved.

Pinpoint Test B: DTC B1342 - ECU (OCSM) is Faulted

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The occupant classification sensor (OCS) system is used to classify the front passenger seat occupant in the event of a deployable impact. Refer to **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.

The occupant classification system module (OCSM) monitors internal circuity, hardware and the 4 OCS weight sensor bolts and external circuitry for faults. If the OCSM detects a fault with the internal circuitry, internal hardware, sensor(s) or external circuitry, it will store DTC B1342 in memory and send a message to the restraints control module (RCM). A short to ground on any one of the sensor reference voltage circuits or an open on 3 or more of the sensor reference voltage circuits may set this DTC. The RCM will store DTC B2290 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

Fault PIDs[a]	Description	Fault Trigger Condition
1342_27_OD and 1342_27_CM	Memory Fault	When the OCSM senses a internal hardware failure, a fault will be indicated.
1342_28_OD and 1342_28_CM	Microcontroller Internal Fault	When the OCSM senses a internal failure, a fault will be indicated.
1342_29_OD and 1342_29_CM	Satellite ASIC Fault	When the OCSM senses a internal hardware failure, a fault will be indicated.
1342_30_OD and 1342_30_CM	Power Supply Internal Fault	When the OCSM senses a failure with the internal power supply, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B1342. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B1342.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Incorrectly wired OCS weight sensors
- OCS system component

PINPOINT TEST B: DTC B1342 - ECU IS FAULTED (OCS)

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of**

serious personal injury or death.

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

B1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - OCSM - View and Record All B1342 Fault PIDs
- **Do any OCSM on-demand B1342 fault PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to B2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to B8.

B2 CHECK THE SEAT WIRING AND CONNECTORS

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Carry out a thorough visual inspection of the OCS system wiring, terminals and connectors and the related seat wiring harness and body wiring harness terminals and connectors.
- **Were any wire problems noted?**

  YES : REPAIR the seat connectors and wiring as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to B9.

NO : Using the fault PIDs recorded in step B1, go to the appropriate pinpoint test step.

For 1342_30_OD (Power Supply Internal Fault), go to B3.

For 1342_28_OD (Microcontroller Internal Fault), 1342_29_OD (Satellite ASIC Fault) or 1342_27_OD (Memory Fault), INSTALL a new OCSM. REFER to **Occupant Classification System Module (OCSM) - Manual Seat Track** or **Occupant Classification System Module (OCSM) - Power Seat Track**. Go to B9.

B3 CHECK THE OCS WEIGHT SENSOR BOLT CIRCUITS FOR A SHORT TO GROUND

- Disconnect: OCS Weight Sensor Bolt C3324, C3325, C3330 and C3331
- Disconnect: OCSM C3159

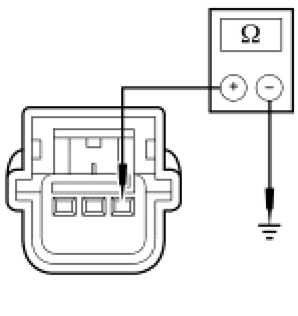


Fig. 2: Measuring Resistance Between OCS Weight Sensor Bolt 1 C3324-1 & Bolt 2 C3325-1, CR140 (YE/GN) & CR141 (BN/YE)
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 1 C3324-1, circuit CR140 (YE/GN), harness side and ground; between OCS weight sensor bolt 2 C3325-1, circuit CR141 (BN/YE), harness side and ground; between OCS weight sensor bolt 3 C3330-1, circuit CR153 (GN/WH), harness side and ground; and between OCS weight sensor bolt 4 C3331-1, circuit CR154 (VT/OG), harness side and ground.
- **Are the resistances greater than 10,000 ohms?**

  YES : Go to B4.

  NO : REPAIR circuit CR140 (YE/GN), circuit CR141 (BN/YE), circuit CR153 (GN/WH) or circuit CR154 (VT/OG). Go to B9.

B4 CHECK FOR A SHORT BETWEEN THE OCS WEIGHT SENSOR BOLT CIRCUITS

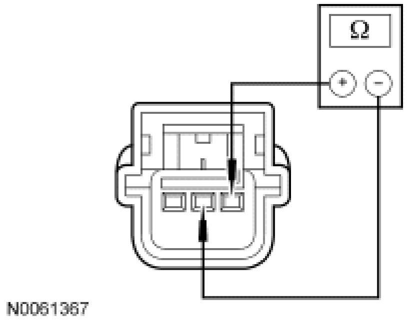


Fig. 3: Measuring Resistance Between OCS Weight Sensor Bolt 1 C3324-1 & Bolt 2 C3325-1, CR140 (YE/GN) & CR141 (BN/YE)
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor:
  - bolt 1 C3324-1, circuit CR140 (YE/GN), harness side and C3324-2, circuit RR140 (GN/OG), harness side; and
  - bolt 2 C3325-1, circuit CR141 (BN/YE), harness side and C3325-2, circuit RR141 (VT/BN), harness side; and
  - bolt 3 C3330-1, circuit CR153 (GN/WH), harness side and C3330-2, circuit RR153 (GY/BU), harness side; and
  - bolt 4 C3331-1, circuit CR154 (VT/OG), harness side and C3331-2, circuit RR154 (VT/WH), harness side.
- **Are the resistances greater than 10,000 ohms?**

  YES : Go to B5.

  NO : REPAIR circuits CR140 (YE/GN) and RR140 (GN/OG), CR141 (BN/YE) and RR141 (VT/BN), CR153 (GN/WH) and RR153 (GY/BU) or CR154 (VT/OG) and RR154 (VT/WH). Go to B9.

B5 CHECK THE OCS WEIGHT SENSOR BOLT CIRCUITS FOR AN OPEN

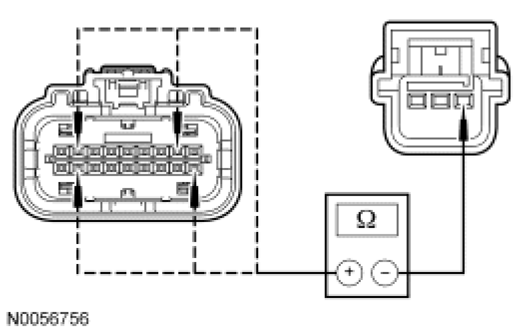


Fig. 4: Measuring Resistance Between OCS Weight Sensor Bolt 1 C3324-1 & Bolt 2 C3325-1, CR140 (YE/GN) & CR141 (BN/YE)
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor:

  - bolt 1 C3324-1, circuit CR140 (YE/GN), harness side and OCSM C3159-2, circuit CR140 (YE/GN), harness side; and
  - bolt 2 C3325-1, circuit CR141 (BN/YE), harness side and OCSM C3159-10, circuit CR141 (BN/YE), harness side; and
  - bolt 3 C3330-1, circuit CR153 (GN/WH), harness side and OCSM C3159-8, circuit CR153 (GN/WH), harness side; and
  - bolt 4 C3331-1, circuit CR154 (VT/OG), harness side and OCSM C3159-17, circuit CR154 (VT/OG), harness side.
- **Are the resistances less than 5 ohms?**

  YES : Go to B6.

  NO : REPAIR circuit CR140 (YE/GN), circuit CR141 (BN/YE), circuit CR153 (GN/WH) or circuit CR154 (VT/OG). Go to B9.

B6 CHECK CIRCUITS CR140 (YE/GN), CR141 (BN/YE), CR153 (GN/WH) AND CR154 (VT/OG) FOR A SHORT TO VOLTAGE

- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

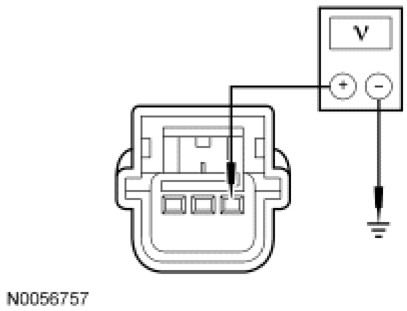


Fig. 5: Measuring Voltage Between OCS Weight Sensor Bolt 1 C3324-1 & Bolt 2 C3325-1, CR140 (YE/GN) & CR141 (BN/YE)
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor:
  - bolt 1 C3324-1, circuit CR140 (YE/GN), harness side and ground; and
  - bolt 2 C3325-1, circuit CR141 (BN/YE), harness side and ground; and
  - bolt 3 C3330-1, circuit CR153 (GN/WH), harness side and ground; and
  - bolt 4 C3331-1, circuit CR154 (VT/OG), harness side and ground.
- **Are the voltages less than 0.2 volt?**

  YES : Go to B7.

  NO : REPAIR circuit CR140 (YE/GN), circuit CR141 (BN/YE), circuit CR153 (GN/WH) or circuit CR154 (VT/OG). Go to B9.

B7 CONFIRM THE OCS MODULE FAULT

NOTE: **Make sure all restraint system sensor electrical connectors and the RCM electrical connector are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: OCSM C3159
- Connect: OCS Weight Sensor Bolt Electrical Connectors
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - OCSM - View and Record All B1342 Fault PIDs
  - Refer to PID list in Normal Operation to view B1342 fault PIDs.
- **Do any OCSM on-demand B1342 fault PIDs indicate a fault?**

  YES : INSTALL a new OCSM. REFER to **Occupant Classification System Module (OCSM) - Manual Seat Track** or **Occupant Classification System Module (OCSM) - Power Seat Track**. Go to B9.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to B8.

B8 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Carry out a thorough visual inspection of the OCS system wiring, terminals and connectors and the related seat wiring harness and body wiring harness terminals and connectors.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - OCSM - View and Record All B1342 Fault PIDs
  - Refer to PID list in Normal Operation to view B1342 fault PIDs.
- **Does the original OCSM on-demand DTC B1342 fault PID indicate a fault?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to B2.

  NO : CHECK for causes of intermittent open or short to ground on circuit CR140 (YE/GN), circuit CR141 (BN/YE), circuit CR153 (GN/WH) or circuit CR154 (VT/OG). ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. **Do not install any new SRS components at this time. SRS**

components should only be installed when directed to do so in the pinpoint test. REPAIR any intermittent wiring, terminal or connector concerns found. Go to B9.

B9 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

> WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test C: DTC B1868 - Lamp Air Bag Warning Indicator Circuit Failure

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

During normal operation, the air bag warning indicator will illuminate continuously for approximately 6 seconds and then go out after the ignition switch is placed in the ON position and no air bag fault exists. Control of the air bag warning indicator is a function of the instrument cluster (IC) module based on information it receives from the restraints control module (RCM) over the high-speed controller area network (HS-CAN). The default mode for the air bag warning indicator is on. The RCM sends a message to the IC module to turn the indicator off if no air bag fault exists. If the RCM detects a problem with the supplemental restraints system (SRS) it will send a message to the IC module to turn the air bag warning indicator on.

If the IC module detects a problem with the air bag warning indicator, it will send a fault message to the RCM and the RCM will store DTC B1868 in memory.

- DTC B1868 Lamp Air Bag Warning Indicator Circuit Failure - If the RCM receives a message from the IC module of an air bag warning indicator failure, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- IC module

- RCM

PINPOINT TEST C: DTC B1868 - LAMP AIR BAG WARNING INDICATOR CIRCUIT FAILURE

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **SRS components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

C1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was on-demand DTC B1868 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to C2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. CHECK the IC module for

DTCs and REPAIR as needed. REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES** article. Go to C4.

C2 CHECK FOR IC MODULE ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - IC Module - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Were any continuous memory or on-demand self test DTCs retrieved?**

  YES : REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES** article to diagnosis IC module concerns. Go to C4.

  NO : If the IC module **does** communicate with the scan tool, go to C3.

  If the IC module **does not** communicate with the scan tool, REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the HS-CAN concern. Go to C4.

C3 CHECK THE IC MODULE

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - IC Module - Active Command - All_Lamp
- **Does the air bag warning indicator turn on and off when using the active command?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to C4.

  NO : INSTALL a new IC module. REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES** article. Go to C4.

C4 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

  WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test D: DTC B1884 - PAD Warning Lamp Circuit Failure

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

When the ignition is in the ON position, the passenger air bag deactivation (PAD) indicator prove-out period is initiated by the restraints control module (RCM). The RCM briefly activates the PAD indicator to verify to the occupants correct functional operation of the PAD indicator. Refer to **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.

If the RCM detects an open or short to ground on the PAD indicator circuit, it will store DTC B1884 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B1884 PAD Warning Lamp Circuit Failure - If the RCM detects an open circuit or short to ground on the PAD indicator circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- PAD indicator
- RCM

PINPOINT TEST D: DTC B1884 - PAD WARNING LAMP CIRCUIT FAILURE

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough**

inspection and verification before proceeding with the pinpoint test.

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

D1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was on-demand DTC B1884 retrieved?**

  YES : If the PAD indicator **does** illuminate, go to D2.

  If the PAD indicator **does not** illuminate, go to D4.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to D8.

D2 CHECK CIRCUIT CR116 (GN/WH) FOR A SHORT TO GROUND

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: PAD Indicator C2355

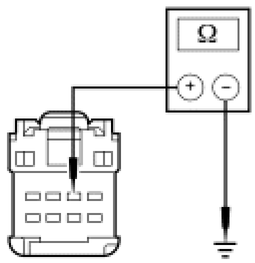


Fig. 6: Measuring Resistance Between PAD Indicator C2355-2, Circuit CR116 (GN/WH) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between PAD indicator C2355-2, circuit CR116 (GN/WH), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to D7.

  NO : Go to D3.

D3 CHECK THE RCM FOR LOW RESISTANCE

- Disconnect: RCM C310a and C310b

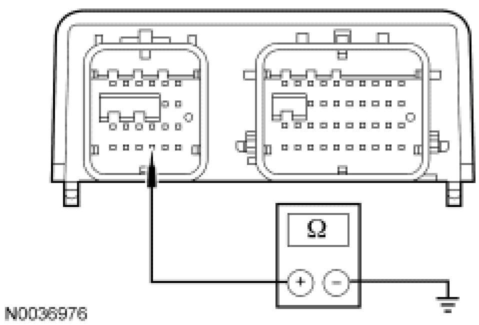


Fig. 7: Checking RCM For Low Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a pin 19, component side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuit CR116 (GN/WH). Go to D9.

  NO : Go to D7.

D4 CHECK CIRCUIT CR116 (GN/WH) FOR AN OPEN

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b
- Disconnect: PAD Indicator C2355

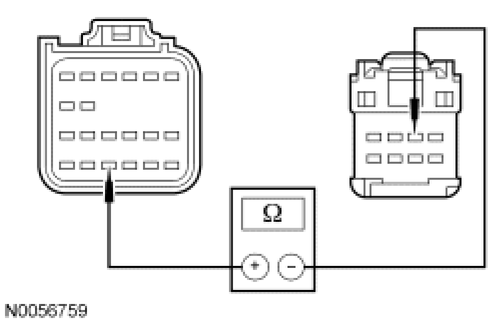


Fig. 8: Measuring Resistance Between PAD Indicator C2355-2, Circuit CR116 (GN/WH) & RCM C310A-19
Courtesy of FORD MOTOR CO.

- Measure the resistance between PAD indicator C2355-2, circuit CR116 (GN/WH), harness side and RCM C310a-19, circuit CR116 (GN/WH), harness side.
- **Is the resistance less than 0.5 ohm?**

  YES : Go to D5.

NO : REPAIR circuit CR116 (GN/WH). Go to D9.

D5 CHECK CIRCUIT CBP46 (WH/BU) FOR AN OPEN

- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

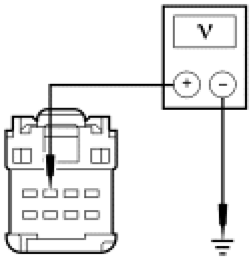


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Fig. 9: Measuring Voltage Between PAD Indicator C2355-3, Circuit CBP46 (WH/BU) & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between PAD indicator C2355-3, circuit CBP46 (WH/BU), harness side and ground.
- **Is the voltage greater than 10 volts?**

  YES : Go to D6.

  NO : REPAIR circuit CBP46 (WH/BU). Go to D9.

D6 CHECK THE PAD INDICATOR LAMP

- Key in OFF position.
- Deactivate the SRS. Refer to **Supplemental Restraint System (SRS) Deactivation and Reactivation**.
- Connect: PAD Indicator C2355
- Key in ON position.

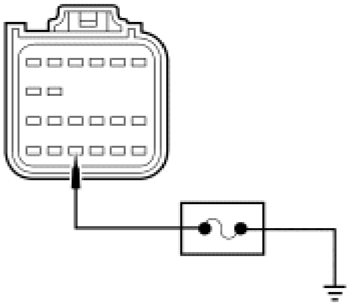

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Fig. 10: Checking PAD Indicator Lamp
Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire between RCM C310a-19, circuit CR116 (GN/WH), harness side and ground.
- **Does the PAD indicator illuminate?**

  YES : Go to D7.

  NO : INSTALL a new PAD indicator. REFER to **Passenger Air Bag Deactivation (PAD) Indicator**. Go to D9.

D7 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system sensor electrical connectors and the RCM electrical connector are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: PAD Indicator C2355 (if previously disconnected)
- Connect: RCM C310a and C310b (if previously disconnected)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was on-demand DTC B1884 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to D9.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to D8.

D8 CHECK FOR AN INTERMITTENT FAULT

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was on-demand DTC B1884 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to D2.

  NO : CHECK for causes of intermittent open or short to ground on circuit CR116 (GN/WH). ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to D9.

D9 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

> WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
    - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
    - If previously directed to deactivate the SRS, reactivate the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Deactivation and Reactivation**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test E: DTC B1890 - PAD Warning Lamp Circuit Short to Battery

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

When the ignition is in the ON position, the passenger air bag deactivation (PAD) indicator prove-out period is initiated by the restraints control module (RCM). The RCM briefly activates the PAD indicator to verify to the occupants correct functional operation of the PAD indicator. Refer to **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.

If the RCM detects a short to battery on the PAD indicator circuit, it will store DTC B1890 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B1890 PAD Warning Lamp Circuit Short to Battery - If the RCM detects a short to battery on the PAD indicator circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- PAD indicator
- RCM

PINPOINT TEST E: DTC B1890 - PAD WARNING LAMP CIRCUIT SHORT TO BATTERY

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

E1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was on-demand DTC B1890 retrieved?**

  YES : Go to E2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to E4.

E2 CHECK CIRCUIT CR116 (GN/WH) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and**

Repowering.

- Disconnect: PAD Indicator C2355
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

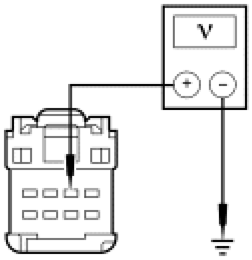

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Fig. 11: Measuring Voltage Between PAD Indicator C2355-2, Circuit CR116 (GN/WH) & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between PAD indicator C2355-2, circuit CR116 (GN/WH), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to E3.

  NO : REPAIR circuit CR116 (GN/WH). Go to E5.

E3 CHECK THE RCM

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs

NOTE:	**DTC B1884 should be retrieved when carrying out the on-demand self**

test due to an open on circuit CR116 (GN/WH). DTC B1890 should not be retrieved at this time.

- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the PAD indicator disconnected, a circuit failure fault would normally be retrieved.
- **Did the PAD warning lamp on-demand DTC change from indicating B1890 to B1884?**

  YES : INSTALL a new PAD indicator. Go to E5.

  NO : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to E5.

E4 CHECK FOR AN INTERMITTENT FAULT

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was on-demand DTC B1890 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to E2.

  NO : CHECK for causes of intermittent short to battery on circuit CR116 (GN/WH). ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to E5.

E5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test F: DTC B2290 - Occupant Classification System Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

NOTE: DTC B2290 reported by the restraint control module (RCM) is informational only and indicates the presence of a fault in the occupant classification system module (OCSM). Refer to DTC Charts when DTC B2290 is reported by the RCM for diagnostic instructions.

NOTE: The following pinpoint test must only be used to diagnose DTC B2290 when reported by the OCSM.

Normal Operation

The occupant classification sensor (OCS) system is used to classify the front passenger seat occupant in the event of a deployable impact. Refer to **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.

The OCSM monitors the 4 OCS weight sensor bolts and circuity for faults. If the OCSM detects a fault with the sensor(s) or circuitry, it will store DTC B2290 in memory and send a message to the RCM. The RCM will store DTC B2290 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

Fault PIDs[a]	Description	Fault Trigger Condition
2290_0_OD and 2290_0_CM	Passenger Seat OCS Sensor No. 4 Circuit Short to Battery	When the OCSM senses a short to voltage on the OCS weight sensor bolt No. 4 circuits, a fault will be indicated.
2290_1_OD and 2290_1_CM	Passenger Seat OCS Sensor No. 4 Circuit Short to Ground	When the OCSM senses a short to ground on the OCS weight sensor bolt No. 4 circuits, a fault will be indicated.
2290_2_OD and 2290_2_CM	Passenger Seat OCS Sensor No. 4 Comm. Fault/Open	When the OCSM senses a communication fault or open circuits on the OCS weight sensor bolt No. 4, a fault will be indicated.
2290_3_OD and 2290_3_CM	Passenger Seat OCS Sensor No. 4 Internal Fault	When the OCSM senses a internal fault with the OCS weight sensor bolt No. 4, a fault will be indicated.
2290_4_OD and 2290_4_CM	Passenger Seat OCS Sensor No. 3 Circuit Short to Battery	When the OCSM senses a short to voltage on the OCS weight sensor bolt No. 3 circuits, a fault will be indicated.
2290_5_OD and 2290_5_CM	Passenger Seat OCS Sensor No. 3 Circuit Short to Ground	When the OCSM senses a short to ground on the OCS weight sensor bolt No. 3 circuits, a fault will be indicated.
	Passenger Seat OCS Sensor	When the OCSM senses a communication fault or open circuits

2290_6_OD and 2290_6_CM	No. 3 Comm. Fault/Open	on the OCS weight sensor bolt No. 3, a fault will be indicated.
2290_7_OD and 2290_7_CM	Passenger Seat OCS Sensor No. 3 Internal Fault	When the OCSM senses a internal fault with the OCS weight sensor bolt No. 3, a fault will be indicated.
2290_8_OD and 2290_8_CM	Passenger Seat OCS Sensor No.2 Circuit Short to Battery	When the OCSM senses a short to voltage on the OCS weight sensor bolt No. 2 circuits, a fault will be indicated.
2290_9_OD and 2290_9_CM	Passenger Seat OCS Sensor No.2 Circuit Short to Ground	When the OCSM senses a short to ground on the OCS weight sensor bolt No. 2 circuits, a fault will be indicated.
2290_10_OD and 2290_10_CM	Passenger Seat OCS Sensor No.2 Comm. Fault/Open	When the OCSM senses a communication fault or open circuits on the OCS weight sensor bolt No. 2, a fault will be indicated.
2290_11_OD and 2290_11_CM	Passenger Seat OCS Sensor No.2 Internal Fault	When the OCSM senses a internal fault with the OCS weight sensor bolt No. 2, a fault will be indicated.
2290_12_OD and 2290_12_CM	Passenger Seat OCS Sensor No.1 Circuit Short to Battery	When the OCSM senses a short to voltage on the OCS weight sensor bolt No. 1 circuits, a fault will be indicated.
2290_13_OD and 2290_13_CM	Passenger Seat OCS Sensor No.1 Circuit Short to Ground	When the OCSM senses a short to ground on the OCS weight sensor bolt No. 1 circuits, a fault will be indicated.
2290_14_OD and 2290_14_CM	Passenger Seat OCS Sensor No.1 Comm. Fault/Open	When the OCSM senses a communication fault or open circuits on the OCS weight sensor bolt No. 1 circuits, a fault will be indicated.
2290_15_OD and 2290_15_CM	Passenger Seat OCS Sensor No.1 Internal Fault	When the OCSM senses a internal fault with the OCS weight sensor bolt No. 1 circuits, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B2290. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B2290.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- OCS system component

PINPOINT TEST F: DTC B2290 - OCCUPANT CLASSIFICATION SYSTEM FAULT

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

F1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - OCSM - View and Record All B2290 Fault PIDs
  - Refer to PID list in Normal Operation to view B2290 fault PIDs.
- **Do any OCSM on-demand DTC B2290 fault PIDs indicate a fault?**

YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to F2.

NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to F36.

F2 CHECK THE SEAT WIRING AND CONNECTORS

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Carry out a thorough visual inspection of the OCS system wiring, terminals and connectors and the related seat wiring harness and body wiring harness terminals and connectors.
- **Were any problems noted?**

YES : REPAIR the seat connectors and wiring as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to F37.

NO : Using the fault PIDs recorded in step F1, go to the appropriate pinpoint test step.

For 2290_3_OD (Passenger Seat OCS Sensor No. 4 Internal Fault), 2290_7_OD (Passenger Seat OCS Sensor No. 3 Internal Fault), 2290_11_OD (Passenger Seat OCS Sensor No.2 Internal Fault) or 2290_15_OD (Passenger Seat OCS Sensor No.1 Internal Fault), INSTALL a new OCS weight sensor bolt. REFER to **Occupant Classification Sensor - Manual Seat Track** or **Occupant Classification Sensor - Power Seat Track**.

For 2290_14_OD (Passenger Seat OCS Sensor No.1 Comm. Fault/Open), go to F3.

For 2290_13_OD (Passenger Seat OCS Sensor No.1 Circuit Short to Ground), go to F7.

For 2290_12_OD (Passenger Seat OCS Sensor No.1 Circuit Short to Battery), go to F9.

For 2290_10_OD (Passenger Seat OCS Sensor No.2 Comm. Fault/Open), go to F11.

For 2290_9_OD (Passenger Seat OCS Sensor No.2 Circuit Short to Ground), go to F15.

For 2290_8_OD (Passenger Seat OCS Sensor No.2 Circuit Short to Battery), go to F17.

For 2290_6_OD (Passenger Seat OCS Sensor No. 3 Comm. Fault/Open), go to F19.

For 2290_5_OD (Passenger Seat OCS Sensor No. 3 Circuit Short to Ground), go to F23.

For 2290_4_OD (Passenger Seat OCS Sensor No. 3 Circuit Short to Battery), go to F25.

For 2290_2_OD (Passenger Seat OCS Sensor No. 4 Comm. Fault/Open), go to F27.

For 2290_1_OD (Passenger Seat OCS Sensor No. 4 Circuit Short to Ground), go to F31.

For 2290_0_OD (Passenger Seat OCS Sensor No. 4 Circuit Short to Battery), go to F33.

F3 CHECK CIRCUITS CR140 (YE/GN), VR211 (WH/BN) AND RR140 (GN/OG) FOR AN OPEN

- Disconnect: OCSM C3159
- Disconnect: OCS Weight Sensor Bolt 1 C3324

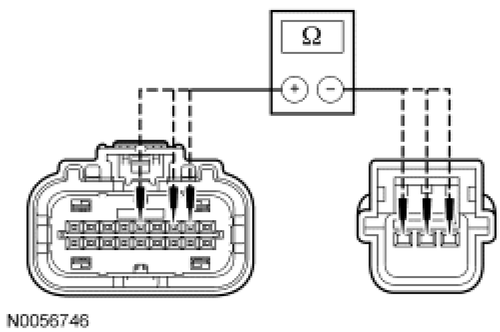

Fig. 12: Measuring Resistance Between OCSM C3159-5, Circuit VR211 (WH/BN) & OCS Weight Sensor Bolt 1 C3324-3
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCSM C3159-5, circuit VR211 (WH/BN), harness side and OCS weight sensor bolt 1 C3324-3, circuit VR211 (WH/BN), harness side; between OCSM C3159-2, circuit CR140 (YE/GN), harness side and OCS weight sensor bolt 1 C3324-1, circuit CR140 (YE/GN), harness side; and between OCSM C3159-5, circuit RR140 (GN/OG), harness side and OCS weight sensor bolt 1 C3324-2, circuit RR140 (GN/OG), harness side.
- **Are the resistances less than 0.5 ohm?**

  YES : Go to F4.

  NO : REPAIR circuit CR140 (YE/GN), circuit VR211 (WH/BN) or circuit RR140 (GN/OG). Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to F37.

F4 CHECK FOR A SHORT BETWEEN CIRCUITS CR140 (YE/GN), VR211 (WH/BN) AND RR140 (GN/OG)

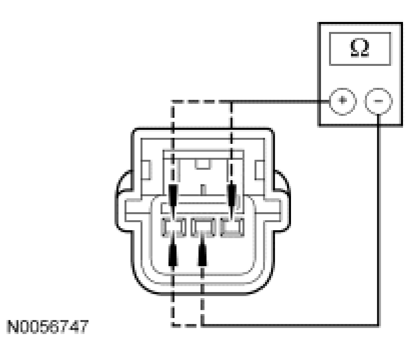

Fig. 13: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 1:
    - C3324-1, circuit CR140 (YE/GN), harness side and C3324-3, circuit VR211 (WH/BN), harness side.
    - C3324-1, circuit CR140 (YE/GN), harness side and C3324-2, circuit RR140 (GN/OG), harness side.
    - C3324-3, circuit VR211 (WH/BN), harness side and C3324-2, circuit RR140 (GN/OG), harness side.
- **Are the resistances greater than 10,000 ohms?**

  YES : Go to F5.

  NO : REPAIR the affected circuits, circuits CR140 (YE/GN), VR211 (WH/BN) and/or RR140 (GN/OG). Go to F37.

F5 CHECK CIRCUIT VR211 (WH/BN) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

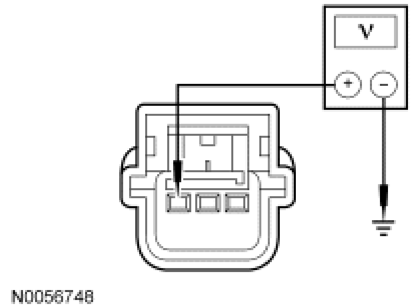

Fig. 14: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 1 C3324-3, circuit VR211 (WH/BN), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to F6.

  NO : REPAIR circuit VR211 (WH/BN). Go to F37.

F6 CHECK CIRCUIT VR211 (WH/BN) FOR A SHORT TO GROUND

- Key in OFF position.

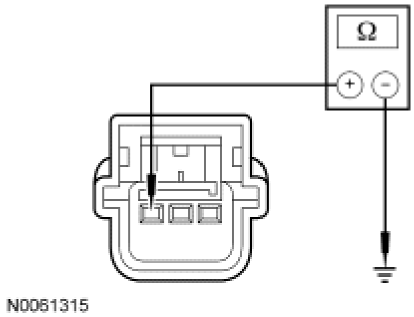

Fig. 15: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 1 C3324-3, circuit VR211 (WH/BN), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : INSTALL a new OCS weight sensor bolt 1. REFER to **Occupant Classification Sensor - Manual Seat Track** or **Occupant Classification Sensor - Power Seat Track**. Go to F37.

  NO : REPAIR circuit VR211 (WH/BN). Go to F37.

F7 CHECK CIRCUIT VR211 (WH/BN) FOR A SHORT TO GROUND AND A SHORT TO CIRCUIT RR140 (GN/OG)

- Disconnect: OCS Weight Sensor Bolt 1 C3324
- Disconnect: OCSM C3159

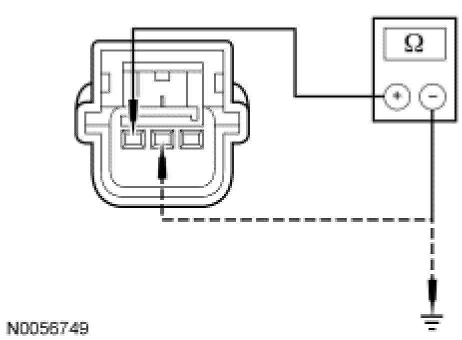

Fig. 16: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 1 C3324-3, circuit VR211 (WH/BN), harness side and ground; and between OCS weight sensor bolt 1 C3324-3, circuit VR211 (WH/BN), harness side and C3324-2, circuit RR140 (GN/OG), harness side.
- **Are the resistances greater than 10,000 ohms?**

  YES : Go to F8.

  NO : REPAIR circuit VR211 (WH/BN) and/or circuit RR140 (GN/OG). Go to F37.

F8 CHECK CIRCUIT CR140 (YE/GN) FOR AN OPEN

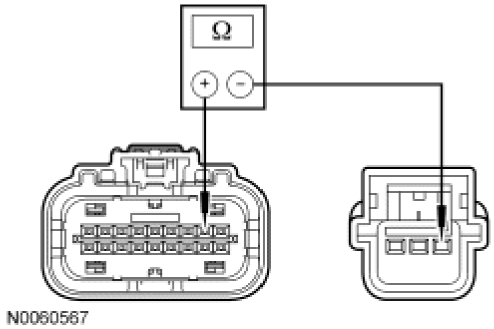

Fig. 17: Measuring Resistance Between OCS Weight Sensor Bolt 1 C3324-1, Circuit CR140 (YE/GN) & OCSM C3159-2
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 1 C3324-1, circuit CR140 (YE/GN), harness side and OCSM C3159-2, circuit CR140 (YE/GN), harness side.
- **Is the resistances less than 5 ohms?**

  YES : Go to F35.

  NO : REPAIR circuit CR140 (YE/GN). Go to F37.

F9 CHECK CIRCUIT RR140 (GN/OG) FOR AN OPEN

- Disconnect: OCS Weight Sensor Bolt 1 C3324

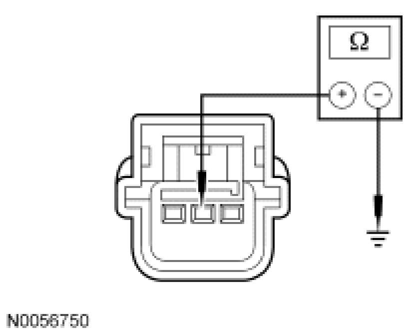

Fig. 18: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 1 C3324-2, circuit RR140 (GN/OG), harness side and ground.
- **Is the resistance less than 5 ohms?**

  YES : Go to F10.

  NO : REPAIR circuit RR140 (GN/OG). Go to F37.

F10 CHECK CIRCUIT VR211 (WH/BN) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Disconnect: OCSM C3159
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint**

System (SRS) Depowering and Repowering.

- Key in ON position.

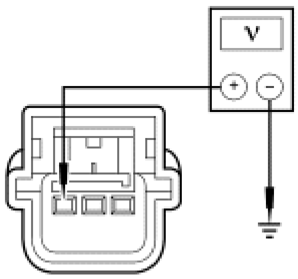

Fig. 19: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 1 C3324-3, circuit VR211 (WH/BN), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to F35.

  NO : REPAIR circuit VR211 (WH/BN). Go to F37.

F11 CHECK CIRCUITS CR141 (BN/YE), VR212 (YE/GN) AND RR141 (VT/BN) FOR AN OPEN

- Disconnect: OCSM C3159
- Disconnect: OCS Weight Sensor Bolt 2 C3325

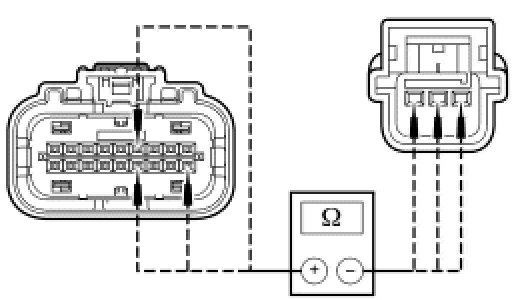

Fig. 20: Measuring Resistance Between OCSM C3159-4, Circuit VR212 (YE/GN) & OCS Sensor 2 C3325-3
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCSM C3159-4, circuit VR212 (YE/GN), harness side and OCS sensor 2 C3325-3, circuit VR212 (YE/GN), harness side; between OCSM C3159-10, circuit CR141 (BN/YE), harness side and OCS sensor 2 C3325-1, circuit CR141 (BN/YE), harness side; and between OCSM C3159-13, circuit RR141 (VT/BN), harness side and OCS sensor 2 C3325-2, circuit RR141 (VT/BN), harness side.
- **Are the resistances less than 0.5 ohm?**

YES : Go to F12.

NO : REPAIR circuit CR141 (BN/YE), circuit VR212 (YE/GN) or circuit RR141 (VT/BN). Go to F37.

F12 CHECK FOR A SHORT BETWEEN CIRCUITS CR141 (BN/YE), VR212 (YE/GN) AND RR141 (VT/BN)

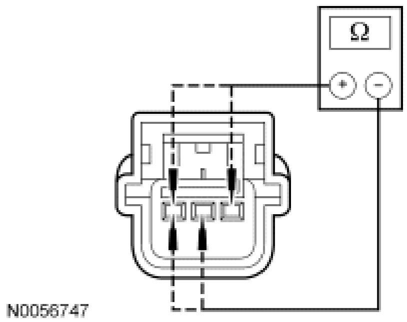

Fig. 21: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 2:
  - C3325-1, circuit CR141 (BN/YE), harness side and C3325-3, circuit VR212 (YE/GN), harness side.
  - C3325-1, circuit CR141 (BN/YE), harness side and C3325-2, circuit RR141 (VT/BN), harness side.
  - C3325-3, circuit VR212 (YE/GN), harness side and C3325-2, circuit RR141 (VT/BN), harness side.
- **Are the resistances greater than 10,0000 ohms?**

  YES : Go to F13.

  NO : REPAIR the affected circuits, circuits CR141 (BN/YE), VR212 (YE/GN) and/or RR141 (VT/BN). Go to F37.

F13 CHECK CIRCUIT VR212 (YE/GN) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

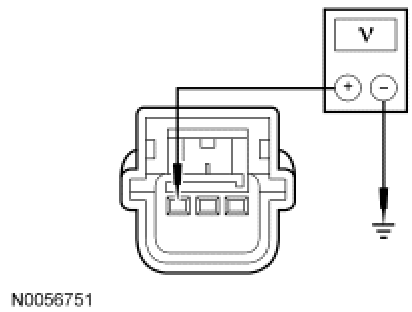

Fig. 22: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 2 C3325-3, circuit VR212 (YE/GN), harness side and ground.
- **Is the voltage less than 0.2 volt?**

YES : Go to F14.

NO : REPAIR circuit VR212 (YE/GN). Go to F37.

F14 CHECK CIRCUIT VR212 (YE/GN) FOR A SHORT TO GROUND

- Key in OFF position.

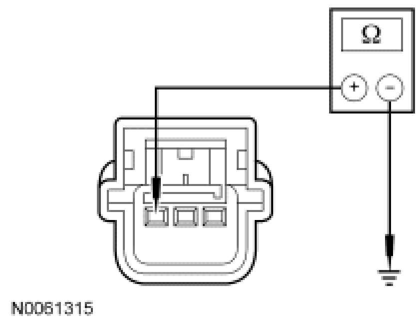

Fig. 23: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 2 C3325-3, circuit VR212 (YE/GN), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

YES : INSTALL a new OCS weight sensor bolt 2. REFER to **Occupant Classification Sensor - Manual Seat Track** or **Occupant Classification Sensor - Power Seat Track**. Go to F37.

NO : REPAIR circuit VR212 (YE/GN). Go to F37.

F15 CHECK CIRCUIT VR212 (YE/GN) FOR A SHORT TO GROUND AND A SHORT TO CIRCUIT RR141 (VT/BN)

- Disconnect: OCS Weight Sensor Bolt 2 C3325
- Disconnect: OCSM C3159

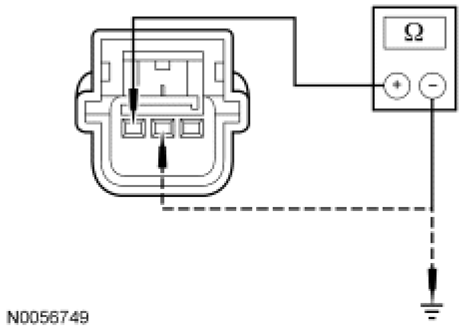

Fig. 24: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 2 C3325-3, circuit VR212 (YE/GN), harness side and ground; and between OCS weight sensor bolt 2 C3325-3, circuit VR212 (YE/GN), harness side and C3325-2, circuit RR141 (VT/BN), harness side.
- **Are the resistances greater than 10,000 ohms?**

  YES : Go to F16.

  NO : REPAIR circuits VR212 (YE/GN) and RR141 (VT/BN). Go to F37.

F16 CHECK CIRCUIT CR141 (BN/YE) FOR AN OPEN

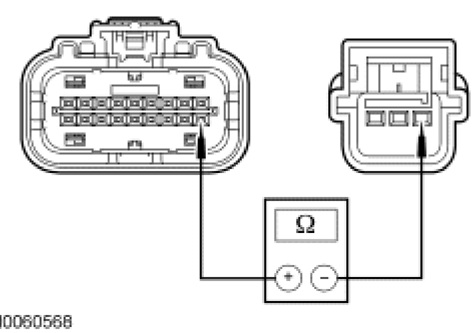

Fig. 25: Measuring Resistance Between OCS Weight Sensor Bolt 2 C3325-1, Circuit CR141 (BN/YE) & OCSM C3159-10
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 2 C3325-1, circuit CR141 (BN/YE), and OCSM C3159-10, circuit CR141 (BN/YE), harness side.
- **Is the resistances less than 5 ohms?**

  YES : Go to F35.

  NO : REPAIR circuit CR141 (BN/YE). Go to F37.

F17 CHECK CIRCUIT RR141 (VT/BN) FOR AN OPEN

- Disconnect: OCS Weight Sensor Bolt 2 C3325

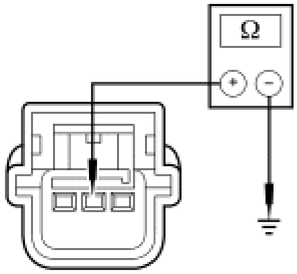

Fig. 26: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 2 C3325-2, circuit RR141 (VT/BN), harness side and ground.
- **Is the resistance less than 5 ohms?**

  YES : Go to F18.

  NO : REPAIR circuit RR141 (VT/BN). Go to F37.

F18 CHECK CIRCUIT VR212 (YE/GN) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Disconnect: OCSM C3159
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

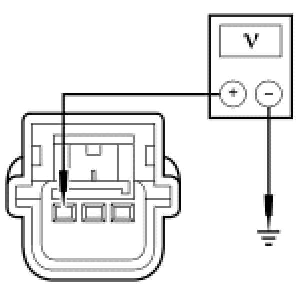

Fig. 27: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 2 C3325-3, circuit VR212 (YE/GN), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to F35.

  NO : REPAIR circuit VR212 (YE/GN). Go to F37.

F19 CHECK CIRCUITS CR153 (GN/WH), VR226 (YE/GY) AND RR153 (GY/BU) FOR AN

OPEN

- Disconnect: OCSM C3159
- Disconnect: OCS Weight Sensor Bolt 3 C3330

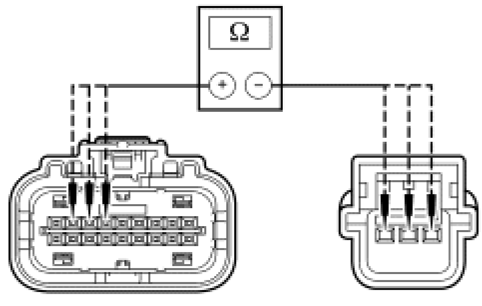

Fig. 28: Measuring Resistance Between OCSM C3159-7, Circuit VR226 (YE/GY) & OCS Weight Sensor Bolt 3 C3330-3
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCSM C3159-7, circuit VR226 (YE/GY), harness side and OCS weight sensor bolt 3 C3330-3, circuit VR226 (YE/GY), harness side; between OCSM C3159-8, circuit CR153 (GN/WH), harness side and OCS sensor 3 C3330-1, circuit CR153 (GN/WH), harness side; and between OCSM C3159-6, circuit RR153 (GY/BU), harness side and OCS sensor 3 C3330-2, circuit RR153 (GY/BU), harness side.
- **Are the resistances less than 0.5 ohm?**

  YES : Go to F20.

  NO : REPAIR circuits CR153 (GN/WH), VR226 (YE/GY) or RR153 (GY/BU). Go to F37.

F20 CHECK FOR A SHORT BETWEEN CIRCUITS CR153 (GN/WH), VR226 (YE/GY) AND RR153 (GY/BU)

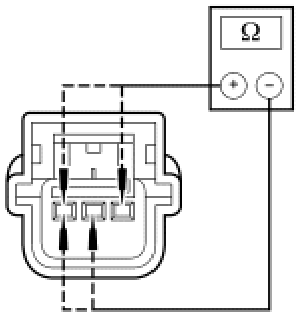

Fig. 29: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 3:
  - C3330-1, circuit CR153 (GN/WH), harness side and C3330-3, circuit VR226 (YE/GY), harness side.

- C3330-1, circuit CR153 (GN/WH), harness side and C3330-2, circuit RR153 (GY/BU), harness side.
- C3330-3, circuit VR226 (YE/GY), harness side and C3330-2, circuit RR153 (GY/BU), harness side.

- **Are the resistances greater than 10,0000 ohms?**

  YES : Go to F21.

  NO : REPAIR the affected circuits, circuits CR153 (GN/WH), VR226 (YE/GY) and/or RR153 (GY/BU). Go to F37.

F21 CHECK CIRCUIT VR226 (YE/GY) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

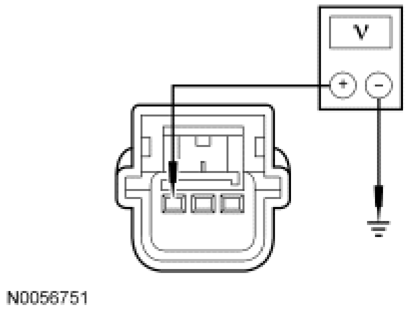

Fig. 30: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 3 C3330-3, circuit VR226 (YE/GY), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to F22.

  NO : REPAIR circuit VR226 (YE/GY). Go to F37.

F22 CHECK CIRCUIT VR226 (YE/GY) FOR A SHORT TO GROUND

- Key in OFF position.

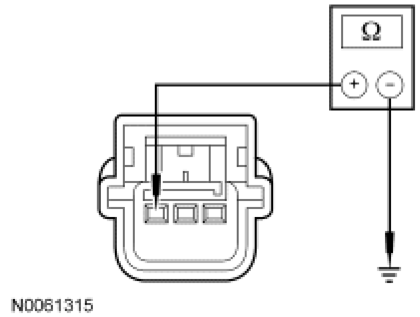

Fig. 31: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 3 C3330-3, circuit VR226 (YE/GY), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : INSTALL a new OCS weight sensor bolt 3. REFER to **Occupant Classification Sensor - Manual Seat Track** or **Occupant Classification Sensor - Power Seat Track**. Go to F37.

  NO : REPAIR circuit VR226 (YE/GY). Go to F37.

F23 CHECK CIRCUIT VR226 (YE/GY) FOR A SHORT TO GROUND AND A SHORT TO CIRCUIT RR153 (GY/BU)

- Disconnect: OCS Weight Sensor Bolt 3 C3330
- Disconnect: OCSM C3159

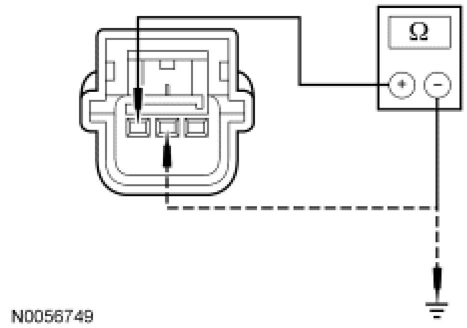

Fig. 32: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 3 C3330-3, circuit VR226 (YE/GY), harness side and ground; and between OCS weight sensor bolt 3 C3330-3, circuit VR226 (YE/GY), harness side and C3291-2, circuit RR153 (GY/BU), harness side.
- **Are the resistances greater than 10,000 ohms?**

  YES : Go to F24.

  NO : REPAIR circuits VR226 (YE/GY) and/or RR153 (GY/BU). Go to F37.

F24 CHECK CIRCUIT CR153 (GN/WH) FOR AN OPEN

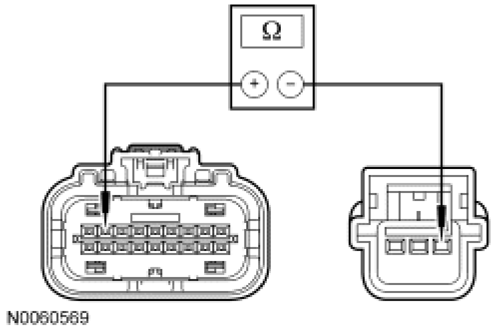

Fig. 33: Measuring Resistance Between OCS Weight Sensor Bolt 3 C3330-1, Circuit CR153 (GN/WH) & OCSM C3159-8
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 3 C3330-1, circuit CR153 (GN/WH), harness side and OCSM C3159-8, circuit CR153 (GN/WH), harness side.
- **Is the resistances less than 5 ohms?**

  YES : Go to F35.

  NO : REPAIR circuit CR153 (GN/WH). Go to F37.

F25 CHECK CIRCUIT RR153 (GY/BU) FOR AN OPEN

- Disconnect: OCS Weight Sensor Bolt 3 C3330

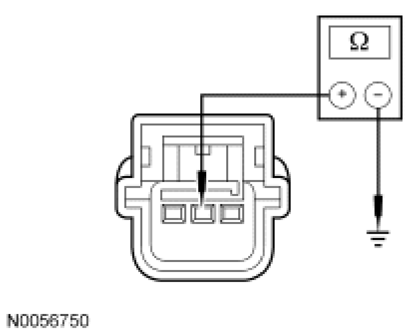

Fig. 34: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 3 C3330-2, circuit RR153 (GY/BU), harness side and ground.
- **Is the resistance less than 5 ohms?**

  YES : Go to F26.

  NO : REPAIR circuit RR153 (GY/BU). Go to F37.

F26 CHECK CIRCUIT VR226 (YE/GY) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Disconnect: OCSM C3159
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint**

System (SRS) Depowering and Repowering.

- Key in ON position.

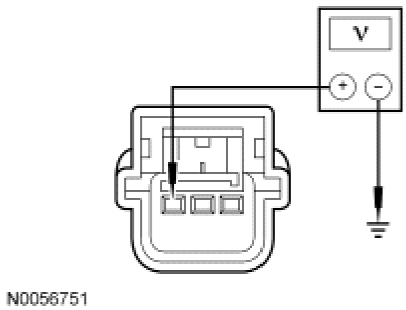

Fig. 35: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 3 C3330-3, circuit VR226 (YE/GY), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to F35.

  NO : REPAIR circuit VR226 (YE/GY). Go to F37.

F27 CHECK CIRCUITS CR154 (VT/OG), VR227 (VT/GN) AND RR154 (VT/WH) FOR AN OPEN

- Disconnect: OCSM C3159
- Disconnect: OCS Weight Sensor Bolt 4 C3331

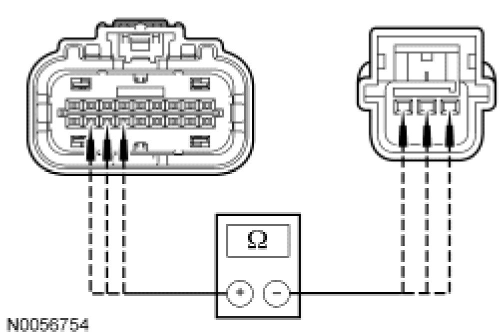

Fig. 36: Measuring Resistance Between OCSM C3159-16, Circuit VR227 (VT/GN) & OCS Weight Sensor Bolt 4 C3331-3
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCSM C3159-16, circuit VR227 (VT/GN), harness side and OCS weight sensor bolt 4 C3331-3, circuit VR227 (VT/GN), harness side; between OCSM C3159-17, circuit CR154 (VT/OG), harness side and OCS sensor 4 C3331-1, circuit CR154 (VT/OG), harness side; and between OCSM C3159-15, circuit RR154 (VT/WH), harness side and OCS sensor 4 C3331-2, circuit RR154 (VT/WH), harness side.

- **Are the resistances less than 0.5 ohm?**

  YES : Go to F28.

  NO : REPAIR circuit CR154 (VT/OG), circuit VR227 (VT/GN) or circuit RR154 (VT/WH). Go to F37.

F28 CHECK FOR A SHORT BETWEEN CIRCUITS CR154 (VT/OG), VR227 (VT/GN) AND RR154 (VT/WH)

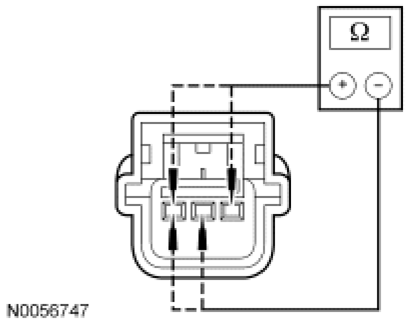

Fig. 37: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 4:
  - C3331-1, circuit CR154 (VT/OG), harness side and C3331-3, circuit VR227 (VT/GN), harness side.
  - C3331-1, circuit CR154 (VT/OG), harness side and C3331-2, circuit RR154 (VT/WH), harness side.
  - C3331-3, circuit VR227 (VT/GN), harness side and C3331-2, circuit RR154 (VT/WH), harness side.
- **Are the resistances greater than 10,0000 ohms?**

  YES : Go to F29.

  NO : REPAIR the affected circuits, circuits CR154 (VT/OG), VR227 (VT/GN) and/or RR154 (VT/WH). Go to F37.

F29 CHECK CIRCUIT VR227 (VT/GN) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

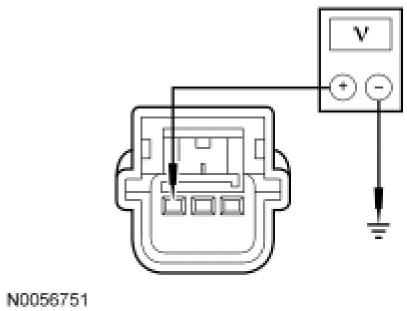

Fig. 38: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 4 C3331-3, circuit VR227 (VT/GN), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to F30.

  NO : REPAIR circuit VR227 (VT/GN). Go to F37.

F30 CHECK CIRCUIT VR227 (VT/GN) FOR A SHORT TO GROUND

- Key in OFF position.

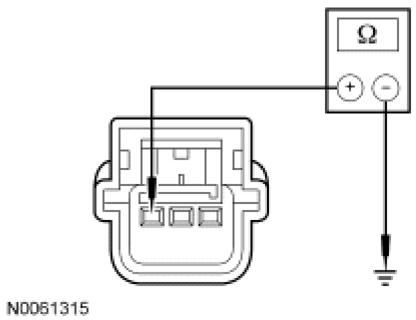

Fig. 39: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 4 C3331-3, circuit VR227 (VT/GN), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : INSTALL a new OCS weight sensor bolt 4. REFER to **Occupant Classification Sensor - Manual Seat Track** or **Occupant Classification Sensor - Power Seat Track**. Go to F37.

  NO : REPAIR circuit VR227 (VT/GN). Go to F37.

F31 CHECK CIRCUIT VR227 (VT/GN) FOR A SHORT TO GROUND AND A SHORT TO CIRCUIT RR154 (VT/WH)

- Disconnect: OCSM C3159
- Disconnect: OCS Weight Sensor Bolt 4 C3331

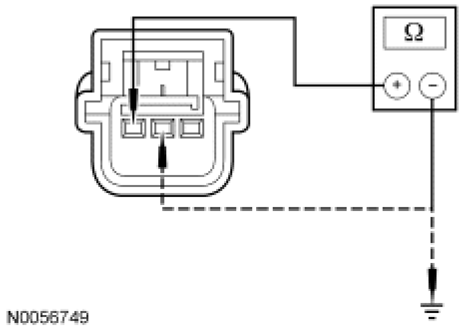

Fig. 40: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 4 C3331-3, circuit VR227 (VT/GN), harness side and ground; and between C3331-3, circuit VR227 (VT/GN), harness side and C3331-2, circuit RR154 (VT/WH), harness side.
- **Are the resistances greater than 10,000 ohms?**

YES : Go to F32.

NO : REPAIR circuits VR227 (VT/GN) and/or RR154 (VT/WH). Go to F37.

F32 CHECK CIRCUIT CR154 (VT/OG) FOR AN OPEN

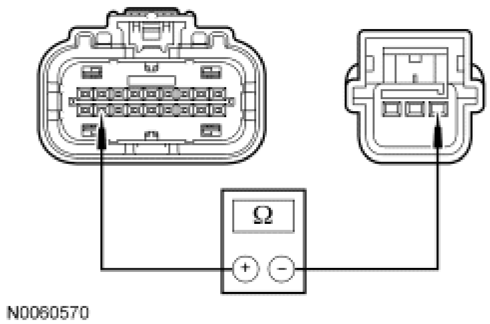

Fig. 41: Measuring Resistance Between OCS Weight Sensor Bolt 4 C3331-1, Circuit CR154 (VT/OG) & OCSM C3159-17
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 4 C3331-1, circuit CR154 (VT/OG), harness side and OCSM C3159-17, circuit CR154 (VT/OG), harness side.
- **Is the resistance less than 5 ohms?**

YES : Go to F35.

NO : REPAIR circuit CR154 (VT/OG). Go to F37.

F33 CHECK CIRCUIT RR154 (VT/WH) FOR AN OPEN

- Disconnect: OCS Weight Sensor Bolt 4 C3331

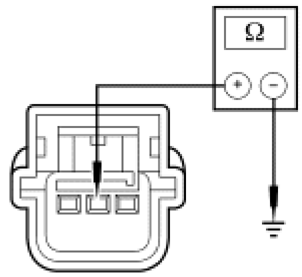

Fig. 42: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between OCS weight sensor bolt 4 C3331-2, circuit RR154 (VT/WH), harness side and ground.
- **Is the resistance less than 5 ohms?**

  YES : Go to F34.

  NO : REPAIR circuit RR154 (VT/WH). Go to F37.

F34 CHECK CIRCUIT VR227 (VT/GN) FOR A SHORT TO VOLTAGE

- Disconnect: Passenger Seat Side Air Bag Module C327
- Disconnect: OCSM C3159
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

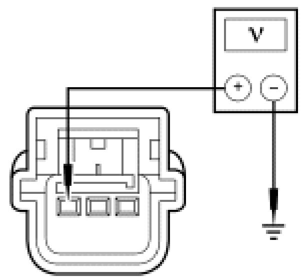

Fig. 43: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between OCS weight sensor bolt 4 C3331-3, circuit VR227 (VT/GN), harness side and ground.
- **Is the voltage less than 0.2 volt?**

  YES : Go to F35.

  NO : REPAIR circuit VR227 (VT/GN). Go to F37.

F35 CONFIRM THE OCS MODULE FAULT

NOTE: Make sure all restraint system sensor electrical connectors and the RCM electrical connector are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: OCSM C3159 (if previously disconnected)
- Connect: Affected OCS Weight Sensor Bolt Electrical Connector(s)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - OCSM - View and Record All B2290 Fault PIDs
  - Refer to PID list in Normal Operation to view B2290 fault PIDs.
- **Does the original OCSM on-demand DTC B2290 fault PID indicate a fault?**

  YES : INSTALL a new OCSM. REFER to **Occupant Classification System Module (OCSM) - Manual Seat Track** or **Occupant Classification System Module (OCSM) - Power Seat Track**. Go to F37.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to F36.

F36 CHECK FOR AN INTERMITTENT FAULT

- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - OCSM - View and Record All B2290 Fault PIDs
  - Refer to PID list in Normal Operation to view B2290 fault PIDs.
- **Do any OCSM on-demand DTC B2290 fault PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test.

  Using the fault PIDs recorded, go to the appropriate pinpoint test step.

  For 2290_3_OD (Passenger Seat OCS Sensor No. 4 Internal Fault), 2290_7_OD (Passenger Seat OCS Sensor No. 3 Internal Fault), 2290_11_OD (Passenger Seat OCS Sensor No.2 Internal Fault) or 2290_15_OD (Passenger Seat OCS Sensor No.1 Internal Fault), INSTALL a new OCS weight sensor bolt. REFER to **Occupant Classification Sensor - Manual Seat Track** or **Occupant Classification Sensor - Power Seat Track**.

For 2290_14_OD (Passenger Seat OCS Sensor No.1 Comm. Fault/Open), go to F3.

For 2290_13_OD (Passenger Seat OCS Sensor No.1 Circuit Short to Ground), go to F7.

For 2290_12_OD (Passenger Seat OCS Sensor No.1 Circuit Short to Battery), go to F9.

For 2290_10_OD (Passenger Seat OCS Sensor No.2 Comm. Fault/Open), go to F11.

For 2290_9_OD (Passenger Seat OCS Sensor No.2 Circuit Short to Ground), go to F15.

For 2290_8_OD (Passenger Seat OCS Sensor No.2 Circuit Short to Battery), go to F17.

For 2290_6_OD (Passenger Seat OCS Sensor No. 3 Comm. Fault/Open), go to F19.

For 2290_5_OD (Passenger Seat OCS Sensor No. 3 Circuit Short to Ground), go to F23.

For 2290_4_OD (Passenger Seat OCS Sensor No. 3 Circuit Short to Battery), go to F25.

For 2290_2_OD (Passenger Seat OCS Sensor No. 4 Comm. Fault/Open), go to F27.

For 2290_1_OD (Passenger Seat OCS Sensor No. 4 Circuit Short to Ground), go to F31.

For 2290_0_OD (Passenger Seat OCS Sensor No. 4 Circuit Short to Battery), go to F33.

NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to F37.

F37 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

> WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test G: DTC B2292 - Restraint System - Seatbelt Pretensioner Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) continuously monitors the safety belt retractor pretensioner circuits for an acceptable resistance range, unacceptable voltage and shorts to ground. The RCM will set on-demand and store DTC B2292 in memory if any of these conditions are detected to be outside of their respectful range. If the RCM detects a fault, it will send a message to the Instrument Cluster (IC) module to illuminate the air bag warning indicator.

As the RCM continuously monitors safety belt retractor pretensioner circuits for resistance, it expects a normal resistance between 1.4 and 3.5 ohms.

If a loop resistance is between 0.9 and 1.4 or between 3.5 and 5 ohms, there exists a strong potential for an intermittent fault, a DTC may or may not set. The RCM will also set an on-demand DTC if the loop resistance is less than 0.9 ohm or greater than 5 ohms.

Fault PIDs[a]	Description	Fault Trigger Condition
2292_24_OD and 2292_24_CM	Pretensioner Circuit Short to Ground, Front Passenger Side	When the RCM senses a short to ground on either of the passenger pretensioner circuits, a fault will be indicated.
2292_25_OD and 2292_25_CM	Pretensioner Circuit Short to Battery, Front Passenger Side	When the RCM senses a short to voltage on either of the passenger pretensioner circuits, a fault will be indicated.
2292_26_OD and 2292_26_CM	Pretensioner Circuit Open, Front Passenger Side	When the RCM measures resistance greater than 5 ohms between the passenger pretensioner circuits, a fault will be indicated.
2292_27_OD and 2292_27_CM	Pretensioner Circuit Resistance Low, Front Passenger Side	When the RCM measures resistance less than 0.9 ohm between the passenger pretensioner circuits, a fault will be indicated.
2292_28_OD and 2292_28_CM	Pretensioner Circuit Short to Ground, Front Driver Side	When the RCM senses a short to ground on either of the driver pretensioner circuits, a fault will be indicated.

2292_29_OD and 2292_29_CM	Pretensioner Circuit Short to Battery, Front Driver Side	When the RCM senses a short to voltage on either of the driver pretensioner circuits, a fault will be indicated.
2292_30_OD and 2292_30_CM	Pretensioner Circuit Open, Front Driver Side	When the RCM measures resistance greater than 5 ohms between the driver pretensioner circuits, a fault will be indicated.
2292_31_OD and 2292_31_CM	Pretensioner Circuit Resistance Low, Front Driver Side	When the RCM measures resistance less than 0.9 ohm between the driver pretensioner circuits, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B2292. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B2292.

NOTE: **When monitoring fault and/or resistance PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.**

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Safety belt retractor pretensioner
- RCM

PINPOINT TEST G: DTC B2292 - RESTRAINT SYSTEM - SEATBELT PRETENSIONER FAULT

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt**

pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

G1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2292 Fault PIDs
  - Refer to PID list in Normal Operation to view B2292 fault PIDs.
- **Was on-demand DTC B2292 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test.

  Using the fault PIDs recorded, go to the appropriate pinpoint test step.

  For 2292_31_OD (Pretensioner Circuit Resistance Low, Front Driver Side), go to G2.

  For 2292_30_OD (Pretensioner Circuit Open, Front Driver Side), go to G2.

  For 2292_28_OD (Pretensioner Circuit Short to Ground, Front Driver Side), go to G8.

  For 2292_29_OD (Pretensioner Circuit Short to Battery, Front Driver Side), go to G10.

  For 2292_27_OD (Pretensioner Circuit Resistance Low, Front Passenger Side), go to G11.

  For 2292_26_OD (Pretensioner Circuit Open, Front Passenger Side), go to G11.

  For 2292_24_OD (Pretensioner Circuit Short to Ground, Front Passenger Side), go to G17.

For 2292_25_OD (Pretensioner Circuit Short to Battery, Front Passenger Side), go to G19.

NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time.

Using the fault PIDs recorded, go to the appropriate pinpoint test step.

For 2292_31_CM (Pretensioner Circuit Resistance Low, Front Driver Side), 2292_30_CM (Pretensioner Circuit Open, Front Driver Side), 2292_27_CM (Pretensioner Circuit Resistance Low, Front Passenger Side) or 2292_26_CM (Pretensioner Circuit Open, Front Passenger Side), go to G22.

For 2292_28_CM (Pretensioner Circuit Short to Ground, Front Driver Side) or 2292_24_CM (Pretensioner Circuit Short to Ground, Front Passenger Side), go to G24.

For 2292_29_CM (Pretensioner Circuit Short to Battery, Front Driver Side) or 2292_25_CM (Pretensioner Circuit Short to Battery, Front Passenger Side), go to G25.

G2 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER RESISTANCE PID (DR_PTENS)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: DR_PTENS (Pretensioner Circuit Resistance, Driver Side) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms**

  YES : Go to G21.

  NO : Go to G3.

G3 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER RESISTANCE PID (DR_PTENS) WHILE CARRYING OUT A HARNESS TEST

- Continue monitoring the DR_PTENS (Pretensioner Circuit Resistance, Driver Side) resistance PID while carrying out a harness test of the driver safety belt retractor pretensioner circuits and accessible connectors (including any inline connectors), by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out a harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to G26.

  NO : If the driver safety belt retractor pretensioner resistance PID is less than 1.4 ohms, go to G4.

  If the driver safety belt retractor pretensioner resistance PID is greater than 3.5 ohms, go to G6.

G4 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER FOR LOW RESISTANCE FAULT

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2292_30_OD (Pretensioner Circuit Open, Front Driver Side)
  - 2292_31_OD (Pretensioner Circuit Resistance Low, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2292 on-demand fault PIDs with the driver pretensioner disconnected, an open circuit fault would normally be retrieved.
- **Did the driver safety belt retractor pretensioner on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to G20.

  NO : Go to G5.

G5 CHECK FOR A SHORT BETWEEN DRIVER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS CR120 (BU/OG) AND RR120 (BN/GN)

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

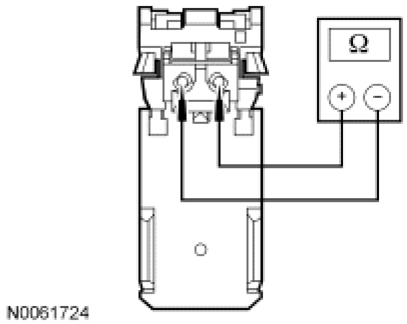

Fig. 44: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver safety belt retractor pretensioner C323-1, circuit CR120 (BU/OG), harness side and C323-2, circuit RR120 (GN/RD), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to G21.

  NO : REPAIR circuits CR120 (BU/OG) and RR120 (BN/GN). Go to G26.

G6 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER FOR AN OPEN

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Connect a fused jumper wire between driver safety belt retractor pretensioner C3201-1, circuit CR120 (BU/OG), harness side and C3201-2, circuit RR120 (BN/GN), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2292_30_OD (Pretensioner Circuit Open, Front Driver Side)
  - 2292_31_OD (Pretensioner Circuit Resistance Low, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2292 on-demand fault PIDs with the driver pretensioner circuits shorted together, a low resistance fault would normally be retrieved.
- **Did the driver safety belt retractor pretensioner on-demand fault PIDs change from indicating an open circuit to a low resistance fault?**

  YES : Go to G20.

  NO : Go to G7.

G7 CHECK CIRCUITS CR120 (BU/OG) AND RR120 (BN/GN) FOR AN OPEN

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the driver safety belt retractor pretensioner C323.
- Disconnect: RCM C310a and C310b

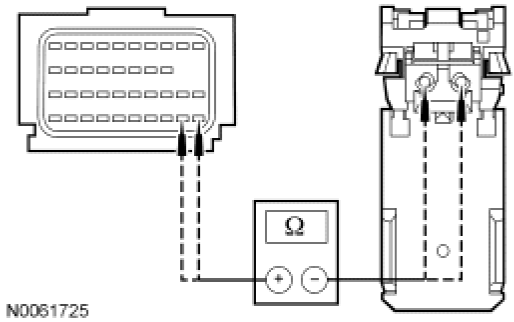

Fig. 45: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-31, circuit CR120 (BU/OG), harness side and driver safety belt retractor pretensioner C323-2, circuit CR120 (BU/OG), harness side; and between RCM C310b-32, circuit RR120 (BN/GN), harness side and driver safety belt retractor pretensioner C323-1, circuit RR120 (GN/RD), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to G21.

  NO : REPAIR circuit CR120 (BU/OG) or circuit RR120 (GN/RD). Go to G26.

G8 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER FOR A SHORT TO GROUND

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2292_28_OD (Pretensioner Circuit Short to Ground, Front Driver Side)
  - 2292_30_OD (Pretensioner Circuit Open, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2292 on-demand fault PIDs with the driver pretensioner disconnected, an open circuit fault would normally be retrieved.

- **Did the driver safety belt retractor pretensioner on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to G20.

  NO : Go to G9.

G9 CHECK CIRCUITS CR120 (BU/OG) AND RR120 (GN/RD) FOR A SHORT TO GROUND

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

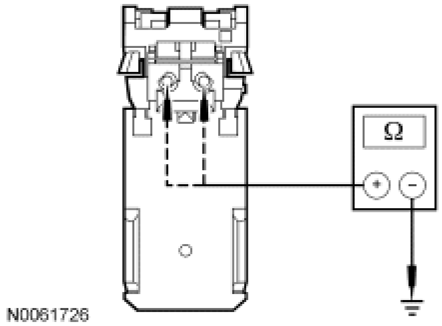

Fig. 46: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver safety belt retractor pretensioner C323-2, circuit CR120 (BU/OG), harness side and ground; and between driver safety belt retractor pretensioner C323-1, circuit RR120 (GN/RD), and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to G21.

  NO : REPAIR circuit CR120 (BU/OG) or circuit RR120 (GN/RD). Go to G26.

G10 CHECK CIRCUITS CR120 (BU/OG) AND RR120 (BN/GN) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

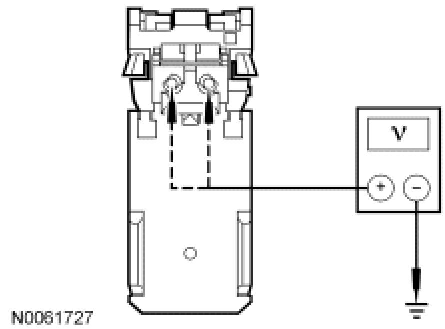

Fig. 47: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between driver safety belt retractor pretensioner C323-2, circuit CR120 (BU/OG), harness side and ground; and between driver safety belt retractor pretensioner C323-1, circuit RR120 (GN/RD), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to G21.

  NO : REPAIR circuit CR120 (BU/OG) or circuit RR120 (BN/GN). Go to G26.

G11 CHECK THE PASSENGER RETRACTOR PRETENSIONER RESISTANCE PID (PS_PTENS)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: PS_TENS (Pretensioner Circuit Resistance, Passenger Side) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms?**

  YES : Go to G21.

  NO : Go to G12.

G12 CHECK THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER RESISTANCE PID (PS_PTENS) WHILE CARRYING OUT A HARNESS TEST

- Continue monitoring the PS_TENS (Pretensioner Circuit Resistance, Passenger Side) resistance PID while carrying out a harness test of the passenger safety belt retractor pretensioner circuits and accessible connectors (including any inline connectors), by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to G26.

  NO : If the passenger safety belt retractor pretensioner resistance PID is less than 1.4 ohms, go to G13.

  If the passenger safety belt retractor pretensioner resistance PID is greater than 3.5 ohms, go to G15.

G13 CHECK THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER FOR LOW RESISTANCE FAULT

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C303
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2292_26_OD (Pretensioner Circuit Open, Front Passenger Side)
  - 2292_27_OD (Pretensioner Circuit Resistance Low, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2292 on-demand fault PIDs with the passenger pretensioner disconnected, an open circuit fault would normally be retrieved.
- **Did the passenger safety belt retractor pretensioner on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to G21.

  NO : Go to G14.

G14 CHECK FOR A SHORT BETWEEN PASSENGER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS RR122 (BN) AND CR122 (WH/OG)

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

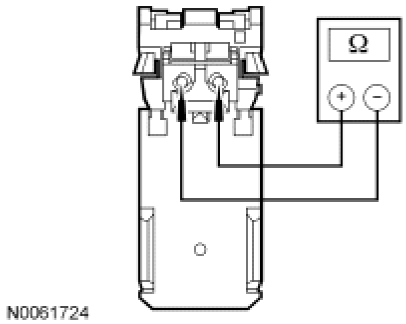

Fig. 48: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger safety belt pretensioner C303-2, circuit CR122 (WH/OG), harness side and C303-1, circuit RR122 (BN), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to G21.

  NO : REPAIR circuits CR122 (WH/OG) and RR122 (BN). Go to G26.

G15 CHECK THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER FOR AN OPEN

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C303
- Connect a fused jumper wire between passenger safety belt retractor pretensioner C303-1, circuit CR122 (WH/OG), harness side and C303-2, circuit RR122 (BN), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2292_26_OD (Pretensioner Circuit Open, Front Passenger Side)
  - 2292_27_OD (Pretensioner Circuit Resistance Low, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2292 on-demand fault PIDs with the passenger pretensioner circuits shorted together, a low resistance fault would normally be retrieved.

- **Did the passenger safety belt retractor pretensioner on-demand fault PIDs change from indicating an open circuit to a low resistance fault?**

  YES : Go to G20.

  NO : Go to G16.

G16 CHECK CIRCUITS CR122 (WH/OG) AND RR122 (BN) FOR AN OPEN

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the passenger safety belt retractor pretensioner fused jumper wire.
- Disconnect: RCM C310a and C310b

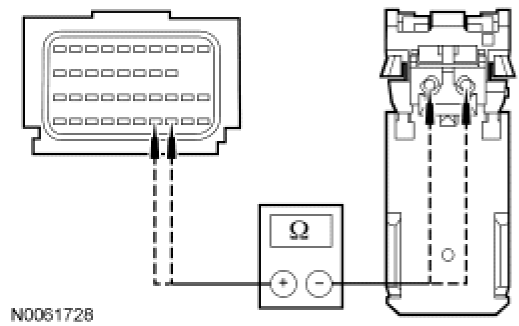

Fig. 49: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-34, circuit RR122 (BN), harness side and passenger safety belt retractor pretensioner C303-1, circuit RR122 (BU/RD), harness side; and between RCM C310b-33, circuit CR122 (WH/OG), harness side and passenger safety belt retractor pretensioner C303-2, circuit CR122 (WH/OG), harness side.
- **Is resistances less than 0.5 ohm?**

  YES : Go to G21.

  NO : REPAIR circuit RR122 (BN) or circuit CR122 (WH/OG). Go to G26.

G17 CHECK THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER FOR A SHORT TO GROUND

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C303

- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2292_24_OD (Pretensioner Circuit Short to Ground, Front Passenger Side)
  - 2292_26_OD (Pretensioner Circuit Open, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2292 on-demand fault PIDs with the passenger pretensioner disconnected, an open circuit fault would normally be retrieved.
- **Did the passenger safety belt retractor pretensioner on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to G20.

  NO : Go to G18.

G18 CHECK CIRCUITS CR122 (WH/OG) AND RR122 (BN) FOR A SHORT TO GROUND

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

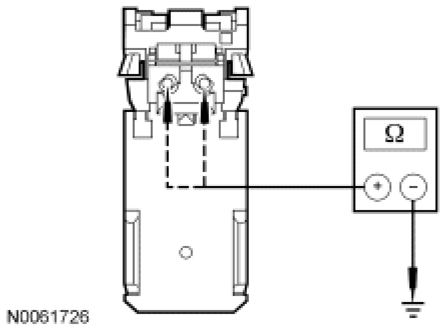

Fig. 50: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger safety belt retractor pretensioner C303-2, circuit CR122 (WH/OG), harness side and ground; and between passenger safety belt retractor pretensioner C303-1, circuit RR122 (BN), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to G21.

  NO : REPAIR circuit CR122 (WH/OG) or circuit RR122 (BN). Go to G26.

G19 CHECK CIRCUITS RR122 (BN) AND CR122 (WH/OG) FOR A SHORT TO VOLTAGE

- Key in OFF position.

- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C303
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

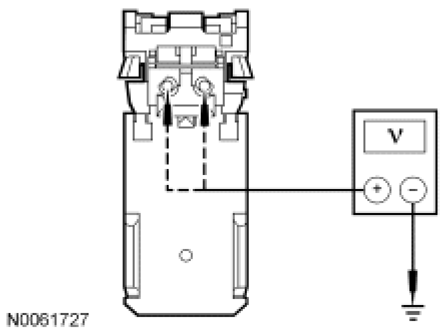

Fig. 51: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between passenger safety belt retractor pretensioner C303-1, circuit RR122 (BN), harness side and ground; and between passenger safety belt retractor pretensioner C303-2, circuit CR122 (WH/OG), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to G21.

  NO : REPAIR circuit RR122 (BN) or circuit CR122 (WH/OG). Go to G26.

G20 CONFIRM THE SAFETY BELT RETRACTOR PRETENSIONER FAULT

NOTE:	**Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Affected Safety Belt Retractor Pretensioner C323 (Driver) C303 (Passenger)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record

All B2292 Fault PIDs

- Refer to PID list in Normal Operation to view B2292 fault PIDs.

- **Does the original RCM on-demand DTC B2292 fault PID indicate a fault?**

  YES : INSTALL a new driver or passenger safety belt retractor pretensioner. REFER to **SAFETY BELT SYSTEM** article. Go to G26.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2292_31_CM (Pretensioner Circuit Resistance Low, Front Driver Side), 2292_30_CM (Pretensioner Circuit Open, Front Driver Side), 2292_27_CM (Pretensioner Circuit Resistance Low, Front Passenger Side) or 2292_26_CM (Pretensioner Circuit Open, Front Passenger Side), go to G22.

  For 2292_28_CM (Pretensioner Circuit Short to Ground, Front Driver Side) or 2292_24_CM (Pretensioner Circuit Short to Ground, Front Passenger Side), go to G24.

  For 2292_29_CM (Pretensioner Circuit Short to Battery, Front Driver Side) or 2292_25_CM (Pretensioner Circuit Short to Battery, Front Passenger Side), go to G25.

G21 CONFIRM THE RCM FAULT

NOTE: **Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: All Restraint System Components (if previously disconnected)
- Connect: RCM C310a and C310b (if previously disconnected)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2292 Fault PIDs
  - Refer to PID list in Normal Operation to view B2292 fault PIDs.
- **Does the original RCM on-demand DTC B2292 fault PID indicate a fault?**

  YES : INSTALL a new RCM and CARRY OUT PMI. REFER to **Restraints Control Module (RCM)**. Go to G26.

NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

For 2292_31_CM (Pretensioner Circuit Resistance Low, Front Driver Side), 2292_30_CM (Pretensioner Circuit Open, Front Driver Side), 2292_27_CM (Pretensioner Circuit Resistance Low, Front Passenger Side) or 2292_26_CM (Pretensioner Circuit Open, Front Passenger Side), go to G22.

For 2292_28_CM (Pretensioner Circuit Short to Ground, Front Driver Side) or 2292_24_CM (Pretensioner Circuit Short to Ground, Front Passenger Side), go to G24.

For 2292_29_CM (Pretensioner Circuit Short to Battery, Front Driver Side) or 2292_25_CM (Pretensioner Circuit Short to Battery, Front Passenger Side), go to G25.

G22 CHECK FOR AN INTERMITTENT PRETENSIONER LOW RESISTANCE OR OPEN CIRCUIT FAULT

- If the fault PID was reported for the driver safety belt retractor pretensioner, monitor the following on the scan tool:
  - DR_PTENS (Pretensioner Circuit Resistance, Driver Side) resistance PID.
  - 2292_30_OD (Pretensioner Circuit Open, Front Driver Side) and 2292_31_OD (Pretensioner Circuit Resistance Low, Front Driver Side) on-demand fault PIDs.
- If the fault PID was reported for the passenger safety belt retractor pretensioner, monitor the following on the scan tool:
  - PS_PTENS (Pretensioner Circuit Resistance, Passenger Side) resistance PID.
  - 2292_26_OD (Pretensioner Circuit Open, Front Passenger Side) and 2292_27_OD (Pretensioner Circuit Resistance Low, Front Passenger Side) on-demand fault PIDs.

NOTE: **A resistance PID that reads between 0.9 and 1.4 ohms or between 3.5 and 5 ohms may or may not set an on-demand DTC, yet indicates a strong potential of an intermittent fault condition. Make sure to thoroughly check wire harness(es), connectors and terminals as they are the likely source of the fault condition.**

- Attempt to recreate the fault by wiggling the connectors (including any inline connectors) and flexing the wire harness.
- **Does any safety belt retractor pretensioner on-demand fault PID indicate a fault is present or is the pretensioner resistance PID less than 1.4 ohms or greater than 3.5 ohms?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to G26.

  NO : Go to G23.

G23 CHECK THE HARNESS AND CONNECTORS

- Key in OFF position.

- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect the affected safety belt retractor pretensioner C323 (driver) or C303 (passenger):
  - inspect connector(s) (including any inline connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
  - inspect wire harness for any damage, pinched, cut or pierced wires.
- **Were any concerns found?**

  YES : REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to G26.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** Go to G26.

G24 CHECK FOR AN INTERMITTENT SAFETY BELT RETRACTOR PRETENSIONER CIRCUIT SHORT TO GROUND FAULT

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2292 Fault PIDs
  - Refer to PID list in Normal Operation to view B2292 fault PIDs.
- If the fault PID was reported for the driver safety belt retractor pretensioner, monitor the 2292_28_OD (Pretensioner Circuit Short to Ground, Front Driver Side) on-demand fault PID on the scan tool.
- If the fault PID was reported for the passenger safety belt retractor pretensioner, monitor the 2292_24_OD (Pretensioner Circuit Short to Ground, Front Passenger Side) on-demand fault PID on the scan tool.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.
- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any safety belt retractor pretensioner on-demand fault PID indicate a fault is present?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to G26.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** Go to G26.

G25 CHECK FOR AN INTERMITTENT SAFETY BELT RETRACTOR PRETENSIONER CIRCUIT SHORT TO BATTERY FAULT

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2292 Fault PIDs
  - Refer to PID list in Normal Operation to view B2292 fault PIDs.

- If the fault PID was reported for the driver safety belt retractor pretensioner, monitor the 2292_29_OD (Pretensioner Circuit Short to Battery, Front Driver Side) on-demand fault PID.
- If the fault PID was reported for the passenger safety belt retractor pretensioner, monitor the 2292_25_OD (Pretensioner Circuit Short to Battery, Front Passenger Side) on-demand fault PID.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.
- Attempt to recreate the fault by wiggling the connectors (including any inline connectors) and flexing the wire harness.
- **Does the safety belt retractor pretensioner short to battery on-demand fault PID indicate a fault?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to G26.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** Go to G26.

G26 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test H: DTC B2293 - Restraint System - Air Bag Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) continuously monitors all front air bag module circuits (loop) for an acceptable resistance range, unacceptable voltage and shorts to ground. The RCM will set on-demand and store DTC B2293 in memory if any of these conditions are detected to be outside of their respectful range. If the RCM detects a fault, it will send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

As the RCM continuously monitors all front air bag module circuits for resistance, it expects a normal resistance for the driver air bag module loop 1 and 2, between 1.7 and 4.4 ohms.

The normal expected resistance for the passenger air bag module loop 1 and 2 is between 1.4 and 3.5 ohms.

If a loop resistance for the driver air bag loop 1 or 2 is between 0.9 and 1.7 or between 4.4 and 5 ohms, there exists a strong potential for an intermittent fault, a DTC may or may not set. The RCM will also set an on-demand DTC if the loop resistance is less than 0.9 ohm or greater than 5 ohms.

If a loop resistance for the passenger air bag loop 1 & 2 is between 0.9 and 1.4 or between 3.5 and 5 ohms, there exists a strong potential for an intermittent fault, a DTC may or may not set. The RCM will also set an on-demand DTC if the loop resistance is less than 0.9 ohm or greater than 5 ohms.

Fault PIDs[a]	Description	Fault Trigger Condition
2293_16_OD and 2293_16_CM	Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side	When the RCM measures resistance less than 0.9 ohm between the passenger air bag loop circuits, a fault will be indicated.
2293_17_OD and 2293_17_CM	Air Bag Circuit Open - Loop No. 2, Front Passenger Side	When the RCM measures resistance greater than 5 ohms between the passenger air bag loop circuits, a fault will be indicated.
2293_18_OD and 2293_18_CM	Air Bag Circuit Short to Battery - Loop No. 2, Front Passenger Side	When the RCM senses a short to voltage on either on the passenger air bag loop circuits, a fault will be indicated.
2293_19_OD and 2293_19_CM	Air Bag Circuit Short to Ground - Loop No. 2, Front Passenger Side	When the RCM senses a short to ground on either on the passenger air bag loop circuits, a fault will be indicated.
2293_20_OD and 2293_20_CM	Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side	When the RCM measures resistance less than 0.9 ohm between the driver air bag loop circuits, a fault will be indicated.
		When the RCM measures resistance greater than 5 ohms

2293_21_OD and 2293_21_CM	Air Bag Circuit Open - Loop No. 2, Front Driver Side	between the driver air bag loop circuits, a fault will be indicated.
2293_22_OD and 2293_22_CM	Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side	When the RCM senses a short to voltage on either on the driver air bag loop circuits, a fault will be indicated.
2293_23_OD and 2293_23_CM	Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side	When the RCM senses a short to ground on either on the driver air bag loop circuits, a fault will be indicated.
2293_24_OD and 2293_24_CM	Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side	When the RCM measures resistance less than 0.9 ohm between the passenger air bag loop circuits, a fault will be indicated.
2293_25_OD and 2293_25_CM	Air Bag Circuit Open - Loop No. 1, Front Passenger Side	When the RCM measures resistance greater than 5 ohms between the passenger air bag loop circuits, a fault will be indicated.
2293_26_OD and 2293_26_CM	Air Bag Circuit Short to Battery - Loop No. 1, Front Passenger Side	When the RCM senses a short to voltage on either on the passenger air bag loop circuits, a fault will be indicated.
2293_27_OD and 2293_27_CM	Air Bag Circuit Short to Ground - Loop No. 1, Front Passenger Side	When the RCM senses a short to ground on either on the passenger air bag loop circuits, a fault will be indicated.
2293_28_OD and 2293_28_CM	Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side	When the RCM measures resistance less than 0.9 ohm between the driver air bag loop circuits, a fault will be indicated.
2293_29_OD and 2293_29_CM	Air Bag Circuit Open - Loop No. 1, Front Driver Side	When the RCM measures resistance greater than 5 ohms between the driver air bag loop circuits, a fault will be indicated.
2293_30_OD and 2293_30_CM	Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side	When the RCM senses a short to voltage on either on the driver air bag loop circuits, a fault will be indicated.
		When the RCM senses a short

2293_31_OD and 2293_31_CM	Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side	to ground on either on the driver air bag loop circuits, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B2293. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B2293.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Clockspring
- Driver air bag module
- Passenger air bag module
- RCM

PINPOINT TEST H: DTC B2293 - RESTRAINT SYSTEM - AIR BAG FAULT

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle

to the customer.

H1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2293 Fault PIDs
  - Refer to PID list in Normal Operation to view B2293 fault PIDs.
- **Do any RCM on-demand DTC B2293 fault PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test.

  Using the fault PIDs recorded, go to the appropriate pinpoint test step.

  For 2293_28_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side), go to H2.

  For 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side), go to H2.

  For 2293_31_OD (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side), go to H10.

  For 2293_30_OD (Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side), go to H13.

  For 2293_20_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side), go to H15.

  For 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side), go to H15.

  For 2293_23_OD (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side), go to H23.

  For 2293_22_OD (Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side), go to H26.

  For 2293_24_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side), go to H28.

  For 2293_25_OD (Air Bag Circuit Open - Loop No. 1, Front Passenger Side), go to H28.

  For 2293_27_OD (Air Bag Circuit Short to Ground - Loop No. 1, Front Passenger Side), go to H34.

  For 2293_26_OD (Air Bag Circuit Short to Battery - Loop No. 1, Front Passenger Side), go to H36.

  For 2293_16_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side), go to H37.

For 2293_17_OD (Air Bag Circuit Open - Loop No. 2, Front Passenger Side), go to H37.

For 2293_19_OD (Air Bag Circuit Short to Ground - Loop No. 2, Front Passenger Side), go to H43.

For 2293_18_OD (Air Bag Circuit Short to Battery - Loop No. 2, Front Passenger Side), go to H45.

NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time.

Using the fault PIDs recorded, go to the appropriate pinpoint test step.

For 2293_28_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side) or 2293_20_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side), go to H50.

For 2293_29_CM (Air Bag Circuit Open - Loop No. 1, Front Driver Side) or 2293_21_CM (Air Bag Circuit Open - Loop No. 2, Front Driver Side), go to H50.

For 2293_31_CM (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side) or 2293_23_CM (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side), go to H51.

For 2293_30_CM (Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side) or 2293_22_CM (Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side), go to H53.

For 2293_24_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side) or 2293_16_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side), go to H50.

For 2293_25_CM (Air Bag Circuit Open - Loop No. 1, Front Passenger Side) or 2293_17_CM (Air Bag Circuit Open - Loop No. 2, Front Passenger Side), go to H50.

For 2293_27_CM (Air Bag Circuit Short to Ground - Loop No. 1, Front Passenger Side) or 2293_19_CM (Air Bag Circuit Short to Ground - Loop No. 2, Front Passenger Side), go to H52.

For 2293_26_CM (Air Bag Circuit Short to Battery - Loop No. 1, Front Passenger Side) or 2293_18_CM (Air Bag Circuit Short to Battery - Loop No. 2, Front Passenger Side), go to H54.

H2 CHECK THE DRIVER AIR BAG MODULE LOOP 1 RESISTANCE PID (D_ABAGR)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: D_ABAGR (Driver Air Bag) Resistance PID
- **Is the resistance PID value between 1.7 and 4.4 ohms?**

  YES : Go to H49.

  NO : Go to H3.

H3 CHECK THE DRIVER AIR BAG MODULE LOOP 1 RESISTANCE PID (D_ABAGR) WHILE CARRYING OUT A HARNESS TEST

NOTE: Do not remove the driver air bag module from its mounted position at this time.

- Continue monitoring the D_ABAGR (driver air bag) resistance PID while carrying out a harness test of the driver air bag circuits and accessible connectors (including any inline connectors) by wiggling and flexing the wire harness, connectors and rotating the steering wheel frequently.
- **Does the resistance PID value read between 1.7 and 4.4 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness or INSTALL a new clockspring as needed. REFER to **Clockspring**. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

  NO : If the driver air bag module loop 1 resistance PID is less than 1.7 ohms, go to H4.

  If the driver air bag module loop 1 resistance PID is greater than 4.4 ohms, go to H7.

H4 CHECK THE DRIVER AIR BAG MODULE LOOP 1 LOW RESISTANCE FAULT

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_28_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side)
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 1 on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

YES : Go to H46.

NO : Go to H5.

H5 CHECK THE DRIVER AIR BAG MODULE LOOP 1 CIRCUITS FOR LOW RESISTANCE BETWEEN THE CLOCKSPRING AND RCM

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Disconnect: Clockspring C218
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_28_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side)
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module/clockspring disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 1 on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to H48.

  NO : Go to H6.

H6 CHECK THE RCM FOR LOW RESISTANCE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

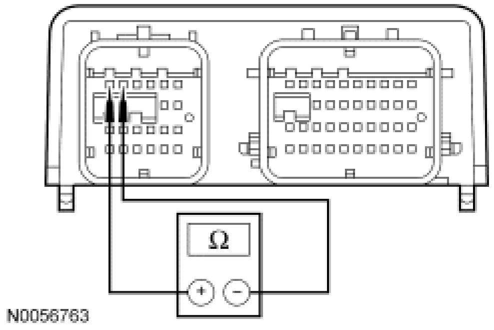

Fig. 52: Measuring Resistance Between RCM C310A Pin 1 & C310A Pin 2, Component Side
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a pin 1 and C310a pin 2, component side.
- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuits CR101 (VT/BN) and RR101 (YE/GN). Go to H56.

  NO : Go to H49.

H7 CHECK THE DRIVER AIR BAG MODULE LOOP 1 FOR AN OPEN

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Connect a fused jumper wire between driver air bag module loop 1 electrical connector pin-1, harness side and pin-2, harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_28_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side)
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module loop 1 circuits shorted together, a low resistance fault would normally be retrieved on loop 1. Loop 2 will show an open circuit fault due to the driver air bag being disconnected.
- **Did the driver air bag module loop 1 on-demand fault PIDs change from indicating a open circuit to a low resistance fault?**

  YES : Go to H46.

  NO : Go to H8.

H8 CHECK THE DRIVER AIR BAG MODULE LOOP 1 CIRCUITS FOR AN OPEN BETWEEN THE CLOCKSPRING AND RCM

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the

fault.

- Key in OFF position.
- Remove the fused jumper wire from driver air bag module loop 1 electrical connector.
- Disconnect: Clockspring C218
- Connect a fused jumper wire between clockspring C218-1, circuit CR101 (VT/BN), harness side and clockspring C218-9, circuit RR101 (YE/GN), harness side.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_28_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side)
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module loop 1 circuits shorted together, a low resistance fault would normally be retrieved on loop 1. Loop 2 will show an open circuit fault due to the driver air bag being disconnected.
- **Did the driver air bag module loop 1 on-demand fault PIDs change from indicating a open circuit to a low resistance fault?**

  YES : Go to H48.

  NO : Go to H9.

H9 CHECK THE DRIVER AIR BAG MODULE LOOP 1 CIRCUITS CR101 (VT/BN) AND RR101 (YE/GN) FOR AN OPEN BETWEEN THE CLOCKSPRING AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the clockspring connector.
- Disconnect: RCM C310a and C310b

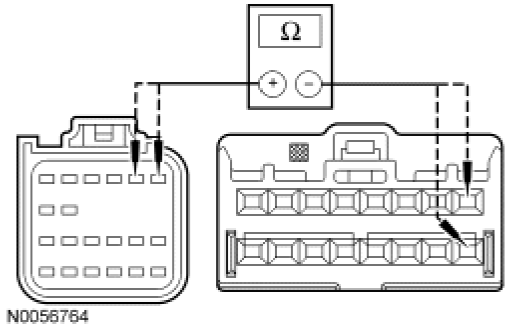

Fig. 53: Measuring Resistance Between RCM C310A-1, Circuit CR101 (VT/BN) & Clockspring C218-1
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a-1, circuit CR101 (VT/BN), harness side and clockspring C218-1, circuit CR101 (VT/BN), harness side; and between RCM C310a-2, circuit RR101 (YE/GN), harness side and clockspring C218-9, circuit RR101 (YE/GN), harness side.
- **Are the resistances less than 0.5 ohm?**

  YES : Go to H49.

  NO : REPAIR circuit CR101 (VT/BN) or circuit RR101 (YE/GN). Go to H56.

H10 CHECK THE DRIVER AIR BAG MODULE LOOP 1 FOR A SHORT TO GROUND

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side)
  - 2293_31_OD (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 1 on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to H46.

  NO : Go to H11.

H11 CHECK DRIVER AIR BAG MODULE LOOP 1 CIRCUITS FOR A SHORT TO GROUND BETWEEN THE RCM AND CLOCKSPRING

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.

- Disconnect: Clockspring C218
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side)
  - 2293_31_OD (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the clockspring disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 1 on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to H48.

  NO : Go to H12.

H12 CHECK CIRCUITS CR101 (VT/BN) AND RR101 (YE/GN) FOR A SHORT TO GROUND BETWEEN THE RCM AND CLOCKSPRING

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

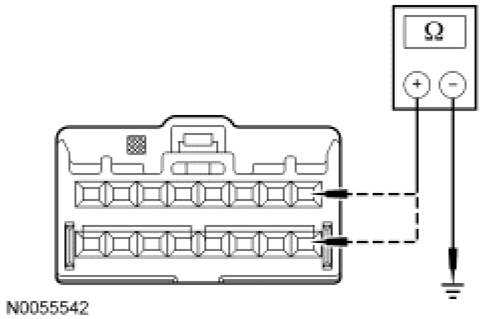

Fig. 54: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between clockspring C218-1, circuit CR101 (VT/BN), harness side and ground; and between clockspring C218-9, circuit RR101 (YE/GN), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR101 (VT/BN) or circuit RR101 (YE/GN). Go to H56.

H13 CHECK THE DRIVER AIR BAG MODULE LOOP 1 CIRCUITS FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND CLOCKSPRING

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Disconnect: Clockspring C218
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side)
  - 2293_30_OD (Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module/clockspring disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 1 on-demand fault PIDs change from indicating a short to battery to an open circuit fault?**

  YES : Go to H48.

  NO : Go to H14.

H14 CHECK CIRCUITS CR101 (VT/BN) AND RR101 (YE/GN) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND CLOCKSPRING

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

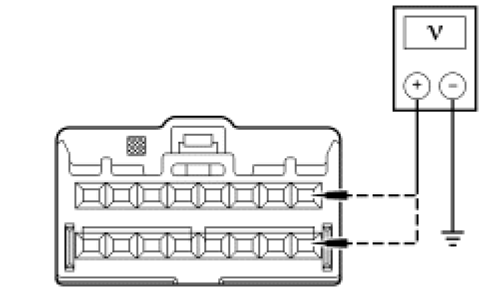

Fig. 55: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between clockspring C218-1, circuit CR101 (VT/BN), harness side and ground; and between clockspring C218-9, circuit RR101 (YE/GN), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR101 (VT/BN) or circuit RR101 (YE/GN). Go to H56.

H15 CHECK THE DRIVER AIR BAG MODULE LOOP 2 RESISTANCE PID (D_ABAGR2)

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: D_ABAGR2 (Driver Air Bag #2 Resistance) Resistance PID
- **Is the resistance PID value between 1.7 and 4.4 ohms?**

  YES : Go to H49.

  NO : Go to H16.

H16 CHECK THE DRIVER AIR BAG MODULE LOOP 2 RESISTANCE PID (D_ABAGR2) WHILE CARRYING OUT A HARNESS TEST

NOTE: Do not remove the driver air bag module from its mounted position at this time.

- With the driver air bag in its mounted position, continue monitoring the D_ABAGR2 (Driver Air Bag #2 Resistance) resistance PID while carrying out a harness test of the driver air bag circuits and accessible connectors (including any inline connectors), by wiggling and flexing the wire harness, connectors, tilting and rotating the steering wheel frequently.
- **Does the resistance PID value read between 1.7 and 4.4 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness or INSTALL a new clockspring as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

NO : If the driver air bag module loop 2 resistance PID is less than 1.7 ohms, go to H17.

If the driver air bag module loop 2 resistance PID is greater than 4.4 ohms, go to H20.

H17 CHECK THE DRIVER AIR BAG MODULE LOOP 2 LOW RESISTANCE FAULT

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_20_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side)
  - 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 2 on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to H46.

  NO : Go to H18.

H18 CHECK THE DRIVER AIR BAG MODULE LOOP 2 CIRCUITS FOR LOW RESISTANCE BETWEEN THE CLOCKSPRING AND RCM

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Disconnect: Clockspring C218
- Key in ON position.

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_20_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side)
  - 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module/clockspring disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 2 on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to H48.

  NO : Go to H19.

H19 CHECK THE RCM FOR LOW RESISTANCE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

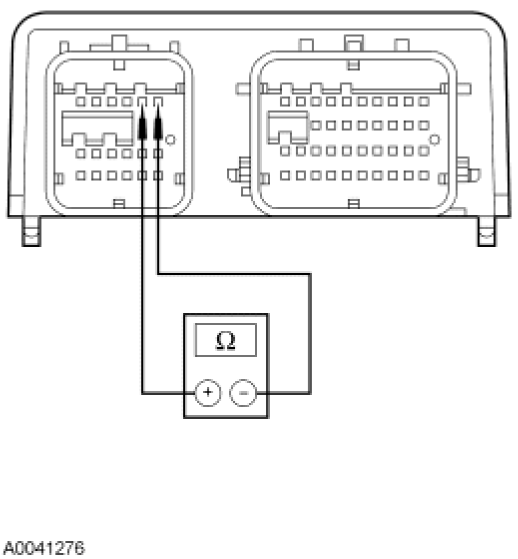

Fig. 56: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a pin 5 and C310a pin 6, component side.

- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuits CR102 (BU) and RR102 (WH). Go to H56.

  NO : Go to H49.

H20 CHECK THE DRIVER AIR BAG MODULE LOOP 2 FOR AN OPEN

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Connect a fused jumper wire between driver air bag module loop 2 electrical connector, pin-1, harness side and pin-2, harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_20_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side)
  - 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module loop 2 circuits shorted together, a low resistance fault would normally be retrieved on loop 2. Loop 1 will show an open circuit fault due to the driver air bag being disconnected.
- **Did the driver air bag module loop 2 on-demand fault PIDs change from indicating an open circuit to a low resistance fault?**

  YES : Go to H46.

  NO : Go to H21.

H21 CHECK THE DRIVER AIR BAG MODULE LOOP 2 CIRCUITS FOR AN OPEN BETWEEN THE CLOCKSPRING AND RCM

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.

- Remove the fused jumper wire from driver air bag module loop 2 electrical connector.
- Disconnect: Clockspring C218
- Connect a fused jumper wire between clockspring C218-2, circuit CR102 (BU), harness side and clockspring C218-10, circuit RR102 (WH), harness side.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
    - 2293_20_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side)
    - 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module loop 2 circuits shorted together, a low resistance fault would normally be retrieved on loop 2. Loop 1 will show an open circuit fault due to the driver air bag being disconnected.
- **Did the driver air bag module loop 2 on-demand fault PIDs change from indicating an open circuit to a low resistance fault?**

  YES : Go to H48.

  NO : Go to H22.

H22 CHECK THE DRIVER AIR BAG MODULE LOOP 2 CIRCUITS CR102 (BU) AND RR102 (WH) FOR AN OPEN BETWEEN THE CLOCKSPRING AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the clockspring connector.
- Disconnect: RCM C310a and C310b

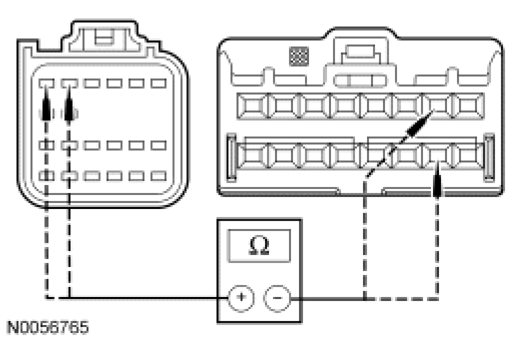

Fig. 57: Measuring Resistance Between RCM C310A-5, Circuit CR102 (BU) & Clockspring C218-2
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a-5, circuit CR102 (BU), harness side and clockspring C218-2, circuit CR102 (BU), harness side; and between RCM C310a-6, circuit RR102 (WH), harness side and clockspring C218-10, circuit RR102 (WH), harness side.

- **Are resistances less than 0.5 ohm?**

  YES : Go to H49.

  NO : REPAIR circuit CR102 (BU) or circuit RR102 (WH). Go to H56.

H23 CHECK THE DRIVER AIR BAG MODULE LOOP 2 CIRCUITS FOR A SHORT TO GROUND

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side)
  - 2293_23_OD (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 2 on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to H46.

  NO : Go to H24.

H24 CHECK THE DRIVER AIR BAG MODULE LOOP 2 CIRCUITS FOR A SHORT TO GROUND BETWEEN THE RCM AND CLOCKSPRING

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Disconnect: Clockspring C218
- Key in ON position.

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side)
  - 2293_23_OD (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 2 on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to H48.

  NO : Go to H25.

H25 CHECK CIRCUITS CR102 (BU) AND RR102 (WH) FOR A SHORT TO GROUND BETWEEN THE RCM AND CLOCKSPRING

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

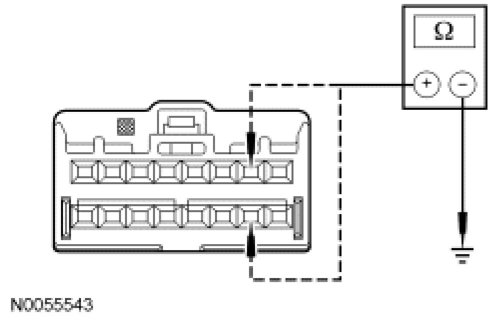

Fig. 58: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between clockspring C218-2, circuit CR102 (BU), harness side and ground; and between clockspring C218-10, circuit RR102 (WH), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR102 (BU) or circuit RR102 (WH). Go to H56.

H26 CHECK THE DRIVER AIR BAG MODULE LOOP 2 CIRCUITS FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND CLOCKSPRING

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Disconnect: Clockspring C218
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side)
  - 2293_22_OD (Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the driver air bag module/clockspring disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the driver air bag module loop 2 on-demand fault PIDs change from indicating a short to battery to an open circuit fault?**

  YES : Go to H48.

  NO : Go to H27.

H27 CHECK CIRCUITS CR102 (BU) AND RR102 (WH) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND CLOCKSPRING

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

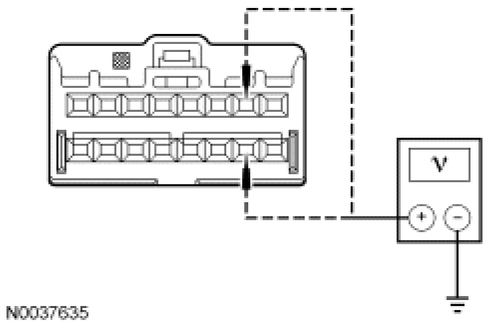

Fig. 59: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between clockspring C218-2, circuit CR102 (BU), harness side and ground; and between clockspring C218-10, circuit RR102 (WH), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR102 (BU) or circuit RR102 (WH). Go to H56.

H28 CHECK THE PASSENGER AIR BAG MODULE LOOP 1 RESISTANCE PID (P_ABAGR)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: P_ABAGR (Passenger Air Bag) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms?**

  YES : Go to H49.

  NO : Go to H29.

H29 CHECK THE PASSENGER AIR BAG MODULE LOOP 1 RESISTANCE PID (P_ABAGR) WHILE CARRYING OUT A HARNESS TEST

NOTE: Do not remove the passenger air bag module from its mounted position at this time.

- With the passenger air bag in its mounted position, continue monitoring the P_ABAGR (Passenger Air Bag) resistance PID while carrying out a harness test of the passenger air bag module circuits and accessible connectors (including any inline connectors), by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

NO : If the passenger air bag module loop 1 resistance PID is less than 1.4 ohms, go to H30.

If the passenger air bag module loop 1 resistance PID is greater than 3.5 ohms, go to H32.

H30 CHECK THE PASSENGER AIR BAG MODULE LOOP 1 FOR LOW RESISTANCE FAULT

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256a
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_24_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side)
  - 2293_25_OD (Air Bag Circuit Open - Loop No. 1, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the passenger air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the passenger air bag module loop 1 on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to H47.

  NO : Go to H31.

H31 CHECK THE RCM FOR LOW RESISTANCE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

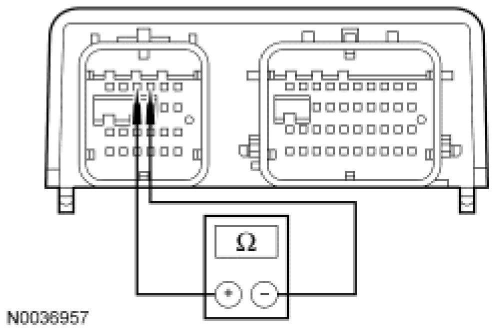

Fig. 60: Checking RCM
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a pin 3 and C310a pin 4, component side.
- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuits CR103 (GY/BU) and RR103 (VT/GN). Go to H56.

  NO : Go to H49.

H32 CHECK THE PASSENGER AIR BAG MODULE LOOP 1 FOR AN OPEN

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256a
- Connect a fused jumper wire between passenger air bag module C256a-1, circuit CR103 (GY/BU), harness side and C256a-2, circuit RR103 (VT/GN), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_24_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side)
  - 2293_25_OD (Air Bag Circuit Open - Loop No. 1, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the passenger air bag module loop 1 circuits shorted together, a low resistance fault would normally be retrieved on loop 1. Loop 2 will show an open circuit fault due to the passenger air bag being disconnected.
- **Did the passenger air bag module loop 1 on-demand fault PIDs change from indicating an**

open circuit to a low resistance fault?

YES : Go to H47.

NO : Go to H33.

H33 CHECK THE PASSENGER AIR BAG MODULE LOOP 1 CIRCUITS FOR AN OPEN BETWEEN THE PASSENGER AIR BAG MODULE AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

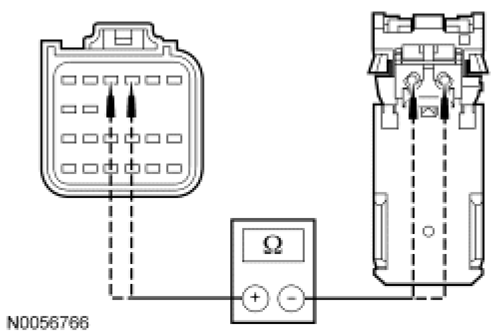

Fig. 61: Measuring Resistance Between RCM C310A-1, Circuit CR103 (GY/BU) & Passenger Air Bag Module C256A-1
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a-1, circuit CR103 (GY/BU), harness side and passenger air bag module C256a-1, circuit CR103 (GY/BU), harness side; and between RCM C310a-2, circuit RR103 (VT/GN), harness side and passenger air bag module C256a-2, circuit RR103 (VT/GN), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to H49.

  NO : REPAIR circuit CR103 (GY/BU) or circuit RR103 (VT/GN). Go to H56.

H34 CHECK THE PASSENGER AIR BAG MODULE LOOP 1 CIRCUITS FOR A SHORT TO GROUND

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256a

- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_25_OD (Air Bag Circuit Open - Loop No. 1, Front Passenger Side)
  - 2293_27_OD (Air Bag Circuit Short to Ground - Loop No. 1, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the passenger air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the passenger air bag module loop 1 on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to H47.

  NO : Go to H35.

H35 CHECK CIRCUITS CR103 (GY/BU) AND RR103 (VT/GN) FOR A SHORT TO GROUND BETWEEN THE RCM AND PASSENGER AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

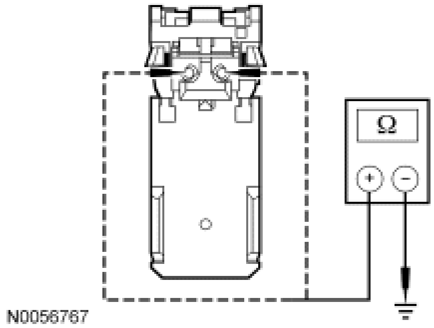

Fig. 62: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger air bag module C256a-1, circuit CR103 (GY/BU), harness side and ground; and between passenger air bag module C256a-2, circuit RR103 (VT/GN), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

REPAIR circuit CR103 (GY/BU) or circuit RR103 (VT/GN). Go to H56.

H36 CHECK CIRCUITS CR103 (GY/BU) AND RR103 (VT/GN) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND PASSENGER AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256a
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

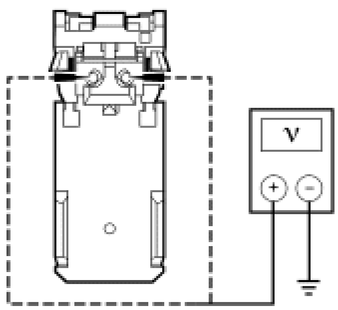

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Fig. 63: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between passenger air bag module C256a-1, circuit CR103 (GY/BU), harness side and ground; and between passenger air bag module C256a-2, circuit RR103 (VT/GN), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR103 (GY/BU) or circuit RR103 (VT/GN). Go to H56.

H37 CHECK THE PASSENGER AIR BAG MODULE LOOP 2 RESISTANCE PID (P_ABAGR2)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: P_ABAGR2 (Passenger Side Air Bag #2 Resistance) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms?**

  YES : Go to H49.

  NO : Go to H38.

H38 CHECK THE PASSENGER AIR BAG MODULE LOOP 2 RESISTANCE PID (P_ABAGR2) WHILE CARRYING OUT A HARNESS TEST

NOTE: Do not remove the passenger air bag module from its mounted position at this time.

- With the passenger air bag in its mounted position, continue monitoring the P_ABAGR2 (Passenger Side Air Bag #2 Resistance) resistance PID while carrying out a harness test of the passenger air bag module circuits and accessible connectors (including any inline connectors), by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

  NO : If the passenger air bag module loop 2 resistance PID is less than 1.4 ohms, go to H39.

  If the passenger air bag module loop 2 resistance PID is greater than 3.5 ohms, go to H41.

H39 CHECK THE PASSENGER AIR BAG MODULE LOOP 2 FOR LOW RESISTANCE FAULT

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_16_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side)
  - 2293_17_OD (Air Bag Circuit Open - Loop No. 2, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the passenger air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the passenger air bag module loop 2 on-demand fault PIDs change from indicating a low**

resistance to an open circuit fault?

YES : Go to H47.

NO : Go to H40.

H40 CHECK THE RCM FOR LOW RESISTANCE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

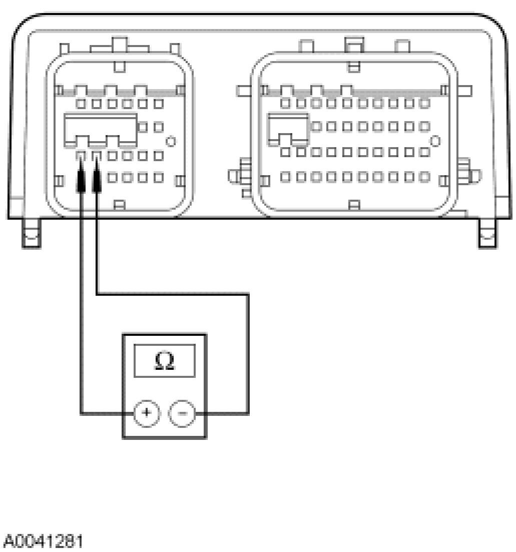


Fig. 64: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a pin 13 and C310a pin 14, component side.
- **Is resistance greater than 10,000 ohms?**

YES : REPAIR circuits CR104 (YE/GY) and RR104 (WH/BU). Go to H56.

NO : Go to H49.

H41 CHECK PASSENGER AIR BAG MODULE LOOP 2 FOR AN OPEN

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256b
- Connect a fused jumper wire between passenger air bag module C256b-1, circuit CR104 (YE/GY), harness side and C256b-2, circuit RR104 (WH/BU), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_16_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side)
  - 2293_17_OD (Air Bag Circuit Open - Loop No. 2, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the passenger air bag module loop 2 circuits shorted together, a low resistance fault would normally be retrieved on loop 2. Loop 1 will show an open circuit fault due to the passenger air bag module being disconnected.
- **Did the passenger air bag module loop 2 on-demand fault PIDs change from indicating an open circuit to a low resistance fault?**

  YES : Go to H47.

  NO : Go to H42.

H42 CHECK PASSENGER AIR BAG MODULE LOOP 2 CIRCUITS FOR AN OPEN BETWEEN THE PASSENGER AIR BAG MODULE AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from passenger air bag module C256b.
- Disconnect: RCM C310a and C310b

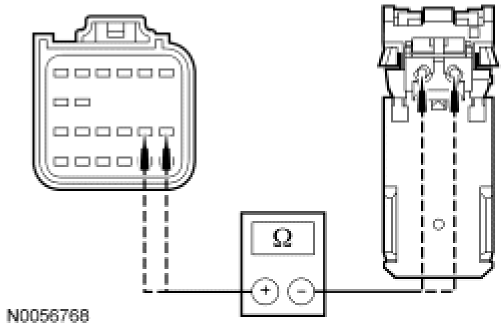


Fig. 65: Measuring Resistance Between RCM C310A-13, Circuit CR104 (YE/GY) & Passenger Air Bag Module C256B-1

Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310a-13, circuit CR104 (YE/GY), harness side and passenger air bag module C256b-1, circuit CR104 (YE/GY), harness side; and between RCM C310a-14, circuit RR104 (WH/BU), harness side and passenger air bag module C256b-2, circuit RR104 (WH/BU), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to H49.

  NO : REPAIR circuit CR104 (YE/GY) or circuit RR104 (WH/BU). Go to H56.

H43 CHECK THE PASSENGER AIR BAG MODULE LOOP 2 CIRCUITS FOR A SHORT TO GROUND

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2293_17_OD (Air Bag Circuit Open - Loop No. 2, Front Passenger Side)
  - 2293_19_OD (Air Bag Circuit Short to Ground - Loop No. 2, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2293 on-demand fault PIDs with the passenger air bag module disconnected, open circuit faults would normally be retrieved on loop 1 and loop 2.
- **Did the passenger air bag module loop 2 on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to H47.

  NO : Go to H44.

H44 CHECK CIRCUITS CR104 (YE/GY) AND RR104 (WH/BU) FOR A SHORT TO GROUND BETWEEN THE RCM AND PASSENGER AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

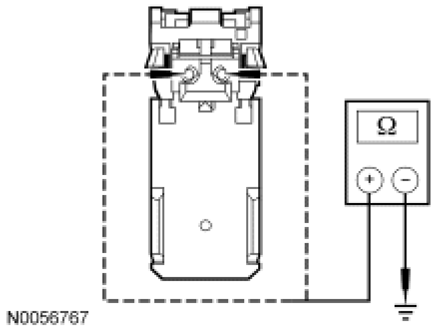


Fig. 66: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger air bag module C256b-1, circuit CR104 (YE/GY), harness side and ground; and between passenger air bag module C256b-2, circuit RR104 (WH/BU), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR104 (YE/GY) or circuit RR104 (WH/BU). Go to H56.

H45 CHECK CIRCUITS CR104 (YE/GY) AND RR104 (WH/BU) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND PASSENGER AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256b
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

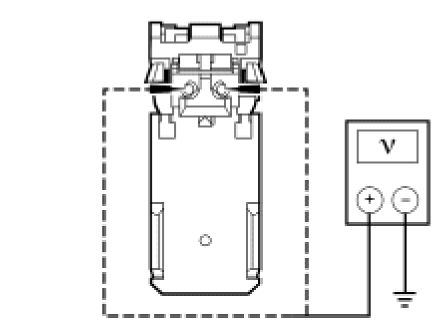


Fig. 67: Measuring Voltage

Courtesy of FORD MOTOR CO.

- Measure the voltage between passenger air bag module C256b-1, circuit CR104 (YE/GY), harness side and ground; and between passenger air bag module C256b-2, circuit RR104 (WH/BU), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to H49.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR104 (YE/GY) or circuit RR104 (WH/BU). Go to H56.

H46 CONFIRM THE DRIVER AIR BAG MODULE FAULT

NOTE:	**Make sure all restraint system components and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Install the driver air bag module. Refer to **Driver Air Bag Module**.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2293 Fault PIDs
  - Refer to PID list in Normal Operation to view B2293 fault PIDs.
- **Does the original RCM on-demand DTC B2293 fault PID indicate a fault?**

  YES : INSTALL a new driver air bag module. REFER to **Driver Air Bag Module**. Go to H56.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2293_28_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side) or 2293_20_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side), go to H50.

  For 2293_29_CM (Air Bag Circuit Open - Loop No. 1, Front Driver Side) or 2293_21_CM (Air Bag Circuit Open - Loop No. 2, Front Driver Side), go to H50.

For 2293_31_CM (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side) or 2293_23_CM (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side), go to H51.

For 2293_30_CM (Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side) or 2293_22_CM (Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side), go to H53.

H47 CONFIRM THE PASSENGER AIR BAG MODULE FAULT

NOTE: **Make sure all restraint system components and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Passenger Air Bag Module C256a and C256b (if previously disconnected)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2293 Fault PIDs
  - Refer to PID list in Normal Operation to view B2293 fault PIDs.
- **Does the original RCM on-demand DTC B2293 fault PID indicate a fault?**

  YES : INSTALL a new passenger air bag module. REFER to **Passenger Air Bag Module - Edge** or **Passenger Air Bag Module - MKX**. Go to H56.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2293_24_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side) or 2293_16_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side), go to H50.

  For 2293_25_CM (Air Bag Circuit Open - Loop No. 1, Front Passenger Side) or 2293_17_CM (Air Bag Circuit Open - Loop No. 2, Front Passenger Side), go to H50.

  For 2293_27_CM (Air Bag Circuit Short to Ground - Loop No. 1, Front Passenger Side) or 2293_19_CM (Air Bag Circuit Short to Ground - Loop No. 2, Front Passenger Side), go to H52.

  For 2293_26_CM (Air Bag Circuit Short to Battery - Loop No. 1, Front Passenger Side) or 2293_18_CM (Air Bag Circuit Short to Battery - Loop No. 2, Front Passenger Side), go to H54.

H48 CONFIRM THE CLOCKSPRING FAULT

NOTE: Make sure all restraint system components and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Clockspring C218
- Install the driver air bag module. Refer to **Driver Air Bag Module**.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2293 Fault PIDs
  - Refer to PID list in Normal Operation to view B2293 fault PIDs.
- **Does the original RCM on-demand DTC B2293 fault PID indicate a fault?**

  YES : INSTALL a new clockspring. REFER to **Clockspring**. Go to H56.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2293_28_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side) or 2293_20_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side), go to H50.

  For 2293_29_CM (Air Bag Circuit Open - Loop No. 1, Front Driver Side) or 2293_21_CM (Air Bag Circuit Open - Loop No. 2, Front Driver Side), go to H50.

  For 2293_31_CM (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side) or 2293_23_CM (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side), go to H51.

  For 2293_30_CM (Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side) or 2293_22_CM (Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side), go to H53.

H49 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the RCM electrical connectors are connected before carrying out the on-demand self test. If

not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- If previously removed, install the driver air bag module. Refer to **Driver Air Bag Module**.
- Connect: Clockspring C218 (if previously disconnected)
- Connect: Passenger Air Bag Module C256a and C256b (if previously disconnected)
- Connect: RCM C310a and C310b (if previously disconnected)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2293 Fault PIDs
  - Refer to PID list in Normal Operation to view B2293 fault PIDs.
- **Does the original RCM on-demand DTC B2293 fault PID indicate a fault?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to H56.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2293_28_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side) or 2293_20_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side), go to H50.

  For 2293_29_CM (Air Bag Circuit Open - Loop No. 1, Front Driver Side) or 2293_21_CM (Air Bag Circuit Open - Loop No. 2, Front Driver Side), go to H50.

  For 2293_31_CM (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side) or 2293_23_CM (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side), go to H51.

  For 2293_30_CM (Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side) or 2293_22_CM (Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side), go to H53.

  For 2293_24_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side) or 2293_16_CM (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side), go to H50.

  For 2293_25_CM (Air Bag Circuit Open - Loop No. 1, Front Passenger Side) or 2293_17_CM (Air Bag Circuit Open - Loop No. 2, Front Passenger Side), go to H50.

For 2293_27_CM (Air Bag Circuit Short to Ground - Loop No. 1, Front Passenger Side) or 2293_19_CM (Air Bag Circuit Short to Ground - Loop No. 2, Front Passenger Side), go to H52.

For 2293_26_CM (Air Bag Circuit Short to Battery - Loop No. 1, Front Passenger Side) or 2293_18_CM (Air Bag Circuit Short to Battery - Loop No. 2, Front Passenger Side), go to H54.

H50 CHECK THE DRIVER OR PASSENGER AIR BAG MODULE FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- If the fault PID was reported for the driver air bag module loop 1 or loop 2, monitor the following on the scan tool:
  - D_ABAGR (Driver Air Bag) resistance PID.
  - D_ABAGR2 (Driver Air Bag #2 Resistance) resistance PID.
  - 2293_28_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Driver Side) and 2293_20_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Driver Side) on-demand fault PID.
  - 2293_29_OD (Air Bag Circuit Open - Loop No. 1, Front Driver Side) and 2293_21_OD (Air Bag Circuit Open - Loop No. 2, Front Driver Side) on-demand fault PID.
- If the fault PID was reported for the passenger air bag module loop 1 or loop 2, monitor the following on the scan tool:
  - P_ABAGR (Passenger Air Bag) resistance PID.
  - P_ABAGR2 (Passenger Side Air Bag #2 Resistance) resistance PID.
  - 2293_24_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 1, Front Passenger Side) and 2293_16_OD (Air Bag Inflator Circuit Resistance Low - Loop No. 2, Front Passenger Side) on-demand fault PID.
  - 2293_25_OD (Air Bag Circuit Open - Loop No. 1, Front Passenger Side) and 2293_17_OD (Air Bag Circuit Open - Loop No. 2, Front Passenger Side) on-demand fault PID.

NOTE: **A resistance PID that reads between 0.9 and 1.7 ohms for driver air bag and between 0.9 and 1.4 ohms for passenger air bag or between 4.4 and 5 ohms for driver air bag or between 3.5 and 5 ohms for the passenger air bag may or may not set an on-demand DTC yet indicates a strong potential of an intermittent fault condition. Make sure to thoroughly check wire harness(es), connectors and terminals as they are the likely source of the fault condition.**

NOTE: **Do not remove the driver or passenger air bag module from its mounted position at this time.**

- With the driver or passenger air bag in its mounted position, attempt to recreate the fault by wiggling connectors (including any inline connectors), and flexing the wire harness, tilting and rotating the steering wheel frequently.
- **Does any driver or passenger air bag on-demand fault PID indicate a fault or is any air bag resistance PID less than 1.7 ohms for driver air bag and less than 1.4 ohms for passenger air bag or greater than 4.4 ohms for driver air bag and greater than 3.5 ohms for passenger air**

bag?

YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

NO : Go to H55.

H51 CHECK THE DRIVER AIR BAG MODULE CIRCUITS FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Monitor the 2293_23_OD (Air Bag Circuit Short to Ground - Loop No. 2, Front Driver Side) and 2293_31_OD (Air Bag Circuit Short to Ground - Loop No. 1, Front Driver Side) on-demand fault PIDs.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.

NOTE: **Do not remove the driver air bag module from its mounted position at this time.**

- With the driver air bag in its mounted position, attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.
- **Does any RCM on-demand B2293 fault PID indicate a fault is present?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

  NO : Go to H55.

H52 CHECK THE PASSENGER AIR BAG MODULE CIRCUITS FOR AN INTERMITTENT SHORT TO GROUND FAULT

NOTE: **When monitoring fault and/or resistance PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.**

- Key in ON position.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.
- Monitor the 2293_27_OD (Air Bag Circuit Short to Ground - Loop No. 1, Front Passenger Side) and 2293_19_OD (Air Bag Circuit Short to Ground - Loop No. 2, Front Passenger Side) on-demand fault PIDs.

NOTE: **Do not remove the passenger air bag module from its mounted position at this time.**

- With the passenger air bag in its mounted position, attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any RCM on-demand B2293 fault PID indicate a fault is present?**

YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

NO : Go to H55.

H53 CHECK THE DRIVER AIR BAG MODULE CIRCUITS FOR AN INTERMITTENT SHORT TO BATTERY FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the driver air bag module. Refer to **Driver Air Bag Module**.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.
- Monitor the 2293_30_OD (Air Bag Circuit Short to Battery - Loop No. 1, Front Driver Side) and 2293_22_OD (Air Bag Circuit Short to Battery - Loop No. 2, Front Driver Side) on-demand fault PIDs.
- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.
- **Does any driver air bag module short to battery fault PID indicate a fault?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

  NO : Go to H55.

H54 CHECK THE PASSENGER AIR BAG MODULE CIRCUITS FOR AN INTERMITTENT SHORT TO VOLTAGE FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Air Bag Module C256
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.
- Monitor the 2293_26_OD (Air Bag Circuit Short to Battery - Loop No. 1, Front Passenger Side) and 2293_18_OD (Air Bag Circuit Short to Battery - Loop No. 2, Front Passenger Side) on-demand fault PIDs.

NOTE: **Do not remove the passenger air bag module from its mounted**

position at this time.

- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any passenger air bag module short to battery fault PID indicate a fault?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

  NO : Go to H55.

H55 CHECK THE HARNESS AND CONNECTORS

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- If the fault PID was reported for the driver air bag:
  - remove the driver air bag module.
  - inspect connector(s) (including any inline connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
  - inspect wire harness for any damage, pinched, cut or pierced wires.
- If the fault PID was reported for the passenger air bag:
  - disconnect the passenger air bag module C256.
  - inspect connector(s) (including any inline connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
  - inspect wire harness for any damage, pinched, cut or pierced wires.
- **Were any concerns found?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to H56.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to H56.

H56 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at

this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test I: DTC B2294 - Restraint System - Curtain Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) continuously monitors the driver and passenger safety canopy module circuits (loop) for an acceptable resistance range, unacceptable voltage and shorts to ground. The RCM will set on-demand and store DTC B2294 in memory if any of these conditions are detected to be outside of their respectful range. If the RCM detects a fault, it will send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

As the RCM continuously monitors safety canopy module circuits for resistance, it expects a normal resistance between 1.4 and 3.5 ohms.

If a loop resistance is between 0.9 and 1.4 or between 3.5 and 5 ohms, there exists a strong potential for an intermittent fault, a DTC may or may not set. The RCM will also set an on-demand DTC if the loop resistance is less than 0.9 ohm or greater than 5 ohms.

Fault PIDs[a]	Description	Fault Trigger Condition
2294_24_OD and 2294_24_CM	A - B or A - C Pillar Curtain Circuit Resistance Low, Passenger Side	When the RCM measures less than 0.9 ohm between the passenger safety canopy circuits, a fault will be indicated.
2294_25_OD and 2294_25_CM	A - B or A - C Pillar Curtain Circuit Open, Passenger Side	When the RCM measures greater than 5 ohms between the passenger safety canopy circuits, a fault will be indicated.
2294_26_OD and 2294_26_CM	A - B or A - C Pillar Curtain Circuit Short to Ground, Passenger Side	When the RCM senses a short to ground on either on the passenger safety canopy circuits, a fault will be indicated.

2294_27_OD and 2294_27_CM	A - B or A - C Pillar Curtain Circuit Short to Battery, Passenger Side	When the RCM senses a short to voltage on either on the passenger safety canopy circuits, a fault will be indicated.
2294_28_OD and 2294_28_CM	A - B or A - C Pillar Curtain Circuit Resistance Low, Driver Side	When the RCM measures less than 0.9 ohm between the driver safety canopy circuits, a fault will be indicated.
2294_29_OD and 2294_29_CM	A - B or A - C Pillar Curtain Circuit Open, Driver Side	When the RCM measures greater than 5 ohms between the driver safety canopy circuits, a fault will be indicated.
2294_30_OD and 2294_30_CM	A - B or A - C Pillar Curtain Circuit Short to Ground, Driver Side	When the RCM senses a short to ground on either on the driver safety canopy circuits, a fault will be indicated.
2294_31_OD and 2294_31_CM	A - B or A - C Pillar Curtain Circuit Short to Battery, Driver Side	When the RCM senses a short to voltage on either on the driver safety canopy circuits, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B2294. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B2294.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Safety canopy module
- RCM

PINPOINT TEST I: DTC B2294 - RESTRAINT SYSTEM - CURTAIN FAULT

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air**

curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

I1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2294 Fault PIDs
  - Refer to PID list in Normal Operation to view B2294 fault PIDs.
- **Do any RCM on-demand DTC B2294 fault PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test.

  Using the fault PIDs recorded, go to the appropriate pinpoint test step.

  For 2294_28_OD (A - B or A - C Pillar Curtain Circuit Resistance Low, Driver Side), go to I2.

  For 2294_29_OD (A - B or A - C Pillar Curtain Circuit Open, Driver Side), go to I2.

  For 2294_30_OD (A - B or A - C Pillar Curtain Circuit Short to Ground, Driver Side), go to I9.

  For 2294_31_OD (A - B or A - C Pillar Curtain Circuit Short to Battery, Driver Side), go to I11.

  For 2294_24_OD (A - B or A - C Pillar Curtain Circuit Resistance Low, Passenger Side), go to I12.

  For 2294_25_OD (A - B or A - C Pillar Curtain Circuit Open, Passenger Side), go to I12.

  For 2294_26_OD (A - B or A - C Pillar Curtain Circuit Short to Ground, Passenger Side), go to

I19.

For 2294_27_OD (A - B or A - C Pillar Curtain Circuit Short to Battery, Passenger Side), go to I21.

NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time.

Using the fault PIDs recorded, go to the appropriate pinpoint test step.

For 2294_28_CM (A - B or A - C Pillar Curtain Circuit Resistance Low, Driver Side), 2294_24_CM (A - B or A - C Pillar Curtain Circuit Resistance Low, Passenger Side), 2294_29_CM (A - B or A - C Pillar Curtain Circuit Open, Driver Side) or 2294_25_CM (A - B or A - C Pillar Curtain Circuit Open, Passenger Side), go to I24.

For 2294_30_CM (A - B or A - C Pillar Curtain Circuit Short to Ground, Driver Side) or 2294_26_CM (A - B or A - C Pillar Curtain Circuit Short to Ground, Passenger Side), go to I26.

For 2294_31_CM (A - B or A - C Pillar Curtain Circuit Short to Battery, Driver Side) or 2294_27_CM (A - B or A - C Pillar Curtain Circuit Short to Battery, Passenger Side), go to I27.

I2 CHECK THE DRIVER SAFETY CANOPY MODULE RESISTANCE PID (DCURTL1)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: DCURTL1 (Curtain Airbag Driver Loop No.1 Resistance) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms?**

  YES : Go to I23.

  NO : Go to I3.

I3 CHECK THE DRIVER SAFETY CANOPY MODULE RESISTANCE PID (DCURTL1) WHILE CARRYING OUT A HARNESS TEST

- Continue monitoring the DCURTL1 (Curtain Airbag Driver Loop No.1 Resistance) resistance PID while carrying out a harness test of the driver safety canopy circuits and accessible connectors (including any inline connectors) by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to I28.

  NO : If the driver safety canopy resistance PID is less than 1.4 ohm, go to I4.

  If the driver safety canopy resistance PID is greater than 3.5 ohms, go to I7.

I4 CHECK THE DRIVER SAFETY CANOPY FOR LOW RESISTANCE FAULT

NOTE: This pinpoint test step will attempt to change the fault reported by the

RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Canopy C9018
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2294_28_OD (A - B or A - C Pillar Curtain Circuit Resistance Low, Driver Side)
  - 2294_29_OD (A - B or A - C Pillar Curtain Circuit Open, Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2294 on-demand fault PIDs with the driver safety canopy disconnected, an open circuit fault would normally be retrieved.
- **Did the driver safety canopy on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to I22.

  NO : Go to I5.

I5 CHECK FOR A SHORT BETWEEN DRIVER SAFETY CANOPY MODULE CIRCUITS CR109 (BN/BU) AND RR109 (BU/GN)

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

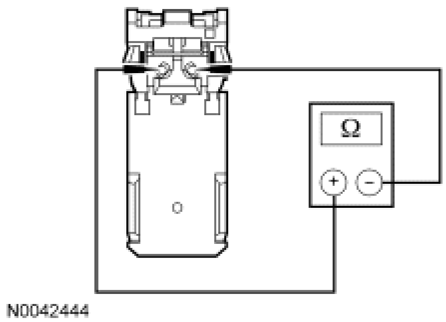


Fig. 68: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver safety canopy C9018-1, circuit CR109 (BN/BU), harness side and C9018-2, circuit RR109 (BU/GN), harness side.

- **Is resistance greater than 10,000 ohms?**

  YES : Go to I23.

  NO : Go to I6.

I6 CHECK THE RCM FOR LOW RESISTANCE

- Disconnect: RCM C310a and C310b

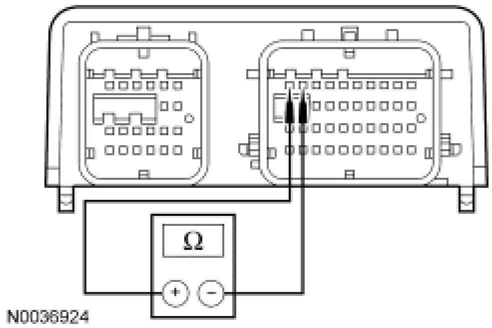


Fig. 69: Checking RCM For Low Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b pin 1, component side, and C310b pin 2, component side.
- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuits CR109 (BN/BU) and RR109 (BU/GN). Go to I28.

  NO : Go to I23.

I7 CHECK THE DRIVER SAFETY CANOPY MODULE FOR AN OPEN

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Canopy C9018
- Connect a fused jumper wire between driver safety canopy C9018-1, circuit CR109 (BN/BU), harness side and C9018-2, circuit RR109 (BU/GN), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
    - 2294_28_OD (A - B or A - C Pillar Curtain Circuit Resistance Low, Driver Side)
    - 2294_29_OD (A - B or A - C Pillar Curtain Circuit Open, Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2294 on-demand fault PIDs with the driver safety canopy circuits shorted together, a low resistance fault would normally be retrieved.
- **Did the driver safety canopy on-demand fault PIDs change from indicating an open circuit to a low resistance fault?**

  YES : Go to I22.

  NO : Go to I8.

I8 CHECK CIRCUITS CR109 (BN/BU) AND RR109 (BU/GN) FOR AN OPEN BETWEEN THE RCM AND DRIVER SAFETY CANOPY MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the driver safety canopy C9018.
- Disconnect: RCM C310a and C310b

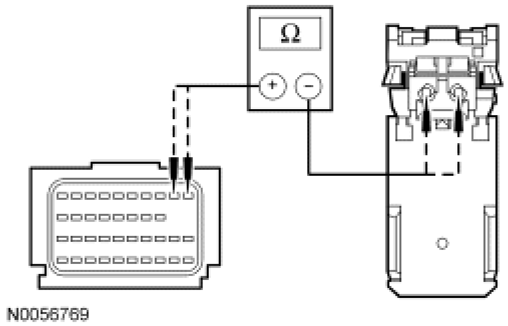


Fig. 70: Measuring Resistance Between RCM C310B-1, Circuit CR109 (BN/BU) & Driver Safety Canopy C9018-1
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-1, circuit CR109 (BN/BU), harness side and driver safety canopy C9018-1, circuit CR109 (BN/BU), harness side; and between RCM C310b-2, circuit RR109 (BU/GN), harness side and driver safety canopy C9018-2, circuit RR109 (BU/GN), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to I23.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR109 (BN/BU) or circuit RR109 (BU/GN). Go to I28.

I9 CHECK THE DRIVER SAFETY CANOPY MODULE FOR A SHORT TO GROUND

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Canopy C9018
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2294_29_OD (A - B or A - C Pillar Curtain Circuit Open, Driver Side)
  - 2294_30_OD (A - B or A - C Pillar Curtain Circuit Short to Ground, Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2294 on-demand fault PIDs with the driver safety canopy disconnected, an open circuit fault would normally be retrieved.
- **Did the driver safety canopy on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to I22.

  NO : Go to I10.

I10 CHECK CIRCUITS CR109 (BN/BU) AND RR109 (BU/GN) FOR A SHORT TO GROUND BETWEEN THE RCM AND DRIVER SAFETY CANOPY MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

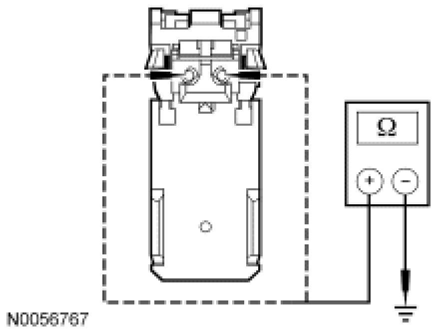


Fig. 71: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver safety canopy C9018-1, circuit CR109 (BN/BU), harness side and ground; and between driver safety canopy C9018-2, circuit RR109 (BU/GN), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to I23.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR109 (BN/BU) or circuit RR109 (BU/GN). Go to I28.

I11 CHECK CIRCUITS CR109 (BN/BU) AND RR109 (BU/GN) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND DRIVER SAFETY CANOPY MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Canopy C9018
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

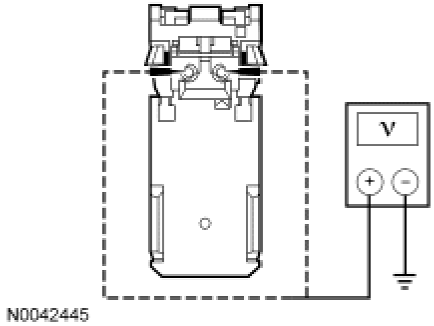


Fig. 72: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between driver safety canopy C9018-1, circuit CR109 (BN/BU), harness side and ground; and between driver safety canopy C9018-2, circuit RR109 (BU/GN), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to I23.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

REPAIR circuit CR109 (BN/BU) or circuit RR109 (BU/GN). Go to I28.

I12 CHECK THE PASSENGER SAFETY CANOPY MODULE RESISTANCE PID

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: PCURTL1 (Curtain Airbag Passenger Loop No.1 Resistance) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms?**

  YES : Go to I23.

  NO : Go to I13.

I13 CHECK THE PASSENGER SAFETY CANOPY MODULE RESISTANCE PID (PCURTL1) WHILE CARRYING OUT A HARNESS TEST

- Continue monitoring the PCURTL1 (Curtain Airbag Passenger Loop No.1 Resistance) resistance PID while carrying out a harness test of the passenger safety canopy circuits and accessible connectors (including any inline connectors), by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to I28.

  NO : If the passenger safety canopy resistance PID is less than 1.4 ohm, go to I14.

  If the passenger safety canopy resistance PID is greater than 3.5 ohms, go to I17.

I14 CHECK THE PASSENGER SAFETY CANOPY MODULE FOR LOW RESISTANCE FAULT

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Canopy C9019
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs

- 2294_24_OD (A - B or A - C Pillar Curtain Circuit Resistance Low, Passenger Side)
- 2294_25_OD (A - B or A - C Pillar Curtain Circuit Open, Passenger Side)

- **DIAGNOSTIC TIP:** When viewing DTC B2294 on-demand fault PIDs with the passenger safety canopy disconnected, an open circuit fault would normally be retrieved.
- **Did the passenger safety canopy on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to I22.

  NO : Go to I15.

I15 CHECK FOR A SHORT BETWEEN PASSENGER SAFETY CANOPY MODULE CIRCUITS CR111 (BN/WH) AND RR111 (YE/VT)

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

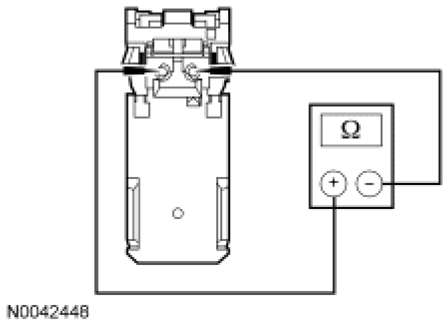


Fig. 73: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger safety canopy module C9019-1, circuit CR111 (BN/WH), harness side and C9019-2, circuit RR111 (YE/VT), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to I23.

  NO : Go to I16.

I16 CHECK THE RCM FOR LOW RESISTANCE

- Disconnect: RCM C310a and C310b

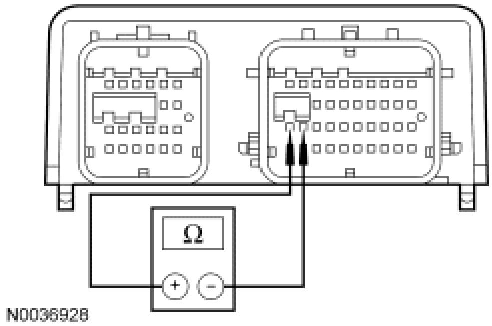


Fig. 74: Checking RCM
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b pin 21, component side, and C310b pin 22, component side.
- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuits CR111 (BN/WH) and RR111 (YE/VT). Go to I28.

  NO : Go to I23.

I17 CHECK THE PASSENGER SAFETY CANOPY MODULE FOR AN OPEN

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Canopy C9007
- Connect a fused jumper wire between passenger safety canopy C9019-1, circuit CR111 (BN/WH), harness side and C9019-3, circuit RR111 (YE/VT), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2294_24_OD (A - B or A - C Pillar Curtain Circuit Resistance Low, Passenger Side)
  - 2294_25_OD (A - B or A - C Pillar Curtain Circuit Open, Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2294 on-demand fault PIDs with the passenger safety canopy circuits shorted together, a low resistance fault would normally be retrieved.
- **Did the passenger safety canopy on-demand fault PIDs change from indicating an open**

circuit to a low resistance fault?

YES : Go to I22.

NO : Go to I18.

I18 CHECK CIRCUITS CR111 (BN/WH) AND RR111 (YE/VT) FOR AN OPEN BETWEEN THE RCM AND PASSENGER SAFETY CANOPY MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the passenger safety canopy C9019.
- Disconnect: RCM C310a and C310b

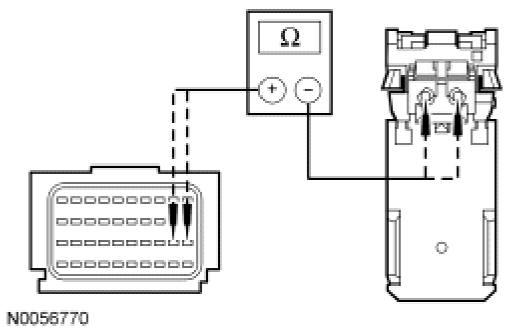


Fig. 75: Measuring Resistance Between RCM C310B-21, Circuit CR111 (BN/WH) & Passenger Safety Canopy Module C9019-1
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-21, circuit CR111 (BN/WH), harness side and passenger safety canopy module C9019-1, circuit CR111 (BN/WH), harness side; and between RCM C310b-22, circuit RR111 (YE/VT), harness side and passenger safety canopy module C9019-2, circuit RR111 (YE/VT), harness side.
- **Are resistances less than 0.5 ohm?**

YES : Go to I23.

NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

REPAIR circuit CR111 (BN/WH) or circuit RR111 (YE/VT). Go to I28.

I19 CHECK THE PASSENGER SAFETY CANOPY MODULE FOR A SHORT TO GROUND

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.

- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Canopy C9019
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2294_25_OD (A - B or A - C Pillar Curtain Circuit Open, Passenger Side)
  - 2294_26_OD (A - B or A - C Pillar Curtain Circuit Short to Ground, Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2294 on-demand fault PIDs with the passenger safety canopy disconnected, an open circuit fault would normally be retrieved.
- **Did the passenger safety canopy on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to I22.

  NO : Go to I20.

I20 CHECK CIRCUITS CR111 (BN/WH) AND RR111 (YE/VT) FOR A SHORT TO GROUND BETWEEN THE RCM AND PASSENGER SAFETY CANOPY MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

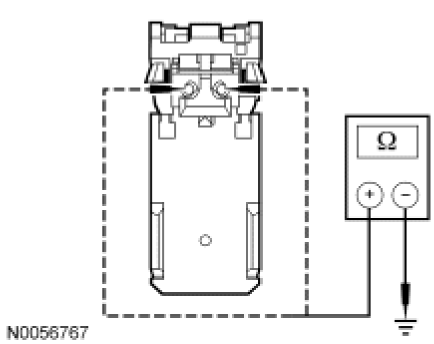


Fig. 76: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger safety canopy C9019-1, circuit CR111 (BN/WH), harness side and ground; and between passenger safety canopy C9019-2, circuit RR111 (YE/VT), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

YES : Go to I23.

NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

REPAIR circuit CR111 (BN/WH) or circuit RR111 (YE/VT). Go to I28.

I21 CHECK CIRCUITS CR111 (BN/WH) AND RR111 (YE/VT) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND PASSENGER SAFETY CANOPY MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Canopy C9019
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

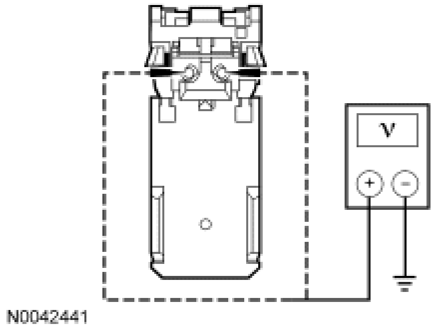


Fig. 77: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between passenger safety canopy C9019-1, circuit CR111 (BN/WH), harness side and ground; and between passenger safety canopy C9019-2, circuit RR111 (YE/VT), harness side and ground.
- **Are voltages less than 0.2 volt?**

YES : Go to I23.

NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

REPAIR circuit CR111 (BN/WH) or circuit RR111 (YE/VT). Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to I28.

I22 CONFIRM THE SAFETY CANOPY MODULE FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Affected Safety Canopy C9018 (Driver) C9019 (Passenger)
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2294 Fault PIDs
    - Refer to PID list in Normal Operation to view B2294 fault PIDs.
- **Does the original RCM on-demand DTC B2294 fault PID indicate a fault?**

  YES : INSTALL a new driver or passenger safety canopy. REFER to **Safety Canopy Module**. Go to I28.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2294_28_CM (A - B or A - C Pillar Curtain Circuit Resistance Low, Driver Side), 2294_24_CM (A - B or A - C Pillar Curtain Circuit Resistance Low, Passenger Side), 2294_29_CM (A - B or A - C Pillar Curtain Circuit Open, Driver Side) or 2294_25_CM (A - B or A - C Pillar Curtain Circuit Open, Passenger Side), go to I24.

  For 2294_30_CM (A - B or A - C Pillar Curtain Circuit Short to Ground, Driver Side) or 2294_26_CM (A - B or A - C Pillar Curtain Circuit Short to Ground, Passenger Side), go to I26.

  For 2294_31_CM (A - B or A - C Pillar Curtain Circuit Short to Battery, Driver Side) or 2294_27_CM (A - B or A - C Pillar Curtain Circuit Short to Battery, Passenger Side), go to I27.

I23 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.

- If previously directed to disconnect any SRS component(s):
  - Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
  - Connect all previously disconnected restraint system component(s), including RCM C310a and C310b.
  - Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2294 Fault PIDs
  - Refer to PID list in Normal Operation to view B2294 fault PIDs.
- **Does the original RCM on-demand DTC B2294 fault PID indicate a fault?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to I28.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2294_28_CM (A - B or A - C Pillar Curtain Circuit Resistance Low, Driver Side), 2294_24_CM (A - B or A - C Pillar Curtain Circuit Resistance Low, Passenger Side), 2294_29_CM (A - B or A - C Pillar Curtain Circuit Open, Driver Side) or 2294_25_CM (A - B or A - C Pillar Curtain Circuit Open, Passenger Side), go to I24.

  For 2294_30_CM (A - B or A - C Pillar Curtain Circuit Short to Ground, Driver Side) or 2294_26_CM (A - B or A - C Pillar Curtain Circuit Short to Ground, Passenger Side), go to I26.

  For 2294_31_CM (A - B or A - C Pillar Curtain Circuit Short to Battery, Driver Side) or 2294_27_CM (A - B or A - C Pillar Curtain Circuit Short to Battery, Passenger Side), go to I27.

I24 CHECK FOR AN INTERMITTENT SAFETY CANOPY LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2294 Fault PIDs
  - Refer to PID list in Normal Operation to view B2294 fault PIDs.
- If the fault PID was reported for the driver safety canopy, monitor the DCURTL1 (Curtain Airbag Driver Loop No.1 Resistance) resistance PID on the scan tool.
- If the fault PID was reported for the passenger safety canopy, monitor the PCURTL1 (Curtain Airbag Passenger Loop No.1 Resistance) resistance PID on the scan tool.

NOTE: **A resistance PID that reads between 0.9 and 1.4 ohms or between 3.5 and 5 ohms may or may not set an on-demand DTC yet indicates a**

strong potential of an intermittent fault condition. Make sure to thoroughly check wire harness(es), connectors and terminals as they are the likely source of the fault condition.

- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any RCM on-demand B2294 fault PID indicate a fault is present or is any safety canopy resistance PID less than 1.4 ohms or greater than 3.5 ohms?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to I28.

  NO : Go to I25.

I25 CHECK HARNESS AND CONNECTORS

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect the affected safety canopy module C9006 (driver) or C9007 (passenger):
  - inspect connector(s) (including any inline connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
  - inspect wire harness for any damage, pinched, cut or pierced wires.
- **Were any concerns found?**

  YES : REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to I28.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to I28.

I26 CHECK FOR AN INTERMITTENT SAFETY CANOPY MODULE SHORT TO GROUND FAULT

NOTE: **When monitoring fault and/or resistance PIDs with the scan tool, up to a 2-second delay for the scan tool display to update should be expected.**

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- If the fault PID was reported for the driver safety canopy, monitor the 2294_30_OD (A - B or A - C Pillar Curtain Circuit Short to Ground, Driver Side) fault PID on the scan tool.
- If the fault PID was reported for the passenger safety canopy, monitor the 2294_26_OD (A - B or A - C Pillar Curtain Circuit Short to Ground, Passenger Side) fault PID on the scan tool.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to a 2-second delay for the scan tool display to update.
- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing

the wire harness.

- **Does any safety canopy on-demand fault PID indicate a fault?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to I28.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to I28.

I27 CHECK FOR AN INTERMITTENT SAFETY CANOPY MODULE SHORT TO BATTERY FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- If the fault PID was reported for the driver safety canopy, disconnect the driver safety canopy module C9018.
- If the fault PID was reported for the passenger safety canopy, disconnect the passenger safety canopy module C9019.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to a 2-second delay for the scan tool display to update.
- Monitor the 2294_31_OD (A - B or A - C Pillar Curtain Circuit Short to Battery, Driver Side) or 2294_27_OD (A - B or A - C Pillar Curtain Circuit Short to Battery, Passenger Side) fault PIDs on the scan tool.
- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any safety canopy on-demand fault PID indicate a fault?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to I28.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to I28.

I28 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an**

accidental deployment.

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test J: DTC B2295 - Restraint System - Side Air Bag Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

A seat side air bag module provides protection of the thorax area (between the neck and abdomen) of the body, working in conjunction with the head protection provided by a safety canopy module. Refer to **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.

The restraints control module (RCM) continuously monitors the driver and passenger side air bag module circuits (loop) for an acceptable resistance range, unacceptable voltage and shorts to ground. The RCM will set on-demand and store DTC B2295 in memory if any of these conditions are detected to be outside of their respectful range. If the RCM detects a fault, it will send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

As the RCM continuously monitors side air bag module circuits for resistance, it expects a normal resistance between 1.4 and 3.5 ohms.

If a loop resistance is between 0.9 and 1.4 or between 3.5 and 5 ohms, there exists a strong potential for an intermittent fault, a DTC may or may not set. The RCM will also set an on-demand DTC if the loop resistance is less than 0.9 ohm or greater than 5 ohms.

Fault PIDs[a]	Description	Fault Trigger Condition
2295_24_OD and 2295_24_CM	Side Air Bag Circuit Resistance Low, Front Passenger Side	When the RCM measures less than 0.9 ohm between the passenger side air bag module circuits, a fault will be indicated.
		When the RCM measures

2295_25_OD and 2295_25_CM	Side Air Bag Circuit Open, Front Passenger Side	greater than 5 ohms between the passenger side air bag module circuits, a fault will be indicated.
2295_26_OD and 2295_26_CM	Side Air Bag Circuit Short to Ground, Front Passenger Side	When the RCM senses a short to ground on either of the passenger side air bag module circuits, a fault will be indicated.
2295_27_OD and 2295_27_CM	Side Air Bag Circuit Short to Battery, Front Passenger Side	When the RCM senses a short to voltage on either of the passenger side air bag module circuits, a fault will be indicated.
2295_28_OD and 2295_28_CM	Side Air Bag Circuit Resistance Low, Front Driver Side	When the RCM measures less than 0.9 ohm between the driver side air bag module circuits, a fault will be indicated.
2295_29_OD and 2295_29_CM	Side Air Bag Circuit Open, Front Driver Side	When the RCM measures greater than 5 ohms between the driver side air bag module circuits, a fault will be indicated.
2295_30_OD and 2295_30_CM	Side Air Bag Circuit Short to Ground, Front Driver Side	When the RCM senses a short to ground on either on the driver side air bag module circuits, a fault will be indicated.
2295_31_OD and 2295_31_CM	Side Air Bag Circuit Short to Battery, Front Driver Side	When the RCM senses a short to voltage on either on the driver side air bag module circuits, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B2295. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B2295.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Side air bag module
- RCM

PINPOINT TEST J: DTC B2295 - RESTRAINT SYSTEM - SIDE AIR BAG FAULT

WARNING: Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.

WARNING: Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

J1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2295 Fault PIDs
  - Refer to PID list in Normal Operation to view B2295 fault PIDs.
- **Do any RCM on-demand DTC B2295 fault PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test.

  Using the on-demand fault PIDs recorded, go to the appropriate pinpoint test step.

  For 2295_28_OD (Side Air Bag Circuit Resistance Low, Front Driver Side), go to J2.

For 2295_29_OD (Side Air Bag Circuit Open, Front Driver Side), go to J2.

For 2295_30_OD (Side Air Bag Circuit Short to Ground, Front Driver Side), go to J9.

For 2295_31_OD (Side Air Bag Circuit Short to Battery, Front Driver Side), go to J11.

For 2295_24_OD (Side Air Bag Circuit Resistance Low, Front Passenger Side), go to J12.

For 2295_25_OD (Side Air Bag Circuit Open, Front Passenger Side), go to J12.

For 2295_26_OD (Side Air Bag Circuit Short to Ground, Front Passenger Side), go to J19.

For 2295_27_OD (Side Air Bag Circuit Short to Battery, Front Passenger Side), go to J21.

NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time.

Using the fault PIDs recorded, go to the appropriate pinpoint test step.

For 2295_28_CM (Side Air Bag Circuit Resistance Low, Front Driver Side), 2295_24_CM (Side Air Bag Circuit Resistance Low, Front Passenger Side), 2295_29_CM (Side Air Bag Circuit Open, Front Driver Side) or 2295_25_CM (Side Air Bag Circuit Open, Front Passenger Side), go to J24.

For 2295_30_CM (Side Air Bag Circuit Short to Ground, Front Driver Side) or 2295_26_CM (Side Air Bag Circuit Short to Ground, Front Passenger Side), go to J26.

For 2295_31_CM (Side Air Bag Circuit Short to Battery, Front Driver Side) or 2295_27_CM (Side Air Bag Circuit Short to Battery, Front Passenger Side), go to J27.

J2 CHECK THE DRIVER SIDE AIR BAG MODULE RESISTANCE PID (DS_AB)

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: DS_AB (Driver Side Impact Air Bag Resistance) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms?**

  YES : Go to J23.

  NO : Go to J3.

J3 CHECK THE DRIVER SIDE AIR BAG RESISTANCE PID (DS_AB) WHILE CARRYING OUT A HARNESS TEST

- Continue monitoring the DS_AB (Driver Side Impact Air Bag Resistance) resistance PID while carrying out a harness test of the driver side air bag circuits and accessible connectors (including any inline connectors) by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out the harness test?**

YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to J28.

NO : If the driver side air bag resistance PID is less than 1.4 ohms, go to J4.

If the driver side air bag resistance PID is greater than 3.5 ohms, go to J7.

J4 CHECK THE DRIVER SIDE AIR BAG MODULE FOR LOW RESISTANCE FAULT

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

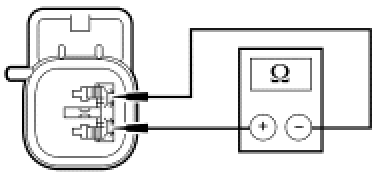
- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Side Air Bag C328
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2295_28_OD (Side Air Bag Circuit Resistance Low, Front Driver Side)
  - 2295_29_OD (Side Air Bag Circuit Open, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2295 on-demand fault PIDs with the driver side air bag disconnected, an open circuit fault would normally be retrieved.
- **Did the driver side air bag on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to J22.

  NO : Go to J5.

J5 CHECK FOR A SHORT BETWEEN DRIVER SIDE AIR BAG MODULE CIRCUITS CR105 (GN/BU) AND RR105 (GY/YE)

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.



N0058102

Fig. 78: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver side air bag C328-1, circuit CR105 (GN/BU), harness side and C328-2, circuit RR105 (GY/YE), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to J22.

  NO : Go to J6.

J6 CHECK THE RCM FOR LOW RESISTANCE

- Disconnect: RCM C310a and C310b

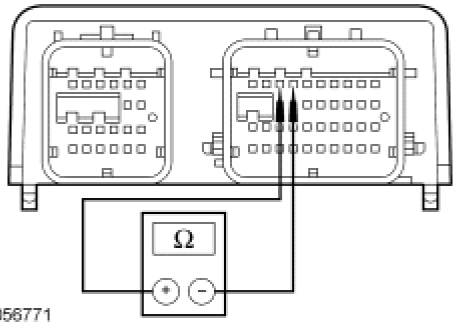


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Fig. 79: Measuring Resistance Between RCM C310B Pin 3, Component Side, & C310B Pin 4
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b pin 3, component side, and C310b pin 4, component side.
- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuits CR105 (GN/BU) and RR105 (GY/YE). Go to J25.

  NO : Go to J23.

J7 CHECK THE DRIVER SIDE AIR BAG MODULE FOR AN OPEN

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Side Air Bag Module C328
- Connect a fused jumper wire between driver side air bag C328-1, circuit CR110 (GN/BN), harness side and C328-2, circuit RR110 (GY/OG), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2295_28_OD (Side Air Bag Circuit Resistance Low, Front Driver Side)
  - 2295_29_OD (Side Air Bag Circuit Open, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2295 on-demand fault PIDs with the driver side air bag module circuits shorted together, a low resistance fault would normally be retrieved.
- **Did the driver side air bag on-demand fault PIDs change from indicating an open circuit to a low resistance fault?**

  YES : Go to J22.

  NO : Go to J8.

J8 CHECK CIRCUITS CR105 (GN/BU) AND RR105 (GY/YE) FOR AN OPEN BETWEEN THE RCM AND DRIVER SIDE AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the driver side air bag C328.
- Disconnect: RCM C310a and C310b

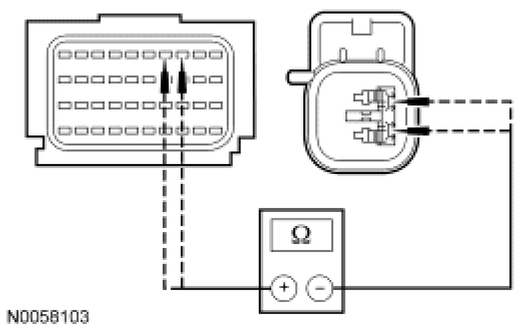

Fig. 80: Measuring Resistance Between RCM C310B-3 & C328-1, Circuit CR110 (GN/BN) & CR105 (GN/BU)
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-3, circuit CR110 (GN/BN), harness side and driver side air bag C328-1, circuit CR105 (GN/BU), harness side; and between RCM C310b-4, circuit RR105 (GY/YE), harness side and driver side air bag C328-2, circuit RR105 (GY/YE), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to J23.

  NO : REPAIR circuit CR105 (GN/BU) or circuit RR105 (GY/YE). Go to J28.

J9 CHECK THE DRIVER SIDE AIR BAG MODULE FOR A SHORT TO GROUND

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Side Air Bag Module C328
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2295_29_OD (Side Air Bag Circuit Open, Front Driver Side)
  - 2295_30_OD (Side Air Bag Circuit Short to Ground, Front Driver Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2295 on-demand fault PIDs with the driver side air bag module disconnected, an open circuit fault would normally be retrieved.
- **Did the driver side air bag on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to J22.

  NO : Go to J10.

J10 CHECK CIRCUITS CR105 (GN/BU) AND RR105 (GY/YE) FOR A SHORT TO GROUND BETWEEN THE RCM AND DRIVER SIDE AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

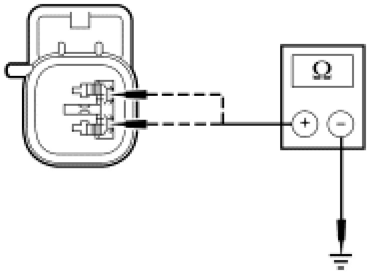

Fig. 81: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver side air bag C328-1, circuit CR105 (GN/BU), harness side and ground; and between driver side air bag C328-2, circuit RR105 (GY/YE), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to J23.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR105 (GN/BU) or circuit RR105 (GY/YE). Go to J28.

J11 CHECK CIRCUITS CR105 (GN/BU) AND RR105 (GY/YE) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND DRIVER SIDE AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Side Air Bag C328
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

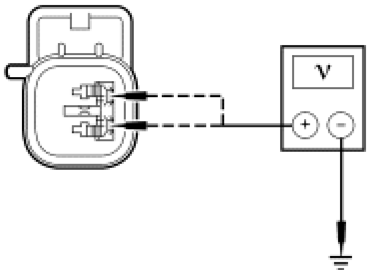

Fig. 82: Measuring Voltage

- Measure the voltage between driver side air bag C328-1, circuit CR105 (GN/BU), harness side and ground; and between driver side air bag C328-2, circuit RR105 (GY/YE), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to J23.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR105 (GN/BU) or circuit RR105 (GY/YE). Go to J28.

J12 CHECK THE PASSENGER SIDE AIR BAG MODULE RESISTANCE PID (PS_AB)

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM
- Enter the following diagnostic mode on the diagnostic tool: PS_AB (Passenger Side Impact Air Bag Resistance) Resistance PID
- **Is the resistance PID value between 1.4 and 3.5 ohms?**

  YES : Go to J23.

  NO : Go to J13.

J13 CHECK THE PASSENGER SIDE AIR BAG RESISTANCE PID (PS_AB) WHILE CARRYING OUT A HARNESS TEST

- Continue monitoring the PS_AB (Passenger Side Impact Air Bag Resistance) resistance PID while carrying out a harness test of the passenger side air bag circuits and accessible connectors (including any inline connectors), by wiggling connectors and flexing the wire harness.
- **Does the resistance PID value read between 1.4 and 3.5 ohms while carrying out the harness test?**

  YES : DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to J28.

  NO : If the passenger side air bag resistance PID is less than 1.4 ohms, go to J14.

  If the passenger side air bag resistance PID is greater than 3.5 ohms, go to J17.

J14 CHECK THE PASSENGER SIDE AIR BAG MODULE FOR LOW RESISTANCE FAULT

NOTE:	**This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Disconnect: Passenger Side Air Bag C327
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2295_24_OD (Side Air Bag Circuit Resistance Low, Front Passenger Side)
  - 2295_25_OD (Side Air Bag Circuit Open, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2295 on-demand fault PIDs with the passenger side air bag module disconnected, an open circuit faults would normally be retrieved.
- **Did the passenger side air bag on-demand fault PIDs change from indicating a low resistance to an open circuit fault?**

  YES : Go to J22.

  NO : Go to J15.

J15 CHECK FOR A SHORT BETWEEN PASSENGER SIDE AIR BAG MODULE CIRCUITS CR106 (VT/GY) AND RR106 (YE/OG)

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

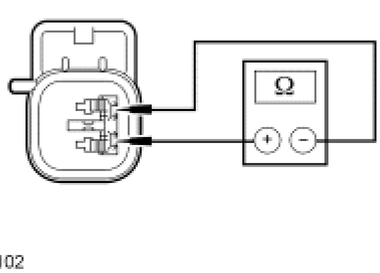

Fig. 83: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger side air bag C327-1, circuit CR106 (VT/GY), harness side and C327-2, circuit RR106 (YE/OG), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to J23.

  NO : Go to J16.

J16 CHECK THE RCM FOR LOW RESISTANCE

- Disconnect: RCM C310a and C310b

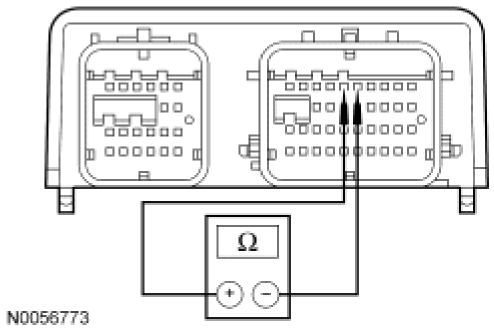

Fig. 84: Measuring Resistance Between RCM C310B Pin 5, Component Side, & C310B Pin 6
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b pin 5, component side, and C310b pin 6, component side.
- **Is resistance greater than 10,000 ohms?**

  YES : REPAIR circuits CR106 (VT/GY) and RR106 (YE/OG). Go to J28.

  NO : Go to J23.

J17 CHECK THE PASSENGER SIDE AIR BAG MODULE FOR AN OPEN

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Side Air Bag C327
- Connect a fused jumper wire between passenger side air bag C327-1, circuit CR106 (VT/GY), harness side and C327-2, circuit RR106 (YE/OG), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2295_24_OD (Side Air Bag Circuit Resistance Low, Front Passenger Side)
  - 2295_25_OD (Side Air Bag Circuit Open, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2295 on-demand fault PIDs with the passenger side air bag circuits shorted together, a low resistance fault would normally be retrieved.
- **Did the passenger side air bag on-demand fault PIDs change from indicating an open circuit**

to a low resistance fault?

YES : Go to J22.

NO : Go to J18.

J18 CHECK CIRCUITS CR106 (VT/GY) AND RR106 (YE/OG) FOR AN OPEN BETWEEN THE RCM AND PASSENGER SIDE AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the passenger side air bag C327.
- Disconnect: RCM C310a and C310b

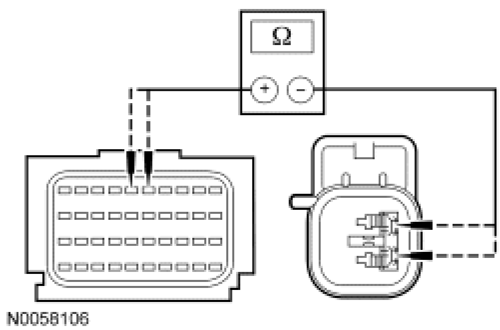

Fig. 85: Measuring Resistance Between RCM C310B-5, Circuit CR106 (VT/GY) & Passenger Side Air Bag Module C327-1
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-5, circuit CR106 (VT/GY), harness side and passenger side air bag module C327-1, circuit CR106 (VT/GY), harness side; and between RCM C310b-6, circuit RR106 (YE/OG), harness side and passenger side air bag module C327-2, circuit RR106 (YE/OG), harness side.
- **Are resistances less than 0.5 ohm?**

YES : Go to J23.

NO : REPAIR circuit CR106 (VT/GY) or circuit RR106 (YE/OG). Go to J28.

J19 CHECK THE PASSENGER SIDE AIR BAG MODULE FOR A SHORT TO GROUND

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Side Air Bag Module C327

- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record the Following Fault PIDs
  - 2295_25_OD (Side Air Bag Circuit Open, Front Passenger Side)
  - 2295_26_OD (Side Air Bag Circuit Short to Ground, Front Passenger Side)
- **DIAGNOSTIC TIP:** When viewing DTC B2295 on-demand fault PIDs with the passenger side air bag module disconnected, an open circuit fault would normally be retrieved.
- **Did the passenger side air bag on-demand fault PIDs change from indicating a short to ground to an open circuit fault?**

  YES : Go to J22.

  NO : Go to J20.

J20 CHECK CIRCUITS CR106 (VT/GY) AND RR106 (YE/OG) FOR A SHORT TO GROUND BETWEEN THE RCM AND PASSENGER SIDE AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

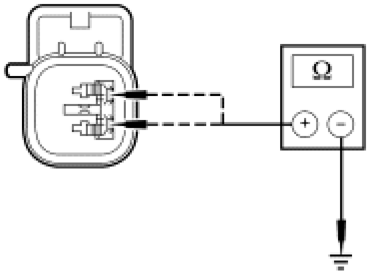

Fig. 86: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger side air bag C327-1, circuit CR106 (VT/GY), harness side and ground; and between passenger side air bag C327-2, circuit RR106 (YE/OG), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to J23.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

REPAIR circuit CR106 (VT/GY) or circuit RR106 (YE/OG). Go to J28.

J21 CHECK CIRCUITS CR106 (VT/GY) AND RR106 (YE/OG) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND PASSENGER SIDE AIR BAG MODULE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Side Air Bag C327
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

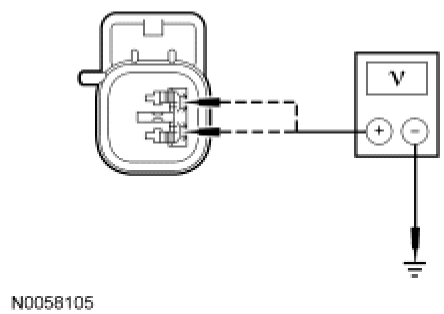

Fig. 87: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between passenger side air bag C327-1, circuit CR106 (VT/GY), harness side and ground; and between passenger side air bag C327-2, circuit RR106 (YE/OG), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to J23.

  NO : Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar.

  REPAIR circuit CR106 (VT/GY) or circuit RR106 (YE/OG). Go to J28.

J22 CONFIRM THE SIDE AIR BAG MODULE FAULT

NOTE:	**Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and**

Repowering.

- Connect: Affected Side Air Bag C328 (Driver) C3327 (Passenger)
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2295 Fault PIDs
  - Refer to PID list in Normal Operation to view B2295 fault PIDs.
- **Does the original RCM on-demand DTC B2295 fault PID indicate a fault?**

  YES : REMOVE and INSPECT the seat side air bag module jumper harness, connector terminals for damage. REFER to **Side Air Bag Module**. If a concern is found, REPAIR or INSTALL a new seat jumper harness. Go to J28.

  If a concern was not found, INSTALL a new driver or passenger side air bag module. Go to J28.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2295_28_CM (Side Air Bag Circuit Resistance Low, Front Driver Side), 2295_24_CM (Side Air Bag Circuit Resistance Low, Front Passenger Side), 2295_29_CM (Side Air Bag Circuit Open, Front Driver Side) or 2295_25_CM (Side Air Bag Circuit Open, Front Passenger Side), go to J24.

  For 2295_30_CM (Side Air Bag Circuit Short to Ground, Front Driver Side) or 2295_26_CM (Side Air Bag Circuit Short to Ground, Front Passenger Side), go to J26.

  For 2295_31_CM (Side Air Bag Circuit Short to Battery, Front Driver Side) or 2295_27_CM (Side Air Bag Circuit Short to Battery, Front Passenger Side), go to J27.

J23 CONFIRM THE RCM FAULT

NOTE:	**Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- If previously directed to disconnect any SRS component(s):
  - Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
  - Connect all previously disconnected restraint system component(s), including RCM C310a

and C310b.

    - Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2295 Fault PIDs
    - Refer to PID list in Normal Operation to view B2295 fault PIDs.
- **Does the original RCM on-demand DTC B2295 fault PID indicate a fault?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to J28.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. Using the fault PIDs recorded, go to the appropriate pinpoint test step and CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected.

  For 2295_28_CM (Side Air Bag Circuit Resistance Low, Front Driver Side), 2295_24_CM (Side Air Bag Circuit Resistance Low, Front Passenger Side), 2295_29_CM (Side Air Bag Circuit Open, Front Driver Side) or 2295_25_CM (Side Air Bag Circuit Open, Front Passenger Side), go to J24.

  For 2295_30_CM (Side Air Bag Circuit Short to Ground, Front Driver Side) or 2295_26_CM (Side Air Bag Circuit Short to Ground, Front Passenger Side), go to J26.

  For 2295_31_CM (Side Air Bag Circuit Short to Battery, Front Driver Side) or 2295_27_CM (Side Air Bag Circuit Short to Battery, Front Passenger Side), go to J27.

J24 CHECK FOR AN INTERMITTENT SIDE AIR BAG MODULE LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2295 Fault PIDs
    - Refer to PID list in Normal Operation to view B2295 fault PIDs.
- If the fault PID was reported for the driver side air bag module, monitor the DS_AB (Driver Side Impact Air Bag Resistance) resistance PID on the scan tool.
- If the fault PID was reported for the passenger side air bag, monitor the PS_AB (Passenger Side Impact Air Bag Resistance) resistance PID on the scan tool.

NOTE:	**A resistance PID that reads between 0.9 and 1.4 ohms or between 3.5 and 5 ohms may or may not set an on-demand DTC yet indicates a strong potential of an intermittent fault condition. Make sure to thoroughly check wire harness(es), connectors and terminals as they are the likely source of the fault condition.**

- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any side air bag on-demand fault PID indicate a fault is present or is the side air bag resistance PID less than 1.4 ohms or greater than 3.5 ohms?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to J28.

  NO : Go to J25.

J25 CHECK THE HARNESS AND CONNECTORS

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect the affected side air bag C328 (driver) C327 (passenger):
  - inspect connector(s) (including any inline connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
  - inspect wire harness for any damage, pinched, cut or pierced wires.
- **Were any concerns found?**

  YES : REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to J28.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to J28.

J26 CHECK FOR AN INTERMITTENT SIDE AIR BAG MODULE SHORT TO GROUND FAULT

- If the fault PID was reported for the driver side air bag module, monitor the driver side air bag short to ground on-demand fault PID 2295_30_OD on the scan tool.
- If the fault PID was reported for the passenger side air bag, monitor the passenger side air bag short to ground on-demand fault PID 2295_26_OD on the scan tool.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.
- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any RCM on-demand B2295 fault PID indicate a fault is present?**

  YES : REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to J28.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to J28.

J27 CHECK FOR AN INTERMITTENT SIDE AIR BAG SHORT TO VOLTAGE FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Affected Safety Canopy Module C328 (Driver) C327 (Passenger)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- **DIAGNOSTIC TIP:** When monitoring fault PIDs with the scan tool, it may take up to 2 seconds for the scan tool display to update.
- Monitor the 2295_31_OD (Side Air Bag Circuit Short to Battery, Front Driver Side) or 2295_27_OD (Side Air Bag Circuit Short to Battery, Front Passenger Side) fault PIDs on the scan tool.
- Attempt to recreate the fault by wiggling connectors (including any inline connectors) and flexing the wire harness.
- **Does any RCM on-demand B2295 fault PID indicate a fault is present?**

  YES : DEPOWER the SRS and REPAIR as necessary. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information. Go to J28.

  NO : The fault is not present and cannot be recreated at this time. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to J28.

J28 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

> WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test K: DTC B2296 - Restraint System - Impact Sensor Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors all of the impact sensor circuits for faults. If the RCM detects one of the following faults on any of the impact sensor circuits, it will store DTC B2296 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

Fault PIDs[a]	Description	Fault Trigger Condition
2296_3_OD and 2296_3_CM	Driver/Center Front Crash Sensor Circuit Short to Battery	When the RCM senses a short to battery on the driver/center front impact sensor circuits, a fault will be indicated.
2296_5_OD and 2296_5_CM	Driver/Center Front Crash Sensor Communication Fault	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the driver/center front impact sensor, a fault will be indicated.
2296_6_OD and 2296_6_CM	Side Crash Sensor Circuit Short to Battery, Row No.2 Passenger Side	When the RCM senses a short to battery on the passenger second row side impact sensor, a fault will be indicated.
2296_7_OD and 2296_7_CM	Side Crash Sensor Circuit Short to Ground, Row No.2 Passenger Side	When the RCM senses a short to ground on the passenger second row side impact sensor, a fault will be indicated.
2296_8_OD and 2296_8_CM	Side Crash Sensor Communication Fault, Row No.2 Passenger Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the passenger second row side impact sensor, a fault will be indicated.
2296_9_OD and 2296_9_CM	Side Crash Sensor Circuit Short to Battery, Row No.2 Driver Side	When the RCM senses a short to battery on the driver second row side impact sensor circuits, a fault will be indicated.
		When the RCM senses a short to ground on the

2296_10_OD and 2296_10_CM	Side Crash Sensor Circuit Short to Ground, Row No.2 Driver Side	driver second row side impact sensor circuits, a fault will be indicated.
2296_11_OD and 2296_11_CM	Side Crash Sensor Communication Fault, Row No.2 Driver Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the driver second row side impact sensor, a fault will be indicated.
2296_12_OD and 2296_12_CM	Side Crash Sensor Circuit Short to Battery, Front Passenger Side	When the RCM senses a short to battery on the passenger first row side impact sensor circuits, a fault will be indicated.
2296_13_OD and 2296_13_CM	Side Crash Sensor Circuit Short to Ground, Front Passenger Side	When the RCM senses a short to ground on the passenger first row side impact sensor circuits, a fault will be indicated.
2296_14_OD and 2296_14_CM	Side Crash Sensor Communication Fault, Front Passenger Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the passenger first row side impact sensor, a fault will be indicated.
2296_15_OD and 2296_15_CM	Side Crash Sensor Circuit Short to Battery, Front Driver Side	When the RCM senses a short to battery on the driver first row side impact sensor circuits, a fault will be indicated.
2296_16_OD and 2296_16_CM	Side Crash Sensor Circuit Short to Ground, Front Driver Side	When the RCM senses a short to ground on the driver first row side impact sensor circuits, a fault will be indicated.
2296_17_OD and 2296_17_CM	Side Crash Sensor Communication Fault, Front Driver Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the driver first row side impact sensor, a fault will be indicated.
2296_18_OD and		When the RCM senses a internal failure with the

2296_18_CM	Driver/Center Front Crash Sensor Internal Fault	driver/center front impact sensor, a fault will be indicated.
2296_19_OD and 2296_19_CM	Driver/Center Front Crash Sensor Mount/Communication Fault	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the driver/center front impact sensor, a fault will be indicated.
2296_24_OD and 2296_24_CM	Side Crash Sensor 2 Internal Fault, Passenger Side	When the RCM senses a internal failure with the passenger second row side impact sensor, a fault will be indicated.
2296_25_OD and 2296_25_CM	Side Crash Sensor Mount or Communication Fault, Row No. 2 Passenger Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the passenger second row side impact sensor, a fault will be indicated.
2296_26_OD and 2296_26_CM	Side Crash Sensor 2 Internal Fault, Driver Side	When the RCM senses a internal failure with the driver second row side impact sensor, a fault will be indicated.
2296_27_OD and 2296_27_CM	Side Crash Sensor Mount or Communication Fault, Row No. 2 Driver Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the driver second row side impact sensor, a fault will be indicated.
2296_28_OD and 2296_28_CM	Side Crash Sensor Internal Fault, Front Passenger Side	When the RCM senses a internal failure with the passenger first row side impact sensor, a fault will be indicated.
2296_29_OD and 2296_29_CM	Side Crash Sensor Mount or Communication Fault, Front Passenger Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the passenger first row side impact sensor, a fault will be indicated.

2296_30_OD and 2296_30_CM	Side Crash Sensor Internal Fault, Front Driver Side	When the RCM senses a internal failure with the driver first row side impact sensor, a fault will be indicated.
2296_31_OD and 2296_31_CM	Side Crash Sensor Mount or Communication Fault, Front Driver Side	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the driver side first row side impact sensor, a fault will be indicated.
2296_55_OD and 2296_55_CM	Passenger Front Crash Sensor Circuit Short to Battery	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the passenger front impact sensor, a fault will be indicated.
2296_56_OD and 2296_56_CM	Passenger Front Crash Sensor Circuit Short to Ground	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the passenger front impact sensor, a fault will be indicated.
2296_57_OD and 2296_57_CM	Passenger Front Crash Sensor Communication Fault	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the passenger front impact sensor, a fault will be indicated.
2296_58_OD and 2296_58_CM	Passenger Front Crash Sensor Internal Fault	When the RCM senses a internal failure with the with the passenger front impact sensor, a fault will be indicated.
2296_59_OD and 2296_59_CM	Passenger Front Crash Sensor Mount/Communication Fault	When the RCM is unable to communicate (circuit open, short to ground or battery on sensor) with the driver/center front impact sensor, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B2296. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B2296.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Impact sensor
- Incorrect sensor mounting
- RCM

PINPOINT TEST K: DTC B2296 - RESTRAINT SYSTEM - IMPACT SENSOR FAULT

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Do not probe any impact sensor. The impact sensor cannot be tested using a multi-meter.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

K1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs
  - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Do any RCM on-demand DTC B2296 fault PIDs indicate a fault?**

YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test.

Using the fault PIDs recorded, go to the appropriate pinpoint test step.

For 2296_30_OD (Side Crash Sensor Internal Fault, Front Driver Side), INSTALL a new first row driver side impact sensor. REFER to **Side Impact Sensor - B-Pillar**. Go to K34.

For 2296_28_OD (Side Crash Sensor Internal Fault, Front Passenger Side), INSTALL a new first row passenger side impact sensor. REFER to **Side Impact Sensor - B-Pillar**. Go to K34.

For 2296_26_OD (Side Crash Sensor 2 Internal Fault, Driver side), INSTALL a new second row driver side impact sensor. REFER to **Side Impact Sensor - C-Pillar**. Go to K34.

For second row passenger side impact sensor with an internal fault (2296_24_OD (Side Crash Sensor 2 Internal Fault, Passenger side), INSTALL a new second row passenger side impact sensor. REFER to **Side Impact Sensor - C-Pillar**. Go to K34.

For 2296_18_OD (Driver/Center Front Crash Sensor Internal Fault), INSTALL a new driver/center front impact severity sensor. REFER to **Front Impact Severity Sensor**. Go to K34.

For 2296_58_OD (Passenger Front Crash Sensor Internal Fault), INSTALL a new passenger front impact severity sensor. REFER to **Front Impact Severity Sensor**. Go to K34.

For 2296_17_OD (Side Crash Sensor Communication Fault, Front Driver Side)/2296_31_OD (Side Crash Sensor Mount or Communication Fault, Front Driver Side), go to K2.

For 2296_16_OD (Side Crash Sensor Circuit Short to Ground, Front Driver Side), go to K4.

For 2296_15_OD (Side Crash Sensor Circuit Short to Battery, Front Driver Side), go to K6.

For 2296_14_OD (Side Crash Sensor Communication Fault, Front Passenger Side), go to K7.

For 2296_13_OD (Side Crash Sensor Circuit Short to Ground, Front Passenger Side), go to K9.

For 2296_12_OD (Side Crash Sensor Circuit Short to Battery, Front Passenger Side), go to K11.

For 2296_11_OD (Side Crash Sensor Communication Fault, Row No.2 Driver Side), go to K12.

For 2296_10_OD (Side Crash Sensor Circuit Short to Ground, Row No.2 Driver Side), go to K14.

For 2296_9_OD (Side Crash Sensor Circuit Short to Battery, Row No.2 Driver Side), go to K16.

For 2296_8_OD (Side Crash Sensor Communication Fault, Row No.2 Passenger Side), go to K17.

For 2296_7_OD (Side Crash Sensor Circuit Short to Ground, Row No.2 Passenger Side), go to

K19.

For 2296_6_OD (Side Crash Sensor Circuit Short to Battery, Row No.2 Passenger Side), go to K21.

For 2296_5_OD (Driver/Center Front Crash Sensor Communication Fault), go to K22.

For 2296_4_OD (Driver/Center Front Crash Sensor Circuit Short to Ground), go to K24.

For 2296_3_OD (Driver/Center Front Crash Sensor Circuit Short to Battery), go to K26.

For 2296_59_OD (Passenger Front Crash Sensor Mount/Communication Fault), go to K27.

For 2296_56_OD (Passenger Front Crash Sensor Circuit Short to Ground), go to K29.

For 2296_55_OD (Passenger Front Crash Sensor Circuit Short to Battery), go to K31.

NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to K33.

K2 CHECK CIRCUITS VR217 (GY/YE) AND RR131 (VT/GY) FOR AN OPEN BETWEEN THE RCM AND FIRST ROW DRIVER SIDE IMPACT SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: First Row Driver Side Impact Sensor C3209
- Disconnect: RCM C310a and C310b

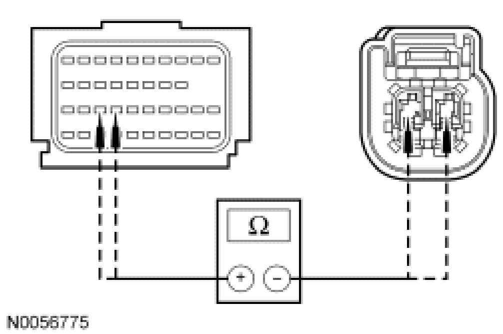

Fig. 88: Measuring Resistance Between RCM C310B-27 & C310B-28, Circuit VR217 (GY/YE) & RR131 (VT/GY)
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-27, circuit VR217 (GY/YE), harness side and first row driver side impact sensor C3209-1, circuit VR217 (GY/YE), harness side; and between RCM C310b-28, circuit RR131 (VT/GY), harness side and first row driver side impact sensor C3209-2, circuit RR131 (VT/GY), harness side.

- **Are resistances less than 0.5 ohm?**

  YES : Go to K3.

  NO : REPAIR circuit VR217 (GY/YE) or circuit RR131 (VT/GY). Go to K34.

K3 CHECK THE FIRST ROW DRIVER SIDE IMPACT SENSOR

- Connect: RCM C310a and C310b
- Install a known good first row driver side impact sensor.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs
  - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Were any on-demand B2296 fault PIDs for the first row driver side impact sensor indicating a fault?**

  YES : Go to K32.

  NO : Fault corrected. Go to K34.

K4 CHECK CIRCUITS VR217 (GY/YE) AND RR131 (VT/GY) FOR A SHORT TO GROUND BETWEEN THE FIRST ROW DRIVER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: First Row Driver Side Impact Sensor C3209
- Disconnect: RCM C310a and C310b

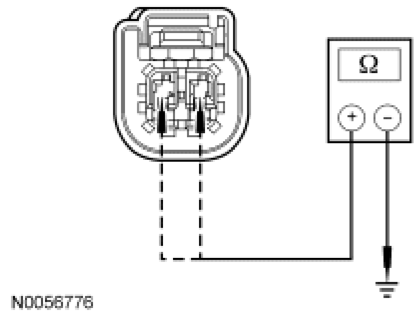

Fig. 89: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between first row driver side impact sensor C3209-2, circuit RR131 (VT/GY), harness side and ground; and between C3209-1, circuit VR217 (GY/YE), harness side and ground.

- **Are resistances greater than 10,000 ohms?**

  YES : Go to K5.

  NO : REPAIR circuit VR217 (GY/YE) or circuit RR131 (VT/GY). Go to K34.

K5 CHECK CIRCUIT VR217 (GY/YE) FOR A SHORT TO CIRCUIT RR131 (VT/GY) BETWEEN THE RCM AND FIRST ROW DRIVER SIDE IMPACT SENSOR

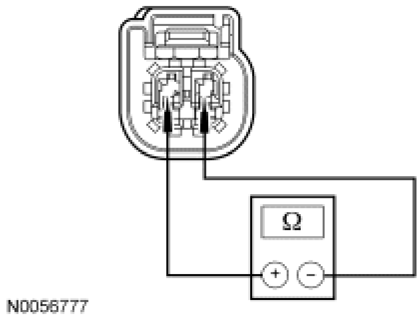

Fig. 90: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between first row driver side impact sensor C3209-1, circuit VR217 (GY/YE), harness side and C3209-2, circuit RR131 (VT/GY), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to K32.

  NO : REPAIR circuits VR217 (GY/YE) and RR131 (VT/GY). Go to K34.

K6 CHECK CIRCUITS VR217 (GY/YE) AND RR131 (VT/GY) FOR A SHORT TO VOLTAGE BETWEEN THE FIRST ROW DRIVER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: First Row Driver Side Impact Sensor C3209
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

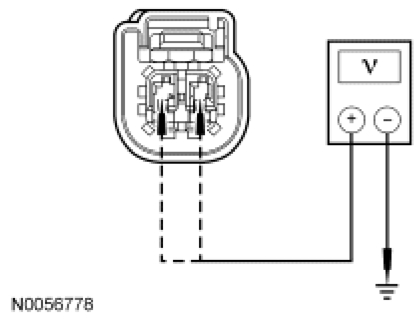

Fig. 91: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between first row driver side impact sensor C3209-2, circuit RR131 (VT/GY), harness side and ground; and between C3209-1, circuit VR217 (GY/YE), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to K32.

  NO : REPAIR circuit VR217 (GY/YE) or circuit RR131 (VT/GY). Go to K34.

K7 CHECK CIRCUITS VR218 (YE/OG) AND RR132 (BU/WH) FOR AN OPEN BETWEEN THE RCM AND FIRST ROW PASSENGER SIDE IMPACT SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: First Row Passenger Side Impact Sensor C3211
- Disconnect: RCM C310a and C310b

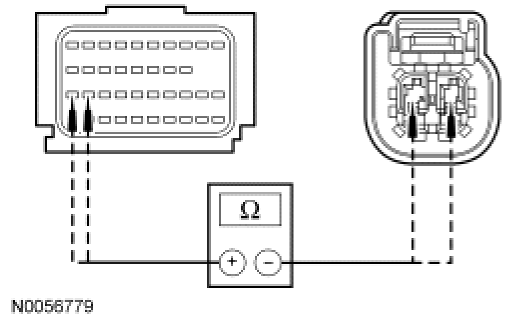

Fig. 92: Measuring Resistance Between RCM C310B-29 & C310B-30, Circuit VR218 (YE/OG) & RR132 (BU/WH)
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-29, circuit VR218 (YE/OG), harness side and first row passenger side impact sensor C3211-1, circuit VR218 (YE/OG), harness side; and between RCM C310b-30, circuit RR132 (BU/WH), harness side and first row passenger side impact sensor C3211-2, circuit RR132 (BU/WH), harness side.
- **Are resistances less than 0.5 ohm?**

YES : Go to K8.

NO : REPAIR circuit VR218 (YE/OG) or circuit RR132 (BU/WH). Go to K34.

K8 CHECK THE FIRST ROW PASSENGER SIDE IMPACT SENSOR

- Connect: RCM C310a and C310b
- Install a known good first row passenger side impact sensor.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs
  - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Were any on-demand B2296 fault PIDs for the first row passenger side impact sensor indicating a fault?**

  YES : Go to K32.

  NO : Fault corrected. Go to K34.

K9 CHECK CIRCUITS VR218 (YE/OG) AND RR132 (BU/WH) FOR A SHORT TO GROUND BETWEEN THE FIRST ROW PASSENGER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: First Row Passenger Side Impact Sensor C3211
- Disconnect: RCM C310a and C310b

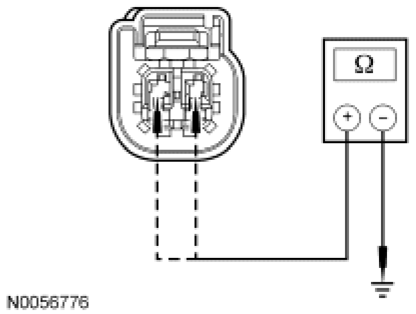

Fig. 93: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between first row passenger side impact sensor C3211-1, circuit VR218 (YE/OG), harness side and ground; and between C3211-2, circuit RR132 (BU/WH), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

YES : Go to K10.

NO : REPAIR circuit VR218 (YE/OG) or circuit RR132 (BU/WH). Go to K34.

K10 CHECK CIRCUIT VR218 (YE/OG) FOR A SHORT TO CIRCUIT RR132 (BU/WH) BETWEEN THE RCM AND FIRST ROW PASSENGER SIDE IMPACT SENSOR

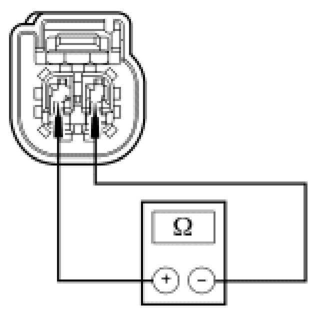

Fig. 94: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between first row passenger side impact sensor C3211-1, circuit VR218 (YE/OG), harness side and C3211-2, circuit RR132 (BU/WH), harness side.
- **Is resistance greater than 10,000 ohms?**

YES : Go to K32.

NO : REPAIR short between circuits VR218 (YE/OG) and RR132 (BU/WH). Go to K34.

K11 CHECK CIRCUITS VR218 (YE/OG) AND RR132 (BU/WH) FOR A SHORT TO VOLTAGE BETWEEN THE FIRST ROW PASSENGER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: First Row Passenger Side Impact Sensor C3211
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

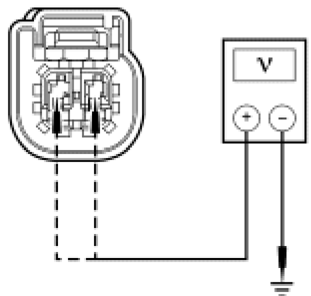

Fig. 95: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between first row passenger side impact sensor C3211-1, circuit VR218 (YE/OG), harness side and ground; and between C3211-2, circuit RR132 (BU/WH), harness side and ground.
- **Are voltages less than 0.2 volt?**

  YES : Go to K32.

  NO : REPAIR circuit VR218 (YE/OG) or circuit RR132 (BU/WH). Go to K34.

K12 CHECK CIRCUITS VR219 (GN/WH) AND RR133 (GY/BN) FOR AN OPEN BETWEEN THE RCM AND SECOND ROW DRIVER SIDE IMPACT SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Second Row Driver Side Impact Sensor C3248
- Disconnect: RCM C310a and C310b

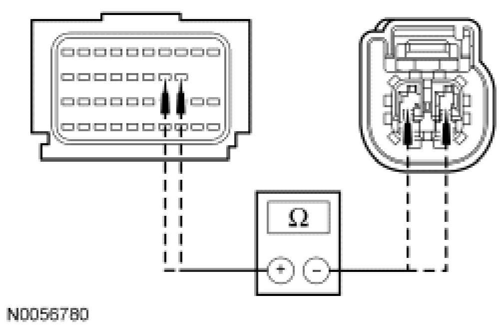

Fig. 96: Measuring Resistance Between RCM C310B-13 & C310B-14, Circuit VR219 (GN/WH) & RR135 (GY/BN)
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-13, circuit VR219 (GN/WH), harness side and second row driver side impact sensor C3248-1, circuit VR219 (GN/WH), harness side; and between RCM C310b-14, circuit RR135 (GY/BN), harness side and second row driver side impact sensor C3248-2, circuit RR133 (GY/BN), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to K13.

  NO : REPAIR circuit VR219 (GN/WH) or circuit RR133 (GY/BN). Go to K34.

K13 CHECK THE SECOND ROW DRIVER SIDE IMPACT SENSOR

- Connect: RCM C310a and C310b
- Install a known good second row driver side impact sensor.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs
    - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Were any on-demand B2296 fault PIDs for the second row driver side impact sensor indicating a fault?**

  YES : Go to K32.

  NO : Fault corrected. Go to K34.

K14 CHECK CIRCUITS VR219 (GN/WH) AND RR133 (GY/BN) FOR A SHORT TO GROUND BETWEEN THE SECOND ROW DRIVER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Second Row Driver Side Impact Sensor C3248
- Disconnect: RCM C310a and C310b

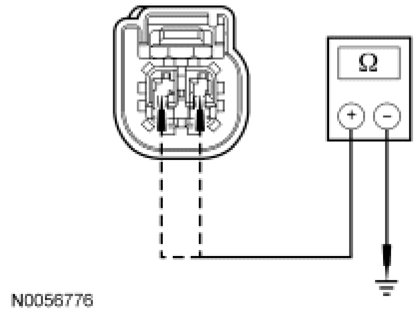

Fig. 97: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between second row driver side impact sensor C3248-1, circuit VR219 (GN/WH), harness side and ground; and between C3248-2, circuit RR133 (GY/BN), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to K15.

  NO : REPAIR circuit VR219 (GN/WH) or circuit RR133 (GY/BN). Go to K34.

K15 CHECK CIRCUIT VR219 (GN/WH) FOR A SHORT TO CIRCUIT RR133 (GY/BN) BETWEEN THE RCM AND SECOND ROW DRIVER SIDE IMPACT SENSOR

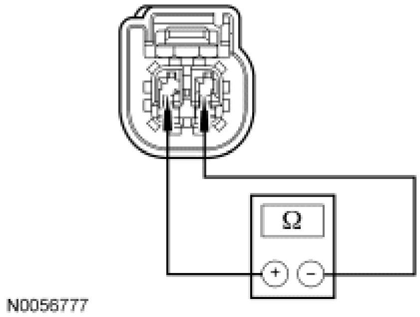

Fig. 98: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between second row driver side impact sensor C3248-1, circuit VR219 (GN/WH), harness side and C3248-2, circuit RR135 (GY/BN), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to K32.

  NO : REPAIR circuits VR219 (GN/WH) and RR133 (GY/BN). Go to K34.

K16 CHECK CIRCUITS VR219 (GN/WH) AND RR133 (GY/BN) FOR A SHORT TO VOLTAGE BETWEEN THE SECOND ROW DRIVER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Second Row Driver Side Impact Sensor C3248
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

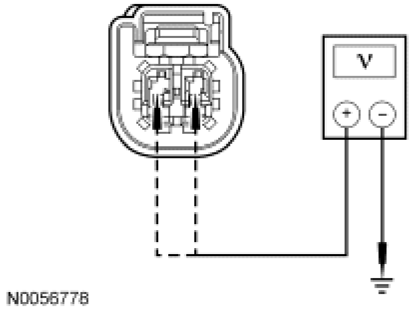

Fig. 99: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between second row driver side impact sensor C3248-1, circuit VR219 (GN/WH), harness side and ground; and between C3248-2, circuit RR133 (GY/BN), harness side and ground.

- **Are voltages less than 0.2 volt?**

  YES : Go to K32.

  NO : REPAIR circuit VR219 (GN/WH) or circuit RR133 (GY/BN). Go to K34.

K17 CHECK CIRCUITS VR220 (VT/OG) AND RR134 (BN/BU) FOR AN OPEN BETWEEN THE RCM AND SECOND ROW PASSENGER SIDE IMPACT SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Second Row Passenger Side Impact Sensor C3249
- Disconnect: RCM C310a and C310b

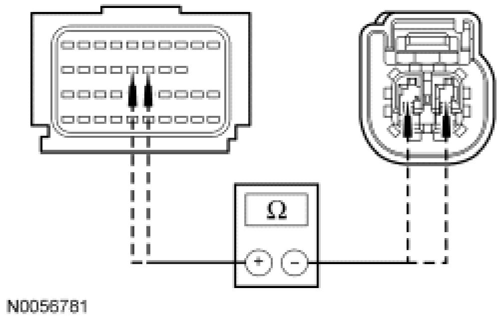

Fig. 100: Measuring Resistance Between RCM C310B-15 & C310B-16, Circuit VR220 (VT/OG) & RR134 (BN/BU)
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-15, circuit VR220 (VT/OG), harness side and second row passenger side impact sensor C3249-1, circuit VR220 (VT/OG), harness side and between RCM C310b-16, circuit RR134 (BN/BU), harness side and second row passenger side impact sensor C3249-2, circuit RR134 (BN/BU), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to K18.

  NO : REPAIR circuit VR220 (VT/OG) or circuit RR134 (BN/BU). Go to K34.

K18 CHECK THE SECOND ROW PASSENGER SIDE IMPACT SENSOR

- Connect: RCM C310a and C310b
- Install a known good second row passenger side impact sensor.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs

  - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Were any on-demand B2296 fault PIDs for the second row passenger side impact sensor indicating a fault?**

  YES : Go to K32.

  NO : Fault corrected. Go to K34.

K19 CHECK CIRCUITS VR220 (VT/OG) AND RR134 (BN/BU) FOR A SHORT TO GROUND BETWEEN THE SECOND ROW PASSENGER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Second Row Passenger Side Impact Sensor C3249
- Disconnect: RCM C310a and C310b

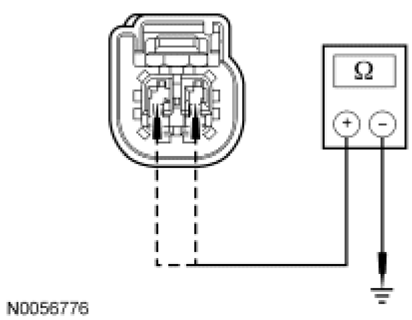

Fig. 101: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between second row passenger side impact sensor C3249-1, circuit VR220 (VT/OG), harness side and ground; and between C3249-2, circuit RR134 (BN/BU), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to K20.

  NO : REPAIR circuit VR220 (VT/OG) or circuit RR134 (BN/BU). Go to K34.

K20 CHECK CIRCUIT VR220 (VT/OG) FOR A SHORT TO CIRCUIT RR134 (BN/BU) BETWEEN THE RCM AND SECOND ROW PASSENGER SIDE IMPACT SENSOR

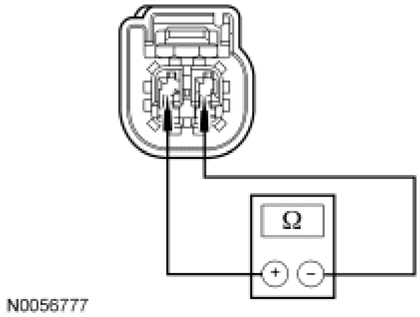

Fig. 102: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between second row passenger side impact sensor C3249-1, circuit VR220 (VT/OG), harness side and C3249-2, circuit RR134 (BN/BU), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to K32.

  NO : REPAIR circuits VR220 (VT/OG) and RR130 (GN/BU). Go to K34.

K21 CHECK CIRCUITS VR220 (VT/OG) AND RR134 (BN/BU) FOR A SHORT TO VOLTAGE BETWEEN THE SECOND ROW PASSENGER SIDE IMPACT SENSOR AND RCM

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Second Row Passenger Side Impact Sensor C3249
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

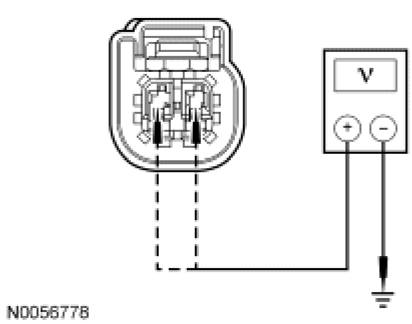

Fig. 103: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between second row passenger side impact sensor C3249-1, circuit VR220 (VT/OG), harness side and ground; and between C3249-2, circuit RR134 (BN/BU), harness side and ground.

- **Are voltages less than 0.2 volt?**

  YES : Go to K32.

  NO : REPAIR circuit VR220 (VT/OG) or circuit RR134 (BN/BU). Go to K34.

K22 CHECK CIRCUITS VR213 (VT/GN) AND RR129 (YE/GY) FOR AN OPEN BETWEEN THE RCM AND DRIVER/CENTER FRONT IMPACT SEVERITY SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver/Center Front Impact Severity Sensor C1465
- Disconnect: RCM C310a and C310b

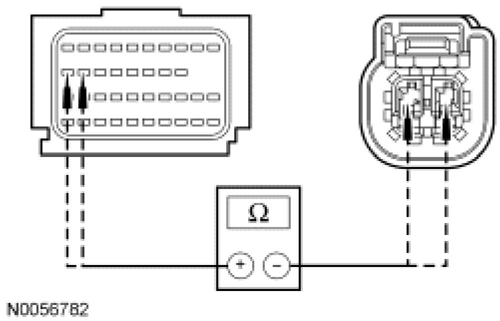

Fig. 104: Measuring Resistance Between RCM C310B-19 & C310B-20, Circuit VR213 (VT/GN) & RR129 (YE/GY)
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-19, circuit VR213 (VT/GN), harness side and driver/center front impact severity sensor C1465-1, circuit VR213 (VT/GN), harness side; and between RCM C310b-20, circuit RR129 (YE/GY), harness side and driver/center front impact severity sensor C1465-2, circuit RR129 (YE/GY), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to K23.

  NO : REPAIR circuit VR213 (VT/GN) or circuit RR129 (YE/GY). Go to K34.

K23 CHECK THE DRIVER/CENTER FRONT IMPACT SEVERITY SENSOR

- Connect: RCM C310a and C310b
- Install a known good driver/center front impact severity sensor.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs

  - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Were any on-demand B2296 fault PIDs for the driver/center front impact severity sensor indicating a fault?**

  YES : Go to K32.

  NO : Fault corrected. Go to K34.

K24 CHECK CIRCUITS VR213 (VT/GN) AND RR129 (YE/GY) FOR A SHORT TO GROUND BETWEEN THE RCM AND DRIVER/CENTER FRONT IMPACT SEVERITY SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver/Center Front Impact Severity Sensor C1465
- Disconnect: RCM C310a and C310b

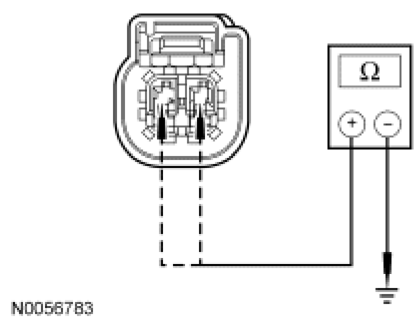

Fig. 105: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver/center front impact severity sensor C1465-1, circuit VR213 (VT/GN), harness side and ground; and between driver/center front impact severity sensor C1465-2, circuit RR129 (YE/GY), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to K25.

  NO : REPAIR circuit VR213 (VT/GN) or circuit RR129 (YE/GY). Go to K34.

K25 CHECK CIRCUIT VR213 (VT/GN) FOR A SHORT TO CIRCUIT RR129 (YE/GY) BETWEEN THE RCM AND DRIVER/CENTER FRONT IMPACT SEVERITY SENSOR

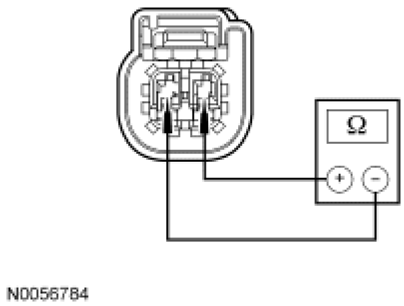

Fig. 106: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver/center front impact severity sensor C1465-1, circuit VR213 (VT/GN), harness side and C1465-2, circuit RR129 (YE/GY), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to K32.

  NO : REPAIR circuits VR213 (VT/GN) and RR129 (YE/GY). Go to K34.

K26 CHECK CIRCUITS VR213 (VT/GN) AND RR129 (YE/GY) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND DRIVER/CENTER FRONT IMPACT SEVERITY SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver/Center Front Impact Severity Sensor C1465
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

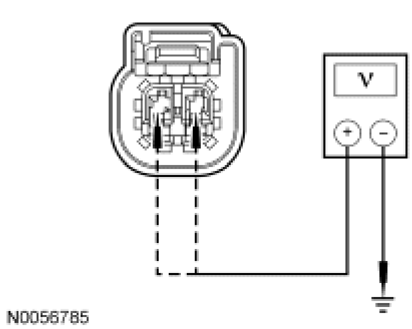

Fig. 107: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between driver/center front impact severity sensor C1465-1, circuit VR213 (VT/GN), harness side and ground; and between driver/center front impact severity sensor C1465-2, circuit RR129 (YE/GY), harness side and ground.

- **Are voltages less than 0.2 volt?**

  YES : Go to K32.

  NO : REPAIR circuit VR213 (VT/GN) or circuit RR129 (YE/GY). Go to K34.

K27 CHECK CIRCUITS VR214 (WH/BU) AND RR130 (GN/BU) FOR AN OPEN BETWEEN THE RCM AND PASSENGER FRONT IMPACT SEVERITY SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Front Impact Severity Sensor C1466
- Disconnect: RCM C310a and C310b

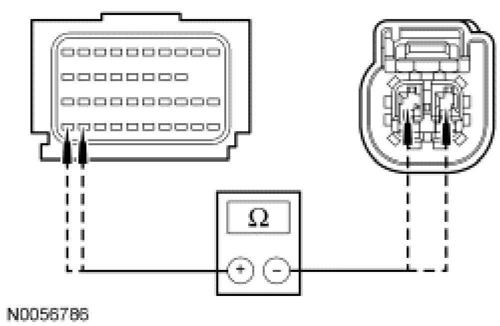

Fig. 108: Measuring Resistance Between RCM C310B-39 & C310B-40, Circuit VR214 (WH/BU) & RR130 (GN/BU)
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-39, circuit VR214 (WH/BU), harness side and driver/center front impact severity sensor C1466-1, circuit VR214 (WH/BU), harness side; and between RCM C310b-40, circuit RR130 (GN/BU), harness side and passenger front impact severity sensor C1466-2, circuit RR130 (GN/BU), harness side.
- **Are resistances less than 0.5 ohm?**

  YES : Go to K28.

  NO : REPAIR circuit VR214 (WH/BU) or circuit RR130 (GN/BU). Go to K34.

K28 CHECK THE PASSENGER FRONT IMPACT SEVERITY SENSOR

- Connect: RCM C310a and C310b
- Install a known good passenger front impact severity sensor.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs

    - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Were any on-demand B2296 fault PIDs for the passenger front impact severity sensor indicating a fault?**

  YES : Go to K32.

  NO : Fault corrected. Go to K34.

K29 CHECK CIRCUITS VR214 (WH/BU) AND RR130 (GN/BU) FOR A SHORT TO GROUND BETWEEN THE RCM AND PASSENGER FRONT IMPACT SEVERITY SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Front Impact Severity Sensor C1466
- Disconnect: RCM C310a and C310b

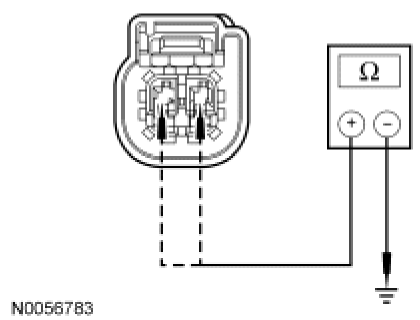

Fig. 109: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger front impact severity sensor C1466-1, circuit VR214 (WH/BU), harness side and ground; and between passenger front impact severity sensor C1466-2, circuit RR130 (GN/BU), harness side and ground.
- **Are resistances greater than 10,000 ohms?**

  YES : Go to K30.

  NO : REPAIR circuit VR214 (WH/BU) or circuit RR130 (GN/BU). Go to K34.

K30 CHECK CIRCUIT VR214 (WH/BU) FOR A SHORT TO CIRCUIT RR130 (GN/BU) BETWEEN THE RCM AND PASSENGER FRONT IMPACT SEVERITY SENSOR

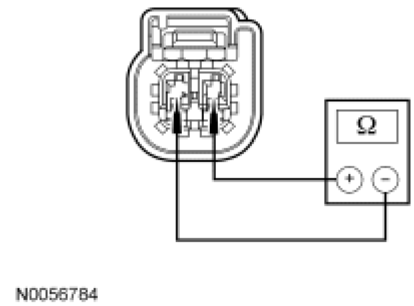

Fig. 110: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger front impact severity sensor C1466-1, circuit VR214 (WH/BU), harness side and C1466-2, circuit RR130 (GN/BU), harness side.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to K32.

  NO : REPAIR circuits VR214 (WH/BU) and RR130 (GN/BU). Go to K34.

K31 CHECK CIRCUITS VR214 (WH/BU) AND RR130 (GN/BU) FOR A SHORT TO VOLTAGE BETWEEN THE RCM AND PASSENGER FRONT IMPACT SEVERITY SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Front Impact Severity Sensor C1466
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

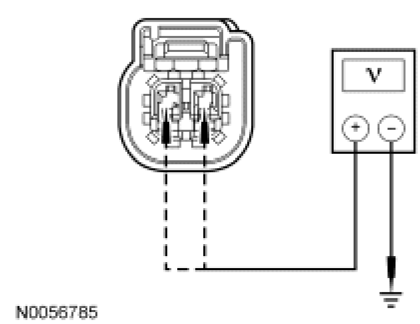

Fig. 111: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between passenger front impact severity sensor C1466-1, circuit VR214 (WH/BU), harness side and ground; and between passenger front impact severity sensor C1466-2, circuit RR130 (GN/BU), harness side and ground.

- **Are voltages less than 0.2 volt?**

  YES : Go to K32.

  NO : REPAIR circuit VR214 (WH/BU) or circuit RR130 (GN/BU). Go to K34.

K32 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Install the original sensor (if known good sensor was installed in a previous step).
- Connect: All Previously Disconnected SRS Components
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs
  - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Does the original RCM on-demand DTC B2296 fault PID indicate a fault?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to K34.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to K33.

K33 CHECK FOR AN INTERMITTENT FAULT

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2296 Fault PIDs
  - Refer to PID list in Normal Operation to view B2296 fault PIDs.
- **Do any on-demand fault DTC B2296 PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is now present. The fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test.

  Using the fault PIDs recorded, go to the appropriate pinpoint test step.

For 2296_30_OD (Side Crash Sensor Internal Fault, Front Driver Side), INSTALL a new first row driver side impact sensor. REFER to **Side Impact Sensor - B-Pillar**. Go to K34.

For 2296_28_OD (Side Crash Sensor Internal Fault, Front Passenger Side), INSTALL a new first row passenger side impact sensor. REFER to **Side Impact Sensor - B-Pillar**. Go to K34.

For 2296_26_OD (Side Crash Sensor 2 Internal Fault, Driver side), INSTALL a new second row driver side impact sensor. REFER to **Side Impact Sensor - C-Pillar**. Go to K34.

For second row passenger side impact sensor with an internal fault (2296_24_OD (Side Crash Sensor 2 Internal Fault, Passenger side), INSTALL a new second row passenger side impact sensor. REFER to **Side Impact Sensor - C-Pillar**. Go to K34.

For 2296_18_OD (Driver/Center Front Crash Sensor Internal Fault), INSTALL a new driver/center front impact severity sensor. REFER to **Front Impact Severity Sensor**. Go to K34.

For 2296_58_OD (Passenger Front Crash Sensor Internal Fault), INSTALL a new passenger front impact severity sensor. REFER to **Front Impact Severity Sensor**. Go to K34.

For 2296_17_OD (Side Crash Sensor Communication Fault, Front Driver Side)/2296_31_OD (Side Crash Sensor Mount or Communication Fault, Front Driver Side), go to K2.

For 2296_16_OD (Side Crash Sensor Circuit Short to Ground, Front Driver Side), go to K4.

For 2296_15_OD (Side Crash Sensor Circuit Short to Battery, Front Driver Side), go to K6.

For 2296_14_OD (Side Crash Sensor Communication Fault, Front Passenger Side), go to K7.

For 2296_13_OD (Side Crash Sensor Circuit Short to Ground, Front Passenger Side), go to K9.

For 2296_12_OD (Side Crash Sensor Circuit Short to Battery, Front Passenger Side), go to K11.

For 2296_11_OD (Side Crash Sensor Communication Fault, Row No.2 Driver Side), go to K12.

For 2296_10_OD (Side Crash Sensor Circuit Short to Ground, Row No.2 Driver Side), go to K14.

For 2296_9_OD (Side Crash Sensor Circuit Short to Battery, Row No.2 Driver Side), go to K16.

For 2296_8_OD (Side Crash Sensor Communication Fault, Row No.2 Passenger Side), go to K17.

For 2296_7_OD (Side Crash Sensor Circuit Short to Ground, Row No.2 Passenger Side), go to K19.

For 2296_6_OD (Side Crash Sensor Circuit Short to Battery, Row No.2 Passenger Side), go to K21.

For 2296_5_OD (Driver/Center Front Crash Sensor Communication Fault), go to K22.

For 2296_4_OD (Driver/Center Front Crash Sensor Circuit Short to Ground), go to K24.

For 2296_3_OD (Driver/Center Front Crash Sensor Circuit Short to Battery), go to K26.

For 2296_59_OD (Passenger Front Crash Sensor Mount/Communication Fault), go to K27.

For 2296_56_OD (Passenger Front Crash Sensor Circuit Short to Ground), go to K29.

For 2296_55_OD (Passenger Front Crash Sensor Circuit Short to Battery), go to K31.

NO : VISUALLY INSPECT the affected impact sensor and mounting surface for damage, corrosion or dirt. INSPECT the wiring, terminals and connectors for damage, corrosion or dirt. CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to K34.

K34 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
    - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test L: DTC B2432 - Drivers Seat Belt Buckle Switch Circuit Open

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) checks the safety belt buckle switch circuits for faults. If the RCM detects an open circuit fault, it will store DTC B2432 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2432 Drivers Seat Belt Buckle Switch Circuit Open - If the RCM detects an open on the driver safety belt buckle circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver safety belt buckle switch
- RCM

PINPOINT TEST L: DTC B2432 - DRIVERS SEAT BELT BUCKLE SWITCH CIRCUIT OPEN

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

L1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2432 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to L2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to L6.

L2 CHECK THE DRIVER SAFETY BELT BUCKLE SWITCH

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Buckle Switch C3065
- Connect a fused jumper wire between driver safety belt buckle switch C3065-1, circuit GD143 (BK/VT), harness side and C3065-2, circuit CR201 (BU/OG).
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the driver safety belt buckle switch circuits shorted together, a short to ground fault would normally be retrieved.
- **Did the driver safety belt buckle switch on-demand DTC change from indicating B2432 to B2434?**

  YES : INSTALL a new driver safety belt buckle. REFER to **SAFETY BELT SYSTEM** article. Go to L7.

  NO : Go to L3.

L3 CHECK CIRCUIT CR201 (BU/OG) FOR AN OPEN BETWEEN THE RCM AND DRIVER SAFETY BELT BUCKLE SWITCH

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

- Remove the fused jumper wire from the driver safety belt buckle switch connector.

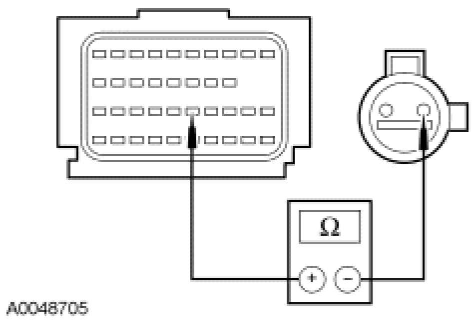

Fig. 112: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-25, circuit CR201 (BU/OG), harness side and driver safety belt buckle switch C3065-2, circuit CR201 (BU/OG), harness side.
- **Is resistance less than 0.5 ohm?**

  YES : Go to L4.

  NO : REPAIR circuit CR201 (BU/OG). Go to L7.

L4 CHECK CIRCUIT GD143 (BK/VT) FOR AN OPEN

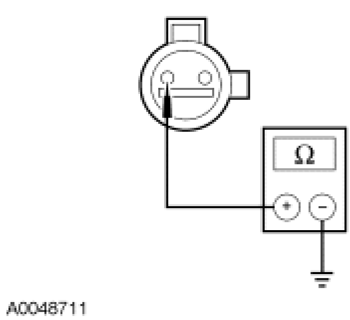

Fig. 113: Measuring Resistance Between Connector And Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver safety belt buckle switch C3065-1, circuit GD143 (BK/VT), harness side and ground.
- **Is resistance less than 0.5 ohm?**

  YES : Go to L5.

  NO : REPAIR circuit GD143 (BK/VT). Go to L7.

L5 CONFIRM THE RCM FAULT

NOTE:	**Make sure all restraint system sensor electrical connectors and the RCM electrical connector are connected before carrying out the on-demand self**

test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Driver Safety Belt Buckle Switch C3065
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2432 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to L7.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to L6.

L6 CHECK FOR AN INTERMITTENT FAULT

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2432 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to L2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to L7.

L7 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints

- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test M: DTC B2433 - Drivers Seat Belt Buckle Switch Circuit Short to Battery

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) checks the safety belt buckle switch circuits for faults. If the RCM detects a short to voltage fault, it will store DTC B2433 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2433 Drivers Seat Belt Buckle Switch Circuit Short to Battery - If the RCM detects a short to battery on the driver safety belt buckle circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver safety belt buckle switch
- RCM

PINPOINT TEST M: DTC B2433 - DRIVERS SEAT BELT BUCKLE SWITCH CIRCUIT SHORT TO BATTERY

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor**

pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

M1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2433 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to M2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to M4.

M2 CHECK CIRCUIT CR201 (BU/OG) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Seat Side Air Bag Module C3628
- Disconnect: Driver Safety Belt Buckle Switch C3065
- Disconnect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

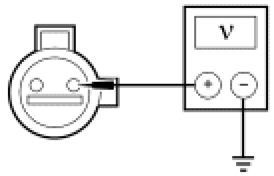

Fig. 114: Measuring Voltage Between connector And Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between driver safety belt buckle switch C3065-2, circuit CR201 (BU/OG), harness side and ground.
- **Is voltage less than 0.2 volt?**

  YES : Go to M3.

  NO : REPAIR circuit CR201 (BU/OG). Go to M5.

M3 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system sensor electrical connectors and the RCM electrical connector are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Driver Safety Belt Buckle Switch C3065
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2433 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to M5.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to M4.

M4 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.

- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Seat Side Air Bag Module C328
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2433 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to M2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to M5.

M5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test N: DTC B2434 - Drivers Seat Belt Buckle Switch Short to Ground

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors the driver safety belt buckle switch circuits for faults. If the RCM detects a short to ground fault, it will store DTC B2434 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2434 Drivers Seat Belt Buckle Switch Circuit Short to Ground - If the RCM detects a short to ground on the driver safety belt buckle circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver safety belt buckle switch
- RCM

PINPOINT TEST N: DTC B2434 - DRIVERS SEAT BELT BUCKLE SWITCH SHORT TO GROUND

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle

to the customer.

N1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2434 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to N2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to N5.

N2 CHECK THE DRIVER SAFETY BELT BUCKLE SWITCH FOR A SHORT TO GROUND

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Buckle Switch C3065
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the driver safety belt buckle switch disconnected, a open circuit fault would normally be retrieved.
- **Did the driver safety belt buckle switch on-demand DTC change from indicating B2434 to B2432?**

  YES : INSTALL a new driver safety belt buckle assembly. REFER to **SAFETY BELT SYSTEM** article. Go to N6.

  NO : Go to N3.

N3 CHECK CIRCUIT CR201 (BU/OG) FOR A SHORT TO GROUND BETWEEN THE RCM AND DRIVER SAFETY BELT BUCKLE SWITCH

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

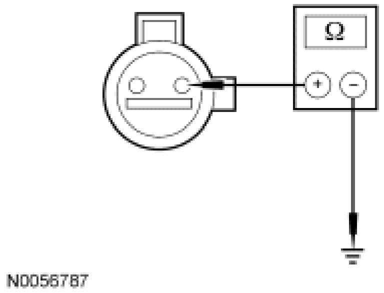

Fig. 115: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver safety belt buckle switch C3065-2, circuit CR201 (BU/OG), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to N4.

  NO : REPAIR circuit CR201 (BU/OG). Go to N6.

N4 CONFIRM THE RCM FAULT

NOTE: **Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Connect: Driver Safety Belt Buckle Switch C3065
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2434 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to N6.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to N5.

N5 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs

- **Was RCM on-demand DTC B2434 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to N2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to N6.

N6 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test O: DTC B2435 - Drivers Seat Belt Buckle Switch Resistance Out of Range

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors the driver safety belt buckle switch circuits for faults. If the RCM detects a current out of range fault, it will store DTC B2435 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2435 Drivers Seat Belt Buckle Switch Resistance Out of Range - If the RCM detects a current out of range between buckled and unbuckled on the driver safety belt buckle switch, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors

- Driver safety belt buckle and pretensioner assembly
- RCM

PINPOINT TEST O: DTC B2435 - DRIVERS SEAT BELT BUCKLE SWITCH RESISTANCE OUT OF RANGE

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

O1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs

- **Was RCM on-demand DTC B2435 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to O2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to O4.

O2 CHECK THE SAFETY BELT BUCKLE SWITCH

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Buckle Switch C3065
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the driver safety belt buckle switch disconnected, a open circuit fault would normally be retrieved.
- **Did the driver safety belt buckle switch on-demand DTC change from indicating B2435 to B2432?**

  YES : INSTALL a new driver safety belt buckle assembly. REFER to **SAFETY BELT SYSTEM** article. Go to O3.

  NO : Go to O5.

O3 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Driver Safety Belt Buckle Switch C3065
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2435 retrieved?**

YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to O5.

NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to O4.

O4 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect the driver safety belt buckle switch C3065:
  - Inspect connector(s) (including any inline connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
  - Inspect wire harness for any damage, pinched, cut or pierced wires.
  - Repair any concerns found. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s)
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2435 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to O2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to O5.

O5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test P: DTC B2436 - Passengers Seat Belt Buckle Switch Circuit Open

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) checks the passenger safety belt buckle switch circuits for faults. If the RCM detects an open circuit fault, it will store DTC B2436 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2436 Passengers Seat Belt Buckle Switch Circuit Open - If the RCM detects an open on the passenger safety belt buckle circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger safety belt buckle switch
- RCM

PINPOINT TEST P: DTC B2436 - PASSENGERS SEAT BELT BUCKLE SWITCH CIRCUIT OPEN

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

P1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2436 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to P2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to P6.

P2 CHECK THE PASSENGER SAFETY BELT BUCKLE SWITCH

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Buckle Switch C3066
- Connect a fused jumper wire between passenger safety belt buckle switch C3066-2, circuit CR203 (GY/VT), harness side and C3066-1, circuit GD143 (BK/VT), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the passenger safety belt buckle switch circuits shorted together, a short to ground fault would normally be retrieved.
- **Did the passenger safety belt buckle switch on-demand DTC change from indicating B2436 to B2438?**

  YES : INSTALL a new passenger safety belt buckle assembly. REFER to **SAFETY BELT SYSTEM** article. Go to P7.

  NO : Go to P3.

P3 CHECK CIRCUIT CR203 (GY/VT) FOR AN OPEN BETWEEN THE RCM AND PASSENGER SAFETY BELT BUCKLE SWITCH

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Remove the fused jumper wire from the passenger safety belt buckle switch C3066.
- Disconnect: RCM C310a and C310b

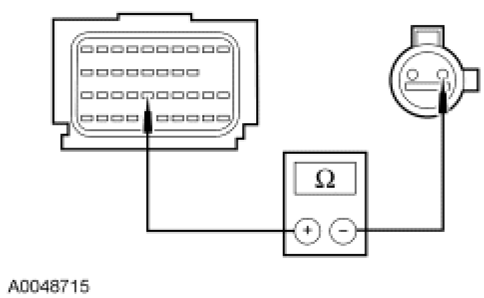

Fig. 116: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-26, circuit CR203 (GY/VT), harness side and the passenger safety belt buckle switch C3066-2, circuit CR203 (GY/VT), harness side.
- **Is resistance less than 0.5 ohm?**

  YES : Go to P4.

  NO : REPAIR circuit CR203 (GY/VT). Go to P7.

P4 CHECK CIRCUIT GD143 (BK/VT) FOR AN OPEN

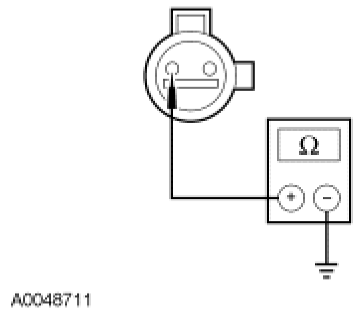

Fig. 117: Measuring Resistance Between Connector And Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger safety belt buckle switch C3066-2, circuit GD143 (BK/VT), harness side and ground.
- **Is resistance less than 0.5 ohm?**

  YES : Go to P5.

  NO : REPAIR circuit GD143 (BK/VT). Go to P7.

P5 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system sensor electrical connectors and the RCM electrical connector are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Passenger Safety Belt Buckle Switch C3066
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2436 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to P7.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to P6.

P6 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2436 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to P2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to P7.

P7 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test Q: DTC B2437 - Passengers Seat Belt Buckle Switch Circuit Short to Battery

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) checks the passenger safety belt buckle switch circuits for faults. If the RCM detects a short to voltage fault, it will store DTC B2437 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2437 Passengers Seat Belt Buckle Switch Circuit Short to Battery - If the RCM detects a short to battery on the passenger safety belt buckle circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger safety belt buckle switch
- RCM

PINPOINT TEST Q: DTC B2437 - PASSENGER'S SEAT BELT BUCKLE SWITCH CIRCUIT SHORT TO BATTERY

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

Q1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2437 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to Q2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to Q4.

Q2 CHECK CIRCUIT CR203 (GY/VT) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b
- Disconnect: Passenger Safety Belt Buckle Switch C3066
- Disconnect: Passenger Seat Side Air Bag Module C327
- Disconnect: Passenger Safety Belt Retractor Pretensioner C303
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

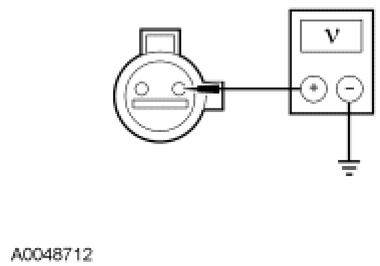

Fig. 118: Measuring Voltage Between connector And Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between passenger seat belt buckle switch C3066-2, circuit CR203 (GY/VT), harness side and ground.
- **Is voltage less than 0.2 volt?**

  YES : Go to Q3.

  NO : REPAIR circuit CR203 (GY/VT). Go to Q5.

Q3 CONFIRM THE RCM FAULT

NOTE: **Make sure all restraint system sensor electrical connectors and the RCM electrical connector are connected before carrying out the on-demand self**

test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: RCM C310a and C310b
- Connect: Passenger Safety Belt Buckle Switch C3066
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2437 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to Q5.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to Q4.

Q4 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C303
- Disconnect: Passenger Seat Side Air Bag Module C327
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2437 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to Q2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to Q5.

Q5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: Turn the ignition OFF and wait one minute to deplete the

backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test R: DTC B2438 - Passengers Seat Belt Buckle Switch Short to Ground

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors the passenger safety belt buckle switch circuits for faults. If the RCM detects a short to ground fault, it will store DTC B2438 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2438 Passenger's Seat Belt Buckle Switch Circuit Short to Ground - If the RCM detects a short to ground on the passenger safety belt buckle circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger safety belt buckle switch
- RCM

PINPOINT TEST R: DTC B2438 - PASSENGERS SEAT BELT BUCKLE SWITCH SHORT TO GROUND

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and**

the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

R1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2438 retrieved?**

YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to R2.

NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to R5.

R2 CHECK THE PASSENGER SAFETY BELT BUCKLE SWITCH

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Buckle Switch C3066
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the passenger safety belt buckle switch disconnected, a open circuit fault would normally be retrieved.
- **Did the passenger safety belt buckle switch on-demand DTC change from indicating B2438 to B2436?**

  YES : INSTALL a new passenger safety belt buckle assembly. REFER to **SAFETY BELT SYSTEM** article. Go to R6.

  NO : Go to R3.

R3 CHECK CIRCUIT CR203 (GY/VT) FOR A SHORT TO GROUND BETWEEN THE RCM AND PASSENGER SAFETY BELT BUCKLE SWITCH

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

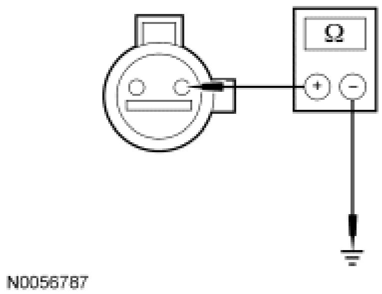

Fig. 119: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between passenger seat belt buckle switch C3066-2, circuit CR203 (GY/VT), harness side and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to R4.

  NO : REPAIR circuit CR203 (GY/VT). Go to R6.

R4 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Connect: Passenger Safety Belt Buckle Switch C3066
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2438 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to R6.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to R5.

R5 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2438 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to R2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to R6.

R6 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.

- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test S: DTC B2439 - Passengers Seat Belt Buckle Switch Resistance Out of Range

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors the passenger safety belt buckle switch circuits for faults. If the RCM detects a current out of range fault, it will store DTC B2439 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B2439 Passengers Seat Belt Buckle Switch Resistance Out of Range - If the RCM detects a current out of range between buckled and unbuckled on the passenger safety belt buckle switch, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger safety belt buckle switch
- RCM

PINPOINT TEST S: DTC B2439 - PASSENGERS SEAT BELT BUCKLE SWITCH RESISTANCE OUT OF RANGE

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

S1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2439 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to S2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to S4.

S2 CHECK THE SAFETY BELT BUCKLE SWITCH

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Passenger Safety Belt Buckle Switch C3066
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs

- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the passenger safety belt buckle switch disconnected, a open circuit fault would normally be retrieved.
- **Did the passenger safety belt buckle switch on-demand DTC change from indicating B2439 to B2436?**

  YES : INSTALL a new passenger safety belt buckle assembly. REFER to **SAFETY BELT SYSTEM** article. Go to S5.

  NO : Go to S3.

S3 CONFIRM THE RCM FAULT

NOTE: **Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Passenger Safety Belt Buckle Switch C3066
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2439 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to S5.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to S4.

S4 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect the passenger safety belt buckle switch C3066:
  - Inspect connector(s) (including any inline connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
  - Inspect wire harness for any damage, pinched, cut or pierced wires.
  - Repair any concerns found. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s)

- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC B2439 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to S2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to S5.

S5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test T: DTC B2792 - Cross Link Between Firing Loops

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors all the deployable devices for a cross link between the circuits of another deployable device. If the RCM detects a short between the circuits of the deployable devices, it will store DTC B2792 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

Fault PIDs[a]	Description	Fault Trigger Condition
2792_8_OD and 2792_8_CM	A-B or A-C pillar Curtain, Passenger Side	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_9_OD and 2792_9_CM	A-B or A-C pillar Curtain, Driver Side	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_12_OD and 2792_12_CM	Side Airbag Passenger	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_13_OD and 2792_13_CM	Side Airbag Driver	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_26_OD and 2792_26_CM	Pretensioner Passenger	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_27_OD and 2792_27_CM	Pretensioner Driver	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_28_OD and 2792_28_CM	Airbag Passenger Front Loop #2	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_29_OD and 2792_29_CM	Airbag Passenger Front Loop #1	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_30_OD and 2792_30_CM	Airbag Driver Front Loop #2	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.
2792_31_OD and 2792_31_CM	Airbag Driver Front Loop #1	When the RCM detects a short between the circuits of 2 deployable devices, a fault will be indicated.

[a] Fault PIDs that end in OD indicate an on-demand status and are associated with on-demand DTC B2792. Fault PIDs that end in CM indicate continuous memory status and are associated with continuous DTC B2792.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RCM

PINPOINT TEST T: DTC B2792 - CROSS LINK BETWEEN FIRING LOOPS

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

T1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record

On-Demand and Continuous Memory DTCs

- Enter the following diagnostic mode on the diagnostic tool: Datalogger - RCM - View and Record All B2792 Fault PIDs
- **Do any RCM on-demand B2792 fault PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to T2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to T4.

T2 CHECK DEPLOYABLE CIRCUITS FOR A CROSS LINK FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b
- Disconnect **ALL of the affected** deployable devices.
- Using the following chart, measure the resistance between the circuits of the affected deployable devices.

DEPLOYABLE DEVICES

Squib/Loop	Connector - Pin	Circuit
Driver air bag module loop 1	• Driver air bag module loop 1 electrical connector • Driver air bag module loop 1 electrical connector	• CR101 (VT/BN) • RR101 (YE/GN)
Driver air bag module loop 2	• Driver air bag module loop 2 electrical connector • Driver air bag module loop 2 electrical	• CR102 (BU) • RR102 (WH)

	connector	
Passenger air bag module loop 1	• C256a-1 • C256a-2	• CR103 (GY/BU) • RR103 (VT/GN)
Passenger air bag module loop 2	• C256b-1 • C256b-2	• CR104 (YE/GY) • RR104 (WH/BU)
Driver side air bag module (seat harness connector)	• C328-1 • C328-2	• CR105 (GN/BN) • RR105 (GY/YE)
Passenger side air bag module (seat harness connector)	• C327-1 • C327-2	• CR106 (VT/GY) • RR106 (YE/OG)
Driver safety canopy module	• C9018-1 • C9018-2	• CR109 (BN/BU) • RR109 (BU/GN)
Passenger safety canopy module	• C9019-1 • C9019-2	• CR111 (BN/WH) • RR111 (YE/VT)
Driver safety belt retractor pretensioner	• C3201-2 • C3201-1	• CR120 (BU/OG) • RR120 (BN/GN)
Passenger safety belt retractor pretensioner	• C3202-2 • C3202-1	• CR122 (WH/OG) • RR122 (BN)

- **Are resistances greater than 10,000 ohms between the affected circuits?**

  YES : Go to T3.

  NO : REPAIR the affected circuits. Go to T5.

T3 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Connect: All Previously Disconnected Restraint System Component(s)
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2792 Fault PIDs
- **Do any RCM on-demand B2792 fault PIDs indicate a fault?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to T5.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to T4.

T4 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- If a cross link fault for driver air bag loop 1 or loop 2 is present continuous, remove the driver air bag module. Refer to **Driver Air Bag Module**.
- If a cross link fault for passenger air bag module loop 1 or loop 2 is present continuous, disconnect the passenger air bag module C256a and C256b.
- If a cross link fault for driver side air bag is present continuous, disconnect the driver side air bag module C328.
- If a cross link fault for passenger side air bag is present continuous, disconnect the passenger side air bag module C327.
- If a cross link fault for driver safety canopy module is present continuous, disconnect the driver safety canopy module C9018.
- If a cross link fault for passenger safety canopy module is present continuous, disconnect the passenger safety canopy module C9019.
- If a cross link fault for driver safety belt retractor pretensioner is present continuous, disconnect the driver safety belt retractor pretensioner C323.
- If a cross link fault for passenger safety belt retractor pretensioner is present continuous, disconnect the passenger safety belt retractor pretensioner C303.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint**

System (SRS) Depowering and Repowering.

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - RCM - View and Record All B2792 Fault PIDs
- **Do any RCM on-demand B2792 fault PIDs indicate a fault?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to T2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to T5.

T5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test U: DTC B229B - Occupant Classification System Obstruction

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The occupant classification sensor (OCS) system is used to classify the front passenger seat occupant in the event of a deployable impact. Refer to **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.

The occupant classification system module (OCSM) monitors the 4 OCS weight sensors bolts and circuity for faults. If the OCSM detects an obstruction, an upward force on the passenger seat for 60 seconds, it will store DTC B229B in memory and send a message to the restraints control module (RCM). The RCM will store DTC B2290 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC B229B Occupant Classification System Obstruction - If the OCSM detects a negative force on one or more OCS weight sensor bolt(s) (weight sensors), it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Foreign object placed under or contacting the passenger seat

PINPOINT TEST U: DTC B229B - OCCUPANT CLASSIFICATION SYSTEM OBSTRUCTION

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

U1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record Continuous Memory DTCs
- **Was OCSM on-demand DTC B229B retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to U2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to U4.

U2 INSPECT UNDER THE PASSENGER SEAT FOR FOREIGN OBJECTS AND WIRE HARNESS

- Key in OFF position.
- Position the seat and inspect under and on the sides of the seat for the presence of any foreign objects contacting the under side and side of the passenger seat. Carry out a thorough visual inspection of the passenger seat wire harness at or near the OCS weight sensor bolts and the related seat wiring harness and body wiring harness for correct harness routing.
- **Were any problems noted?**

  YES : REMOVE obstruction and/or REPAIR as necessary. Go to U5.

  NO : Go to U3.

U3 CONFIRM THE OCS MODULE FAULT

- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record Continuous Memory DTCs
- **Was OCSM on-demand DTC B229B retrieved?**

  YES : INSTALL a new OCSM. REFER to **Occupant Classification System Module (OCSM) - Manual Seat Track** or **Occupant Classification System Module (OCSM) - Power Seat Track**. Go to U5.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to U5.

U4 CHECK FOR AN INTERMITTENT FAULT

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - OCSM - Retrieve and Record Continuous Memory DTCs
- **Was OCSM on-demand DTC B229B retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to U2.

NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to U5.

U5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test V: DTC C1946 - Front Driver's Seat Track Position Switch Circuit Open

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors the driver seat track position sensor circuits. If the RCM detects an open circuit, it will store DTC C1946 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC C1946 Front Driver's Seat Track Position Switch Circuit Open - If the RCM detects a open on the driver seat track position sensor circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver seat track position sensor
- RCM

PINPOINT TEST V: DTC C1946 - FRONT DRIVER'S SEAT TRACK POSITION SWITCH CIRCUIT

OPEN

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

V1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1946 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to V2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to V6.

V2 CHECK THE DRIVER SEAT TRACK POSITION SENSOR

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Seat Track Position Sensor C356
- Connect a fused jumper wire between driver seat track position sensor C356-2, circuit VR215 (YE/VT), harness side and C356-1, circuit GD143 (BK/VT), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the driver seat track position sensor circuits shorted together, a short to ground fault would normally be retrieved.
- **Did the driver seat track position switch on-demand DTC change from indicating C1946 to C1947?**

  YES : INSTALL a new driver seat track position sensor. REFER to **Seat Position Sensor**. Go to V7.

  NO : Go to V3.

V3 CHECK CIRCUIT VR215 (YE/VT) FOR AN OPEN BETWEEN THE RCM AND DRIVER SEAT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b
- Remover the fused jumper wire from the driver seat track position sensor C356.

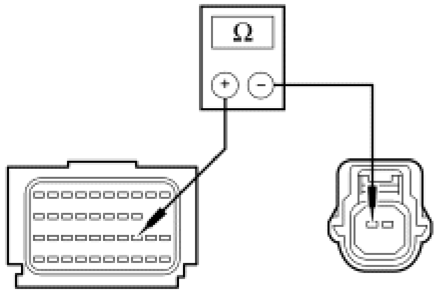

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Fig. 120: Measuring Resistance Between RCM C310B-23, Circuit VR215 (YE/VT) & Driver Seat Track Position Sensor C356-2
Courtesy of FORD MOTOR CO.

- Measure the resistance between RCM C310b-23, circuit VR215 (YE/VT), harness side and driver seat track position sensor C356-2, circuit VR215 (YE/VT).
- **Is resistance less than 0.5 ohm?**

  YES : Go to V4.

  NO : REPAIR circuit VR215 (YE/VT). Go to V7.

V4 CHECK CIRCUIT GD143 (BK/VT) FOR AN OPEN TO GROUND

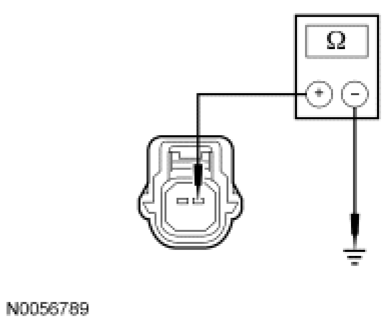

Fig. 121: Measuring Resistance Between Driver Seat Track Position Sensor C356-1, Circuit GD143 (BK/VT) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver seat track position sensor C356-1, circuit GD143 (BK/VT) and ground.
- **Is resistance less than 0.5 ohm?**

  YES : Go to V5.

  NO : REPAIR circuit GD143 (BK/VT). Go to V7.

V5 CONFIRM THE RCM FAULT

NOTE: **Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Driver Seat Track Position Sensor C356
- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1946 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to V7.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to V6.

V6 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Disconnect: Driver Side Air Bag Module C328
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1946 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to V2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to V7.

V7 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints

- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test W: DTC C1947 - Front Driver's Seat Track Position Switch Circuit Short to Ground

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors the driver seat track position sensor circuits. If the RCM detects a short to ground, it will store DTC C1947 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC C1947 Front Driver's Seat Track Position Switch Circuit Short to Ground - If the RCM detects a short to ground on the driver seat track position sensor circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver seat track position sensor
- RCM

PINPOINT TEST W: DTC C1947 - FRONT DRIVER'S SEAT TRACK POSITION SWITCH CIRCUIT SHORT TO GROUND

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor**

pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.

NOTE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

W1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1947 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to W2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to W5.

W2 CHECK THE SEAT TRACK POSITION SENSOR

NOTE: This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Seat Track Position Sensor C356
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs

- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the driver seat track position sensor disconnected, an open circuit fault would normally be retrieved.
- **Did the driver seat track position switch on-demand DTC change from indicating C1947 to C1946?**

  YES : INSTALL a new driver seat track position sensor. REFER to **Seat Position Sensor**. Go to W6.

  NO : Go to W3.

W3 CHECK CIRCUIT VR215 (YE/VT) FOR A SHORT TO GROUND BETWEEN THE RCM AND SEAT TRACK POSITION SENSOR

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b

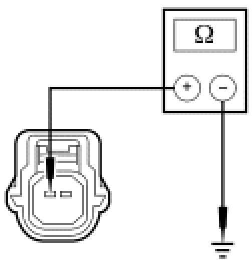

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Fig. 122: Measuring Resistance Between Driver Seat Track Position Sensor C356-2, Circuit VR215 (YE/VT) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between driver seat track position sensor C356-2, circuit VR215 (YE/VT) and ground.
- **Is resistance greater than 10,000 ohms?**

  YES : Go to W4.

  NO : REPAIR circuit VR215 (YE/VT). Go to W6.

W4 CONFIRM THE RCM FAULT

NOTE:	**Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Seat Track Position Sensor C356

- Connect: RCM C310a and C310b
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1947 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to W6.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to W5.

W5 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Disconnect: Driver Side Air Bag Module C3051
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1947 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to W2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to W6.

W6 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at

this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test X: DTC C1948 - Front Driver's Seat Track Position Switch Circuit Resistance Out of Range

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

NOTE: **Due to the seat track position sensor being a Hall-effect type sensor, this pinpoint test will be diagnosing a current out of range fault instead of the current DTC definition for a resistance out of range fault.**

Normal Operation

The restraints control module (RCM) monitors the driver seat track position sensor circuits. If the RCM detects a current out of range condition, it will store DTC C1948 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC C1948 Front Driver's Seat Track Position Switch Circuit Resistance Out of Range - If the RCM detects a current out of range between forward and rearward on the driver seat track position switch, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver seat track position sensor
- RCM

PINPOINT TEST X: DTC C1948 - FRONT DRIVER'S SEAT TRACK POSITION SWITCH CIRCUIT RESISTANCE OUT OF RANGE

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in**

the accidental deployment of the safety canopy and cause serious personal injury or death.

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.**

NOTE: **The SRS must be fully operational and free of faults before releasing the vehicle to the customer.**

X1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1948 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to X2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to X4.

X2 CHECK THE DRIVER SEAT TRACK POSITION SENSOR

NOTE: **This pinpoint test step will attempt to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Seat Track Position Sensor C356
- Connect a fused jumper wire between driver seat track position sensor C356-2, circuit VR215 (YE/VT), harness side and C356-1, circuit GD143 (BK/VT), harness side.
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **DIAGNOSTIC TIP:** When viewing on-demand DTCs with the driver seat track position sensor circuits shorted together, a short to ground fault would normally be retrieved.
- **Did the driver seat track position sensor on-demand DTC change from indicating C1948 to C1947?**

  YES : INSTALL a new driver seat track position sensor. REFER to **Seat Position Sensor**. Go to X5.

  NO : Go to X3.

X3 CONFIRM THE RCM FAULT

NOTE: **Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.**

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: Driver Seat Track Position Sensor C356
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1948 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to X5.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of electrical connectors that were disconnected. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to X4.

X4 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Disconnect: Driver Side Air Bag Module C328
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1948 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to X2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to X5.

X5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to**DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test Y: DTC C1982 - Front Driver's Seat Track Position Switch Circuit Short to Battery

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The restraints control module (RCM) monitors the driver seat track position sensor circuits. If the RCM detects a short to battery, it will store DTC C1982 in memory and send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

- DTC C1982 Front Driver's Seat Track Position Switch Circuit Short to Battery - If the RCM detects a short to battery on the driver seat track position sensor circuit, it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RCM

PINPOINT TEST Y: C1982 - FRONT DRIVER'S SEAT TRACK POSITION SWITCH CIRCUIT SHORT TO BATTERY

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the safety canopy and cause serious personal injury or death.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

NOTE: **Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

NOTE: **Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE: **Supplemental restraint system (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to**

follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

Y1 CHECK FOR CONTINUOUS AND ON-DEMAND DTCs

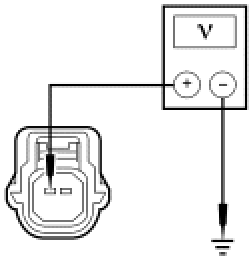
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1982 retrieved?**

  YES : This is a hard fault. The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to Y2.

  NO : This is an intermittent fault when present as a continuous memory DTC only (DTC not retrieved on demand). The fault condition is not present at this time. Go to Y4.

Y2 CHECK CIRCUITS VR215 (YE/VT) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: RCM C310a and C310b
- Disconnect: Driver Seat Track Position Sensor C356
- Disconnect: Driver Side Air Bag Module C328
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.



N0056791

Fig. 123: Measuring Voltage Between Driver Seat Track Position Sensor C356-2, Circuit VR215 (YE/VT) & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between driver seat track position sensor C356-2, circuit VR215 (YE/VT), harness side and ground.
- **Is voltage less than 0.2 volt?**

  YES : Go to Y3.

NO : REPAIR circuit VR215 (YE/VT). Go to Y5.

Y3 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the RCM electrical connectors are connected before carrying out the on-demand self test. If not, erroneous DTCs will be recorded.

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Connect: RCM C310a and C310b
- Connect: Driver Seat Track Position Sensor C356
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1982 retrieved?**

  YES : INSTALL a new RCM. REFER to **Restraints Control Module (RCM)**. Go to Y5.

  NO : In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connector that were disconnected. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test.** REPAIR any intermittent wiring, terminal or connector concerns found. Go to Y4.

Y4 CHECK FOR AN INTERMITTENT FAULT

- Key in OFF position.
- Depower the SRS. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Disconnect: Driver Safety Belt Retractor Pretensioner C323
- Disconnect: Driver Side Air Bag Module C328
- Repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - RCM - Retrieve and Record On-Demand and Continuous Memory DTCs
- **Was RCM on-demand DTC C1982 retrieved?**

  YES : This is a hard fault. The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. Go to Y2.

  NO : CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. **Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint**

test. REPAIR any intermittent wiring, terminal or connector concerns found. Go to Y5.

Y5 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

> WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

  YES : Do not clear any DTCs until all DTCs have been resolved. Go to **DTC Charts** for pinpoint test direction.

  NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

Pinpoint Test Z: DTC B1013 - Occupant Classification Calibration Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Supplemental Restraint System for schematic and connector information.

Normal Operation

The occupant classification sensor (OCS) system is used to classify the front passenger seat occupant in the event of a deployable impact. Refer to **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.

The occupant classification system module (OCSM) monitors the OCS weight sensor bolts for a correct calibration/weight and ID from each sensor. If the OCSM detects an incorrect weight or ID from any of the 4 seat weight sensors, a seat track that is mis-aligned/bent, wire harness connected to the incorrect OCS weight sensor bolt, an unsuccessful system reset or if an OCS system component has been installed new, it will store DTC B1013 in memory and send a message to the restraints control module (RCM) which stores DTC B2290 in memory. The RCM will then send a message to the instrument cluster (IC) module to illuminate the air bag warning indicator.

This pinpoint test is intended to diagnose the following:

- Wiring of OCS weight sensor bolts
- Wiring, terminals or connectors
- OCS system component installed new

- Passenger seat track
- Misrouted harness
- Seat track mounting and alignment
- Seat track fastener torque

PINPOINT TEST Z: DTC B1013 - OCCUPANT CLASSIFICATION CALIBRATION FAULT

NOTE:	**Do not install any new OCS system components until directed to do so during the pinpoint test.**

NOTE:	**The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.**

NOTE:	**Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.**

NOTE:	**If a new OCS system component has been installed, the OCSM will store DTC B1013, causing the air bag warning indicator to illuminate during the prove out. This DTC will clear after a successful system reset.**

Z1 CHECK THE SEAT HARNESS WIRING AND CONNECTORS

- Key in OFF position.
- Carry out a thorough visual inspection of the OCS system harness wiring, terminals and connectors and the related seat wiring harness and body wiring harness terminals and connectors for any damage, misrouting or pulling on any OCS weight sensor bolt.

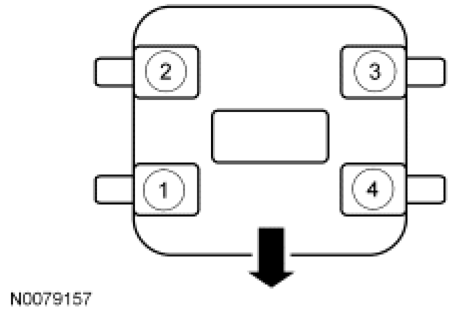


Fig. 124: Identifying OCS System Harness Connector
Courtesy of FORD MOTOR CO.

NOTE:	**The OCS weight sensor bolts do not have differently keyed connectors. Connecting the incorrect connector to a sensor will cause DTC B1013 to set and may cause other erroneous DTCs to set.**

- Using the illustration, identify that the sensors are connected correctly. Refer to appropriate

SYSTEM WIRING DIAGRAMS article for schematic and connector information.

OCS Weight Sensor Bolt Location

- OCS weight sensor bolt 1, located at the front outboard
- OCS weight sensor bolt 2, located at the rear outboard
- OCS weight sensor bolt 3, located at the rear inboard
- OCS weight sensor bolt 4, located at the front inboard

- **Were any wiring or connector concerns noted?**

  YES : REPAIR the seat connectors and wiring as needed. CARRY OUT the OCS system reset. REFER to **Occupant Classification Sensor (OCS) System Reset**. Go to Z3.

  NO : Go to Z2.

Z2 CHECK THE SEAT TRACK

NOTE: **Correct seat track alignment on a passenger front manual seat (if equipped) is critical before removing the seat from the vehicle. During seat installation, the inboard and outboard tracks must be in alignment with each other. Failure to follow this instruction may result in an unsuccessful system reset and incorrect operation of the OCS system.**

NOTE: **Check for a bent or binding power seat track assembly (if equipped). Position the seat in all positions and check for correct operation after installation of the seat into the vehicle. Failure to follow this instruction may result in an unsuccessful system reset and incorrect operation of the OCS system.**

NOTE: **The passenger seat bolts must be tightened to the correct specifications and in the correct sequence. Failure to follow this instruction may result in an unsuccessful system reset and incorrect operation of the OCS system. Refer to SEATING article Seat Removal and Installation.**

- Loosen the 4 seat track to floor mounting bolts.
- Tighten the seat track-to-floor bolts in the following sequence.
  A. Tighten the front inboard seat track-to-floor bolt to 55 Nm (41 lb-ft).
  B. Tighten the front outboard seat track-to-floor bolt to 55 Nm (41 lb-ft).
  C. Tighten the rear inboard seat track-to-floor bolt to 55 Nm (41 lb-ft).
  D. Tighten the rear outboard seat track-to-floor bolt to 55 Nm (41 lb-ft).
- Carry out the OCS system reset. Refer to **Occupant Classification Sensor (OCS) System Reset**.
- **Did the system reset pass?**

  YES : Fault corrected. Go to Z3.

NO : INSTALL a new seat track with OCS weight sensor bolts. REFER to **SEATING** article. Go to Z3.

Z3 CHECK FOR ADDITIONAL SRS DTCs

- Key in OFF position.

WARNING: **Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all SRS components (if previously disconnected).
  - If previously directed to depower the SRS, repower the SRS. **Do not** prove out the SRS at this time. Refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self Test - Restraints
- **Are any RCM and/or OCSM DTCs retrieved indicating a fault?**

YES : Do not clear any DTCs until all DTCs have been resolved. Go to the Supplemental Restraint System (SRS) DTC Chart for pinpoint test direction.

NO : CLEAR all RCM and OCSM continuous memory DTCs. PROVE OUT the SRS. REPAIR is complete. RETURN the vehicle to the customer.

GENERAL PROCEDURES

SUPPLEMENTAL RESTRAINT SYSTEM (SRS) DEPOWERING AND REPOWERING

Depowering Procedure

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **To reduce the risk of accidental deployment, do not use any memory saver devices. Failure to follow this instruction may result in serious personal injury or death.**

NOTE: **The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.**

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Turn all vehicle accessories OFF.
2. Turn the ignition switch to OFF.
3. At the smart junction box (SJB), located in the LH lower kick panel, remove the lower kick panel fuse cover and the RCM fuse 46 (7.5A) from the SJB. For additional information, refer to the appropriate Wiring service information.
4. Turn the ignition ON and visually monitor the air bag warning indicator for at least 30 seconds. The air bag warning indicator will remain lit continuously (no flashing) if the correct RCM fuse has been removed. If the air bag warning indicator does not remain lit continuously, remove the correct RCM fuse before proceeding.
5. Turn the ignition switch to OFF.

WARNING: **Always deplete the backup power supply before repairing or installing any new front or side air bag supplemental restraint system (SRS) component and before servicing, removing, installing, adjusting or striking components near the front or side impact sensors or the restraints control module (RCM). Nearby components include doors, instrument panel, console, door latches, strikers, seats and hood latches.**

See DESCRIPTION AND OPERATION for location of the RCM and impact sensor(s).

To deplete the backup power supply energy, disconnect the battery ground cable and wait at least 1 minute. Be sure to disconnect auxiliary batteries and power supplies (if equipped).

Failure to follow these instructions may result in serious personal injury or death in the event of an accidental deployment.

6. Disconnect the battery ground cable and wait at least one minute. For additional information, refer to **BATTERY, MOUNTING & CABLES** article.

Repowering Procedure

1. Turn the ignition switch from OFF to ON.
2. Install RCM fuse 46 (7.5A) to the SJB and install the lower kick panel fuse cover.

WARNING: **Make sure no one is in the vehicle and there is nothing blocking or placed in front of any air bag module when the battery is connected. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.**

3. Connect the battery ground cable.
4. Prove out the SRS as follows:

   Turn the ignition switch from ON to OFF. Wait 10 seconds, then turn the ignition switch back to ON and visually monitor the air bag warning indicator with the air bag modules installed. The air bag warning indicator will light continuously for approximately 6 seconds and then turn OFF. If an air bag SRS fault is present, the air bag warning indicator will:

   - fail to light.

   - remain lit continuously.

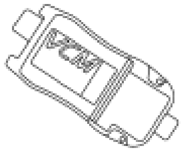
   - flash at a 5 Hz rate (RCM not configured).

   The air bag warning indicator might not light until approximately 30 seconds after the ignition switch has been turned from the OFF to the ON position. This is the time required for the RCM to complete the testing of the SRS. If the air bag warning indicator is inoperative and a SRS fault exists, a chime will sound in a pattern of 5 sets of 5 beeps. If this occurs, the air bag warning indicator and any SRS fault discovered must be diagnosed and repaired.

   Clear all continuous DTCs from the RCM and occupant classification system module (OCSM) using a scan tool.

SUPPLEMENTAL RESTRAINT SYSTEM (SRS) DEACTIVATION AND REACTIVATION

Special Tools

Illustration	Tool Name	Tool Number
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS)	software with appropriate hardware, or equivalent scan tool

Deactivation

WARNING: Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.

WARNING: Always carry or place a live air bag module with the air bag and deployment door/trim cover/tear seam pointed away from the body. Do not set a live air bag module down with the deployment door/trim cover/tear seam face down. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

WARNING: **To reduce the risk of accidental deployment, do not use any memory saver devices. Failure to follow this instruction may result in serious personal injury or death.**

NOTE: The air bag warning indicator illuminates when the restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

All vehicles

1. Turn all vehicle accessories OFF.
2. Turn the ignition switch to OFF.
3. At the smart junction box (SJB), located in the LH lower kick panel, remove the lower kick panel fuse cover and the RCM fuse 46 (7.5A) from the SJB. For additional information, refer to the appropriate Wiring service information.
4. Turn the ignition ON and visually monitor the air bag warning indicator for at least 30 seconds. The air bag warning indicator will remain lit continuously (no flashing) if the correct RCM fuse has been removed. If the air bag warning indicator does not remain lit continuously, remove the correct RCM fuse before proceeding.
5. Turn the ignition OFF.

WARNING: **Always deplete the backup power supply before repairing or installing any new front or side air bag supplemental restraint system (SRS) component and before servicing, removing, installing, adjusting or striking components near the front or side impact sensors or the restraints control module (RCM). Nearby components include doors, instrument panel, console, door latches, strikers, seats and hood latches.**

See <u>DESCRIPTION AND OPERATION</u> for location of the RCM and impact sensor(s).

To deplete the backup power supply energy, disconnect the battery ground cable and wait at least 1 minute. Be sure to disconnect auxiliary batteries and power supplies (if equipped).

Failure to follow these instructions may result in serious personal injury or death in the event of an accidental deployment.

6. Disconnect the battery ground cable and wait at least one minute. For additional information, refer to **BATTERY, MOUNTING & CABLES** article.

NOTE: **Repeat this step for both locking pins.**

7. Using a 3-mm Allen wrench or a suitable tool through the access hole on the backside of the steering wheel, position the tool against the spring clip and push in, disengaging the clip from the locking pin. With the spring clip disengaged from the locking pin, gently pull back on that side of the driver air bag module to release it from the steering wheel.

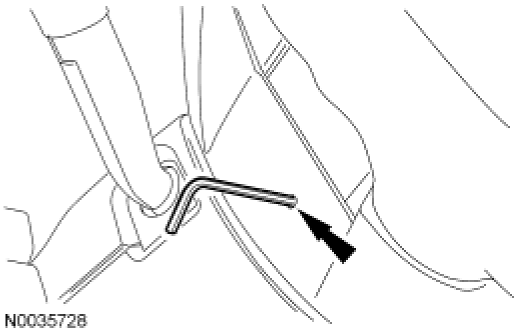


Fig. 125: Locating Allen Wrench
Courtesy of FORD MOTOR CO.

NOTE: **Do not pull the driver air bag module electrical connectors out by the locking buttons. Damage to the locking buttons may occur.**

8. Using a small screwdriver as shown in illustration, lift up and release the locking buttons on the driver air bag module electrical connectors. With the locking buttons released, remove the electrical connectors and the driver air bag module.

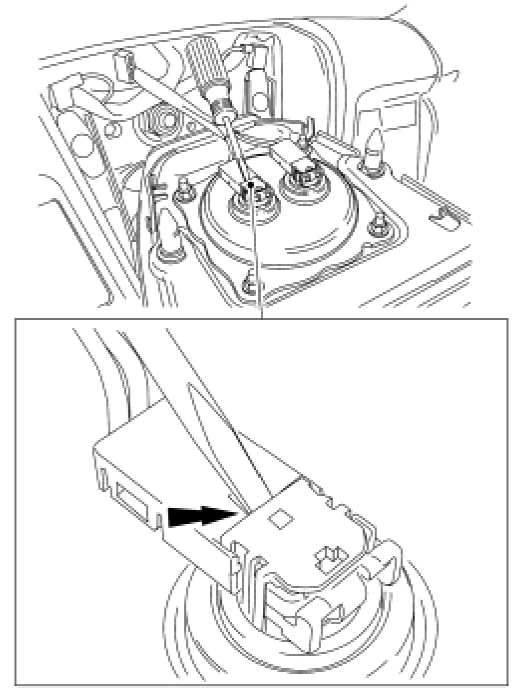


Fig. 126: Releasing Locking Buttons On Driver Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

9. Open and lower the glove compartment door. Then unhook the glove compartment door from the instrument panel.
    - If equipped, detach the glove compartment door dampener.

Edge

10. Remove the 2 passenger air bag module bolts from the cross vehicle beam bracket.

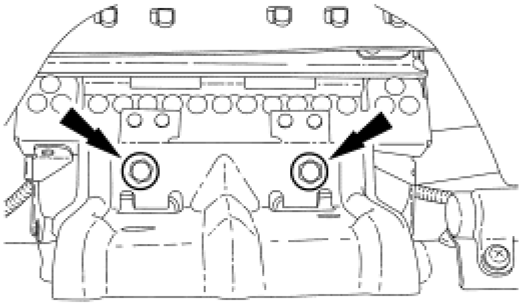


Fig. 127: Locating Passenger Air Bag Module Bolts From Cross Vehicle Beam Bracket
Courtesy of FORD MOTOR CO.

NOTE: **Do not handle the passenger air bag module by grabbing the edges of the deployment doors. Damage to the air bag module may occur.**

11. Through the glove compartment opening, release the 4 LH side deployment door clips. Then release the foreward-most 5 clips and separate the deployment door and passenger air bag module from the instrument panel.

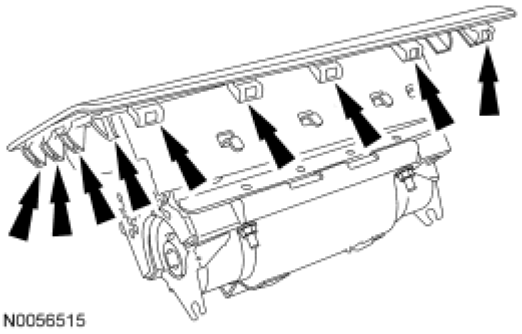


Fig. 128: Locating LH Side Deployment Door Clip
Courtesy of FORD MOTOR CO.

12. Using a small screwdriver as shown in illustration, lift up and release the locking button on the RH passenger air bag module electrical connector.

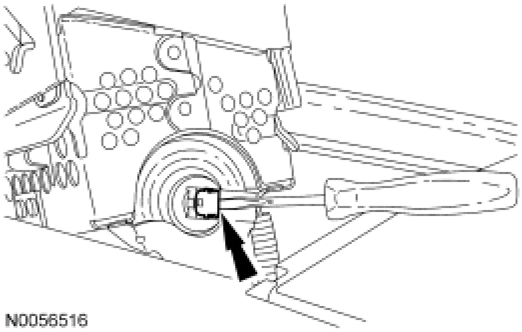


Fig. 129: Releasing Locking Button On RH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE: Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.

13. With the RH locking button released, remove the RH electrical connector.
14. Using a small screwdriver as shown in illustration, lift up and release the locking button on the LH passenger air bag module electrical connector.

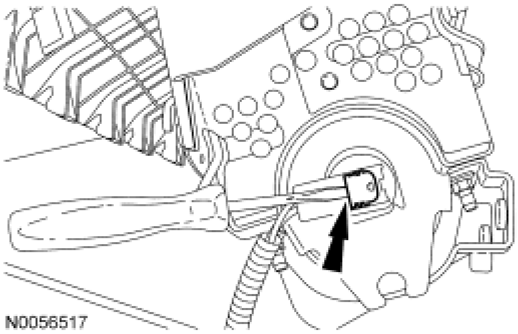


Fig. 130: Releasing Locking Button On LH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE: **Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.**

15. With the LH locking button released, remove the LH electrical connector and remove the passenger air bag module.

MKX

16. Pull straight out to release the 5 retaining clips and remove the RH instrument panel finish panel.

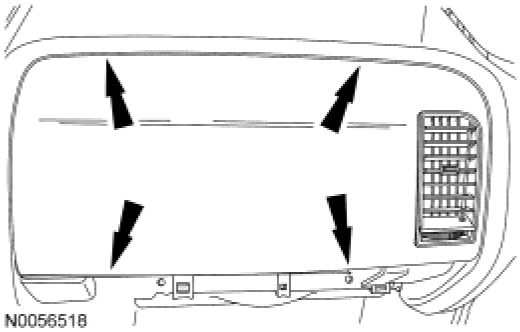


Fig. 131: Locating Retaining Clips On RH I/P Finish Panel
Courtesy of FORD MOTOR CO.

17. Remove the screw for the LH duct assembly bracket from the front of the instrument panel.

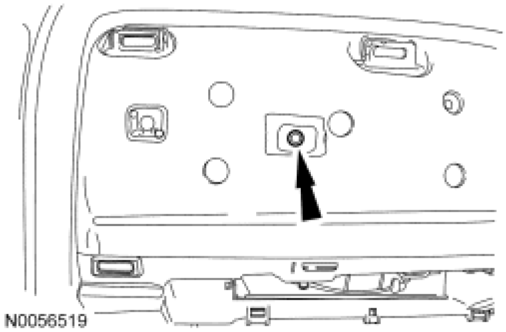


Fig. 132: Locating Screw On LH Duct Assembly Bracket
Courtesy of FORD MOTOR CO.

18. Through the glove compartment opening, remove the screw and the LH duct assembly.

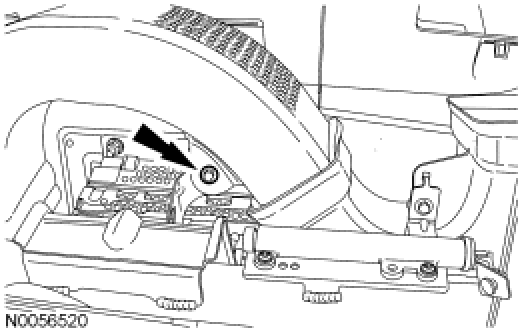


Fig. 133: Locating Screw On LH Duct Assembly
Courtesy of FORD MOTOR CO.

19. Through the glove compartment opening, remove the screw and the RH duct assembly.

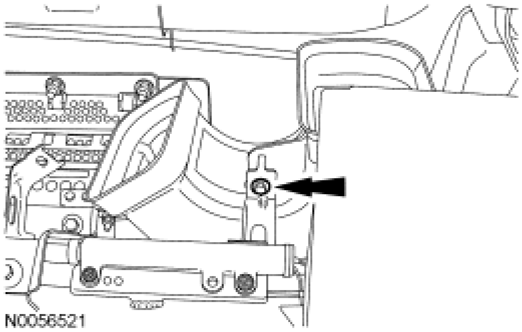


Fig. 134: Locating Screw On RH Duct Assembly
Courtesy of FORD MOTOR CO.

20. Remove the 4 bolts and the passenger air bag module support bracket.

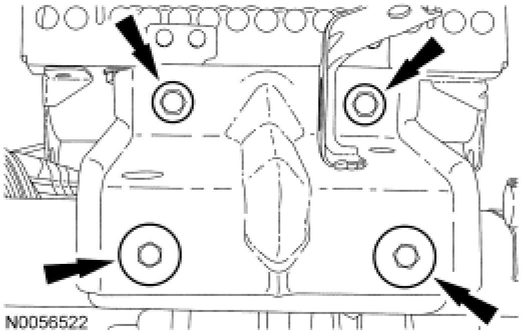


Fig. 135: Locating Bolts On Passenger Air Bag Module Support Bracket
Courtesy of FORD MOTOR CO.

21. Remove the 3 nuts from the passenger air bag module.

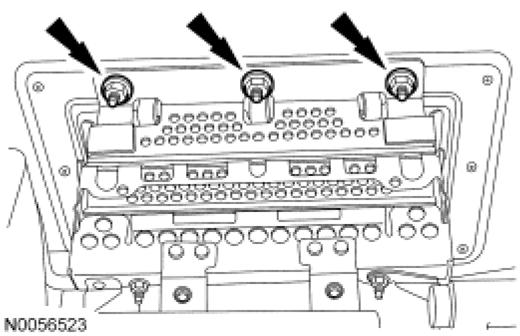


Fig. 136: Locating Nuts On Passenger Air Bag Module
Courtesy of FORD MOTOR CO.

NOTE: Do not allow the passenger air bag module to hang from the wiring harness. Damage to the wiring harness and connectors may occur.

22. Lower the rear of the passenger air bag module off the 3 mounting studs. Move the passenger air bag module rearward to release the 7 hooks from the instrument panel and rest the module on the cross vehicle beam.

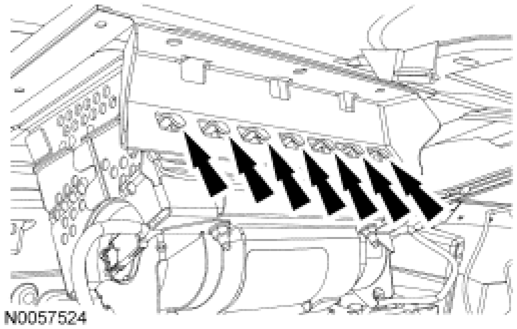


Fig. 137: Identifying Hooks On I/P
Courtesy of FORD MOTOR CO.

23. Using a small screwdriver as shown in illustration, lift up and release the locking button on the LH passenger air bag module electrical connector.

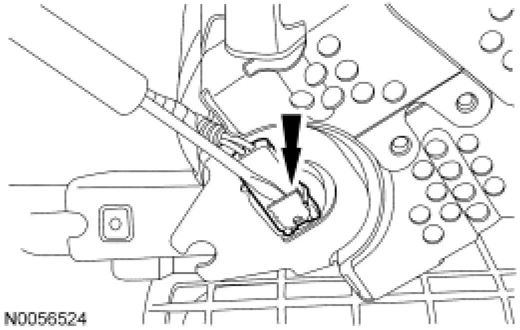


Fig. 138: Releasing Locking Button On LH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE:	**Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.**

24. With the LH locking button released, remove the LH electrical connector from the passenger air bag module.
25. Using a small screwdriver as shown in illustration, lift up and release the locking button on the RH passenger air bag module electrical connector.

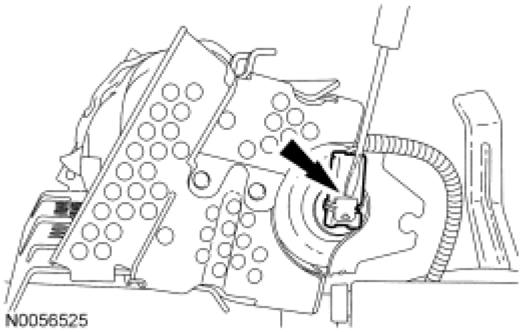


Fig. 139: Releasing Locking Button On RH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE:	**Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.**

26. With the RH locking button released, remove the RH electrical connector. Remove the passenger air bag module from the glove compartment opening.

All vehicles

27. From under the rear of the front passenger seat, slide and disengage the passenger seat side air bag module electrical connector locking clip, and then release the tab and disconnect the passenger seat side air bag module electrical connector.

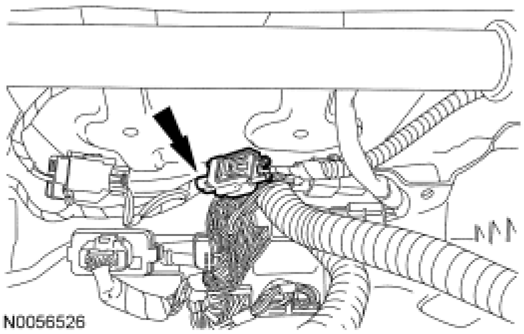


Fig. 140: Identifying Passenger Seat Side Air Bag Module Electrical Connector
Courtesy of FORD MOTOR CO.

28. Remove the passenger side D-pillar trim panel. For additional information, refer to **INTERIOR TRIM & ORNAMENTATION** article.

NOTE: Do not pull the passenger safety canopy module electrical connector out by the locking button. Damage to the locking button may occur.

29. With the locking button released, remove the electrical connector from the passenger safety canopy module.

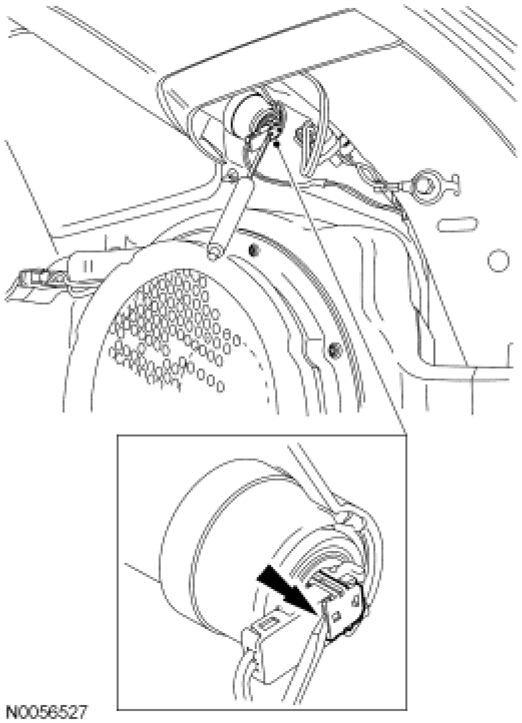


Fig. 141: Locating Electrical Connector From Passenger Safety Canopy Module
Courtesy of FORD MOTOR CO.

30. Remove the driver side D-pillar trim panel. For additional information, refer to **INTERIOR TRIM & ORNAMENTATION** article.

NOTE: Do not pull the driver safety canopy module electrical connector out by the locking button. Damage to the locking button may occur.

31. With the locking button released, remove the electrical connector from the driver safety canopy module.

Fig. 142: Locating Electrical Connector From Driver Safety Canopy Module
Courtesy of FORD MOTOR CO.

32. From under the rear of the driver seat, detach the driver seat side air bag module electrical connector from the seat cushion frame. Then slide and disengage the driver seat side air bag module electrical connector locking clip, and then release the tab and disconnect the driver seat side air bag module electrical connector.

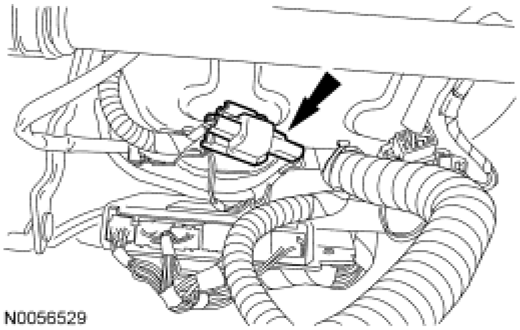


Fig. 143: Identifying Driver Seat Side Air Bag Module Electrical Connector Locking Clip
Courtesy of FORD MOTOR CO.

33. Install RCM fuse 46 (7.5A) to the SJB.
34. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING & CABLES** article.

Reactivation

All vehicles

1. Remove RCM fuse 46 (7.5A) from the SJB.
2. Disconnect the battery ground cable and wait at least one minute.
3. Connect the driver seat side air bag module electrical connector and then slide and engage the seat side air bag electrical connector locking clip. Then attach the driver seat side air bag module electrical connector to the seat cushion frame.

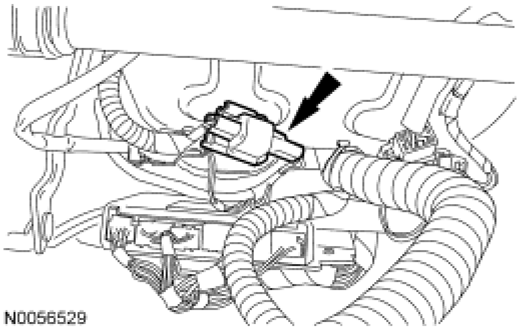


Fig. 144: Identifying Driver Seat Side Air Bag Module Electrical Connector Locking Clip
Courtesy of FORD MOTOR CO.

NOTE: **Do not install the safety canopy module electrical connector by the locking button. Damage to the locking button may occur.**

NOTE: **The safety canopy module electrical connector locking button must be in the released position when the connector is being installed or connector damage may occur.**

NOTE: **The safety canopy module electrical connector is unique and cannot be reversed when connected to the safety canopy module. Match the electrical connector key to the keyway in the safety canopy module. Do not force the electrical connector into the safety canopy module. Damage to the connector or component may occur.**

4. With the locking button released, install the driver safety canopy module electrical connector fully into the driver safety canopy module and seat the locking button.

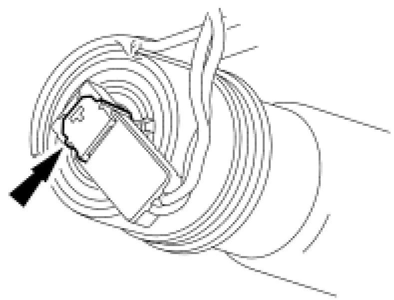

N0056530

Fig. 145: Identifying Driver Safety Canopy Module Electrical Connector
Courtesy of FORD MOTOR CO.

5. Install the driver side D-pillar trim panel. For additional information, refer to **INTERIOR TRIM & ORNAMENTATION** article.

NOTE: **Do not install the safety canopy module electrical connector by the locking button. Damage to the locking button may occur.**

NOTE: **The safety canopy module electrical connector locking button must be in the released position when the connector is being installed or connector damage may occur.**

NOTE: **The safety canopy module electrical connector is unique and cannot be reversed when connected to the safety canopy module. Match the electrical connector key to the keyway in the safety canopy module. Do not force the electrical connector into the safety canopy module. Damage to the connector or component may occur.**

6. With the locking button released, install the passenger safety canopy module electrical connector fully into the passenger safety canopy module and seat the locking button.

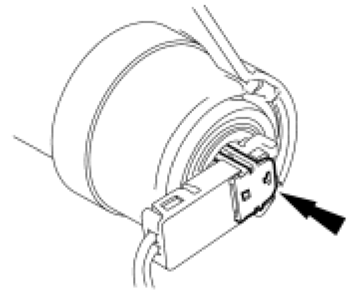


Fig. 146: Identifying Passenger Safety Canopy Module Electrical Connector
Courtesy of FORD MOTOR CO.

7. Install the passenger side D-pillar trim panel. For additional information, refer to **INTERIOR TRIM & ORNAMENTATION** article.
8. Connect the passenger seat side air bag module electrical connector and then slide and engage the seat side air bag electrical connector locking clip.

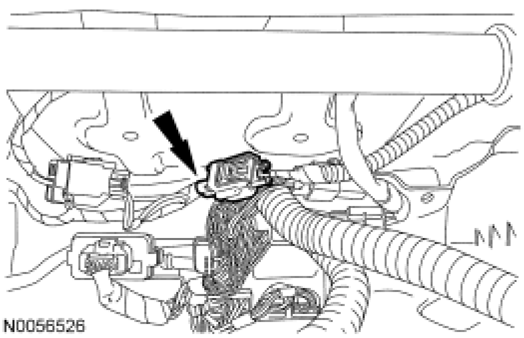


Fig. 147: Identifying Passenger Seat Side Air Bag Module Electrical Connector
Courtesy of FORD MOTOR CO.

MKX

NOTE:	**Do not install the passenger air bag module electrical connectors by the locking buttons. Damage to the locking buttons may occur.**
NOTE:	**The passenger air bag module electrical connector locking buttons must be in the released position when the connector is being installed or connector damage may occur.**
NOTE:	**The passenger air bag module electrical connectors are unique and cannot be reversed when connected to the passenger air bag module. Match the electrical connector key to the keyway in the passenger air bag module. Do not force the electrical connectors into the passenger air bag module. Damage to the connector or component may occur.**

NOTE: RH side shown in illustration, LH side similar.

9. Position the passenger air bag module on the cross vehicle beam. With the locking buttons released, install the 2 passenger air bag module electrical connectors fully into the passenger air bag module and seat the locking buttons.

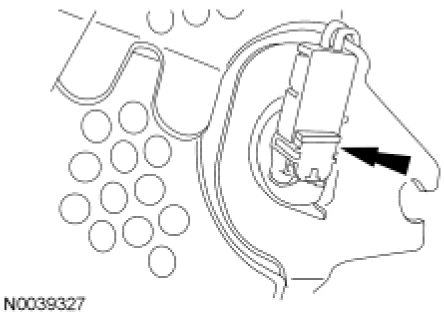


Fig. 148: Locating Passenger Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

NOTE: During the passenger air bag module installation, make sure all 7 of the hooks are fully seated in the instrument panel.

10. Install the passenger air bag module 7 hooks into the instrument panel. Raise the rear of the passenger air bag module onto the 3 mounting studs.

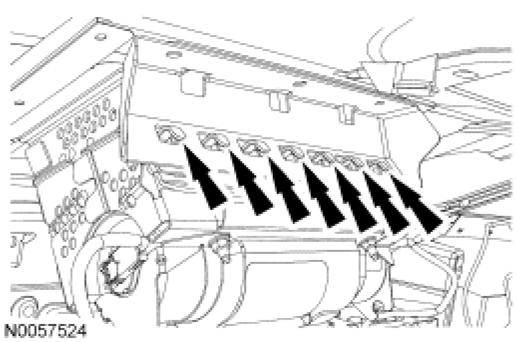


Fig. 149: Identifying Hooks On I/P
Courtesy of FORD MOTOR CO.

11. Install the 3 passenger air bag module nuts.
    - Tighten to 7 Nm (62 lb-in).

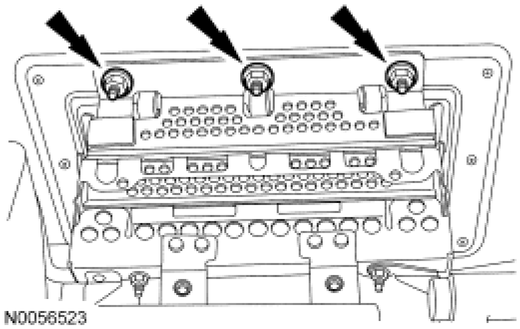


Fig. 150: Locating Nuts On Passenger Air Bag Module
Courtesy of FORD MOTOR CO.

NOTE: To avoid a squeak or rattle condition, install the passenger air bag module bracket and passenger air bag module bolts in this order.

12. Install the passenger air bag module support bracket and the 4 bolts.
    1. Install the passenger air bag module support bracket and 2 bolts to the cross vehicle beam.
        - Tighten to 8 Nm (71 lb-in).
    2. Install the inboard passenger air bag module support bracket-to-passenger air bag module bolt.
        - Tighten to 8 Nm (71 lb-in).
    3. Install the outboard passenger air bag module support bracket-to-passenger air bag module bolt.
        - Tighten to 8 Nm (71 lb-in).

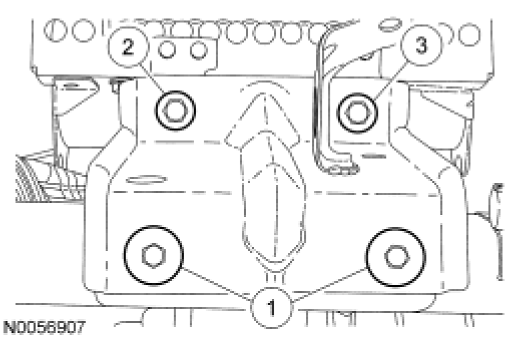


Fig. 151: Installing Passenger Air Bag Module Support Bracket Bolts In Sequence
Courtesy of FORD MOTOR CO.

13. Install the screw and the RH duct assembly.

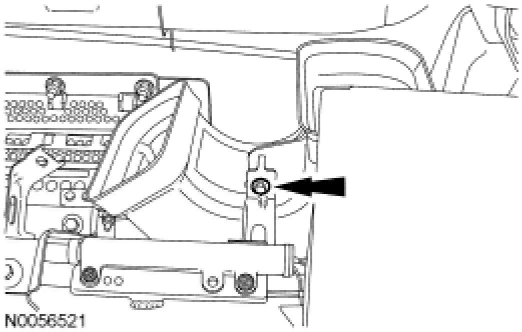


Fig. 152: Locating Screw On RH Duct Assembly
Courtesy of FORD MOTOR CO.

14. Install the screw and the LH duct assembly.

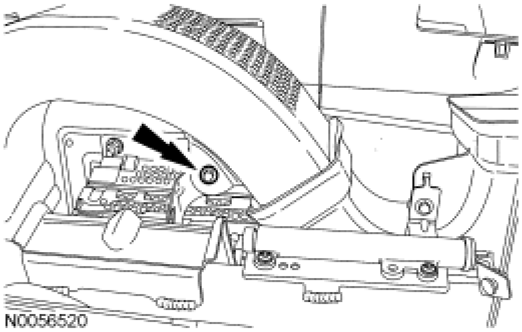


Fig. 153: Locating Screw On LH Duct Assembly
Courtesy of FORD MOTOR CO.

15. Install the screw for the LH duct assembly bracket in the front of the instrument panel.

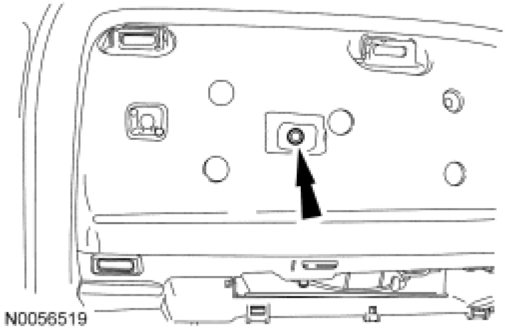


Fig. 154: Locating Screw On LH Duct Assembly Bracket
Courtesy of FORD MOTOR CO.

NOTE: **Make sure the RH instrument panel finish panel 5 retaining clips are fully seated in the instrument panel.**

16. Install the RH instrument panel finish panel.

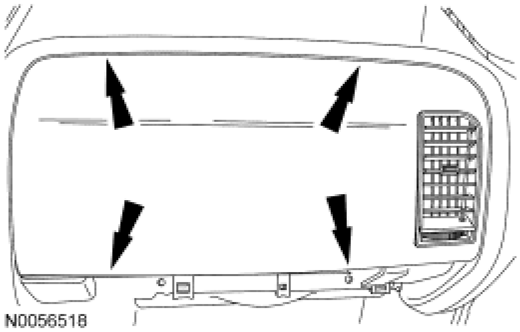


Fig. 155: Locating Retaining Clips On RH I/P Finish Panel
Courtesy of FORD MOTOR CO.

Edge

NOTE:	**Do not install the passenger air bag module electrical connectors by the locking buttons. Damage to the locking buttons may occur.**
NOTE:	**The passenger air bag module electrical connector locking buttons must be in the released position when the connector is being installed or connector damage may occur.**
NOTE:	**The passenger air bag module electrical connectors are unique and cannot be reversed when connected to the passenger air bag module. Match the electrical connector key to the keyway in the passenger air bag module. Do not force the electrical connectors into the passenger air bag module. Damage to the connector or component may occur.**
NOTE:	**RH side shown in illustration, LH side similar.**

17. With the locking buttons released, install the 2 passenger air bag module electrical connectors fully into the passenger air bag module and seat the locking buttons.

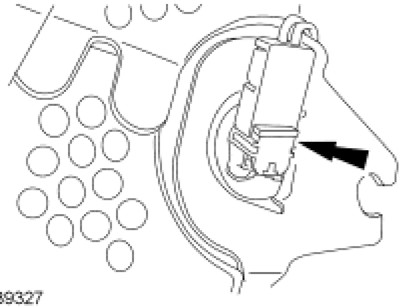


Fig. 156: Locating Passenger Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

NOTE: **Do not handle the passenger air bag module by grabbing the edges of the deployment doors. Damage to the air bag module may occur.**

NOTE: **During air bag module installation, make sure all the deployment door clips are fully seated in the instrument panel.**

18. Install the passenger air bag module deployment door rear-most clips into the instrument panel first. Then install the side deployment door clips working around to the forward-most clips seating each of them fully into the instrument panel.

NOTE: **To avoid a squeak or rattle condition, install the 2 passenger air bag module bolts in this order.**

19. Install the passenger air bag module 2 bolts.
    1. Install the inboard passenger air bag module bolt.
        - Tighten to 8 Nm (71 lb-in).
    2. Install the outboard passenger air bag module bolt.
        - Tighten to 8 Nm (71 lb-in).

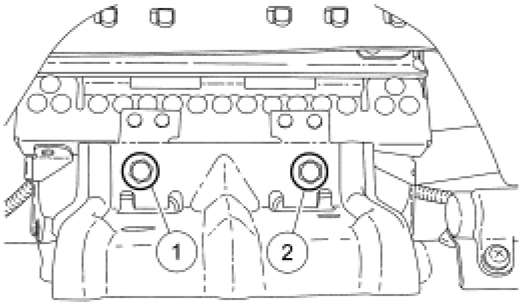


Fig. 157: Installing Passenger Air Bag Module Bolts In Sequence
Courtesy of FORD MOTOR CO.

All vehicles

20. Hook the glove compartment door to the instrument panel. Close the glove compartment door.
    - If equipped, attach the glove compartment door dampener.

NOTE: **Do not install the driver air bag module electrical connectors by the locking buttons. Damage to the locking buttons may occur.**

NOTE: **The driver air bag module electrical connector locking buttons must be in the released position when the connector is being installed or connector damage may occur.**

NOTE: **The passenger air bag module electrical connectors are unique and cannot be reversed when connected to the passenger air bag module. Match the electrical connector key to the keyway in the passenger air bag module. Do not force the electrical connectors into the passenger air bag module. Damage to the connector or component may occur.**

21. With the locking buttons released, install the driver air bag module electrical connectors fully into the driver air bag module and seat the locking buttons.

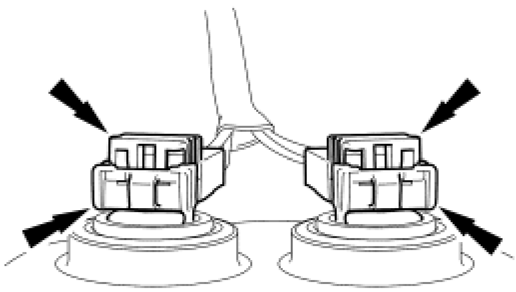


Fig. 158: Locating Driver Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

NOTE: **Audible clicks will be heard when both wire clips are seated in the driver air bag module.**

22. Align the driver air bag module locking pins to the steering wheel and, while pushing inward, seat the 2 driver air bag module locking pins to the steering wheel wire clips.
    - When the 2 locking pins are seated in place, there should be an even gap between the driver air bag module trim cover and the steering wheel.
23. Turn the ignition switch from OFF to ON.
24. Install RCM fuse 46 (7.5A) to the SJB and install the lower kick panel fuse cover.

WARNING: **Make sure no one is in the vehicle and there is nothing blocking or placed in front of any air bag module when the battery is connected. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.**

25. Connect the battery ground cable.
26. Prove out the SRS as follows:

    Turn the ignition key from ON to OFF. Wait 10 seconds, then turn the key back to ON and visually monitor the air bag warning indicator with the air bag modules installed. The air bag warning indicator will light continuously for approximately 6 seconds and then turn off. If an air bag SRS fault is present, the air bag warning indicator will:

- fail to light.

- remain lit continuously.

- flash at a 5Hz rate (RCM not configured).

The air bag warning indicator may not illuminate until approximately 30 seconds after the ignition switch has been turned from the OFF to the ON position. This is the time required for the RCM to complete the testing of the SRS. If the air bag warning indicator is inoperative and an SRS fault exists, a chime will sound in a pattern of 5 sets of 5 beeps. If this occurs, the air bag warning indicator will need to be repaired before diagnosis can continue.

Clear all continuous DTCs from the RCM and occupant classification system module (OCSM) using a scan tool.

INSPECTION AND REPAIR AFTER A SUPPLEMENTAL RESTRAINT SYSTEM (SRS) DEPLOYMENT

WARNING: **Remove restraint system diagnostic tools from the vehicle prior to road testing. If tools are not removed, the supplemental restraint system (SRS) device may not deploy in a crash. Failure to follow this instruction may result in serious personal injury or death in a crash and possibly violate vehicle safety standards.**

NOTE: After diagnosing or repairing a Supplemental Restraint System (SRS), the restraint system diagnostic tools (if required) must be removed before operating the vehicle over the road.

NOTE: Deployable devices (such as air bag modules, pretensioners) may deploy alone or in various combinations depending on the impact event.

NOTE: Always refer to the appropriate service information procedures prior to carrying out vehicle repairs affecting the SRS and safety belt system.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

All vehicles

NOTE: Refer to the correct removal and installation procedure for all SRS components being installed.

1. When any deployable device or combination of devices are deployed and/or the Restraints Control Module (RCM) has the DTC B1231 (Event Threshold Exceeded) in memory, the repair of the vehicle SRS is to include the removal of all deployed devices and the installation of new deployable devices, the

removal and installation of new impact sensors, and the removal and installation of a new RCM. DTCs must be cleared from all required modules after repairs are carried out.

Vehicles with Occupant Classification Sensor (OCS) system

NOTE: After installation of new Occupant Classification Sensor (OCS) system components, carry out the Occupant Classification Sensor (OCS) System Reset procedure as instructed in the service information. Refer to the appropriate service information for OCS system removal and installation procedure.

2. When a vehicle has been involved in a collision and the Occupant Classification System Module (OCSM) has DTC B1231 stored in memory, the repair of the OCS system is to include the following procedures for the specified system:
    - For rail-type OCS system, inspect the passenger side floorpan for damage and repair as necessary. Install new OCS system rails.
    - For weight sensor bolt-type OCS system, inspect the passenger side floorpan for damage and repair as necessary. Install a new seat track with OCS system weight sensor bolts. DTC must be cleared from the OCSM before carrying out Occupant Classification Sensor (OCS) System Reset. Do not install a new OCSM unless DTC B1231 cannot be cleared.

    NOTE: Most bladder-type OCSM do not store a DTC B1231 in memory after deployment. The DTC B1231 is stored only by the RCM.

    - For bladder-type OCS system, inspect for damage and repair as necessary. If installation of an OCS system component is required, an OCS system service kit must be installed.

All vehicles

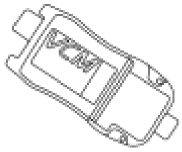
3. When any damage to the impact sensor mounting points or mounting hardware has occurred, repair or install new mounting points and mounting hardware as needed.
4. When the driver air bag module has deployed, a new clockspring must be installed.
5. New driver and/or front passenger safety belt systems (including retractors, buckles and height adjusters) must be installed if the vehicle is involved in a collision that results in deployment of the driver and/or front passenger safety belt pretensioners. For additional information, refer to **SAFETY BELT SYSTEM** article.
6. Inspect the entire vehicle for damage, including the following components:
    - Steering column (deployable column if equipped)
    - Instrument panel knee bolsters and mounting points
    - Instrument panel braces and brackets
    - Instrument panel and mounting points
    - Seats and seat mounting points
    - Safety belts, safety belt buckles and safety belt retractors. For additional information, refer to **SAFETY BELT SYSTEM** article.

- SRS wiring, wiring harnesses and connectors

7. After carrying out the review and inspection of the entire vehicle for damage, repair or install new components as needed.

OCCUPANT CLASSIFICATION SENSOR (OCS) SYSTEM RESET

Special Tools

Illustration	Tool Name	Tool Number
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS)	software with appropriate hardware, or equivalent scan tool

NOTE: If DTC B2290 is retrieved on-demand from the occupant classification system module (OCSM) and restraints control module (RCM):

- **There is a fault with the occupant classification sensor (OCS) weight sensor bolt(s) and/or wiring.**
- **The cause for the DTC must be corrected before carrying out the OCS system reset.**
- **Refer to DTC Charts for diagnostic direction.**
- **The RCM stores DTC B2290 in memory ONLY to identify a problem with the OCS system.**

If DTC B2290 is only present in the RCM and no on-demand or continuous DTC are present in the OCSM, CLEAR all DTCs from the RCM.

1. **Diagnostic Instruction:** The Occupant Classification Sensor (OCS) System Reset procedure **should only** be carried out if instructed to do so by the service information.

   If the concern is with the passenger air bag deactivation (PAD) indicator operation, refer to Passenger Air Bag Deactivation (PAD) Indicator and Occupant Classification Sensor (OCS) System in **DESCRIPTION AND OPERATION** .

   WARNING: **The following precautions must be taken before carrying out the following procedure on the front passenger seat:**

   - **Park the vehicle on level ground.**
   - **Make sure the occupant classification sensor (OCS) system is stabilized at a temperature in the range of 5°C to 48°C (41°F to 118°F) and is not exposed to temperatures outside that range.**
   - **Make sure the front passenger seat is empty and dry.**

- **Make sure the front passenger seat repair is complete and all attached components (head restraint, seat side shield) are correctly installed.**
- **Make sure nothing is placed in the map pockets (if equipped) of the front passenger seat.**
- **Make sure nothing is underneath the front passenger seat.**
- **Make sure no non-factory installed items (seat covers, DVD screens) are on or contacting the front passenger seat. If any non-factory items were installed to the seat, remove the item(s) and communicate to the customer that any alteration/modification to the front passenger seat may affect performance of the OCS system. Refer to the Owner's Literature.**
- **Make sure the seat is correctly installed in the vehicle.**

Failure to follow these instructions may result in incorrect operation of the OCS system and increases the risk of serious personal injury or death in a crash.

NOTE: **Correct seat track alignment of the inboard and outboard seat tracks on a manual seat is critical for a successful system reset. Before removing a front passenger seat equipped with a manual seat track, mark both tracks to indicate correct alignment between the inboard and outboard tracks. Failure to follow this instruction will result in an unsuccessful system reset and incorrect operation of the occupant classification sensor (OCS) system.**

NOTE: **Make sure the occupant classification sensor (OCS) system weight sensor bolt protection brackets are correctly installed to the seat track and that they not contact the cushion side shield, wire harness or any other surrounding parts. Failure to follow this instruction may result in the incorrect operation of the OCS system.**

NOTE: **In the event of a deployable crash and OCSM DTC B1231 is retrieved on-demand, the DTC must be cleared before carrying out the Occupant Classification Sensor (OCS) System Reset.**

NOTE: **If a new OCS system component has been installed, the RCM and OCSM will:**

- **store DTC B1013 and B2290 in memory.**
- **cause the air bag warning indicator to illuminate.**
- **clear after a successful system reset.**

- **the RCM stores DTC B2290 in memory.**

2. Turn the ignition switch to the off position. Wait 10 seconds, then turn the ignition switch back to ON and visually monitor the air bag warning indicator. The air bag warning indicator will light continuously for approximately 6 seconds and then turn OFF. If no SRS DTCs are present **and no new OCS system components were installed** , the air bag warning indicator will remain off. If the SRS indicator illuminates, indicating a fault is present, use a scan tool to retrieve DTCs and refer to **DTC Charts** for diagnostic direction before carrying out the system reset procedure.
3. Cycle the ignition to the OFF position.

NOTE: **The following seat checks must be made before attempting to carry out an occupant classification sensor (OCS) system reset after the seat has been removed and re-installed for any reason:**

- **Make sure the seat is correctly installed in the vehicle.**
- **Make sure the seat has been tightened to specification as identified in the appropriate service information.**
- **Make sure the seat track is correctly aligned (manual seat track).**
- **Make sure the OCS weight sensor bolt harness connections are correct for each sensor at their correct location. Refer to appropriate SYSTEM WIRING DIAGRAMS article for schematic and connector information. Refer to the illustration for correct sensor location on the seat.**

OCS Weight Sensor Bolt Location

- **OCS weight sensor bolt 1, front outboard**
- **OCS weight sensor bolt 2, rear outboard**
- **OCS weight sensor bolt 3, rear inboard**
- **OCS weight sensor bolt 4, front inboard**

4. Position the front passenger seat.
   - Position the seat backrest in the upright (vertical) position.
   - Position the seat in the middle position of seat track travel.
   - Position the seat height to its middle position (power seat only).

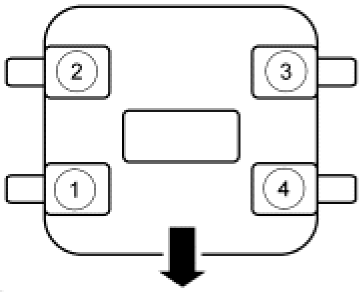


Fig. 159: Identifying OCS System Harness Connector
Courtesy of FORD MOTOR CO.

5. Remove the hysteresis from the front passenger seat by firmly hitting the back of the seat head restraint 3 times.

NOTE: **Make sure a minimum 8-second time period has passed after cycling the ignition switch ON before carrying out the System Reset active command.**

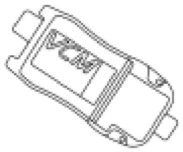
6. Turn the ignition switch ON, wait 8 seconds and carry out the System Reset using the scan tool.
    - If the system passed, go to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.

    NOTE: **Do not install any new OCS system components at this time. OCS system components should only be installed when directed to do so.**

    - If the system did not pass, **proceed to Step 7.**
7. Carry out the OCSM self-test.
    - If DTCs B2290 and B1013 are retrieved on-demand in the OCSM, refer to **DTC Charts** for diagnostic direction. Carry out diagnostics for DTC B2290 before any other DTCs.
    - If any other DTC is retrieved on-demand in the OCSM, refer to **DTC Charts** for diagnostic direction.
    - If only DTC B1013 is retrieved on-demand in the OCSM, refer to **DTC Charts** for diagnostic direction.

OCCUPANT CLASSIFICATION SENSOR (OCS) SYSTEM ZERO SEAT WEIGHT TEST

Special Tools

Illustration	Tool Name	Tool Number
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS)	software with appropriate hardware, or equivalent scan tool

WARNING: **The following precautions must be taken before carrying out the following procedure on the front passenger seat:**

- **Park the vehicle on level ground.**
- **Make sure the occupant classification sensor (OCS) system is stabilized at a temperature in the range of 5°C to 48°C (41°F to 118°F) and is not exposed to temperatures outside that range.**
- **Make sure the front passenger seat is empty and dry.**
- **Make sure the front passenger seat repair is complete and all attached components (head restraint, seat side shield) are correctly installed.**
- **Make sure nothing is placed in the map pockets (if equipped) of the front passenger seat.**
- **Make sure nothing is underneath the front passenger seat.**
- **Make sure no non-factory installed items (seat covers, DVD screens) are on or contacting the front passenger seat. If any non-factory items were installed to the seat, remove the item(s) and communicate to the customer that any alteration/modification to the front passenger seat may affect performance of the OCS system. Refer to the Owner's Literature.**
- **Make sure the seat is correctly installed in the vehicle.**

Failure to follow these instructions may result in incorrect operation of the OCS system and increases the risk of serious personal injury or death in a crash.

NOTE: Make sure the occupant classification sensor (OCS) weight sensor bolt protection brackets are correctly installed to the seat track and that it does not contact the cushion side shield, wire harness or any other surrounding parts. Failure to follow this instruction may result in the incorrect operation of the OCS system.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

NOTE: The Zero Seat Weight Test is done with the passenger seat positioned in 2 different positions to make sure that there is nothing to affect the performance and operation of the occupant classification sensor (OCS) system.

NOTE: The Zero Seat Weight Test verifies the OCS system measures zero weight for an empty seat. It is necessary to carry out the Zero Seat Weight Test procedure after a successful System Reset or as directed to do so in the appropriate service information procedures.

NOTE: **If the occupant classification system module (OCSM) has any on-demand DTCs except C1941 present, the fault condition(s) must be repaired and the DTCs cleared before carrying out the Zero Seat Weight Test on the front passenger seat. Failure to follow this instruction will result in an unsuccessful Zero Seat Weight Test and the incorrect operation of the OCS system.**

WARNING: **The following precautions must be taken before carrying out the following procedure on the front passenger seat:**

- **Park the vehicle on level ground.**
- **Make sure the occupant classification sensor (OCS) system is stabilized at a temperature in the range of 5°C to 48°C (41°F to 118°F) and is not exposed to temperatures outside that range.**
- **Make sure the front passenger seat is empty and dry.**
- **Make sure the front passenger seat repair is complete and all attached components (head restraint, seat side shield) are correctly installed.**
- **Make sure nothing is placed in the map pockets (if equipped) of the front passenger seat.**
- **Make sure nothing is underneath the front passenger seat.**
- **Make sure no non-factory installed items (seat covers, DVD screens) are on or contacting the front passenger seat. If any non-factory items were installed to the seat, remove the item(s) and communicate to the customer that any alteration/modification to the front passenger seat may affect performance of the OCS system. Refer to the Owner's Literature.**
- **Make sure the seat is correctly installed in the vehicle.**

Failure to follow these instructions may result in incorrect operation of the OCS system and increases the risk of serious personal injury or death in a crash.

1. Make sure the front passenger seat is empty and there is nothing underneath the seat. Remove any items from the seat that were not factory installed and make sure that the seat is correctly installed in the vehicle. For additional information, refer to Occupant Classification Sensor (OCS) System in **Air Bag and Safety Belt Pretensioner Supplemental Restraint System (SRS)**.
2. Position the front passenger seat in the first position.
    - Position the seat backrest in the upright position.
    - Position the seat in the full forward position of seat track travel.
    - Position the seat height to its highest position (power seat only).

WARNING: **Remove binding or resistance between all components of the front**

passenger seat as directed in the following procedure. Failure to follow this instruction may result in incorrect operation of the occupant classification sensor (OCS) system and increases the risk of serious personal injury or death in a crash.

3. Remove hysteresis from the front passenger seat by firmly hitting the back of the seat head restraint 3 times.
4. Turn the ignition to the OFF position.

WARNING: **Make sure nothing is placed on the front passenger seat during the following procedure. Do not press, touch or lay anything on the front passenger seat until the following procedure has been completed. Failure to follow these instructions may result in incorrect operation of the occupant classification sensor (OCS) system and increase the risk of serious personal injury or death in a crash.**

5. Turn the ignition to the ON position.

NOTE: Make sure a minimum 8-second time period has passed after cycling the ignition switch ON before carrying out the Zero Seat Weight Test active command.

6. Carry out the Zero Seat Weight Test using the scan tool.
   - If the system passed, **proceed to Step 7.**
   - If the system did not pass (DTC C1941 Zero Seat Weight Test Failure reported), **proceed to Step 9.**
7. Position the front passenger seat to the second position.
   - Position the seat in the full rearward position of seat track travel.
   - Position the seat height to its lowest position (power seat only).
8. Carry out the Zero Seat Weight Test using the scan tool.
   - If the system passed, repair is complete, **Proceed to Step 10.**
   - If the system did not pass (DTC C1941 Zero Seat Weight Test Failure reported), **proceed to Step 9.**

NOTE: Check the repair history of the vehicle for any previous repair that required the removal or service of the passenger seat. The repair history may lead to an area of concern.

9. Inspect the following for causes of a unsuccessful Zero Seat Weight Test.
   - Make sure the front passenger seat is empty and there is nothing underneath the seat. Remove any items from the seat that were not factory installed and make sure that the seat is correctly installed in the vehicle.
   - Make sure the seat wiring harnesses are routed and secured correctly and are not twisted, pinched,

pulling down or pushing up on any seat component. Refer to Seat Cushion Wire Harness Routing - Front Passenger Cushion in the General Procedures portion of **SEATING** article.

- Make sure the OCS weight sensor bolt protection brackets are correctly installed to the seat track and are not contacting the cushion side shield, wire harness or any other surrounding parts.
- Make sure the seat track and backrest frame are not bent, twisted or damaged.
- Move the seat and backrest to all seating positions. Make sure the seat track does not bind or have any resistance when moved to all seating positions. Make sure that there are no obstructions contacting any seat component when the seat is moved to all seating positions.
- Make sure the seat is assembled correctly and all fasteners are tightened to specifications.
  - If a concern was found, repair as necessary, **proceed to Step 1.**
  - If no concern was found, install a new passenger seat track assembly with OCS weight sensor bolts. Refer to **SEATING** article.

10. Prove out the SRS as follows:

    Turn the ignition switch from ON to OFF. Wait 10 seconds, then turn the ignition switch back to ON and visually monitor the air bag warning indicator with the air bag modules installed. The air bag warning indicator will light continuously for approximately 6 seconds and then turn off. If an air bag SRS fault is present, the air bag warning indicator will:

    - fail to light.

    - remain lit continuously.

    - flash at a 5 Hz rate (RCM not configured).

    The air bag warning indicator might not light until approximately 30 seconds after the ignition switch has been turned from the OFF to the ON position. This is the time required for the RCM to complete the testing of the SRS. If the air bag warning indicator is inoperative and a SRS fault exists, a chime will sound in a pattern of 5 sets of 5 beeps. If this occurs, the air bag warning indicator and any SRS fault discovered must be diagnosed and repaired.

    Repair any unresolved DTCs.

    Clear all continuous DTCs from the RCM and OCSM using a scan tool.

PYROTECHNIC DEVICE DISPOSAL

Disposal of Deployable Devices and Pyrotechnic Devices That are Undeployed/Inoperative

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

NOTE: **All inoperative air bag modules and safety belt pretensioners have been placed on the Mandatory Return List. All discolored or damaged air bag modules must**

be treated the same as any inoperative live air bag being returned.

1. Depower the Supplemental Restraint System (SRS). For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Remove the undeployed/inoperative device. For additional information, refer to the appropriate procedure in this service information or **SAFETY BELT SYSTEM** article.

NOTE: **When installing a new air bag module, a prepaid return postcard is provided with the replacement air bag module. The serial number for the new part and the Vehicle Identification Number (VIN) must be recorded and sent to Ford Motor Company.**

3. If installing a new air bag module, record the necessary information and return the inoperative air bag module to Ford Motor Company.

Disposal of Deployable Devices and Pyrotechnic Devices That Are Deployed

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Remove the deployed device. For additional information, refer to the appropriate procedure in this service information or **SAFETY BELT SYSTEM** article.

NOTE: **If a dual stage driver or passenger air bag module has deployed due to a crash event, the air bag module requires manual deployment to make sure both stages have deployed before scrapping the vehicle or disposing of the air bag module. To determine if a vehicle is equipped with dual stage driver or passenger air bag modules, refer to DESCRIPTION AND OPERATION .**

3. Dispose of the deployed device in the same manner as any other part to be scrapped.

Disposal of Deployable Devices and Pyrotechnic Devices That Require Manual Deployment

1. Safety and environmental concerns require consideration and treatment of restraints system deployable and pyrotechnic devices when disposing of vehicles, deployable devices or pyrotechnic devices. Deploying deployable and pyrotechnic devices before scrapping a vehicle or the device eliminates the potential for hazardous exposures or reactions during processing. If special handling procedures are followed, deployable and pyrotechnic devices can be deployed safely and recycled with the vehicle, shipped separately to a recycling facility or disposed of safely.

NOTE: **To determine the deployable devices a vehicle is equipped with, refer to DESCRIPTION AND OPERATION .**

A vehicle equipped with any of the following deployable devices requires manual deployment of the devices before scrapping the vehicle or component. For additional information, refer to the appropriate portion of this procedure.

- Driver air bag module
- Passenger air bag module
- Seat side air bag modules
- Safety canopy modules
- Side air curtain modules

NOTE: To determine the pyrotechnic devices a vehicle is equipped with, refer to DESCRIPTION AND OPERATION .

2. A vehicle equipped with any of the following pyrotechnic devices requires manual deployment of the devices before scrapping the vehicle or component. For additional information, refer to the appropriate portion of this procedure.
   - Safety belt buckle pretensioners
   - Safety belt retractor pretensioners
   - Adaptive load limiting retractors
   - Deployable steering column

NOTE: To determine if a vehicle is equipped with dual stage driver or passenger air bag modules, refer toDESCRIPTION AND OPERATION .

3. If a dual stage driver or passenger air bag module has deployed due to a crash event, the air bag module requires manual deployment to make sure both stages have deployed before scrapping the vehicle or disposing of the air bag module. For additional information, refer to Driver Air Bag Module, Passenger Air Bag Module and Seat Side Air Bag Modules - Remote Deployment in this procedure.

Driver Air Bag Module, Passenger Air Bag Module and Seat Side Air Bag Modules - Remote Deployment

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **Always carry or place a live air bag module with the air bag and deployment door/trim cover/tear seam pointed away from the body. Do not set a live air bag module down with the deployment door/trim cover/tear seam face down. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.**

WARNING: **Always carry or place a live safety canopy, or side air curtain module, with the module and tear seam pointed away from your body. Failure to follow this instruction may result in serious personal injury or death in the**

event of an accidental deployment.

WARNING: **Deploy all supplemental restraint system (SRS) devices (air bags, pretensioners, load limiters, etc.) outdoors with all personnel at least 9.14 meters (30 feet) away to make sure of personal safety. Due to the loud report which occurs when an SRS device is deployed, hearing protection is required. Failure to follow these instructions may result in serious personal injury.**

NOTE: For air bag modules with multiple loops, all the loops on the air bag module must be deployed.

NOTE: Some driver and passenger front air bags have 2 deployment stages. After a collision it is possible that Stage 1 has deployed and Stage 2 has not.

If a front air bag module has deployed, it is mandatory that the front air bag module be remotely deployed using the appropriate air bag disposal procedure.

NOTE: A typical air bag disposal is shown in illustration that is similar for all vehicles.

All driver, passenger and seat side air bag modules

1. Make a container to house the air bag module for deployment.

   NOTE: The tires must be of sufficient size to accommodate the air bag module.

   - Obtain a tire and wheel assembly and an additional 4 tires (without wheels) of the same size.
   - With the tire and wheel assembly on the bottom, stack the tires.
   - Securely tie all of the tires together.

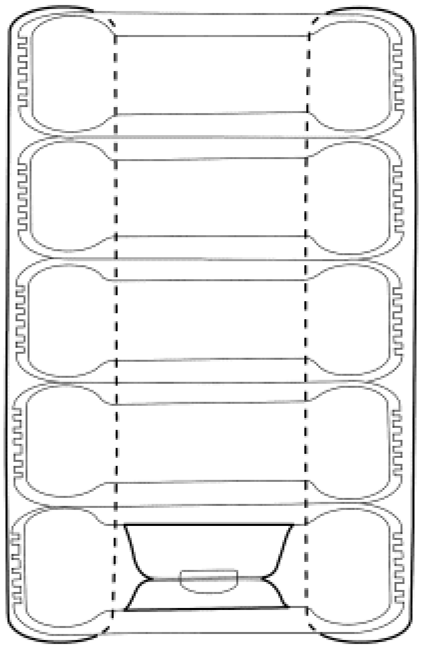


Fig. 160: Placing Air Bag Modules Inside Tire For Deployment
Courtesy of FORD MOTOR CO.

2. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
3. Remove the air bag module. For additional information, refer to the appropriate procedure.

NOTE: If the air bag module does not have a hardwired pigtail, it will be necessary to cut the wires and connector(s) from the vehicle wire harness and reconnect to the air bag module.

4. Cut each of the air bag module wires near the electrical connector that connects to the vehicle wire harness.
5. Remove any sheathing (if present) and strip the insulation from the ends of the cut wires.

NOTE: Typical driver air bag module with 2 loops shown in illustration, other air bag modules with multiple loops similar.

6. For air bag modules with multiple loops, twist together a wire from each loop then repeat for the remaining wires from each loop.

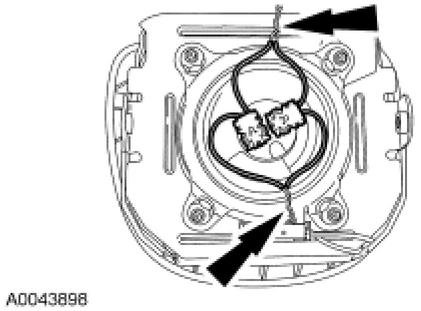


Fig. 161: Locating Air Bag Module Wire Squibs
Courtesy of FORD MOTOR CO.

7. Make a jumper harness to deploy the air bag module.
   - Obtain 2 wires (20 gauge minimum) at least 9.14 m (30 ft) long and strip both ends of each wire.
   - At one end of the jumper harness, connect the wires together.
8. Using the end of the jumper harness that the wires are not connected together, attach each wire of the jumper harness to each wire of the air bag module or to the twisted-together wires if multiple loops. Use tape or other insulating material to make sure that the leads do not make contact with each other.

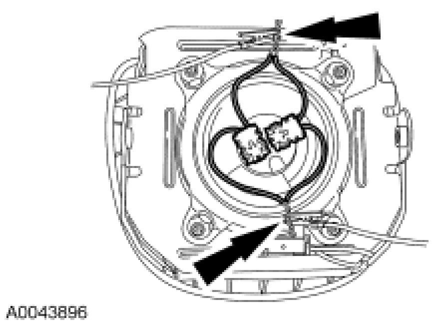


Fig. 162: Connecting Jumper Wires To Squibs
Courtesy of FORD MOTOR CO.

Driver air bag modules

NOTE: **Make sure to maintain the connections to the air bag module.**

9. With the stack of tires upright and the wheel on the bottom, carefully place the driver air bag module, with the trim cover facing up, on the wheel.

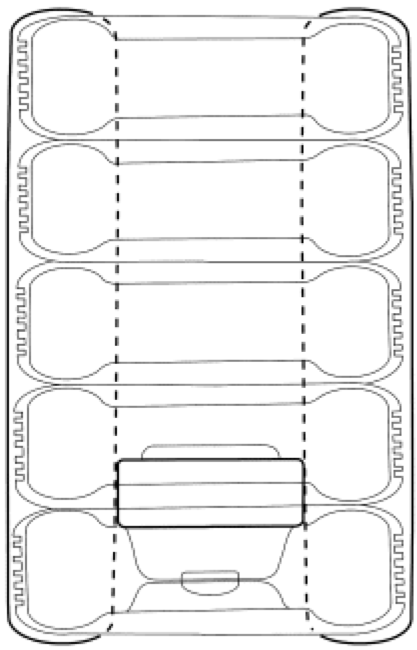


Fig. 163: Placing Driver Air Bag Modules Inside Tires For Deployment
Courtesy of FORD MOTOR CO.

Passenger and seat side air bag modules

NOTE: **Make sure to maintain the connections to the air bag module.**

10. Tip the stack of tires on its side and place the air bag module inside the center tire, making sure that there are 2 tires beneath the tire containing the air bag module and 2 tires (including the tire and wheel assembly) above the tire containing the air bag module.
11. Place the tire stack upright, with the wheel on top.

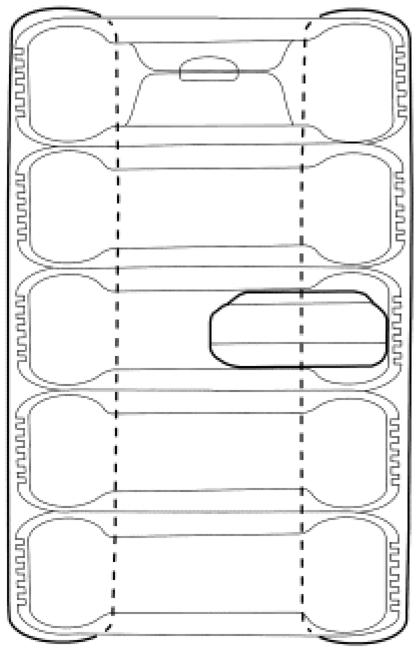


Fig. 164: Identifying Passenger Or Seat Side Air Bag Modules Inside Center Tire
Courtesy of FORD MOTOR CO.

All driver, passenger and seat side air bag modules

12. Remain at least 9.14 m (30 ft) away from the air bag module.
13. From the end of the jumper harness that is not connected to the air bag module, disconnect the 2 wires of the jumper harness from each other.
14. Deploy the air bag module by touching the ends of the 2 wires of the jumper harness to the terminals of a 12-volt battery.
15. To allow for cooling, wait at least 10 minutes before approaching the deployed air bag module.
16. Dispose of the deployed air bag module in the same manner as any other part to be scrapped.

Safety Belt Buckle Pretensioners, Safety Belt Retractor Pretensioners and Adaptive Load Limiting Safety Belt Retractors - Remote Deployment

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

WARNING: **Deploy all supplemental restraint system (SRS) devices (air bags, pretensioners, load limiters, etc.) outdoors with all personnel at least 9.14 meters (30 feet) away to make sure of personal safety. Due to the loud**

report which occurs when an SRS device is deployed, hearing protection is required. Failure to follow these instructions may result in serious personal injury.

NOTE: A typical safety belt buckle and retractor disposal is shown in illustration that is similar for all vehicles.

1. Make a container to house the safety belt buckle or retractor for deployment.

   NOTE: The tires must be of sufficient size to accommodate the safety belt buckle or retractor.

   - Obtain a tire and wheel assembly and an additional 4 tires (without wheels) of the same size.
   - With the tire and wheel assembly on the bottom, stack the tires.
   - Securely tie all of the tires together.

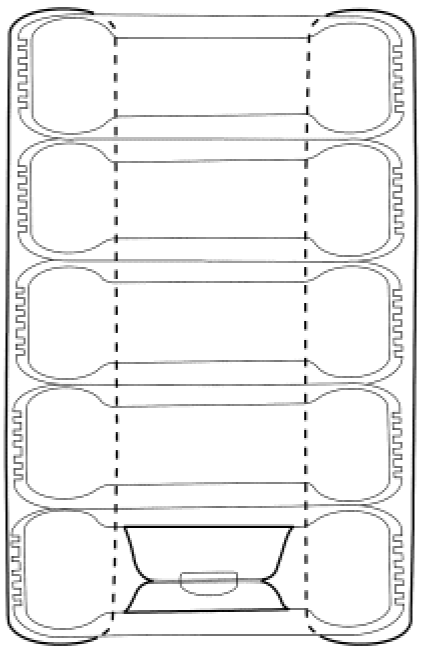


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Fig. 165: Placing Air Bag Modules Inside Tire For Deployment
Courtesy of FORD MOTOR CO.

2. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
3. Remove the safety belt buckle or retractor. For additional information, refer to the appropriate procedure in **SAFETY BELT SYSTEM** article.
   - When deploying a safety belt buckle pretensioner, install a nut and bolt of sufficient length and of

the same diameter as was used to retain it to the seat.

NOTE: If the safety belt buckle or retractor does not have a hardwired pigtail, it will be necessary to cut the wires and connector(s) from the vehicle wire harness and reconnect to the safety belt buckle or retractor.

4. Cut each of the safety belt buckle or retractor wires near the electrical connector that connects to the vehicle wire harness.
5. Remove any sheathing (if present) and strip the insulation from the ends of the cut wires.
6. Make a jumper harness to deploy the safety belt buckle or retractor.
    - Obtain 2 wires (20 gauge minimum) at least 9.14 m (30 ft) long and strip both ends of each wire.
    - At one end of the jumper harness, connect the wires together.

NOTE: Typical safety belt retractor pretensioner shown in illustration, other safety belt buckle pretensioners and load limiting retractors similar.

7. Using the end of the jumper harness that the wires are not connected together, attach each wire of the jumper harness to each wire of the safety belt buckle or retractor. Use tape or other insulating material to make sure that the leads do not make contact with each other.

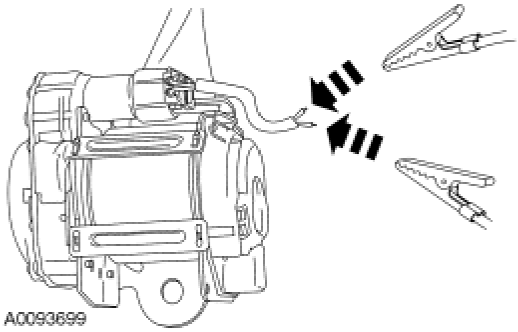

Fig. 166: Attaching Jumper Wires To Safety Belt Retractor Pretensioner
Courtesy of FORD MOTOR CO.

NOTE: Make sure to maintain the connections to the safety belt buckle or retractor.

8. Tip the stack of tires on its side and place the safety belt buckle or retractor inside the center tire, making sure that there are 2 tires beneath the tire containing the safety belt buckle or retractor and 2 tires (including the tire and wheel assembly) above the tire containing the safety belt buckle or retractor.
9. Place the tire stack upright, with the wheel on top.

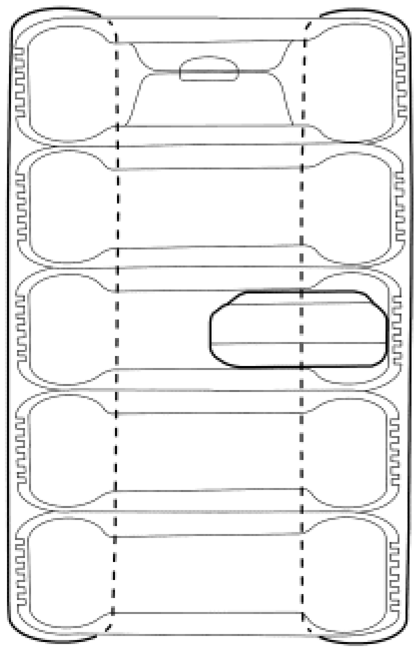

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Fig. 167: Identifying Passenger Or Seat Side Air Bag Modules Inside Center Tire
Courtesy of FORD MOTOR CO.

10. Remain at least 9.14 m (30 ft) away from the safety belt buckle or retractor.
11. From the end of the jumper harness that is not connected to the safety belt buckle or retractor, disconnect the 2 wires of the jumper harness from each other.
12. Deploy the safety belt buckle or retractor by touching the ends of the 2 wires of the jumper harness to the terminals of a 12-volt battery.
13. To allow for cooling, wait at least 10 minutes before approaching the deployed safety belt buckle or retractor.
14. Dispose of the deployed safety belt buckle or retractor in the same manner as any other part to be scrapped.

Safety Belt Buckle Pretensioners, Safety Belt Retractor Pretensioners and Load Limiting Safety Belt Retractors - In-Vehicle Deployment

WARNING: **Never probe the electrical connectors on safety belt buckle/retractor pretensioners or adaptive load limiting retractors. Failure to follow this instruction may result in the accidental deployment of the safety belt pretensioners or adaptive load limiting retractors, which increases the risk of serious personal injury or death.**

WARNING: **Deploy all supplemental restraint system (SRS) devices (air bags, pretensioners, load limiters, etc.) outdoors with all personnel at least 9.14 meters (30 feet) away to make sure of personal safety. Due to the loud**

report which occurs when an SRS device is deployed, hearing protection is required. Failure to follow these instructions may result in serious personal injury.

NOTE: **A typical safety belt buckle and retractor disposal is shown in illustration that is similar for all vehicles.**

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Access the safety belt buckle or retractor electrical connectors. For additional information, refer to **SAFETY BELT SYSTEM** article.
3. Cut each of the safety belt buckle or retractor wires, leaving at least 101.6 mm (4 in) to work with.
4. Remove any sheathing (if present) and strip the insulation from the ends of the cut wires.
5. Make a jumper harness to deploy the safety belt buckle or retractor.
   - Obtain 2 wires (20 gauge minimum) at least 9.14 m (30 ft) long and strip both ends of each wire.
   - At one end of the jumper harness, connect the wires together.

   NOTE: **Typical safety belt retractor pretensioner shown in illustration, other safety belt buckle pretensioners and load limiting retractors are similar.**

6. Using the end of the jumper harness that the wires are not connected together, attach each wire of the jumper harness to each wire of the safety belt buckle or retractor. Use tape or other insulating material to make sure that the leads do not make contact with each other.

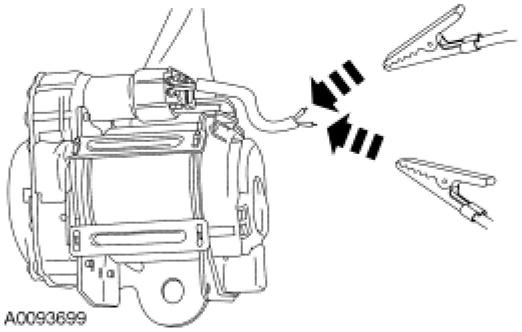


Fig. 168: Attaching Jumper Wires To Safety Belt Retractor Pretensioner
Courtesy of FORD MOTOR CO.

7. Remain at least 9.14 m (30 ft) away from the safety belt buckle or retractor.
8. From the end of the jumper harness that is not connected to the safety belt buckle or retractor, disconnect the 2 wires of the jumper harness from each other.
9. Deploy the safety belt buckle or retractor by touching the ends of the 2 wires of the jumper harness to the terminals of a 12-volt battery.
10. To allow for cooling, wait at least 10 minutes before approaching the deployed safety belt buckle or retractor.

11. Dispose of the deployed safety belt buckle or retractor in the same manner as any other part to be scrapped.

Safety Canopy Modules and Side Air Curtain Modules - In-Vehicle Deployment

WARNING: **Deploy all supplemental restraint system (SRS) devices (air bags, pretensioners, load limiters, etc.) outdoors with all personnel at least 9.14 meters (30 feet) away to make sure of personal safety. Due to the loud report which occurs when an SRS device is deployed, hearing protection is required. Failure to follow these instructions may result in serious personal injury.**

NOTE: The safety canopy module deployment for a scrapped vehicle will occur in its installed position in the vehicle.

NOTE: A typical safety canopy module disposal is shown in illustration that is similar for all vehicles.

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Access the safety canopy/side air curtain module electrical connectors. For additional information, refer to the appropriate procedure.
3. Cut each of the safety canopy/side air curtain module wires leaving at least 101.6 mm (4 in) to work with.
4. Remove any sheathing (if present) and strip the insulation from the ends of the cut wires.

NOTE: Typical safety canopy/side air curtain module with 2 loops shown in illustration, other safety canopy/side air curtain modules with 2 loops are similar.

5. For safety canopy/side air curtain modules with multiple loops, twist together a wire from each loop then repeat for the remaining wires from each loop.

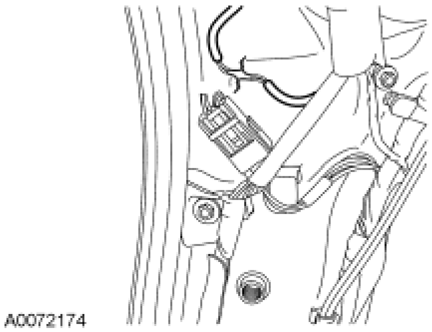


Fig. 169: Identifying Safety Canopy/Side Air Curtain Module Wire Squibs
Courtesy of FORD MOTOR CO.

6. Make a jumper harness to deploy the safety canopy/side air curtain module.
   - Obtain 2 wires (20 gauge minimum) at least 9.14 m (30 ft) long and strip both ends of each wire.
   - At one end of the jumper harness, connect the wires together.
7. Using the end of the jumper harness that the wires are not connected together, attach each wire of the jumper harness to each wire of the safety canopy/side air curtain module or to the twisted-together wires if multiple loops. Use tape or other insulating material to make sure that the leads do not make contact with each other.

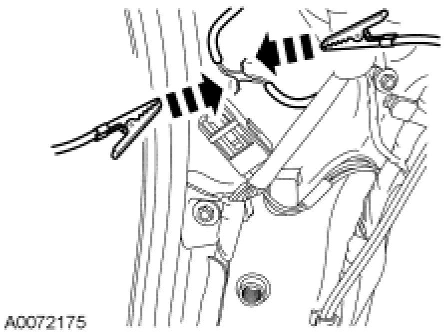


Fig. 170: Connecting Jumper Harness To Twisted Wires
Courtesy of FORD MOTOR CO.

8. From the end of the jumper harness that is not connected to the safety canopy/side air curtain module, disconnect the 2 wires of the jumper harness from each other.
9. Deploy the safety canopy/side air curtain module by touching the ends of the 2 wires of the jumper harness to the terminals of a 12-volt battery.
10. To allow for cooling, wait at least 10 minutes before approaching the deployed safety canopy/side air curtain module.
11. Dispose of the deployed safety canopy/side air curtain module in the same manner as any other part to be scrapped.

Deployable Steering Column - In-Vehicle Deployment

WARNING: **Deploy all supplemental restraint system (SRS) devices (air bags, pretensioners, load limiters, etc.) outdoors with all personnel at least 9.14 meters (30 feet) away to make sure of personal safety. Due to the loud report which occurs when an SRS device is deployed, hearing protection is required. Failure to follow these instructions may result in serious personal injury.**

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

NOTE: **It may be necessary to lower or remove the deployable steering column from the instrument panel to access the deployable steering column electrical connector.**

2. Access the deployable steering column electrical connector.

NOTE: If the deployable steering column does not have a hardwired pigtail, it will be necessary to cut the wires and connector(s) from the vehicle wire harness and reconnect to the deployable steering column.

3. Cut each of the deployable steering column wires, leaving at least 101.6 mm (4 in) to work with.
4. Remove any sheathing (if present) and strip the insulation from the ends of the cut wires.
5. Make a jumper harness to deploy the deployable steering column.
   - Obtain 2 wires (20 gauge minimum) at least 9.14 m (30 ft) long and strip both ends of each wire.
   - At one end of the jumper harness, connect the wires together.
6. Using the end of the jumper harness that the wires are not connected together, attach each wire of the jumper harness to each wire of the deployable steering column. Use tape or other insulating material to make sure that the leads do not make contact with each other.

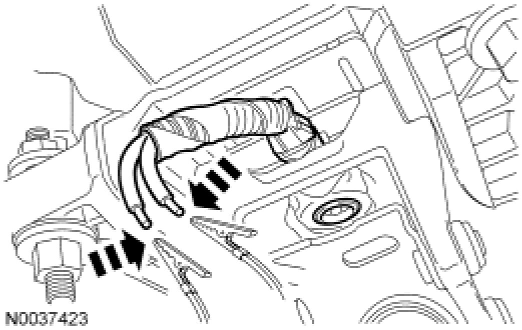


Fig. 171: Connecting Jumper Harness To Wires Of Deployable Steering Column
Courtesy of FORD MOTOR CO.

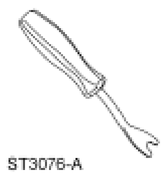
7. Remain at least 9.14 m (30 ft) away from the deployable steering column.
8. From the end of the jumper harness that is not connected to the deployable steering column, disconnect the 2 wires of the jumper harness from each other.
9. Deploy the deployable steering column by touching the ends of the 2 wires of the jumper harness to the terminals of a 12-volt battery.
10. To allow for cooling, wait at least 10 minutes before approaching the deployed steering column.
11. Dispose of the deployed steering column in the same manner as any other part to be scrapped.

REMOVAL AND INSTALLATION

AIR BAG TRIM COVER - PASSENGER, EDGE

Special Tools

Illustration	Tool Name	Tool Number

	Upholstery Clip Remover	SGT87810 (or equivalent)

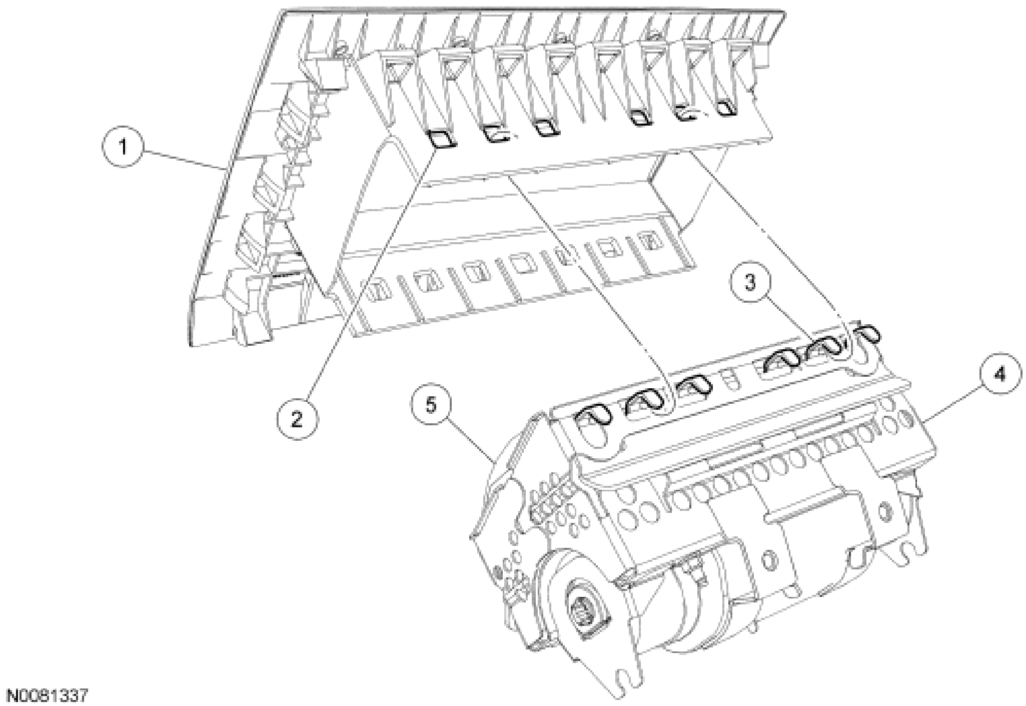


Fig. 172: Identifying Air Bag Trim Cover - Passenger, Edge
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	04338	Passenger air bag cover
2	-	Passenger air bag cover window (part of passenger air bag cover)
3	-	Passenger air bag canister hook (part of canister)
4	-	Passenger air bag canister (part of 044A74)
5	-	Passenger air bag soft pack (part of 044A74)

REMOVAL

WARNING: **Always carry or place a live air bag module with the air bag and deployment door/trim cover/tear seam pointed away from the body. Do**

not set a live air bag module down with the deployment door/trim cover/tear seam face down. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.

WARNING: **Do not use any sharp tools to separate the passenger air bag module trim cover from the passenger air bag module canister. Sharp tools may damage the passenger air bag module. Failure to follow this instruction may result in the passenger air bag module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

WARNING: **Do not unpack or unroll the passenger air bag module soft pack. If the soft pack becomes unpacked or unrolled, install a new passenger air bag module assembly. Failure to follow these instructions may result in the passenger air bag module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

WARNING: **Do not manipulate or compromise the passenger air bag module hooks during the removal or installation procedure. Failure to follow this instruction may result in the passenger air bag module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

1. Remove the passenger air bag module. Refer to **Passenger Air Bag Module - Edge** or **Passenger Air Bag Module - MKX**.
2. Place a mark on top of the passenger air bag module canister for correct installation.

WARNING: **Use a non-marring tool to separate the passenger air bag module cover from the passenger air bag module assembly. Sharp tools may damage the passenger air bag module. Failure to follow this instruction may result in the passenger air bag module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

3. Using an Upholstery Clip Remover or equivalent tool, carefully lift the passenger air bag module cover from the bottom of the air bag module canister, separating the windows from the canister hooks.

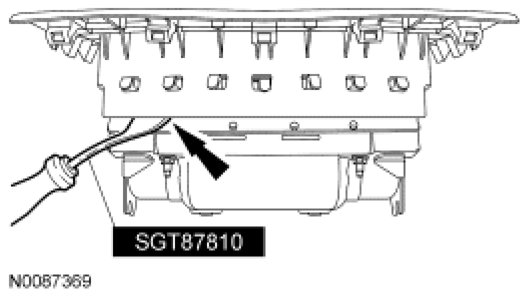


Fig. 173: View Of Passenger Air Bag Module
Courtesy of FORD MOTOR CO.

NOTE: Place an X mark on the old air bag cover to avoid reinstalling.

4. Remove the passenger air bag cover from the canister, separating the top windows from the canister hooks.

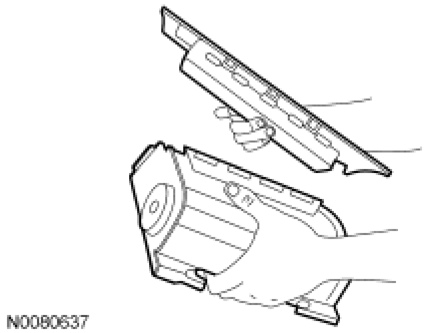


Fig. 174: Identifying Passenger Air Bag Cover
Courtesy of FORD MOTOR CO.

INSTALLATION

1. Place the passenger air bag cover on a clean work surface once it is removed from the shipping package.

WARNING: Carefully inspect the passenger air bag trim cover, canister and soft pack before assembly. If any foreign objects are found, remove them before attaching the passenger air bag trim cover to the canister. If the canister or soft pack is damaged, install a new passenger air bag module assembly. Failure to follow this instruction may result in the passenger air bag module deploying incorrectly, which increases the risk of serious personal injury or death in a crash.

2. Match the number of cover windows to the number of canister hooks before attaching the passenger air bag cover to the canister.
3. Match the top of the passenger air bag cover with the canister mark from the removal procedure.

4. Position the canister onto the passenger air bag cover and engage the bottom hooks into the air bag cover windows.
5. Push the top of the passenger air bag cover to engage the top canister hooks.

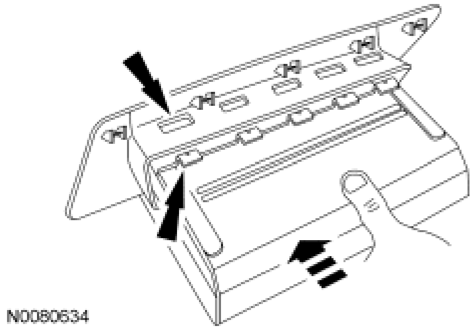


Fig. 175: Pushing Top Of Passenger Air Bag Cover To Engage Top Canister Hooks
Courtesy of FORD MOTOR CO.

6. Inspect the passenger air bag cover windows-to-canister hooks to verify that every window has a hook installed and the sides are not tucked or folded against the air bag canister.

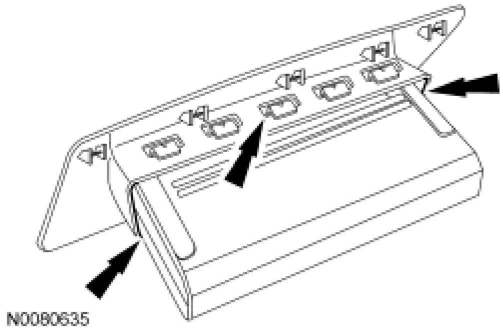


Fig. 176: Inspecting Passenger Air Bag Cover Windows-To-Canister Hooks To Verify Every Window Has A Hook Installed
Courtesy of FORD MOTOR CO.

7. Install the passenger air bag module. Refer to**Passenger Air Bag Module - Edge** or **Passenger Air Bag Module - MKX**.
8.

CLOCKSPRING

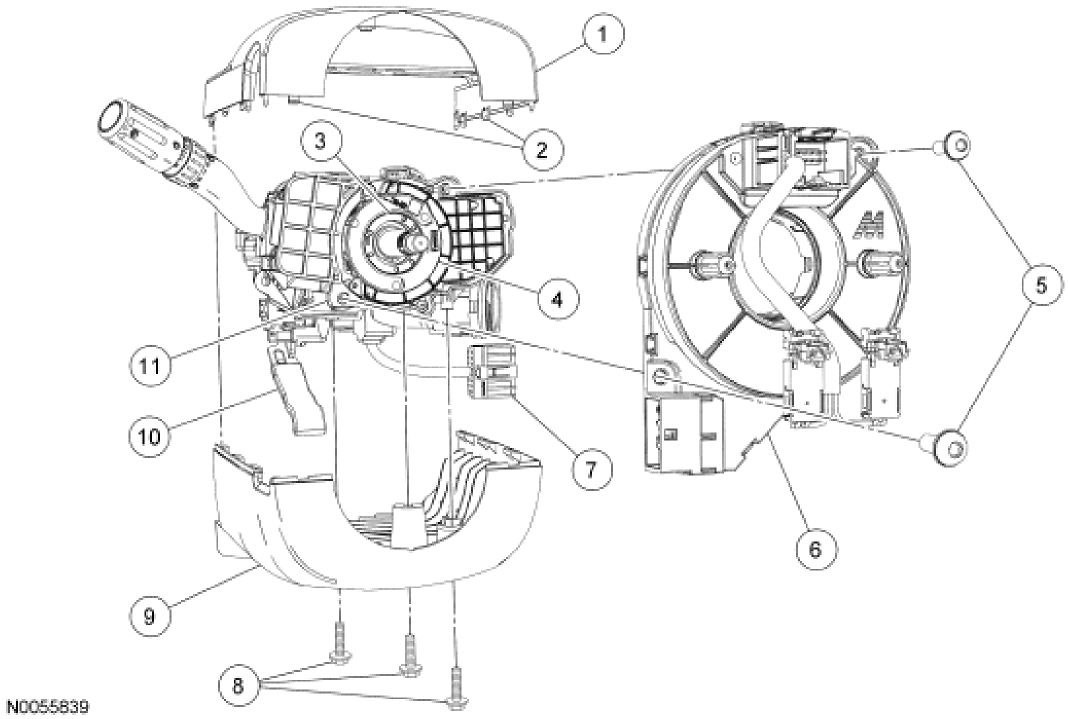


Fig. 177: Exploded View Of Clockspring
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	3530A	Upper steering column shroud
2	-	Upper steering column shroud tabs (2 required) (part of 3530)
3	3F818	Steering wheel rotation sensor (if equipped)
4	-	Steering wheel rotation sensor ring (part of 3F818) (if equipped)
5	-	Clockspring screws (2 required) (part of 13K359)
6	14A644	Clockspring
7	-	Clockspring electrical connector (part of 14401)
8	-	Lower steering column shroud screws (3 required) (part of 3530B)
9	3530B	Lower steering column shroud
10	-	Steering column tilt lock/unlock handle
11	-	Multi-function switch housing

REMOVAL

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Remove the driver air bag module. For additional information, refer to **Driver Air Bag Module**.

   NOTE: Vehicles equipped with a steering wheel rotation sensor and/or adaptive headlamps, do not allow the clockspring rotor to turn from the straight-ahead position after the steering wheel is removed. Failure to follow this instruction may result in component damage and/or system failure.

   NOTE: Make sure the road wheels are in the straight-ahead position.

2. Remove the steering wheel. For additional information, refer to **STEERING COLUMN** article.
3. Tape the clockspring rotor to the steering column shaft to prevent the clockspring rotor from moving out of center.
4. Release the 2 tabs and position the upper steering column shroud upward.
5. Remove the 3 screws and the lower steering column shroud.
6. Disconnect the clockspring electrical connector.
7. Remove the tape from the clockspring rotor to the steering column shaft. Do not allow the clockspring rotor to move from center after tape is removed.

   NOTE: For vehicles equipped a with steering wheel rotation sensor and/or adaptive headlamps, do not allow the clockspring rotor to turn from the straight-ahead position after the steering wheel is removed. Failure to follow this instruction may result in component damage and/or system failure.

   NOTE: For vehicles equipped a with steering wheel rotation sensor and/or adaptive headlamps, after the clockspring has been removed, make sure the arrow on the steering wheel rotation sensor ring is lined up with the arrow on the steering wheel rotation sensor housing as shown in illustration.

8. Remove the 2 clockspring screws and remove the clockspring.

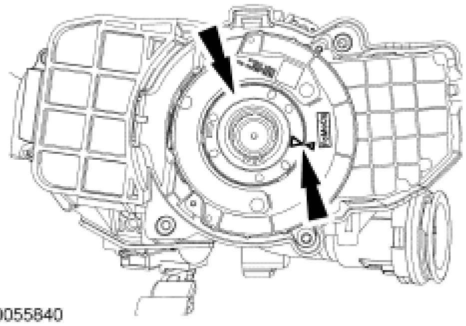


Fig. 178: Locating Clockspring Screws & Clockspring
Courtesy of FORD MOTOR CO.

INSTALLATION

Vehicle repairs reusing the same clockspring

WARNING: **If the clockspring is not correctly centralized, it may fail prematurely. If in doubt, repeat the centralizing procedure. Failure to follow these instructions may increase the risk of serious personal injury or death in a crash.**

NOTE: Make sure the road wheels are in the straight-ahead position. Failure to follow this instruction may result in component damage and/or system failure.

1. If the vehicle's clockspring has rotated out of center, follow these steps to center the clockspring.
    1. Hold the clockspring outer housing stationary.

       NOTE: Overturning will destroy the clockspring. The internal ribbon wire acts as the stop and can be broken from its internal connection.

    2. While turning the rotor counterclockwise, carefully feel for the ribbon wire to run out of length and for a slight resistance. Stop turning at this point.
    3. Turn the clockspring clockwise (approximately 2.25 turns) until the clockspring rotor wiring and connector are in the 12 o'clock position. Clockspring is now centered.
        - Do not allow the rotor to turn from this position.

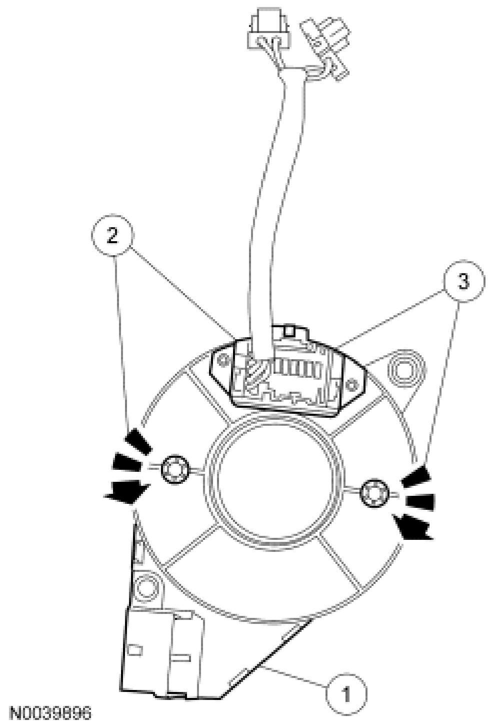


Fig. 179: Centering Clockspring
Courtesy of FORD MOTOR CO.

All vehicles

NOTE:	**Slight rotation of the steering wheel rotation sensor ring is allowed to align the 2 arrows.**

2. Make sure the steering wheel rotation sensor ring arrow is lined up with the steering wheel rotation sensor housing arrow as shown in illustration.

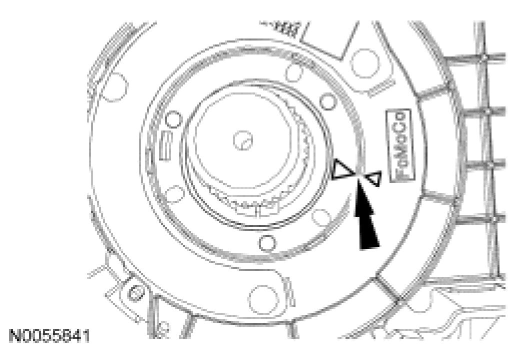


Fig. 180: Making Sure Sensor Ring Arrow Is Lined Up With Absolute Steering Angle Sensor Housing Arrow
Courtesy of FORD MOTOR CO.

NOTE:	**If the clockspring is left unattended by the technician between centralizing**

the clockspring and installing it to the multi-function switch housing, the centralizing procedure must be repeated. Failure to follow this instruction may result in component damage and/or system failure.

NOTE: **For vehicles equipped with a steering wheel rotation sensor and/or adaptive headlamps, slight rotation of the clockspring rotor might be needed to seat the clockspring 3 locator pins into the steering wheel rotation sensor and or adaptive headlamps sensor ring. Very slight rotation is possible on a new clockspring with the sealing key installed.**

NOTE: **Make sure the clockspring is fully seated into the multi-function switch housing before installing the clockspring screws.**

3. Install the clockspring and the 2 screws.

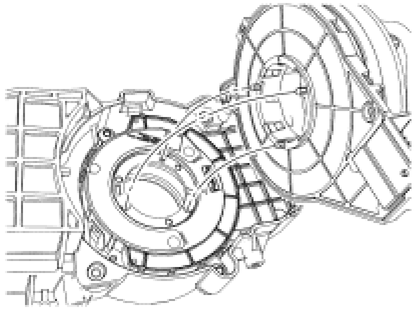


Fig. 181: Installing Clockspring & Screws
Courtesy of FORD MOTOR CO.

4. Connect the clockspring electrical connector.
5. Install the lower steering column shroud and the 3 screws.
6. Attach the upper steering column shroud to the lower steering column shroud.

NOTE: **If not installing a new clockspring, and the vehicle is left unattended by the technician between the installation of the clockspring to the multi-function switch housing and installing the steering wheel, the centralizing procedure can be repeated at this time with the clockspring being installed in the multi-function switch housing. Failure to follow this instruction may result in component damage and/or system failure.**

7. Install the steering wheel. For additional information, refer to **STEERING SYSTEM - GENERAL INFORMATION** article.
    - If a new clockspring is being installed, and after the steering wheel installation, remove the clockspring sealing key.
8. Install the driver air bag module. For additional information, refer to **Driver Air Bag Module**.

DRIVER AIR BAG MODULE

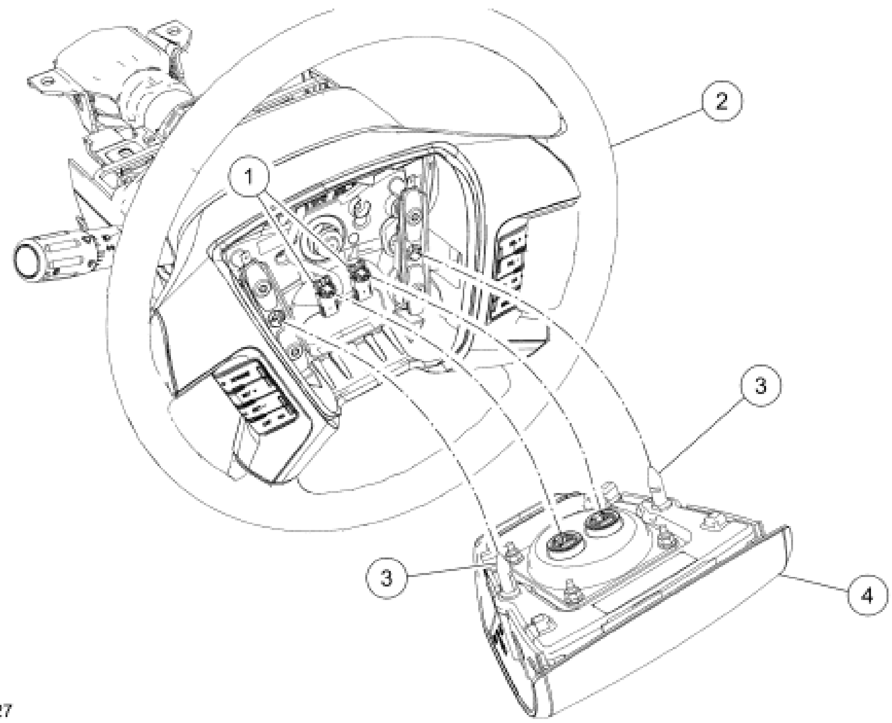


Fig. 182: Exploded View Of Driver Air Bag Module
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Driver air bag module electrical connectors (part of 14A664)
2	3600	Steering wheel
3	-	Driver air bag module locking pins (part of 43B13)
4	43B13	Driver air bag module

REMOVAL

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Always carry or place a live air bag module with the air bag and deployment door/trim cover/tear seam pointed away from the body. Do not set a live air bag module down with the deployment door/trim cover/tear seam face down. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.**

WARNING: Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.

WARNING: Do not repaint air bag modules with discolored or damaged trim covers or deployment doors; new air bag modules must be installed. Failure to follow this instruction may result in the air bag deploying incorrectly, which increases the risk of serious personal injury or death in a crash.

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

   NOTE: Repeat this step for both locking pins.

2. Using a 3 mm (0.11 in) Allen wrench or a suitable tool through the access hole on the backside of the steering wheel, position the tool against the spring clip and push in, disengaging the clip from the locking pin. With the spring clip disengaged from the locking pin, gently pull back on that side of the driver air bag module to release it from the steering wheel.

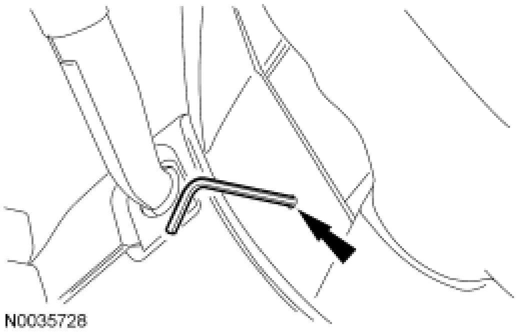


Fig. 183: Locating Allen Wrench
Courtesy of FORD MOTOR CO.

   NOTE: Do not pull the driver air bag module electrical connectors out by the locking buttons. Damage to the locking buttons can occur.

3. Using a small screwdriver as shown in illustration, lift up and release the locking buttons on the driver air bag module electrical connectors. With the locking buttons released, remove the electrical connectors and the driver air bag module.

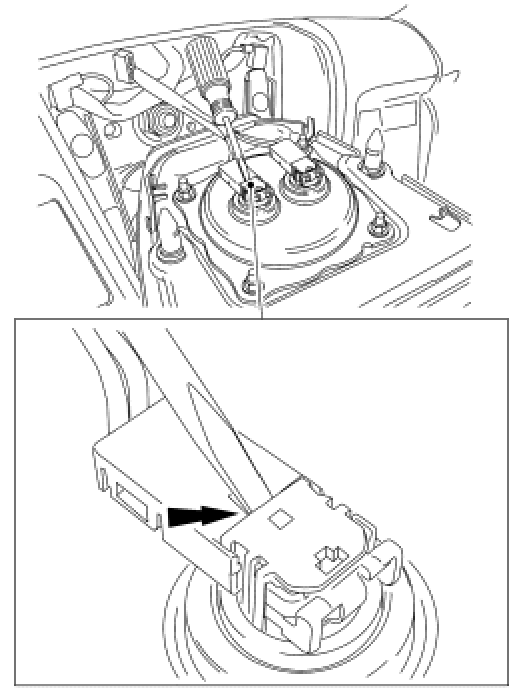


Fig. 184: Releasing Locking Buttons On Driver Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

INSTALLATION

NOTE: **Do not install the driver air bag module electrical connectors by the locking buttons. Damage to the locking buttons can occur.**

NOTE: **The driver air bag module electrical connector locking buttons must be in the released position when the connector is being installed or connector damage may occur.**

NOTE: **The driver air bag module electrical connectors are unique and cannot be reversed when connected to the driver air bag module. Match the electrical connector key to the keyway in the driver air bag module. Do not force the electrical connectors into the driver air bag module. Damage to the connector or component may occur.**

1. With the locking buttons released, install the driver air bag module electrical connectors fully into the driver air bag module and seat the locking buttons.

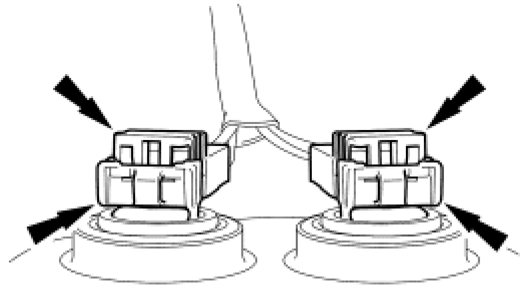


Fig. 185: Locating Driver Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

NOTE: Audible clicks will be heard when both wire clips are seated in the driver air bag module.

2. Align the driver air bag module locking pins to the steering wheel and, while pushing inward, seat the 2 driver air bag module locking pins to the steering wheel wire clips.
   - When the 2 locking pins are seated in place, there should be an even gap between the driver air bag module trim cover and the steering wheel.
3. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

FRONT IMPACT SEVERITY SENSOR

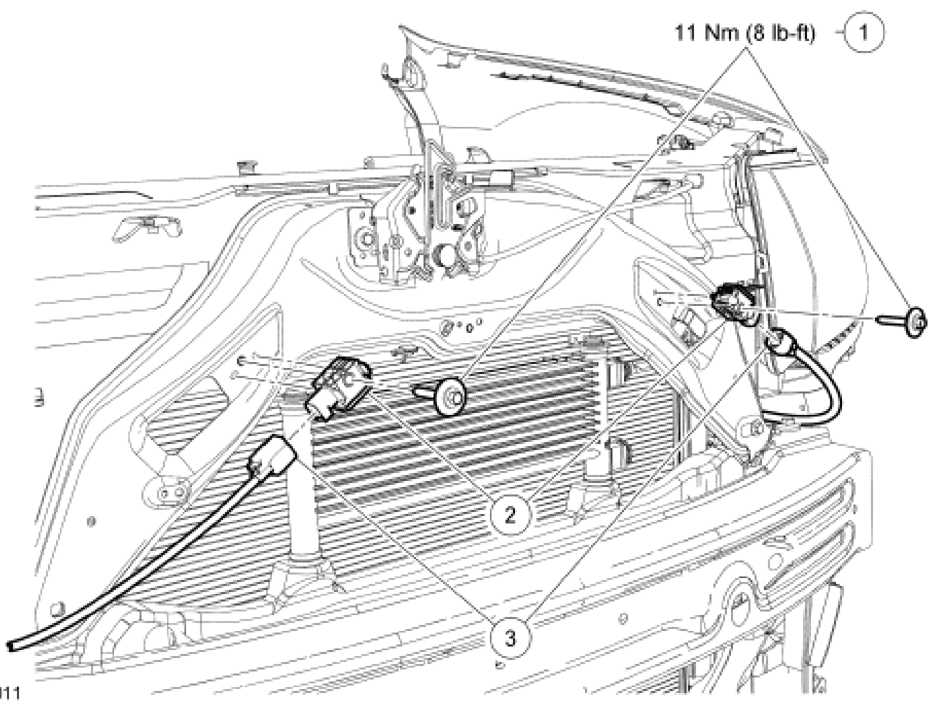


Fig. 186: Exploded View Of Front Impact Severity Sensor With Torque Specification

Item	Part Number	Description
1	W712191	LH and RH front impact severity sensor bolts
2	14B006	LH and RH front impact severity sensors
3	-	LH and RH front impact severity sensor electrical connectors (part of 14290)

WARNING: **If a vehicle has been in a crash, inspect the restraints control module (RCM) and the impact sensor (if equipped) mounting areas for deformation. If damaged, restore the mounting areas to the original production configuration. A new RCM and sensors must be installed whether or not the air bags have deployed. Failure to follow these instructions may result in serious personal injury or death in a crash.**

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Remove the front bumper cover. For additional information, refer to **BUMPERS** article.
3. Disconnect the front impact severity sensor electrical connector.

   WARNING: **Always tighten the fasteners of the restraints control module (RCM) and impact sensor (if equipped) to the specified torque. Failure to do so may result in incorrect restraint system operation, which increases the risk of personal injury or death in a crash.**

   NOTE: Make sure the hood latch bracket and front impact severity sensor mating surfaces are clean and free of foreign material.

   NOTE: Note position of the 2 locator tabs on the front impact severity sensor for installation.

4. Remove the bolt and the front impact severity sensor.
   - To install, tighten to 11 Nm (8 lb-ft).
5. To install, reverse the removal procedure.
6. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

OCCUPANT CLASSIFICATION SYSTEM MODULE (OCSM) - MANUAL SEAT TRACK

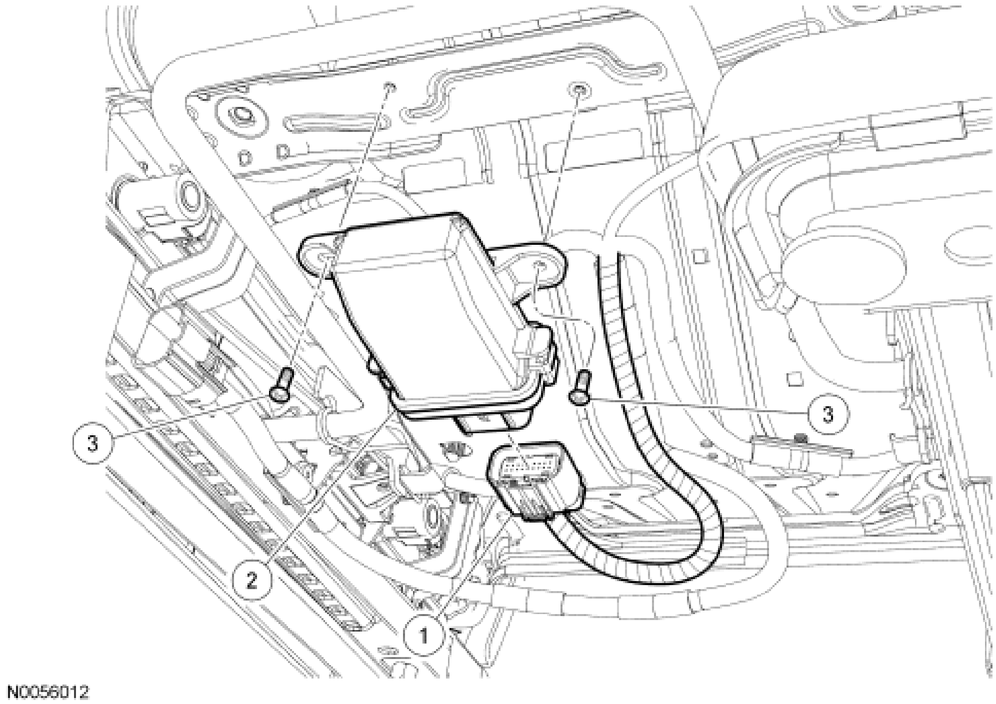


Fig. 187: Exploded View Of Occupant Classification Module - Manual Seat Track
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Occupant classification system module (OCSM) electrical connector (part of 14A698)
2	14B422	OCSM
3	-	OCSM screws

REMOVAL

WARNING: **Use care when handling the front passenger seat and track assembly. Dropping the assembly, placing excessive weight on or sitting on a front passenger seat that is not secured in the vehicle may result in damaged seat components. Failure to follow these instructions may result in incorrect operation of the occupant classification sensor (OCS) system and increases the risk of serious personal injury or death in a crash.**

NOTE: **The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.**

NOTE: **The supplemental restraint system (SRS) must be fully operational and free of**

faults before releasing the vehicle to the customer.

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Disconnect the occupant classification system module (OCSM) electrical connector.
3. Remove the 2 screws and OCSM.
4. To install, reverse the removal procedure.
5. Install the passenger seat and repower the SRS. **Do not prove out the SRS at this time.** For additional information, refer to **SEATING** article.
6. Carry out the appropriate procedure after installation of an OCSM.
    - If installing the original OCSM, carry out the Zero Seat Weight Test and prove out the SRS. For additional information, refer to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.
    - If installing a new OCSM, carry out the System Reset procedure and prove out the SRS. For additional information, refer to **Occupant Classification Sensor (OCS) System Reset**.

OCCUPANT CLASSIFICATION SYSTEM MODULE (OCSM) - POWER SEAT TRACK

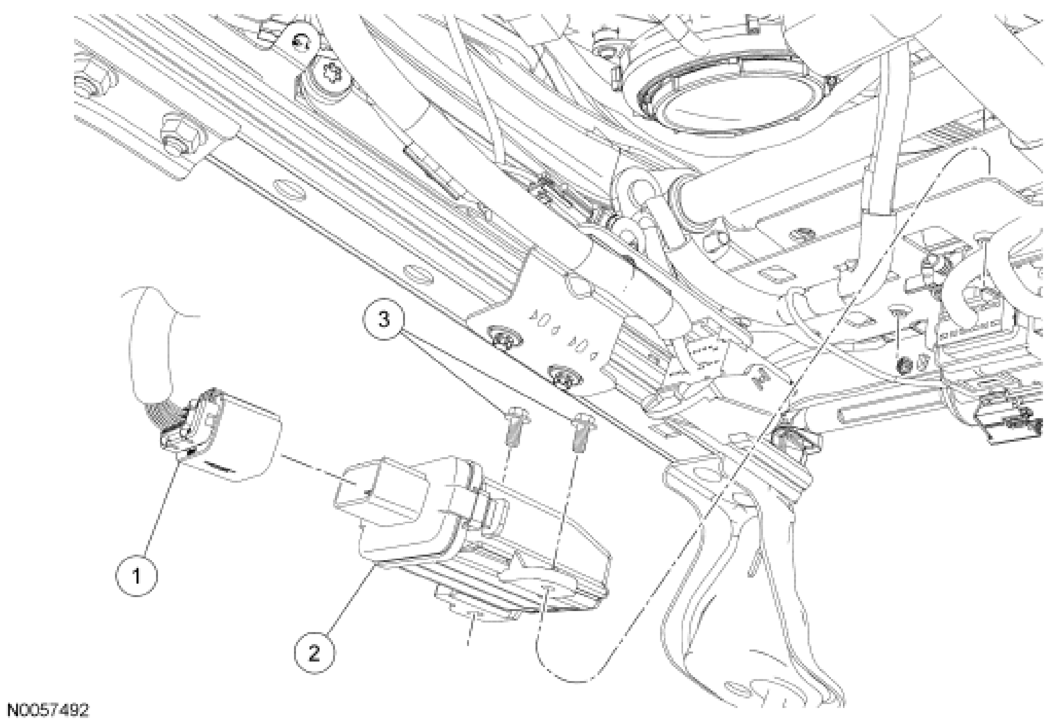


Fig. 188: Exploded View Of Occupant Classification Module - Power Seat Track
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Occupant classification system module (OCSM) electrical connector (part of

		14A698)
2	14B422	OCSM
3	-	OCSM screws

REMOVAL

WARNING: **Use care when handling the front passenger seat and track assembly. Dropping the assembly, placing excessive weight on or sitting on a front passenger seat that is not secured in the vehicle may result in damaged seat components. Failure to follow these instructions may result in incorrect operation of the occupant classification sensor (OCS) system and increases the risk of serious personal injury or death in a crash.**

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Remove the front passenger seat. For additional information, refer to **SEATING** article.
2. Disconnect the occupant classification system module (OCSM) electrical connector.
3. Remove the 2 screws and OCSM.
4. To install, reverse the removal procedure.
5. Install the front passenger seat. **Do not prove out the SRS at this time.** For additional information, refer to **SEATING** article.
6. Repower the SRS. **Do not prove out the SRS at this time.** For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
7. Carry out the appropriate procedure after installation of an OCSM.
   - If installing the original OCSM, carry out the Zero Seat Weight Test. For additional information, refer to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.
   - If installing a new OCSM, carry out the System Reset procedure. For additional information, refer to **Occupant Classification Sensor (OCS) System Reset**.

OCCUPANT CLASSIFICATION SENSOR - MANUAL SEAT TRACK

REMOVAL

WARNING: **Use care when handling the front passenger seat and track assembly. Dropping the assembly, placing excessive weight on or sitting on a front passenger seat that is not secured in the vehicle may result in damaged seat components. Failure to follow these instructions may result in incorrect operation of the occupant classification sensor (OCS) system and increases the risk of serious personal injury or death in a crash.**

NOTE: The air bag warning indicator illuminates when the restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

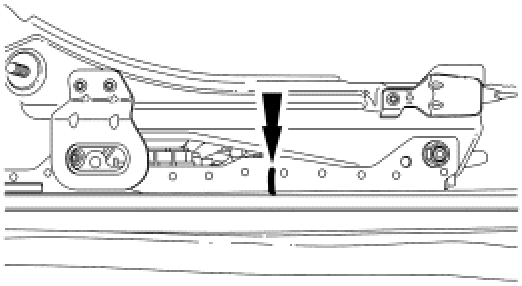
NOTE: Repair is made by installing a new part only. If the new part does not correct the condition, install the original part and carry out the diagnostic procedure again.

All occupant classification sensor (OCS) weight sensor bolt(s)

1. Remove the manual seat track. For additional information, refer to **SEATING** article.

   NOTE: After marking the seat track rails, do not change the position of the seat track.

2. Mark the position of both seat track rails.

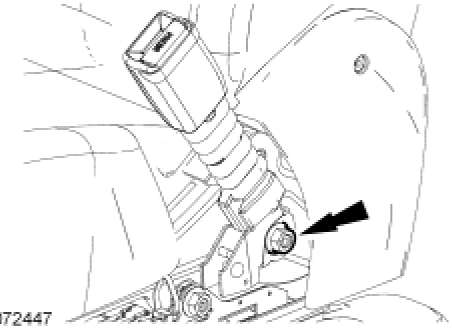


N0060415

Fig. 189: Marking Position Of Both Seat Track Rails
Courtesy of FORD MOTOR CO.

Inboard OCS weight sensor bolt(s)

3. Remove the nut and safety belt buckle.



N0072447

Fig. 190: Identifying Nut And Safety Belt Buckle

Courtesy of FORD MOTOR CO.

All OCS weight sensor bolt(s)

NOTE: Outboard front cross brace screws shown in illustration, inboard similar.

4. On the side with the affected OCS I-bolt(s), remove the 2 front cross brace screws.

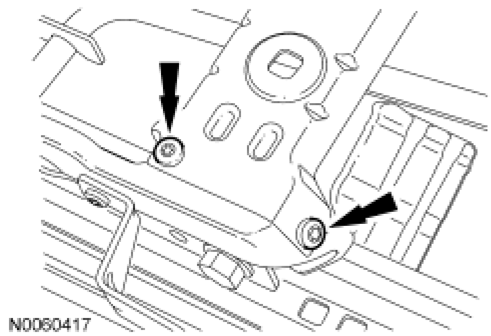


Fig. 191: Identifying Front Cross Brace Screws
Courtesy of FORD MOTOR CO.

5. On the side with the affected OCS weight sensor bolt(s), disengage the seat adjuster handle spring and position it aside.

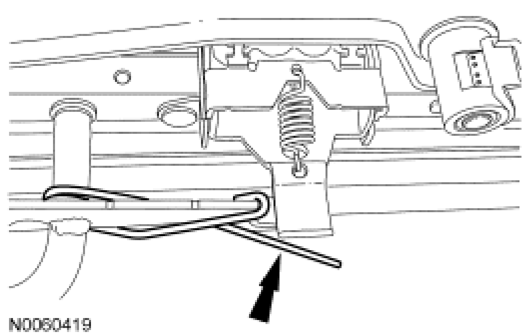


Fig. 192: Identifying Seat Adjuster Handle Spring
Courtesy of FORD MOTOR CO.

6. Separate the seat adjuster handle from the affected side of the seat track rail with the OCS weight sensor bolt(s).

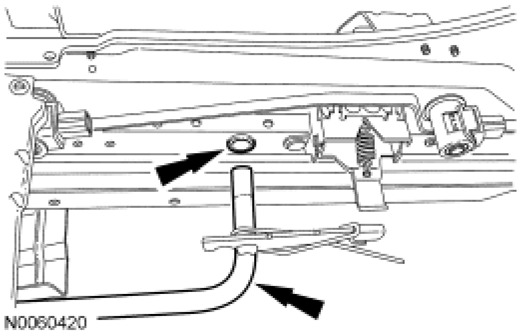


Fig. 193: Identifying Seat Track Rail & OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

NOTE: Outboard rear cross brace screws shown in illustration, inboard similar.

7. On the side with the affected OCS weight sensor bolt(s), remove the 2 rear cross brace screws.

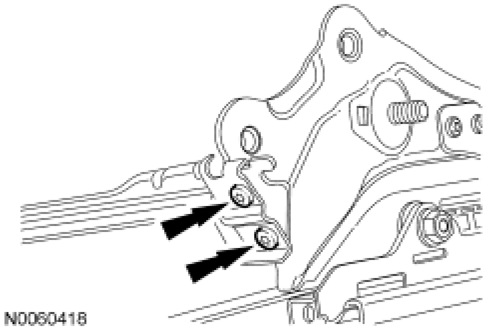


Fig. 194: Identifying Rear Cross Brace Screws
Courtesy of FORD MOTOR CO.

NOTE: If any occupant classification sensor (OCS) weight sensor bolt nuts are removed, new OCS weight sensor bolt nuts must be installed.

8. Remove and discard the 2 OCS weight sensor bolt nuts from the affected side of the seat track rail with the OCS weight sensor bolt(s).

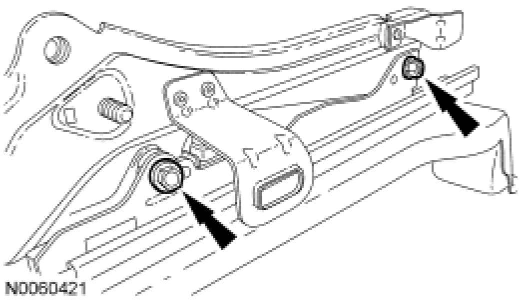


Fig. 195: Identifying OCS Weight Sensor Bolt Nuts

Courtesy of FORD MOTOR CO.

9. Remove the side plate with the OCS weight sensor bolt(s) from the seat track rail.

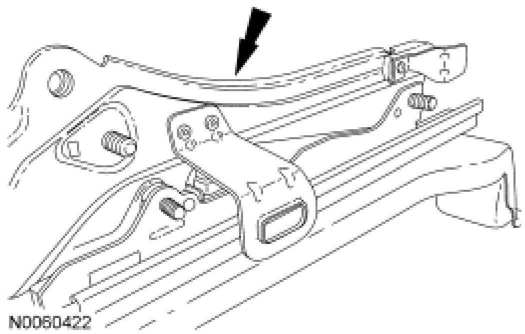


Fig. 196: Identifying Side Plate
Courtesy of FORD MOTOR CO.

NOTE: Do not overtighten the side plate with the occupant classification sensor (OCS) weight sensor bolts to the bench or table. Tighten only enough to compress the spring washer. Overtightening may damage the side plate or OCS weight sensor bolts.

10. Using a suitable tool, clamp the side plate to a bench or table to compress the spring washer on the affected OCS weight sensor bolt as shown in illustration.

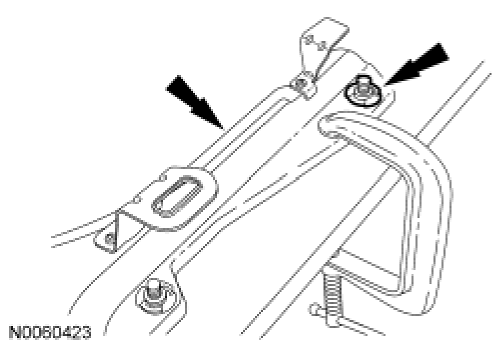


Fig. 197: Clamping Side Plate To A Bench Or Table Using A Suitable Tool
Courtesy of FORD MOTOR CO.

NOTE: Note the position of the OCS weight sensor bolt, spring washer and retaining clip for installation.

11. Remove and discard the retaining clip from the affected OCS weight sensor bolt.

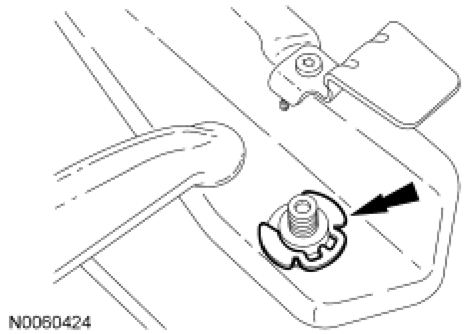


Fig. 198: Identifying Retaining Clip From Affected OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

12. Release the clamp and remove the OCS weight sensor bolt and spring washer from the side plate.

INSTALLATION

All OCS weight sensor bolt(s)

NOTE: **Do not overtighten the side plate with the occupant classification sensor (OCS) weight sensor bolts to the bench or table. Tighten only enough to compress the spring washer. Overtightening may damage the side plate or OCS weight sensor bolts.**

1. Position the spring washer and OCS weight sensor bolt in the side plate. Then clamp the side plate to a bench or table to compress the OCS weight sensor bolt spring washer.

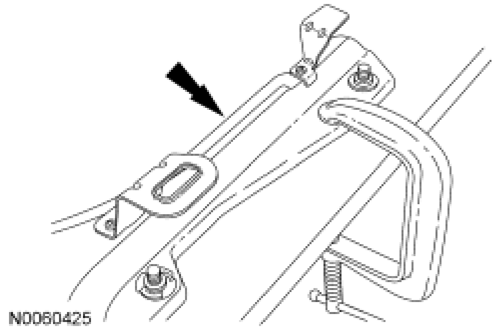


Fig. 199: Clamping Side Plate To A Bench Or Table Using A Suitable Tool
Courtesy of FORD MOTOR CO.

NOTE: **Make sure the OCS weight sensor bolt retaining clip is seated fully in the OCS weight sensor bolt groove underneath the OCS weight sensor O-ring seal.**

2. Install the 2 new OCS weight sensor bolt retaining clips in the OCS weight sensor groove underneath the O-ring seal.

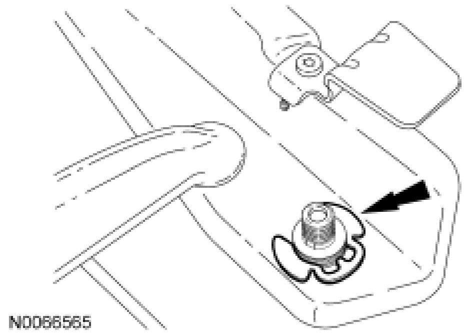


Fig. 200: Identifying Retaining Clip From Affected OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

3. Unclamp the side plate with the installed OCS weight sensor bolt from the bench or table.

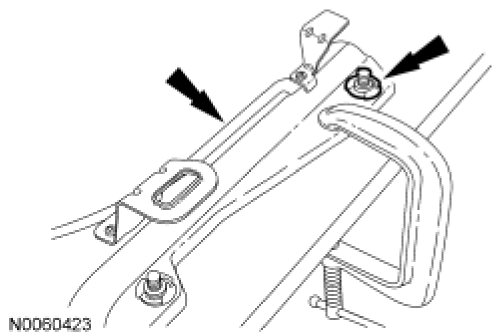


Fig. 201: Clamping Side Plate To A Bench Or Table Using A Suitable Tool
Courtesy of FORD MOTOR CO.

NOTE: **Slight rotational adjustment of the OCS weight sensors bolts may be necessary to install the OCS weight sensor bolts and side plate to the seat track rail.**

4. Install the side plate and OCS weight sensor bolts to the seat track rail.

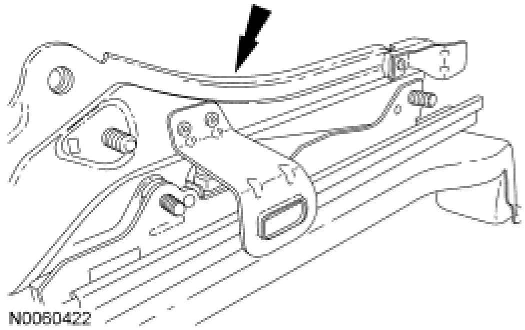


Fig. 202: Identifying Side Plate
Courtesy of FORD MOTOR CO.

NOTE: When installing the side plate to the seat track rail, 2 new OCS weight sensor nuts must be installed.

5. Install the 2 new OCS weight sensor bolt nuts to the seat track rail.
    - Tighten to 33 Nm (24 lb-ft).

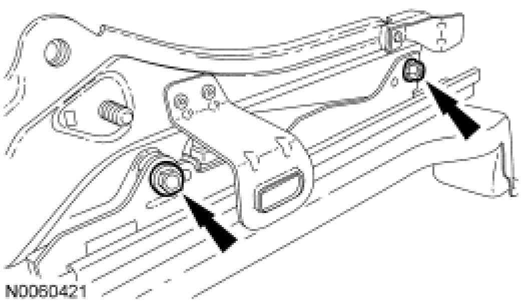


Fig. 203: Identifying OCS Weight Sensor Bolt Nuts
Courtesy of FORD MOTOR CO.

6. Install the seat adjuster handle to the seat track rail.

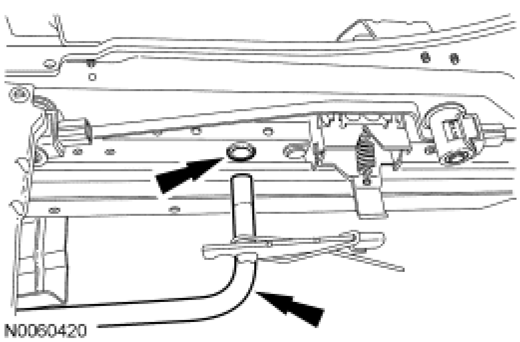


Fig. 204: Identifying Seat Track Rail & OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

7. Engage the seat adjuster handle spring.

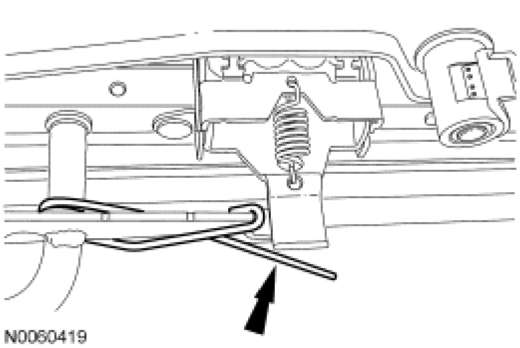


Fig. 205: Identifying Seat Adjuster Handle Spring

Courtesy of FORD MOTOR CO.

8. Install the 2 rear cross brace screws.
    - Tighten to 5 Nm (44 lb-in).

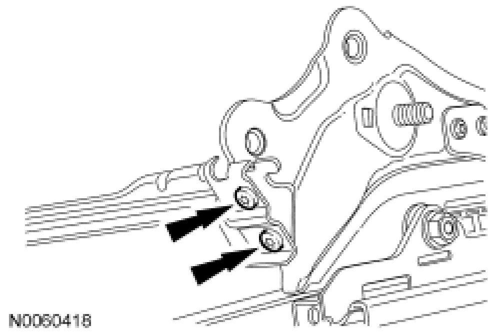


Fig. 206: Identifying Rear Cross Brace Screws
Courtesy of FORD MOTOR CO.

9. Install the 2 front cross brace screws.
    - Tighten to 5 Nm (44 lb-in).

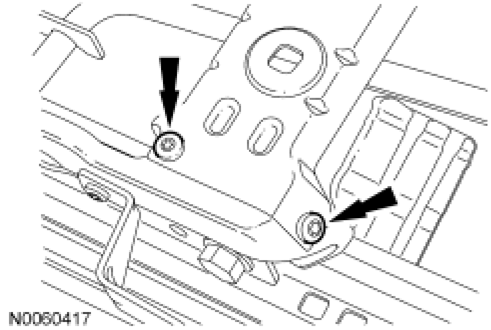


Fig. 207: Identifying Front Cross Brace Screws
Courtesy of FORD MOTOR CO.

Inboard OCS weight sensor bolt(s)

10. Install the nut and safety belt buckle.
    - Tighten to 40 Nm (30 lb-ft).

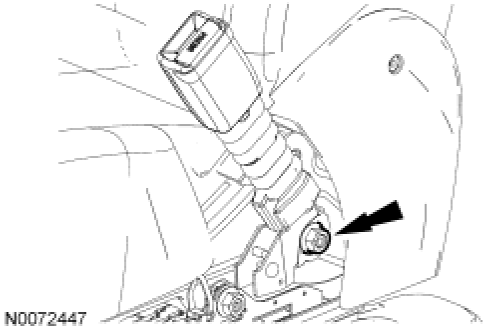


Fig. 208: Identifying Safety Belt Buckle Bolt
Courtesy of FORD MOTOR CO.

All OCS weight sensor bolt(s)

11. Install the manual seat track. For additional information, refer to **SEATING** article.
12. Install the passenger seat and repower the SRS. **Do not prove out the SRS at this time.** For additional information, refer to **SEATING** article.
13. Carry out the appropriate procedure after installation of an OCS system component.
    - If installing the original OCS system component, carry out the Zero Seat Weight Test and prove out the SRS. For additional information, refer to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.
    - If installing a new OCS system component, carry out the System Reset procedure and prove out the SRS. For additional information, refer to **Occupant Classification Sensor (OCS) System Reset**.

OCCUPANT CLASSIFICATION SENSOR - POWER SEAT TRACK

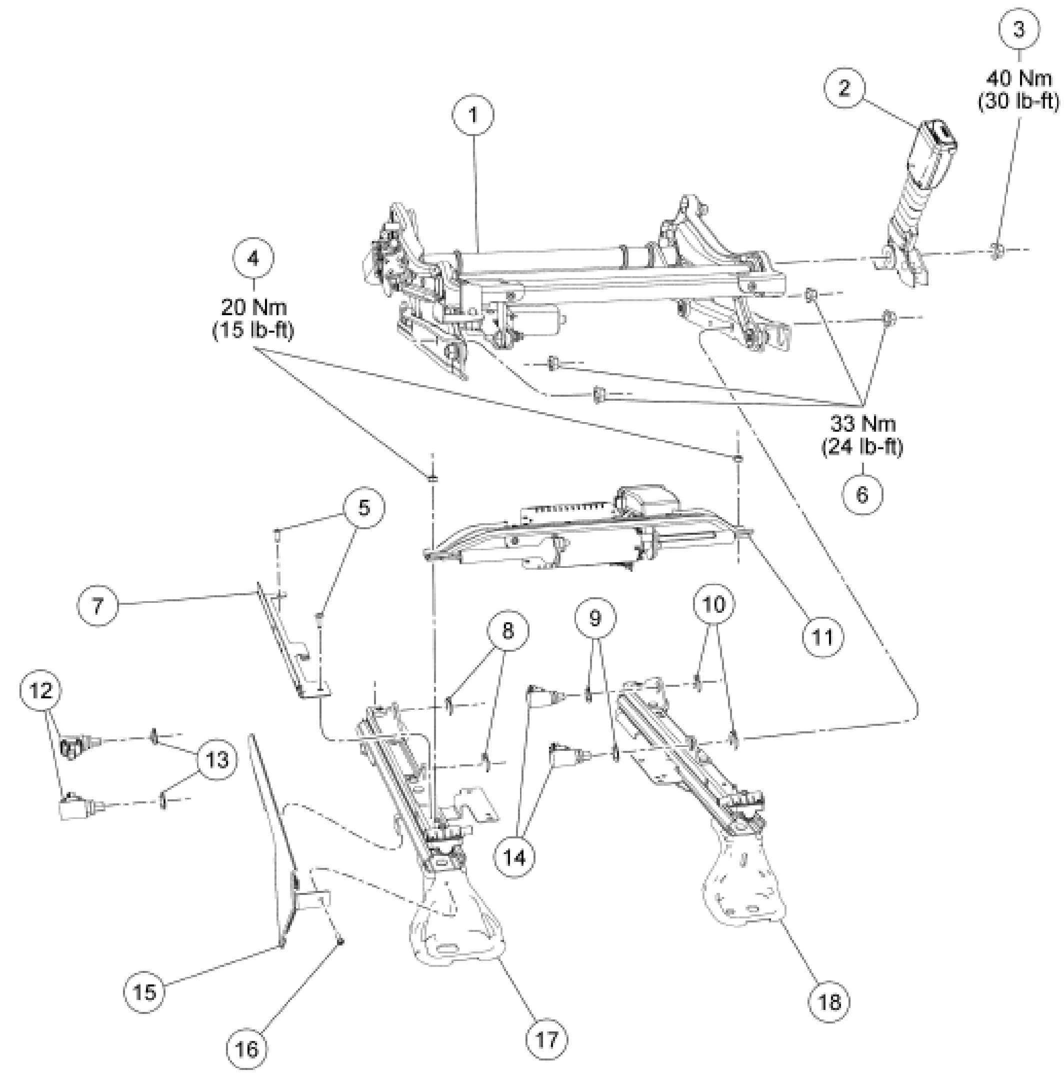


Fig. 209: Identifying Power Seat Track With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	61710	Power seat track assembly
2	61202	Safety belt buckle
3	-	Safety belt buckle nut (part of 61202)
4	-	Seat track front cross brace nuts (2 required) (part of 61710)
5	-	Occupant classification sensor (OCS) weight sensor bolt protection bracket screws (2 required) (part of 61710)
6	-	OCS weight sensor bolt nuts (4 required) (part of 61710)

7	-	OCS weight sensor bolt protection bracket (part of 61710)
8	-	Outboard seat track rail OCS weight sensor bolt retaining clips (2 required) (part of 61710)
9	-	Inboard seat track rail OCS weight sensor bolt spring washers (2 required) (part of 61710)
10	-	Inboard seat track rail OCS weight sensor bolt retaining clips (2 required) (part of 61710)
11	-	Seat track front cross brace (part of 61710)
12	-	Outboard seat track rail OCS weight sensor bolts (2 required) (part of 61710)
13	-	Outboard seat track rail OCS weight sensor bolt spring washers (2 required) (part of 61710)
14	-	Inboard seat track rail OCS weight sensor bolts (2 required) (part of 61710)
15	-	Outboard seat track riser cover shield (part of 61710)
16	-	Outboard seat track riser cover shield retainer (part of 61710)
17	-	Outboard seat track rail (part of 61710)
18	-	Inboard seat track rail (part of 61710)

REMOVAL

WARNING: **Use care when handling the front passenger seat and track assembly. Dropping the assembly, placing excessive weight on or sitting on a front passenger seat that is not secured in the vehicle may result in damaged seat components. Failure to follow these instructions may result in incorrect operation of the occupant classification sensor (OCS) system and increases the risk of serious personal injury or death in a crash.**

NOTE: **The air bag warning indicator illuminates when the restraints control module (RCM) fuse is removed and the ignition switch is ON.**

NOTE: **The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.**

NOTE: **Repair is made by installing a new part only. If the new part does not correct the condition, install the original part and carry out the diagnostic procedure again.**

All occupant classification sensor (OCS) weight sensor bolt(s)

1. Remove the power seat track. For additional information, refer to **SEATING** article.

NOTE: Inboard seat track rail shown in illustration, outboard similar.

NOTE: After marking the seat track rails, do not change the position of the seat track.

2. Mark the position of both seat track rails.

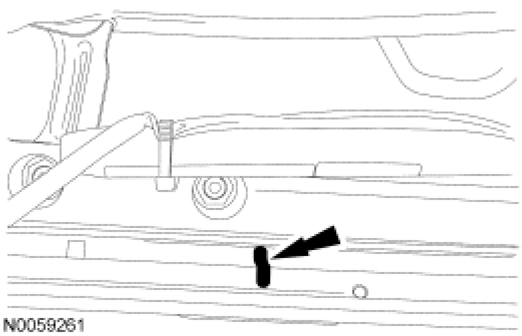


Fig. 210: Marking Position Of Both Seat Track Rails
Courtesy of FORD MOTOR CO.

Inboard OCS weight sensor bolt(s)

3. Remove the nut and safety belt buckle.

Outboard OCS weight sensor bolt(s)

4. Remove the front retainer and outboard seat track riser cover shield.
5. Remove the 2 retainers and occupant classification sensor (OCS) weight sensor bolt protection bracket.

All OCS weight sensor bolt(s)

6. On the side with the affected OCS weight sensor bolt(s), remove the nut from the seat track front cross brace.

NOTE: If any OCS weight sensor bolt nuts are removed, new OCS weight sensor bolt nuts must be installed.

7. Remove and discard the 2 OCS weight sensor bolt nuts from the affected side of the seat track rail with the OCS weight sensor(s).

NOTE: Outboard side shown in illustration, inboard similar.

8. Separate the front seat track motor cable from the affected seat track rail. Then remove the side plate with the OCS weight sensor bolts from the seat track rail.

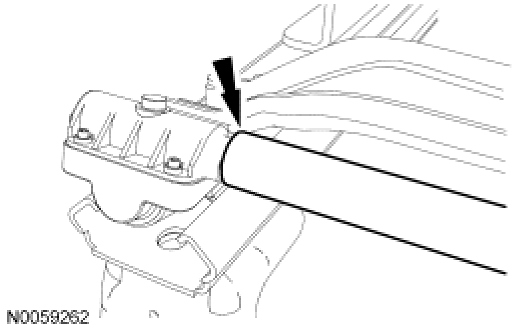


Fig. 211: Identifying Front Seat Track Motor Cable
Courtesy of FORD MOTOR CO.

NOTE: **Do not overtighten the side plate with the occupant classification sensor (OCS) weight sensor bolts to the bench or table. Tighten only enough to compress the spring washer. Overtightening may damage the side plate or OCS weight sensor bolts.**

9. Using a suitable tool, clamp the side plate to a bench or table to compress the spring washer on the affected OCS weight sensor bolt as shown in illustration.

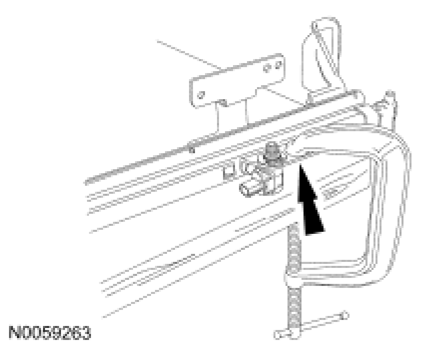


Fig. 212: Clamping Side Plate To A Bench Or Table To Compress Spring Washer On Affected OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

NOTE: **Note the position of the OCS weight sensor bolt, spring washer and retaining clip for installation.**

10. Remove and discard the retaining clip from the affected OCS weight sensor bolt.

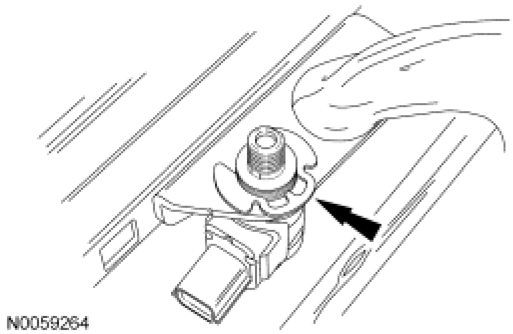


Fig. 213: Identifying Retaining Clip From Affected OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

11. Release the clamp and remove the OCS weight sensor bolt and spring washer from the side plate.

INSTALLATION

All OCS weight sensor bolt(s)

NOTE: Do not overtighten the side plate with the occupant classification sensor (OCS) weight sensor bolts to the bench or table. Tighten only enough to compress the spring washer. Overtightening may damage the side plate or OCS weight sensor bolts.

1. Position the spring washer and OCS weight sensor bolt in the side plate. Then clamp the side plate to a bench or table to compress the OCS weight sensor bolt spring washer.

NOTE: Make sure the OCS weight sensor bolt retaining clip is seated fully in the OCS weight sensor bolt groove underneath the OCS weight sensor O-ring seal.

2. Install the OCS weight sensor bolt retaining clip.

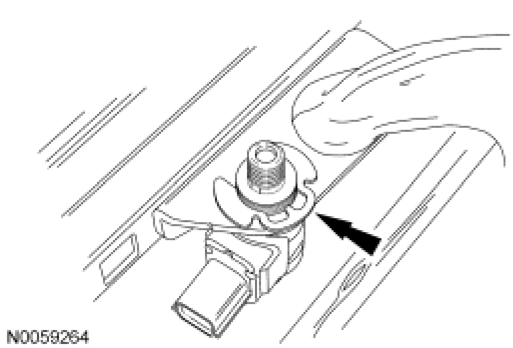


Fig. 214: Identifying Retaining Clip From Affected OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

3. Remove the clamp and side plate from the bench or table.

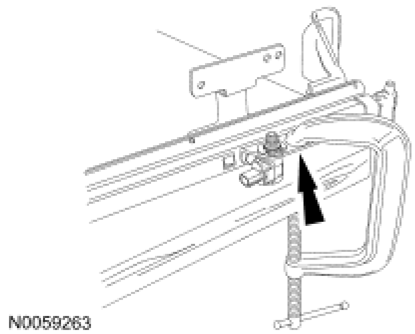


Fig. 215: Clamping Side Plate To A Bench Or Table To Compress Spring Washer On Affected OCS Weight Sensor Bolt
Courtesy of FORD MOTOR CO.

NOTE: **Slight rotational adjustment of the front seat track motor cable and OCS weight sensor bolts may be necessary to install the OCS weight sensor bolts and side plate to the seat track rail.**

4. Install the front seat track motor cable to the seat track rail. Then install the side plate with the OCS weight sensor bolts to the seat track rail.

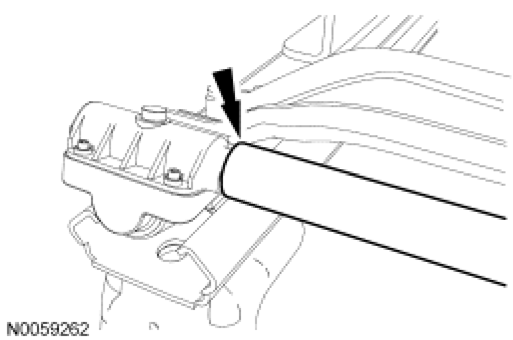


Fig. 216: Identifying Front Seat Track Motor Cable
Courtesy of FORD MOTOR CO.

NOTE: **When installing the side plate to the seat track rail, 2 new OCS weight sensor nuts must be installed.**

5. Install the 2 new OCS weight sensor bolt nuts to the seat track rail.
   - Tighten to 33 Nm (24 lb-ft).

NOTE: **Outboard side shown in illustration, inboard similar.**

6. Install the nut to the seat track front cross brace.
   - Tighten to 20 Nm (15 lb-ft).

Outboard OCS weight sensor bolt(s)

7. Install the 2 retainers and OCS weight sensor bolt protection bracket.
8. Install the front retainer and outboard seat track riser cover shield.

Inboard OCS weight sensor bolt(s)

9. Install the nut and safety belt buckle.
   - Tighten to 40 Nm (30 lb-ft).

All OCS weight sensor bolt(s)

10. Install the power seat track. For additional information, refer to **SEATING** article.
11. Install the passenger seat and repower the SRS. **Do not prove out the SRS at this time.** For additional information, refer to **SEATING** article.
12. Carry out the appropriate procedure after installation of an OCS system component.
    - If installing the original OCS system component, carry out the Zero Seat Weight Test and prove out the SRS. For additional information, refer to **Occupant Classification Sensor (OCS) System Zero Seat Weight Test**.
    - If installing a new OCS system component, carry out the System Reset procedure and prove out the SRS. For additional information, refer to **Occupant Classification Sensor (OCS) System Reset**n.

PASSENGER AIR BAG DEACTIVATION (PAD) INDICATOR

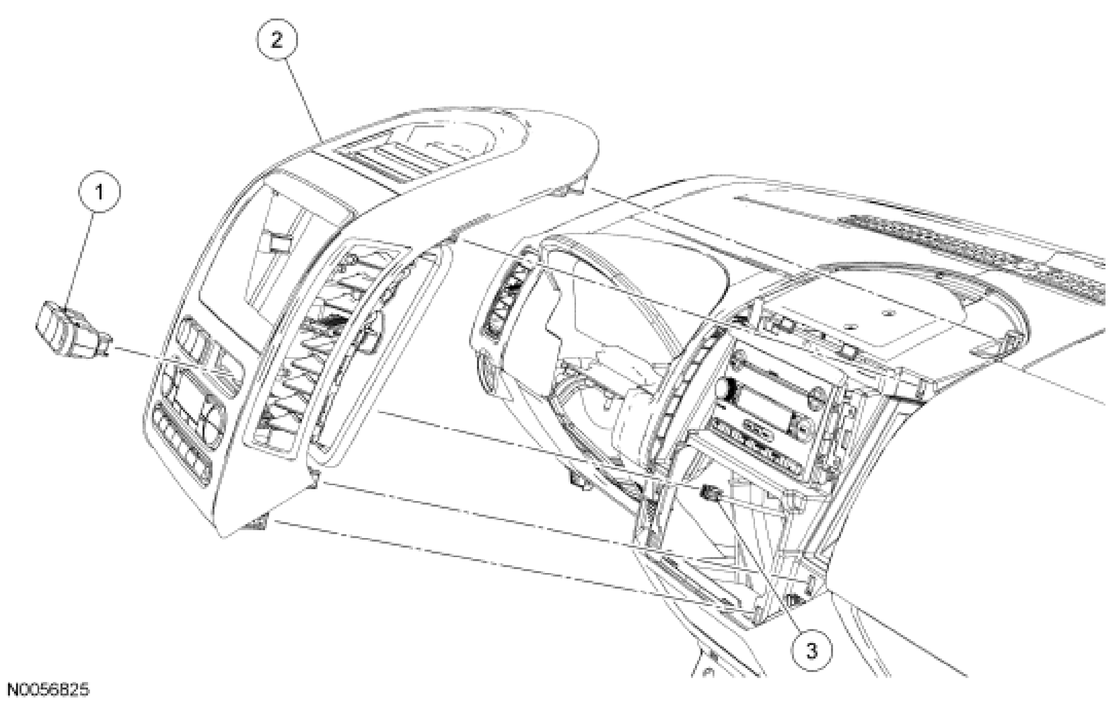


Fig. 217: Exploded View Of Passenger Air Bag Deactivation (PAD) Indicator - Edge
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	13D730	Passenger airbag deactivation (PAD) indicator
2	04302	Instrument panel (I/P) center finish panel
3	-	Passenger airbag deactivation indicator electrical connector (part of 14401)

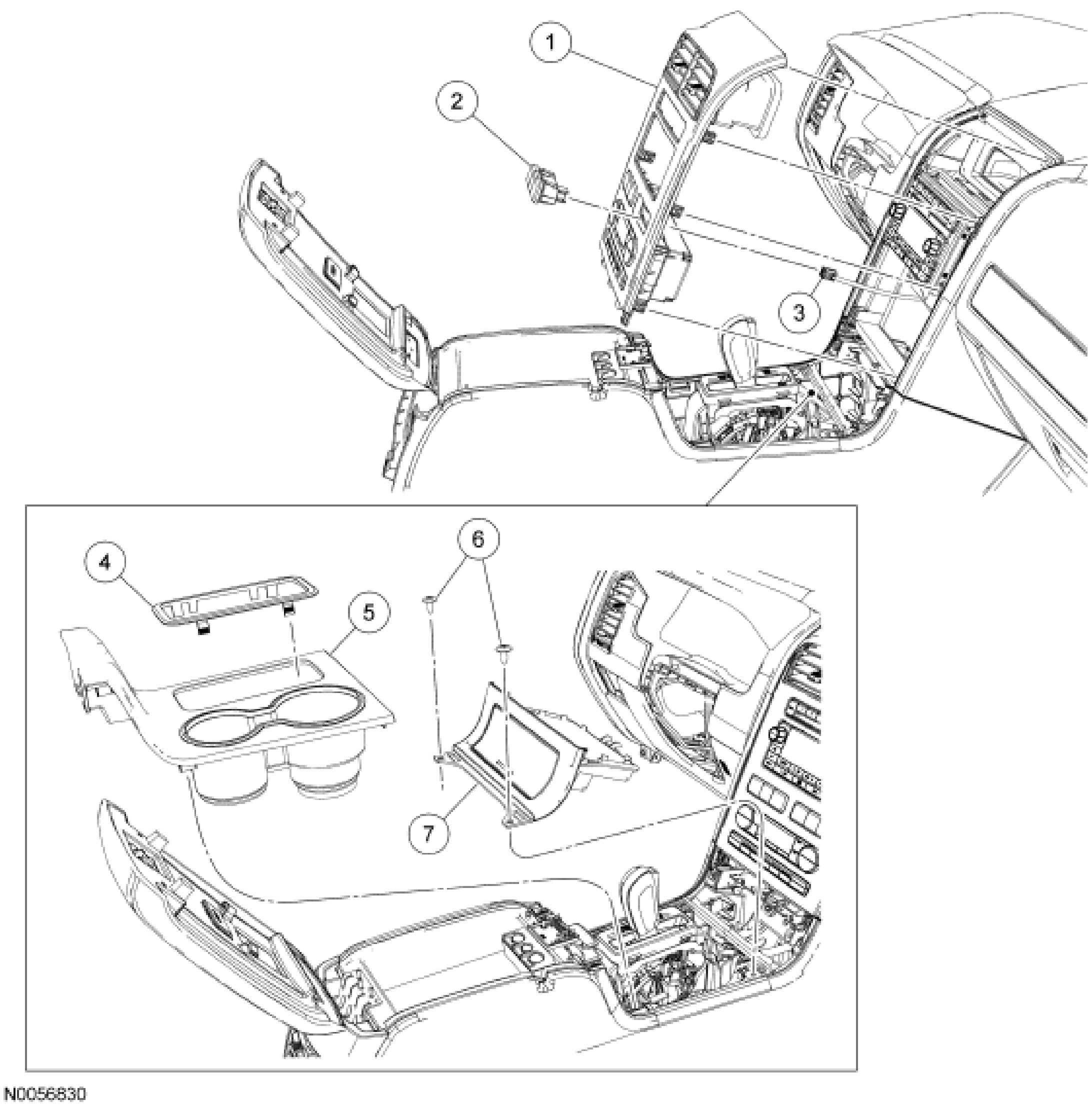


Fig. 218: Exploded View Of Passenger Air Bag Deactivation (PAD) Indicator - MKX
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	04302	Instrument panel center finish panel
2	13D730	Passenger air bag deactivation (PAD) indicator

3	-	PAD indicator electrical connector (part of 14401)
4	7E391	Console shifter bezel
5	045A76	Center finish console panel
6	W706092	Instrument panel power point/storage bin retainers
7	047A04	Instrument panel power point/storage bin

NOTE: **The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.**

NOTE: **The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.**

NOTE: **The passenger air bag deactivation (PAD) indicator is part of the hazard switch/traction control switch assembly.**

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Remove the console shifter bezel (MKX only).
3. Remove the center finish console panel (MKX only).
4. Remove the 2 retainers from the power point/storage bin (MKX only).
5. Pull and release the instrument panel center finish panel.
6. Disconnect the PAD indicator electrical connector.
7. Release the 4 tabs (2 upper and 2 lower) on the PAD indicator and remove it from the instrument panel center finish panel.

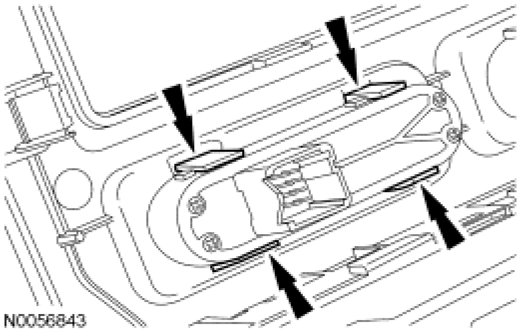


Fig. 219: Locating Tabs On Passenger Air Bag Deactivation Indicator
Courtesy of FORD MOTOR CO.

8. To install, reverse the removal procedure.
9. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

PASSENGER AIR BAG MODULE - EDGE

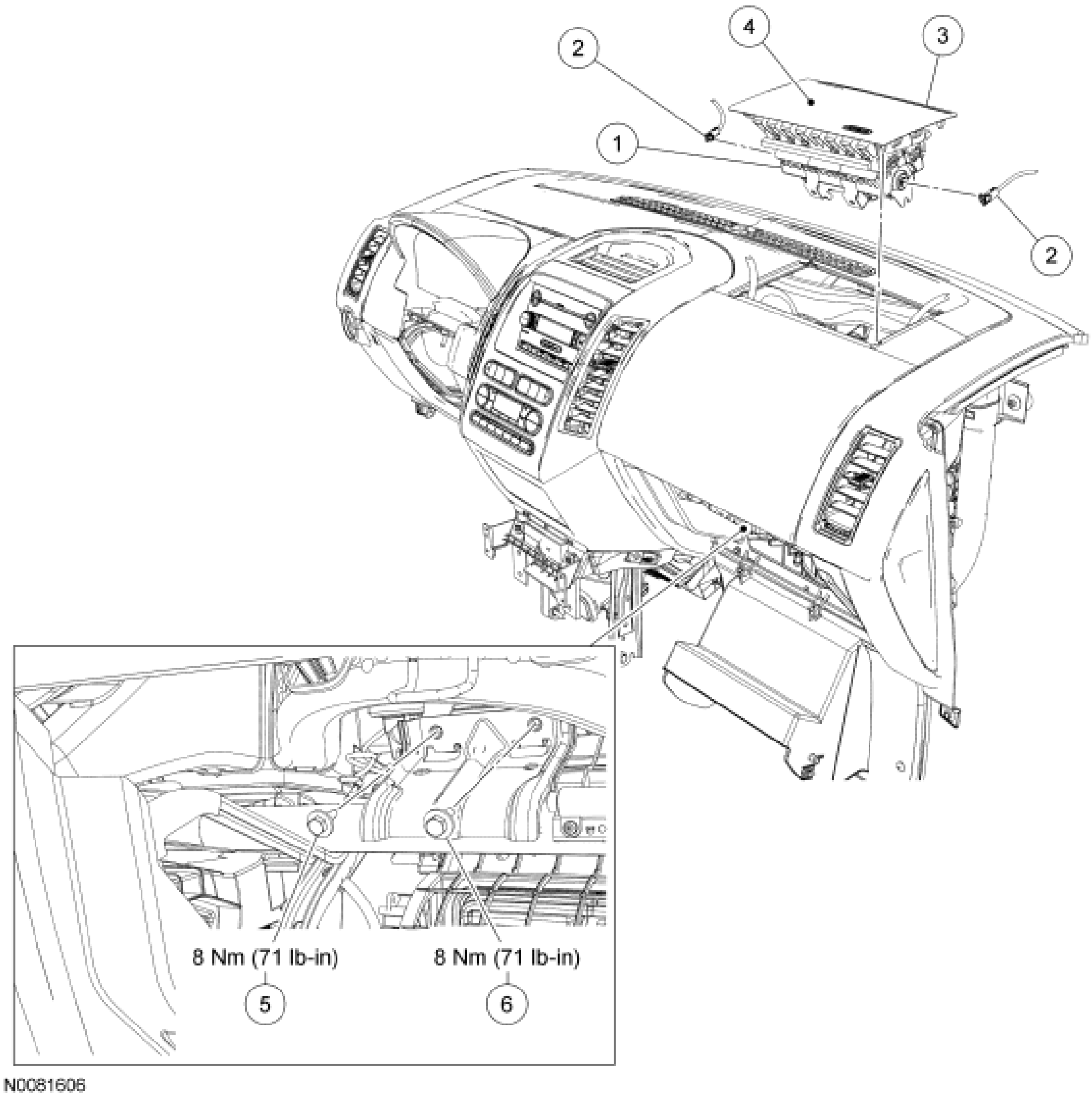


Fig. 220: Identifying Passenger Air Bag Module - Edge With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	044A74	Passenger air bag module assembly
2	-	Passenger air bag module electrical connectors (part of 14401)
3	-	Passenger air bag cover clips (part of 04338)
4	04338	Passenger air bag cover
5	W505422	Passenger air bag module inboard bolt
6	W505422	Passenger air bag module outboard bolt

REMOVAL

WARNING: **Always carry or place a live air bag module with the air bag and deployment door/trim cover/tear seam pointed away from the body. Do not set a live air bag module down with the deployment door/trim cover/tear seam face down. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

NOTE: Do no install a new passenger air bag module assembly due to a damaged or discolored passenger air bag cover. The passenger air bag cover can be serviced separately from the passenger air bag module assembly. For additional information, refer to Air Bag Trim Cover - Passenger, Edge.

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Open and lower the glove compartment door.
   - If equipped, detach the glove compartment door dampener.
3. Remove the 2 passenger air bag module bolts from the cross-car beam bracket.

   NOTE: Do not handle the passenger air bag module by grabbing the edges of the deployment doors. Damage to the air bag module may occur.

4. Through the glove compartment opening, release the LH side passenger air bag cover clips. Then release the foreward-most passenger air bag cover clips and separate the passenger air bag cover and passenger air bag module from the instrument panel.
5. Using a small screwdriver as shown in illustration, lift up and release the locking button on the RH passenger air bag module electrical connector.

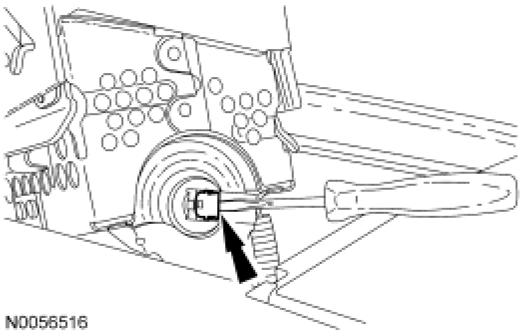


Fig. 221: Releasing Locking Button On RH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE: **Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.**

6. With the RH locking button released, remove the RH electrical connector.
7. Using a small screwdriver as shown in illustration, lift up and release the locking button on the LH passenger air bag module electrical connector.

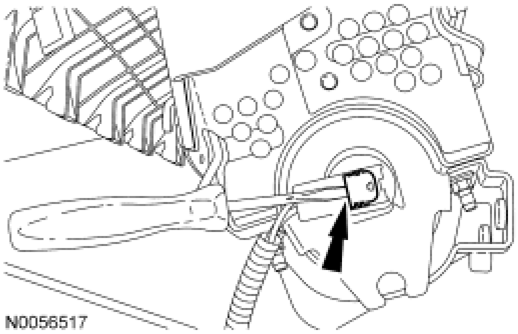


Fig. 222: Releasing Locking Button On LH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE: **Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.**

8. With the LH locking button released, remove the LH electrical connector and remove the passenger air bag module.

INSTALLATION

NOTE: **Do not install the passenger air bag module electrical connectors by the locking buttons. Damage to the locking buttons may occur.**

NOTE: **The passenger air bag module electrical connector locking buttons must be in the released position when the connector is being installed or connector damage may occur.**

NOTE: **The passenger air bag module electrical connectors are unique and cannot be reversed when connected to the passenger air bag module. Match the electrical connector key to the keyway in the passenger air bag module. Do not force the electrical connectors into the passenger air bag module. Damage to the connector or component may occur.**

NOTE: **RH shown in illustration, LH similar.**

1. With the locking buttons released, install the 2 passenger air bag module electrical connectors fully into the passenger air bag module and seat the locking buttons.

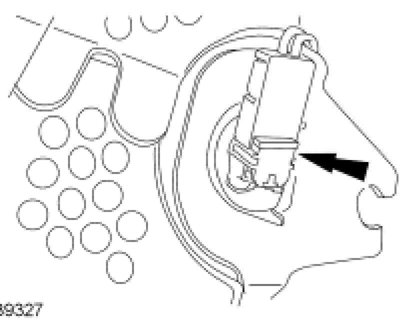


Fig. 223: Locating Passenger Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

NOTE: **Do not handle the passenger air bag module by grabbing the edges of the passenger air bag cover. Damage to the air bag module may occur.**

NOTE: **During passenger air bag module installation, make sure all the passenger air bag cover clips are fully seated in the instrument panel.**

2. Install the passenger air bag module passenger air bag cover rear-most clips into the instrument panel first. Then install the side passenger air bag cover clips working around to the forward-most clips seating each of them fully into the instrument panel.

NOTE: **To avoid a squeak or rattle condition, install the 2 passenger air bag module bolts in this order.**

3. Install the passenger air bag module bolts.
    1. Install the inboard passenger air bag module bolt.
        - Tighten to 8 Nm (71 lb-in).

2. Install the outboard passenger air bag module bolt.
    - Tighten to 8 Nm (71 lb-in).

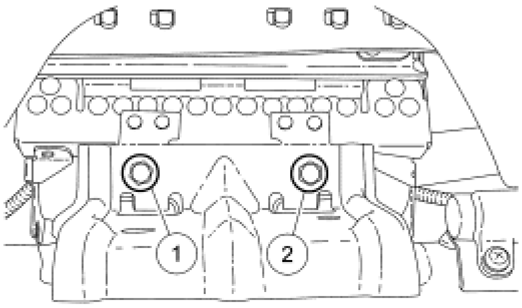


Fig. 224: Installing Passenger Air Bag Module Bolts In Sequence
Courtesy of FORD MOTOR CO.

4. Close the glove compartment door.
    - If equipped, attach the glove compartment door dampener.
5. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

PASSENGER AIR BAG MODULE - MKX

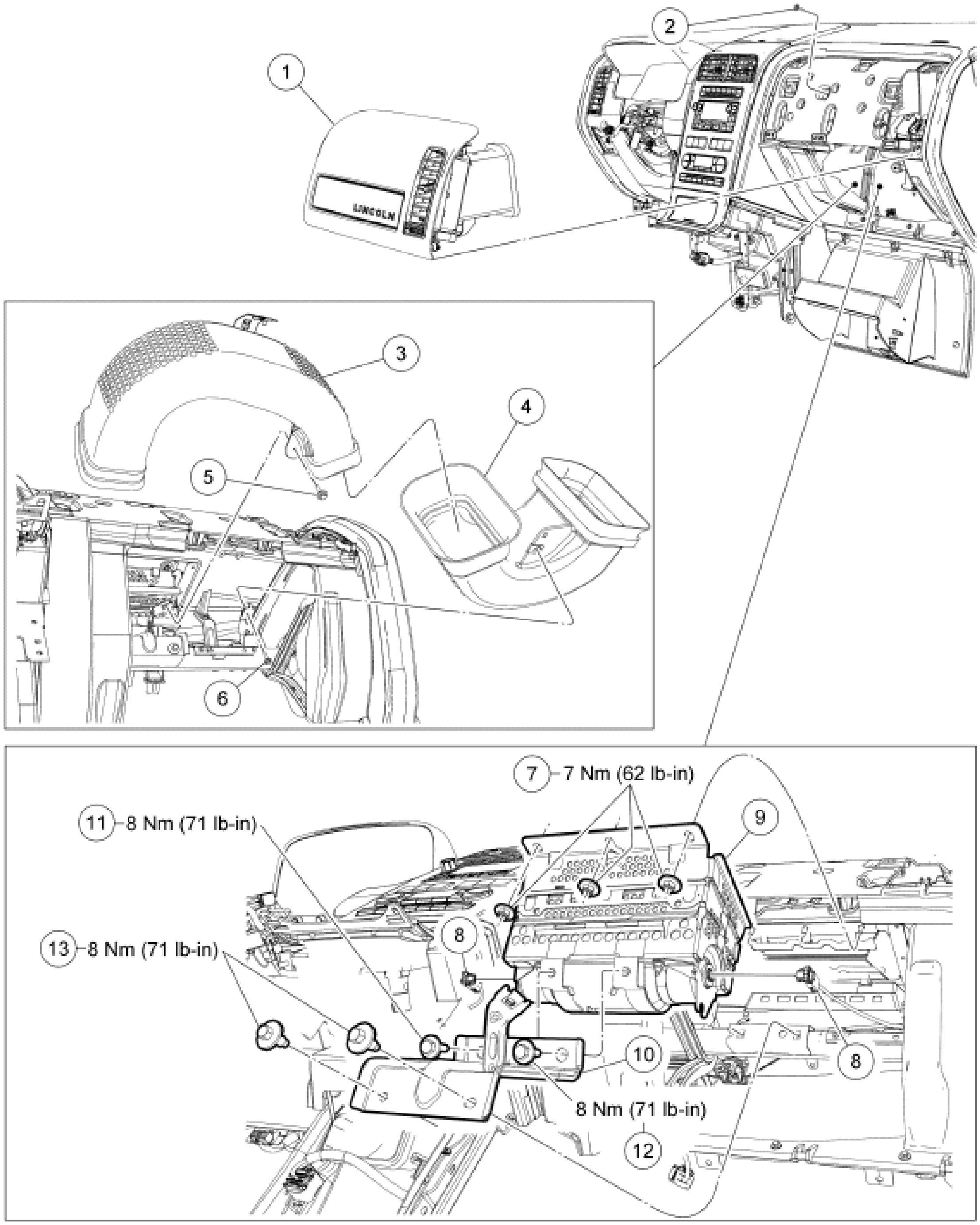


Fig. 225: Exploded View Of Passenger Air Bag Module With Torque Specifications - MKX
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	04338	RH instrument panel finish panel
2	W707628	LH duct assembly bracket screw

3	19B680	LH duct assembly
4	19B680	RH duct assembly
5	W707628	LH duct assembly screw
6	W707628	RH duct assembly screw
7	W701014	Passenger air bag module nuts
8	-	Passenger air bag module electrical connectors part of 14401)
9	044A74	Passenger air bag module
10	-	Passenger air bag module support bracket
11	W505422	Inboard passenger air bag module-to-support bracket bolt
12	W505422	Outboard passenger air bag module-to-support bracket bolt
13	W503922	Passenger air bag module bracket-to-cross vehicle beam bolts

REMOVAL

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Always carry or place a live air bag module with the air bag and deployment door/trim cover/tear seam pointed away from the body. Do not set a live air bag module down with the deployment door/trim cover/tear seam face down. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Open and lower the glove compartment door. Then unhook the glove compartment door from the instrument panel.
   - If equipped, detach the glove compartment door dampener.

3. Pull straight out to release the 5 retaining clips and remove the RH instrument panel finish panel.
4. Remove the screw from the LH duct assembly bracket from the front of the instrument panel.

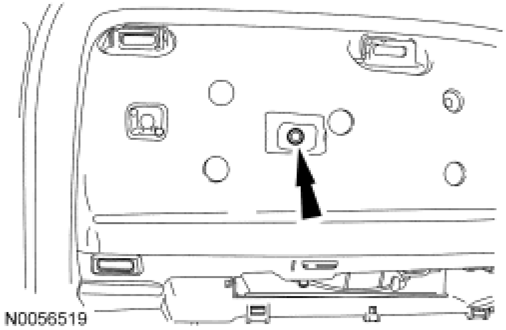


Fig. 226: Locating Screw On LH Duct Assembly Bracket
Courtesy of FORD MOTOR CO.

5. Through the glove compartment opening, remove the screw and the LH duct assembly.

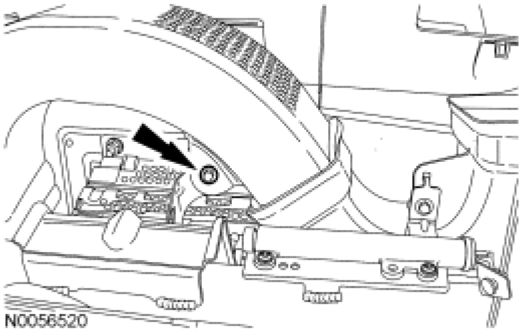


Fig. 227: Locating Screw On LH Duct Assembly
Courtesy of FORD MOTOR CO.

6. Through the glove compartment opening, remove the screw and the RH duct assembly.

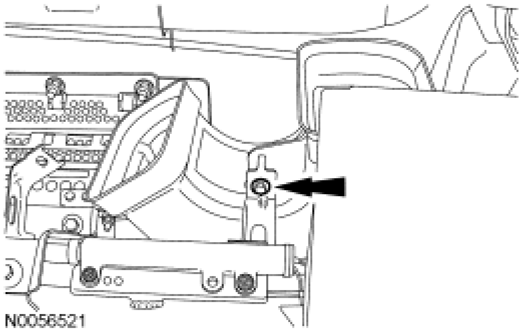


Fig. 228: Locating Screw On RH Duct Assembly
Courtesy of FORD MOTOR CO.

7. Remove the 4 bolts and the passenger air bag module support bracket.

8. Remove the 3 nuts from the passenger air bag module.

NOTE: Do not allow the passenger air bag module to hang from the wiring harness. Damage to the wiring harness and connectors may occur.

9. Lower the rear of the passenger air bag module off the 3 mounting studs. Move the passenger air bag module rearward to release the 7 hooks from the instrument panel and rest the module on the cross vehicle beam.

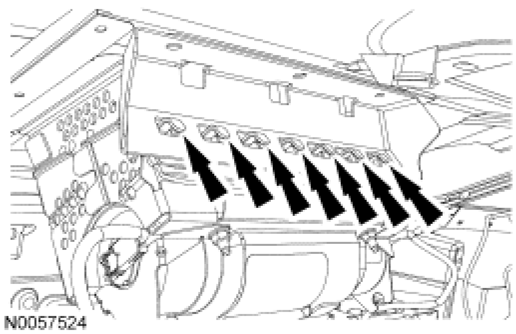


Fig. 229: Identifying Hooks On I/P
Courtesy of FORD MOTOR CO.

10. Using a small screwdriver as shown in illustration, lift up and release the locking button on the LH passenger air bag module electrical connector.

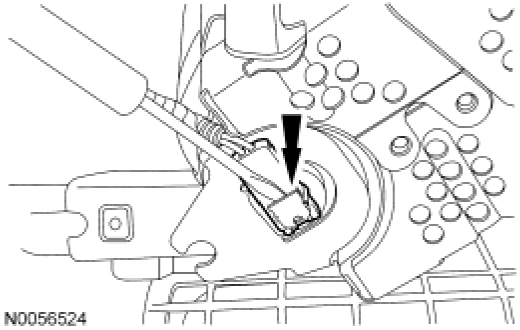


Fig. 230: Releasing Locking Button On LH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE: Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.

11. With the LH locking button released, remove the LH electrical connector from the passenger air bag module.
12. Using a small screwdriver as shown in illustration, lift up and release the locking button on the RH passenger air bag module electrical connector.

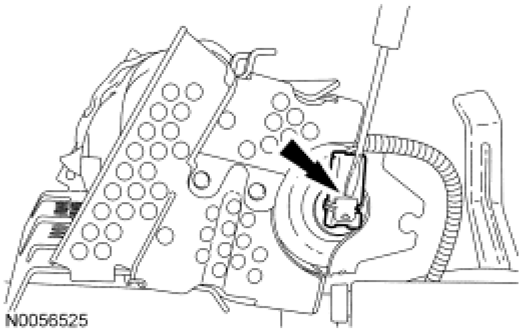


Fig. 231: Releasing Locking Button On RH Passenger Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE: Do not pull the passenger air bag module electrical connector out by the locking button. Damage to the locking button may occur.

13. With the RH locking button released, remove the RH electrical connector. Remove the passenger air bag module from the glove compartment opening.

INSTALLATION

NOTE: Do not install the passenger air bag module electrical connectors by the locking buttons. Damage to the locking buttons may occur.

NOTE: The passenger air bag module electrical connector locking buttons must be in the released position when the connector is being installed or connector damage may occur.

NOTE: The passenger air bag module electrical connectors are unique and cannot be reversed when connected to the passenger air bag module. Match the electrical connector key to the keyway in the passenger air bag module. Do not force the electrical connectors into the passenger air bag module. Damage to the connector or component may occur.

NOTE: RH shown in illustration, LH similar.

1. Position the passenger air bag module on the cross vehicle beam. With the locking buttons released, install the 2 passenger air bag module electrical connectors fully into the passenger air bag module and seat the locking buttons.

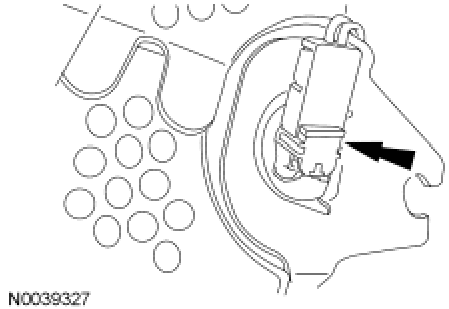


Fig. 232: Locating Passenger Air Bag Module Electrical Connectors
Courtesy of FORD MOTOR CO.

NOTE: During the passenger air bag module installation, make sure all 7 of the module hooks are fully seated in the instrument panel.

2. Install the passenger air bag module 7 hooks into the instrument panel. Raise the rear of the passenger air bag module onto the 3 mounting studs.

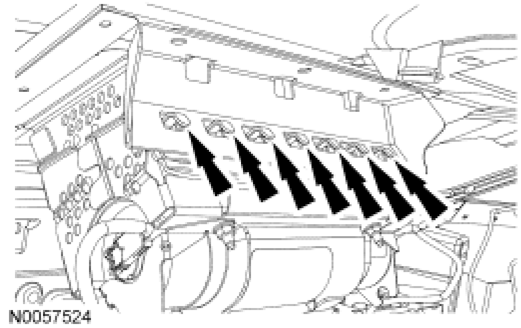


Fig. 233: Identifying Hooks On I/P
Courtesy of FORD MOTOR CO.

3. Install the 3 passenger air bag module nuts.
    - Tighten to 7 Nm (62 lb-in).

NOTE: To avoid a squeak or rattle condition, install the passenger air bag module bracket and passenger air bag module bolts in this order.

4. Install the passenger air bag module support bracket.
    1. Install the passenger air bag module support bracket and 2 bolts to the cross vehicle beam.
        - Tighten to 8 Nm (71 lb-in).
    2. Install the inboard passenger air bag module support bracket-to-passenger air bag module bolt.
        - Tighten to 8 Nm (71 lb-in).
    3. Install the outboard passenger air bag module support bracket-to-passenger air bag module bolt.
        - Tighten to 8 Nm (71 lb-in).

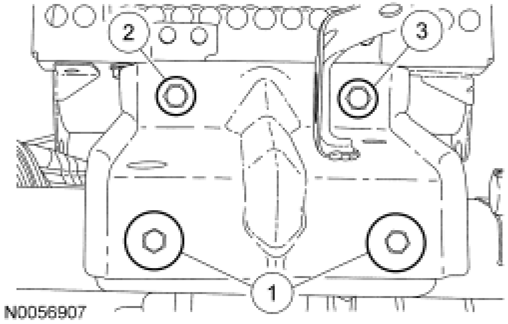


Fig. 234: Installing Passenger Air Bag Module Support Bracket Bolts In Sequence
Courtesy of FORD MOTOR CO.

5. Install the screw and the RH duct assembly.
6. Install the screw and the LH duct assembly.
7. Install the screw for the LH duct assembly bracket in the front of the instrument panel.

NOTE: Make sure the RH instrument panel finish panel 5 retaining clips are fully seated in the instrument panel.

8. Install the RH instrument panel finish panel.
9. Install and close the glove compartment door.
   - If equipped, attach the glove compartment door dampener.
10. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

RESTRAINTS CONTROL MODULE (RCM)

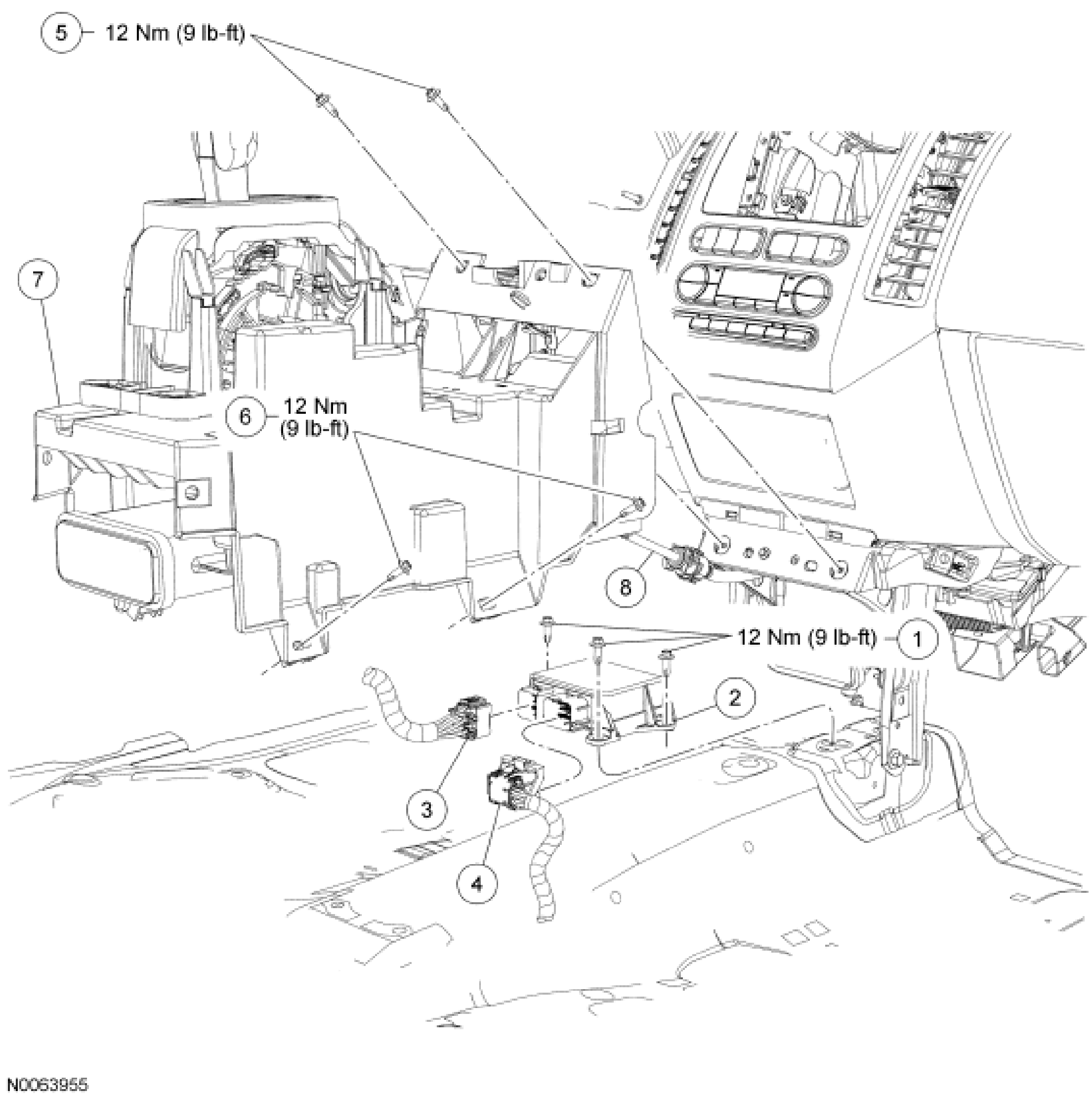


Fig. 235: Exploded View Of Restraints Control Module (RCM) With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W707935	Restraints control module (RCM) bolts
2	14B321	RCM
3	-	RCM small connector (part of 14401)
4	-	RCM large connector (part of 14A005)
5	W707935	Inner floor console support-to-instrument panel bolts (2 required)
6	W505157	Inner floor console support bolts (4 required)
7	45B34	Inner floor console support
8	7E395	Selector lever cable

REMOVAL

WARNING: If a vehicle has been in a crash, inspect the restraints control module (RCM) and the impact sensor (if equipped) mounting areas for deformation. If damaged, restore the mounting areas to the original production configuration. A new RCM and sensors must be installed whether or not the air bags have deployed. Failure to follow these instructions may result in serious personal injury or death in a crash.

NOTE: When installing a new restraints control module (RCM), it is necessary to carry out programmable module installation (PMI). For additional information, refer to MODULE CONFIGURATION article. Failure to follow this instruction may result in system failure.

NOTE: When installing a new restraints control module (RCM), always make sure the correct RCM is being installed. If an incorrect RCM is installed, erroneous DTCs will result.

NOTE: The air bag warning indicator illuminates when the correct RCM fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. If installing a new RCM, carry out the steps necessary to prepare for programmable module installation (PMI). For additional information, refer to **MODULE CONFIGURATION** article.
2. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
3. Remove the floor console. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
4. Disconnect the selector lever cable from the ball stud located on the shifter and remove the cable from the retaining clips.

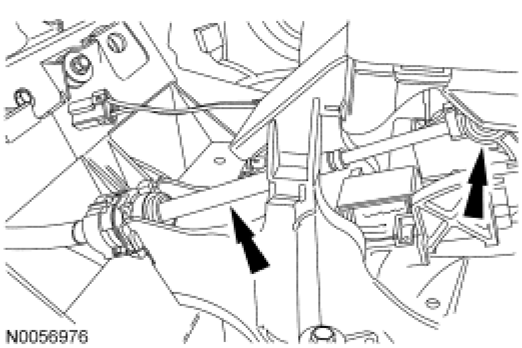


Fig. 236: Locating Shift Cable From Ball Stud On Shifter
Courtesy of FORD MOTOR CO.

5. Remove the 4 bolts (2 each side) from the inner floor console support.
6. Remove the bolts from the inner floor console support-to-instrument panel.
7. Disconnect the electrical connectors from the front inner floor console support.
8. Remove the inner floor console support.
9. Press to release the locking tab and disconnect the small RCM electrical connector.

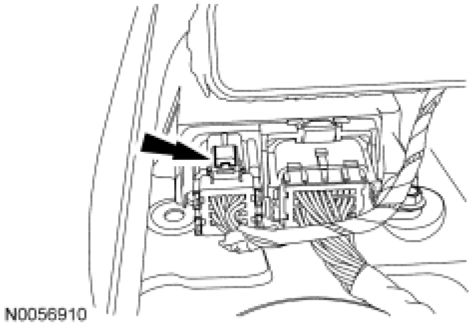


Fig. 237: Locating Locking Tab On Small RCM Electrical Connector
Courtesy of FORD MOTOR CO.

10. Disconnect the large RCM electrical connector.
    1. Pinch the thumb tab and pivot the connector position assurance lever all the way back until it stops.
    2. Pull out and disconnect the RCM electrical connector.

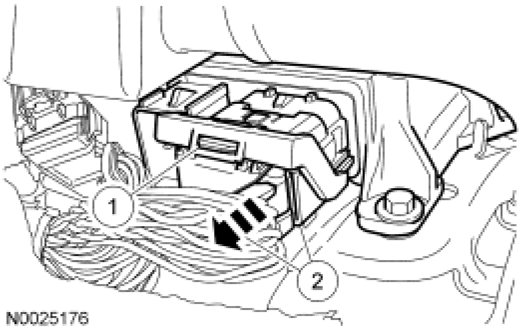


Fig. 238: Disconnecting RCM Electrical Connector
Courtesy of FORD MOTOR CO.

11. Remove the 3 bolts and RCM.

INSTALLATION

WARNING: **Always tighten the fasteners of the restraints control module (RCM) and impact sensor (if equipped) to the specified torque. Failure to do so may result in incorrect restraint system operation, which increases the risk of personal injury or death in a crash.**

1. Install the RCM and the 3 bolts.
   - Tighten to 12 Nm (9 lb-ft).
2. Make sure the large RCM connector position assurance lever is in the full release position before attempting to connect the connector.

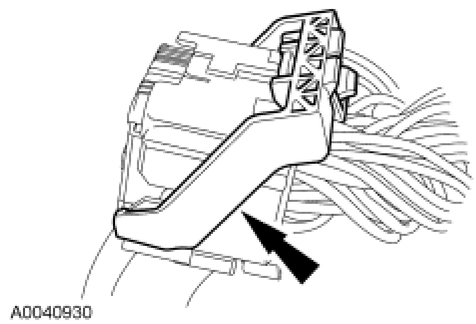


Fig. 239: View Of Connector Position Assurance Lever
Courtesy of FORD MOTOR CO.

NOTE:	**Placing the large restraints control module (RCM) electrical wiring connector into the RCM on an angle, may cause bad electrical connections and damaged components.**

3. Position the large RCM electrical connector into the RCM.

NOTE:	**Do not push the connector to the point where the lever pivots and seats itself. Light pressure is needed to get the connector into position on the restraints control module (RCM), before using the lever to fully seat the connector. Failure to follow these instructions may cause component or connector damage.**

   - With the large RCM electrical connector uniformly aligned to the RCM, lightly push in until a subtle audible click is heard and slight resistance is felt.

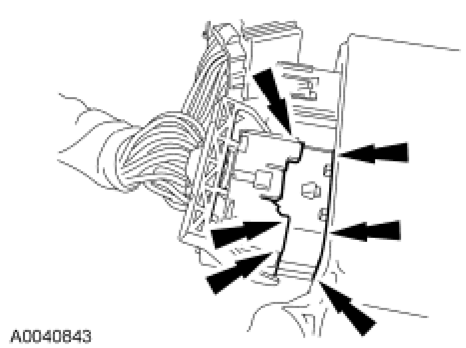


Fig. 240: Large RCM Electrical Wiring Connector
Courtesy of FORD MOTOR CO.

4. Connect the large RCM electrical connector.
    - Using the connector position assurance lever, pivot it toward the RCM, drawing the connector into the RCM.
        - Make sure the thumb tab is engaged to the retainer on the RCM and locked in place.

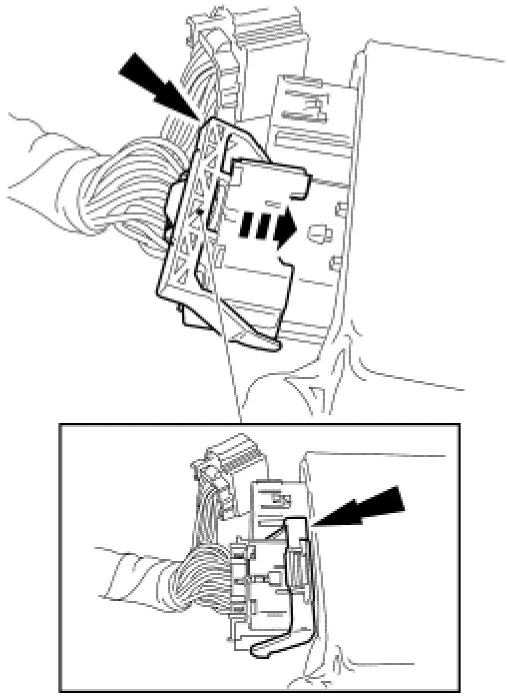

A0040931

Fig. 241: Locating RCM Electrical Connector
Courtesy of FORD MOTOR CO.

5. Connect the small RCM electrical connector.
6. Install the inner floor console support.
7. Install the 4 bolts (2 each side) to the inner floor console support.
    - Tighten to 12 Nm (9 lb-ft).
8. Install the screws to the inner floor console support bracket-to-floor instrument panel.
    - Tighten to 12 Nm (9 lb-ft).
9. Install the selector lever cable in the selector lever housing.

NOTE: **When installing the selector lever cable, make sure that the selector lever cable locking tabs are locked in place and the cable end is snapped onto the ball stud. Press the selector lever cable into the bracket and listen for the cable to click in place. Pull back on the selector lever cable to make sure that it is locked into the bracket. Also make sure that the selector lever cable end is correctly installed onto the ball stud. Pull back on the selector lever cable end to make sure that the cable end is correctly installed.**

10. Connect the selector lever cable end to the floor shifter.
11. Attach the electrical connectors to the front inner floor console support bracket.
12. Install the floor console. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
13. Repower the SRS. **Do not prove out the SRS at this time.** For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
14. When installing a new RCM, carry out the steps necessary to complete PMI. For additional information, refer to **MODULE CONFIGURATION** article.
15. Prove out the SRS as follows:

    Turn the ignition switch from ON to OFF. Wait 10 seconds, then turn the ignition switch back to ON and visually monitor the air bag indicator with the air bag modules installed. The air bag warning indicator will light continuously for approximately 6 seconds and then turn OFF. If an air bag SRS fault is present, the air bag indicator will:

    - fail to light.

    - remain lit continuously.

    - flash at a 5 Hz rate (RCM not configured).

    The air bag warning indicator might not light until approximately 30 seconds after the ignition switch has been turned from the OFF to the ON position. This is the time required for the RCM to complete the testing of the SRS. If the air bag warning indicator is inoperative and a SRS fault exists, a chime will sound in a pattern of 5 sets of 5 beeps. If this occurs, the air bag warning indicator and any SRS fault discovered must be diagnosed and repaired.

    Clear all continuous DTCs from the RCM and occupant classification system module (OCSM) using a scan tool.

SAFETY CANOPY MODULE

NOTE: **RH shown in illustration, LH similar.**

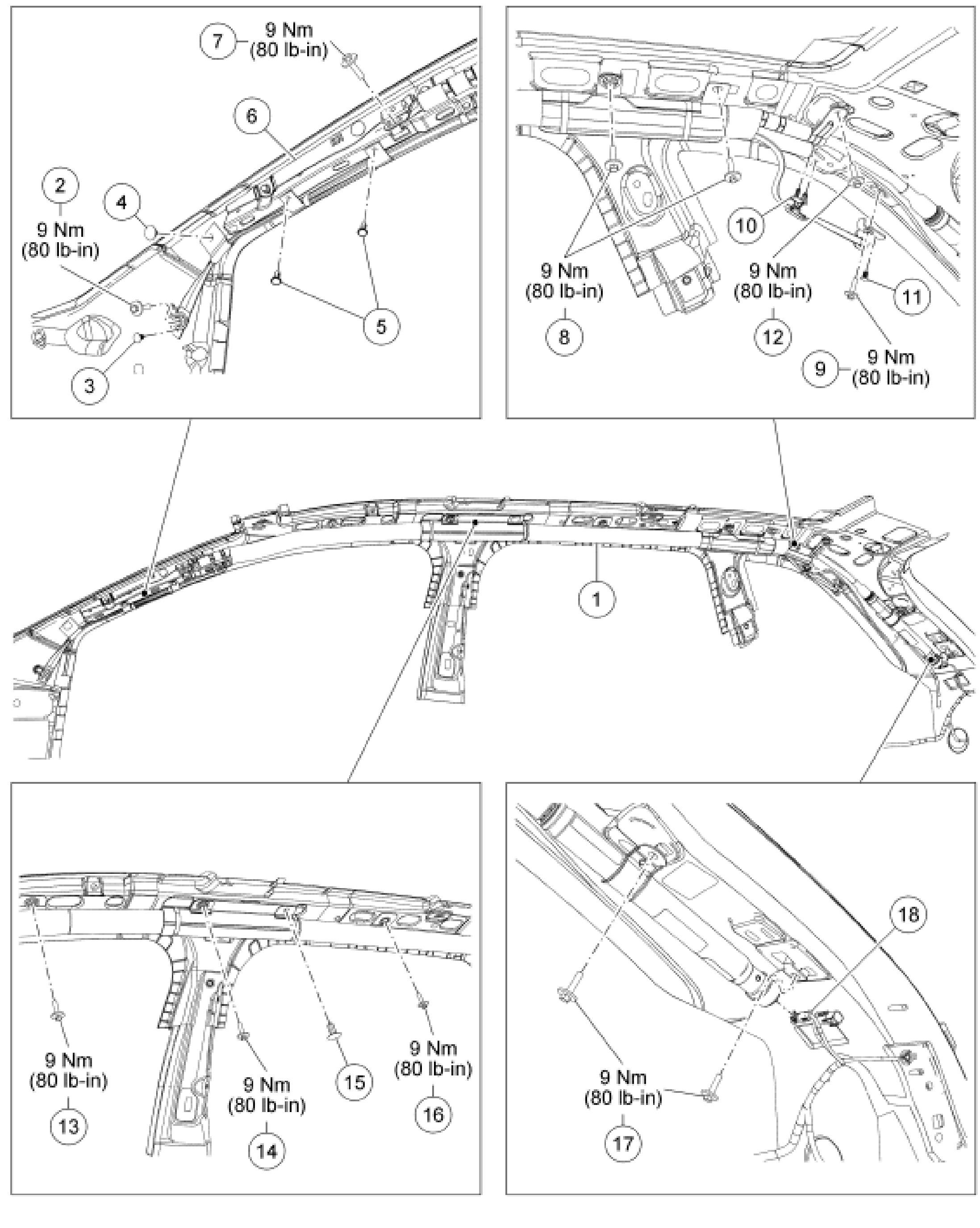


Fig. 242: Identifying Safety Canopy Module With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	042D94 RH/042D95 LH	Safety canopy module
2	W713437	Safety canopy module front tether cord anchor bolt

3	-	Safety canopy module front tether cord anchor pin-type retainer (part of 042D94)
4	-	Safety canopy module front tether cord A-pillar routing pin-type retainer (part of 042D94)
5	-	Safety canopy module front tether cord A-pillar trim bracket routing pin-type retainers (2 required) (part of 042D94)
6	-	A-pillar trim bracket
7	W713437	Safety canopy module A-pillar trim bracket bolt
8	W713437	Safety canopy module C-pillar bolts (2 required)
9	W713437	Safety canopy module rear tether cord anchor bracket bolt
10	-	Safety canopy module rear tether cord routing clip pin-type retainer (part of 042D94)
11	-	Safety canopy module rear tether cord anchor bracket pin-type retainer (part of 042D94)
12	W713437	Safety canopy module D-pillar bolt
13	W713437	Safety canopy module front door opening bolt
14	W713437	Safety canopy module B-pillar bolt
15	-	Safety canopy module B-pillar pin-type retainer (part of 042D94)
16	W713437	Safety canopy module rear door opening bolt
17	W713437	Safety canopy module rear bolts (2 required)
18	-	Safety canopy module electrical connector (part of 14A005)

REMOVAL

WARNING: **Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.**

WARNING: **Always carry or place a live safety canopy, or side air curtain module, with the module and tear seam pointed away from your body. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

WARNING: **Anytime the safety canopy or side air curtain module has deployed, a new headliner and new A-, B-, C- and D-pillar upper trim panels and attaching hardware must be installed. Remove any other damaged components and hardware and install new components and hardware as needed. Failure to follow these instructions may result in the safety canopy or side air curtain module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

WARNING: **When installing a new headliner on a vehicle equipped with safety canopy or side air curtain modules, make sure the headliner has the word AIRBAG on the headliner where it meets each B-pillar trim panel. Otherwise, you have the wrong headliner. Failure to follow this instruction may result in the safety canopy or side air curtain module not deploying or deploying incorrectly, increasing the risk of personal injury or death in a crash.**

WARNING: **Before installing a safety canopy or side air curtain module, inspect the roofline for any damage. If necessary, the sheet metal must be reworked to its original condition and structural integrity. Install new fasteners if damaged and remove foreign material. Failure to follow these instructions may result in the safety canopy or side air curtain deploying incorrectly, which increases the risk of serious personal injury or death in a crash.**

WARNING: **Inspect a safety canopy or side air curtain module before installation. If the module is damaged, the cover has separated or the safety canopy/side air curtain material has been exposed, install a new module. Do not attempt to repair the module. Failure to follow these instructions may result in the safety canopy or side air curtain deploying incorrectly, which increases the risk of serious personal injury or death in a crash.**

WARNING: **Do not obstruct or place objects in the deployment path of the safety canopy or side air curtain module. Failure to follow this instruction may result in the safety canopy or side air curtain module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

WARNING: **Never put any type of fastener or tie strap around any part of a safety canopy module, side air curtain module or interior trim panel. This will prevent the safety canopy or side air curtain module from deploying correctly. Failure to follow this instruction may increase the risk of serious personal injury or death in a crash.**

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of**

serious personal injury or death.

WARNING: **Always use new torque-prevailing type J-nuts when installing the safety canopy in the vehicle. Use of these J-nuts is mandatory to reduce the risk of loss of fastener effectiveness. Failure to follow this instruction may result in the safety canopy module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

NOTE: The air bag warning indictor illuminates when the restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraints system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

NOTE: Repair is made by installing a new part only. If the new part does not correct the condition, install the original part and carry out the diagnostic procedure again.

1. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
2. Remove the headliner. For additional information, refer to **INTERIOR TRIM & ORNAMENTATION** article.

   NOTE: Do not pull the safety canopy module electrical connector out by the locking button. Damage to the locking button can occur.

3. Using a small screwdriver as shown in illustration, lift up and release the locking button on the safety canopy module electrical connector. With the locking button released, remove the electrical connector.

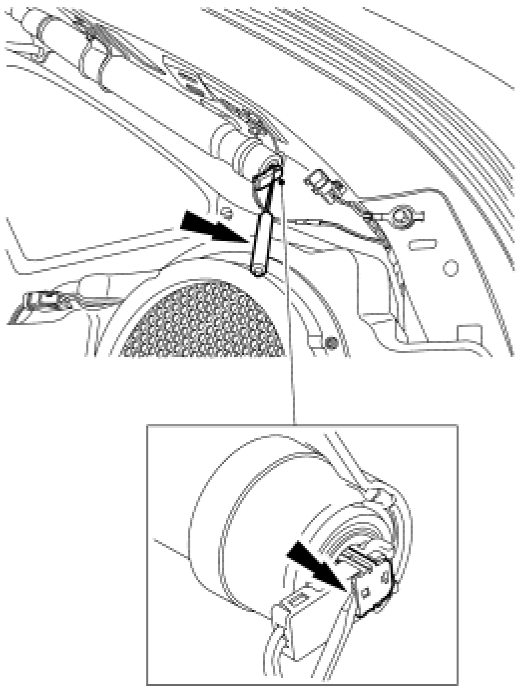


Fig. 243: Releasing Locking Button On Safety Canopy Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

NOTE:	**Note the pin-type retainer on the safety canopy module front tether cord to the A-pillar sheet metal for installation.**

4. Remove the anchor bolt, then detach the pin-type retainer and the front tether cord anchor from the A-pillar.
5. Remove the safety canopy module front tether cord routing pin-type retainer from the A-pillar sheet metal.
6. Detach the 2 safety canopy module front tether cord A-pillar trim bracket routing pin-type retainers from the A-pillar trim bracket.
7. Remove the safety canopy module A-pillar trim bracket bolt.
8. Remove the safety canopy module front door opening bolt.

NOTE:	**Note the position and the routing of the safety canopy module rear tether cord and routing clip pin-type retainer for installation.**

9. Detach the safety canopy module rear tether cord routing clip pin-type retainer from the upper D-pillar sheet metal.

NOTE:	**Note the position of the safety canopy module rear tether anchor bracket hook and pin-type retainer in the D-pillar sheet metal.**

10. Remove the bolt, then detach the pin-type retainer and remove the safety canopy module rear tether anchor bracket from the D-pillar sheet metal.
11. Remove the safety canopy module rear door opening bolt.
12. Remove the 2 safety canopy module bolts near the C-pillar.
13. Remove the safety canopy module D-pillar bolt.
14. Remove the 2 safety canopy module rear bolts.
15. Remove the safety canopy module bolt near the B-pillar.
16. Remove the safety canopy module B-pillar pin-type retainer.
17. Move the safety canopy module forward to release the rear bracket hook from the sheet metal and remove the safety canopy module.

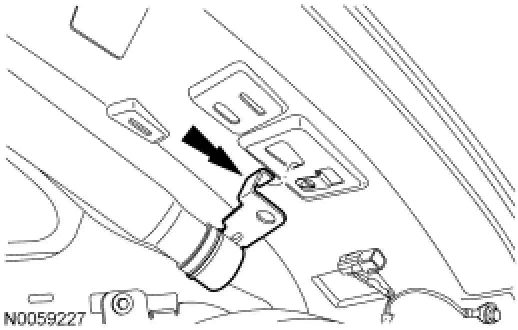


Fig. 244: Moving Safety Canopy Module Forward To Release Rear Bracket Hook From Sheet Metal
Courtesy of FORD MOTOR CO.

INSTALLATION

WARNING: **Before installing a safety canopy or side air curtain module, inspect the roofline for any damage. If necessary, the sheet metal must be reworked to its original condition and structural integrity. Install new fasteners if damaged and remove foreign material. Failure to follow these instructions may result in the safety canopy or side air curtain deploying incorrectly, which increases the risk of serious personal injury or death in a crash.**

WARNING: **Inspect a safety canopy or side air curtain module before installation. If the module is damaged, the cover has separated or the safety canopy/side air curtain material has been exposed, install a new module. Do not attempt to repair the module. Failure to follow these instructions may result in the safety canopy or side air curtain deploying incorrectly, which increases the risk of serious personal injury or death in a crash.**

WARNING: **Never put any type of fastener or tie strap around any part of a safety canopy module, side air curtain module or interior trim panel. This will prevent the safety canopy or side air curtain module from deploying**

correctly. Failure to follow this instruction may increase the risk of serious personal injury or death in a crash.

WARNING: **Always use new torque-prevailing type J-nuts when installing the safety canopy in the vehicle. Use of these J-nuts is mandatory to reduce the risk of loss of fastener effectiveness. Failure to follow this instruction may result in the safety canopy module deploying incorrectly and increases the risk of serious personal injury or death in a crash.**

1. Discard the 2 front J-nuts and install new.

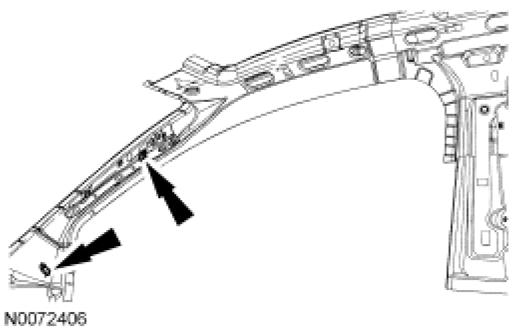


Fig. 245: Identifying 2 Front J-nuts
Courtesy of FORD MOTOR CO.

2. Discard the 2 rear J-nuts and install new.

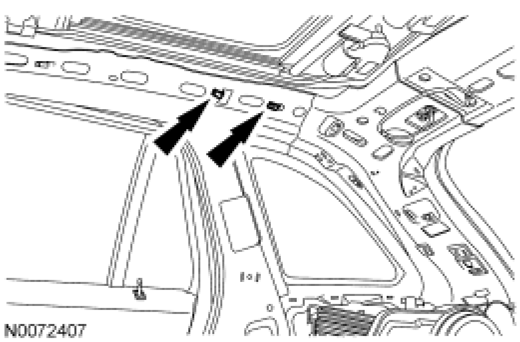


Fig. 246: Identifying 2 Rear J-nuts
Courtesy of FORD MOTOR CO.

3. Position the safety canopy module by moving it rearward, installing the rear bracket hook in the sheet metal.

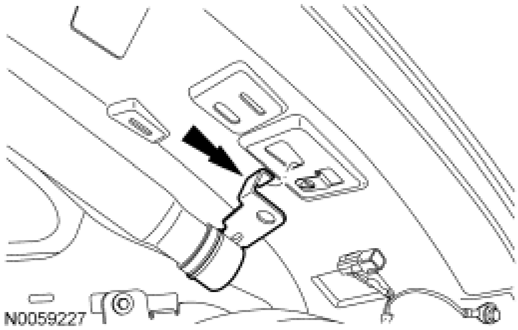


Fig. 247: Moving Safety Canopy Module Forward To Release Rear Bracket Hook From Sheet Metal
Courtesy of FORD MOTOR CO.

4. Install the safety canopy module B-pillar pin-type retainer.
5. Install the front tether cord anchor pin-type retainer.
6. Install the safety canopy module front tether cord routing pin-type retainer to the A-pillar sheet metal.
7. Attach the 2 safety canopy module front tether cord A-pillar trim bracket routing pin-type retainers to the A-pillar trim bracket.
8. Install the safety canopy module rear tether anchor bracket to the D-pillar sheet metal and attach the pin-type retainer.
9. Attach the safety canopy module rear tether cord routing clip pin-type retainer to the upper D-pillar sheet metal.

NOTE: Make sure that all safety canopy module bolt holes line up with all J-nuts.

10. Install the safety canopy module D-pillar bolt.
    - Tighten to 9 Nm (80 lb-in).
11. Install the safety canopy module A-pillar trim bracket bolt.
    - Tighten to 9 Nm (80 lb-in).
12. Install the safety canopy module bolt near the B-pillar.
    - Tighten to 9 Nm (80 lb-in).
13. Install the 2 safety canopy module rear bolts.
    - Tighten to 9 Nm (80 lb-in).
14. Install the 2 safety canopy module bolts near the C-pillar.
    - Tighten to 9 Nm (80 lb-in).
15. Install the safety canopy module rear door opening bolt.
    - Tighten to 9 Nm (80 lb-in).
16. Install the safety canopy module rear tether cord anchor bracket bolt.
    - Tighten to 9 Nm (80 lb-in).
17. Install the safety canopy module front door opening bolt.
    - Tighten to 9 Nm (80 lb-in).

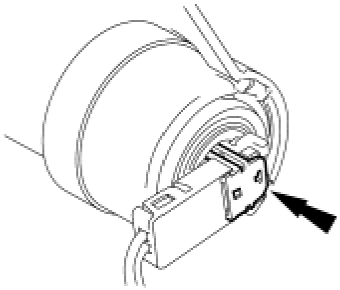
18. Install the safety canopy module front tether cord anchor bolt.
    - Tighten to 9 Nm (80 lb-in).

NOTE: Do not install the safety canopy module electrical connector by the locking button. Damage to the locking button may occur.

NOTE: The safety canopy module electrical connector locking button must be in the released position when the connector is being installed or connector damage may occur.

NOTE: The safety canopy module electrical connector is unique and cannot be reversed when connected to the safety canopy module. Match the electrical connector key to the keyway in the safety canopy module. Do not force the electrical connector into the safety canopy module. Damage to the connector or component may occur.

19. With the locking button released, install the safety canopy module electrical connector fully into the safety canopy module and seat the locking button.



N0056531

Fig. 248: Identifying Passenger Safety Canopy Module Electrical Connector
Courtesy of FORD MOTOR CO.

20. Install the headliner. For additional information, refer to **INTERIOR TRIM & ORNAMENTATION** article.
21. Repower the SRS For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

SEAT POSITION SENSOR

NOTE: Power seat shown in illustration, manual similar.

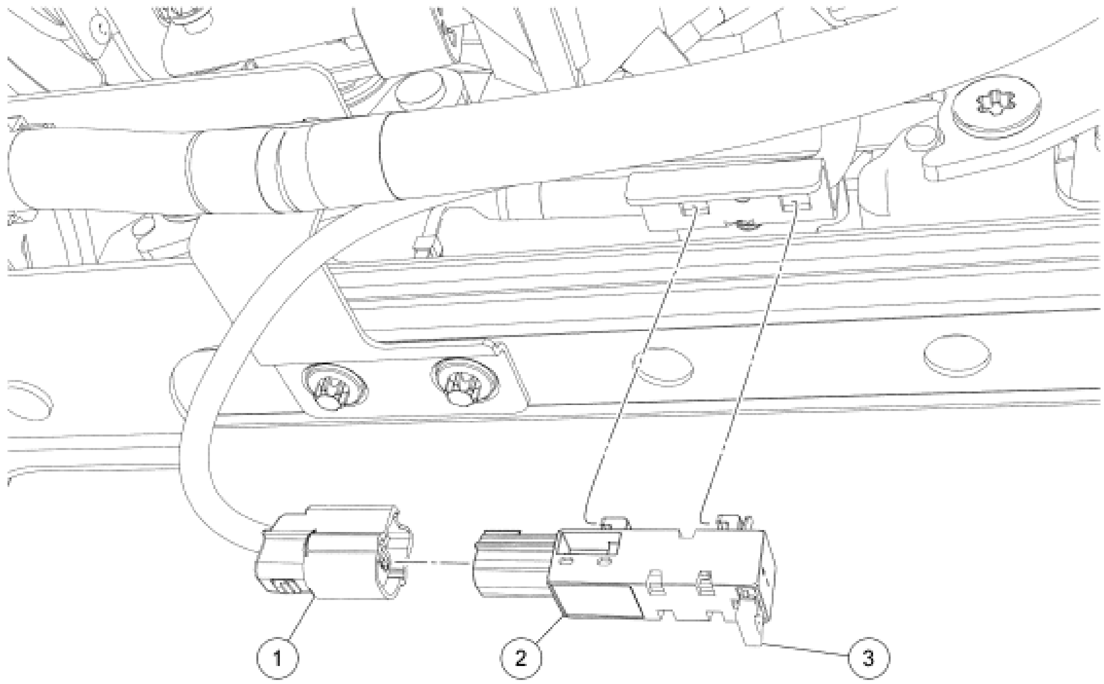


Fig. 249: Exploded View Of Seat Position Sensor
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Seat position sensor electrical connector (part of 14A699)
2	14B416	Seat position sensor
3	-	Seat position sensor retaining clip (part of 14B416)

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

All seats

1. Remove the driver seat. For additional information, refer to **SEATING** article.
2. Disconnect the seat position sensor electrical connector.

Power seats

3. Detach the wire harness routing pin-type retainer from the seat position sensor bracket.

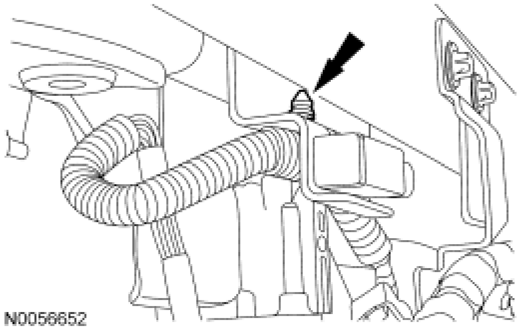


Fig. 250: Locating Wire Harness Routing Pin-Type Retainer From Seat Position Sensor Bracket
Courtesy of FORD MOTOR CO.

All seats

4. Push apart then inward to release the seat position sensor locking clip.

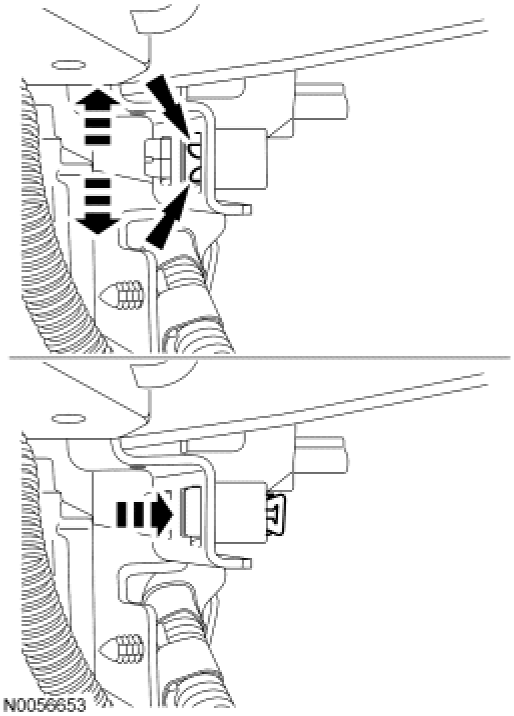


Fig. 251: Pushing Apart Then Inward To Release Seat Position Sensor Locking Clip
Courtesy of FORD MOTOR CO.

5. Slide the seat position sensor rearward to release the 2 hooks from the bracket. Then remove the seat position sensor.
6. To install, reverse the removal procedure.
7. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

SIDE AIR BAG MODULE

NOTE: Passenger shown in illustration, driver similar.

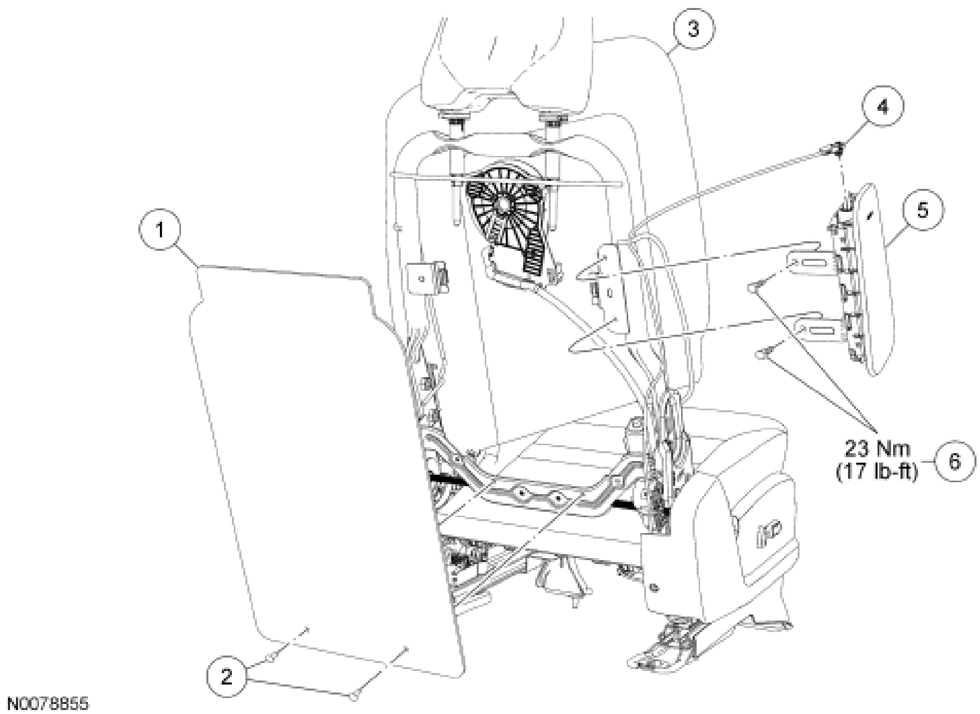


Fig. 252: Identifying Side Air Bag Module With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	60508	Seat backrest trim cover rear panel reinforcement board
2	-	Seat backrest trim cover rear panel reinforcement board pin-type retainers
3	64416/64810	Seat backrest trim cover and foam pad
4	-	Side air bag module electrical connector (part of 14K155)
5	611D10 RH/611D11 LH	Side air bag module
6	W506021	Side air bag module bolts (2 required)

REMOVAL

WARNING: Always wear eye protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.

WARNING: Always carry or place a live air bag module with the air bag and

deployment door/trim cover/tear seam pointed away from the body. Do not set a live air bag module down with the deployment door/trim cover/tear seam face down. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.

WARNING: **Never probe the electrical connectors on air bag, safety canopy or side air curtain modules. Failure to follow this instruction may result in the accidental deployment of these modules, which increases the risk of serious personal injury or death.**

WARNING: **If the seat side air bag cover has been damaged or separated from its mounting, or if the air bag material has been exposed, install a new seat side air bag module. Never try to repair the seat side air bag module. Failure to follow these instructions may result in the seat side air bag deploying incorrectly, which increases the risk of serious personal injury or death in a crash.**

NOTE: If a side air bag deployment took place, a new side air bag module and bolts must be installed. The side air bag module bracket on the seat back frame should be inspected and installed new if necessary.

NOTE: When installing a new side air bag after deployment, refer to SEATING article for additional information.

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

NOTE: Repair is made by installing a new part only. If the new part does not correct the condition, install the original part and carry out the diagnostic procedure again.

1. Position the affected seat and backrest fully up and forward.
2. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
3. Release the seat backrest trim cover rear panel lower J-clip. Then, release the 2 seat backrest trim cover rear panel side J-clips.

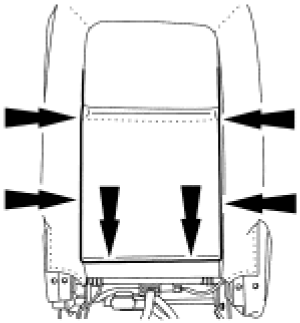


Fig. 253: Identifying Backrest Trim Cover J-Clips
Courtesy of FORD MOTOR CO.

NOTE: **The reinforcement board and seat backrest trim cover rear panel can be positioned aside with the reinforcement board still inside the seat backrest trim cover rear panel.**

4. Detach the 2 pin-type retainers and position aside the seat backrest trim cover rear panel and reinforcement board.
5. Through the rear of the seat remove and discard the 2 side air bag module bolts. Then, separate the side air bag module from the seat backrest.

NOTE: **Do not pull the side air bag module electrical connector out by the locking button. Damage to the locking button may occur.**

6. Using a small screwdriver as shown in illustration, lift up and release the locking button on the side air bag module electrical connector. With the locking button released, remove the electrical connector and the side air bag module.

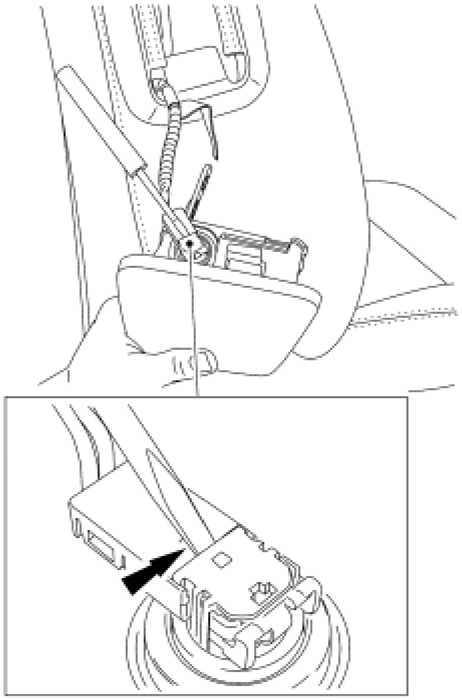


Fig. 254: Lifting Up & Releasing Locking Button On Side Air Bag Module Electrical Connector Using A Small Screwdriver
Courtesy of FORD MOTOR CO.

INSTALLATION

NOTE:	**Do not install the side air bag module electrical connector by the locking button. Damage to the locking button may occur.**
NOTE:	**The side air bag module electrical connector locking button must be in the released position when the connector is being installed or connector damage may occur.**
NOTE:	**The side air bag module electrical connector is unique and cannot be reversed when connected to the side air bag module. Match the electrical connector key to the keyway in the side air bag module. Do not force the electrical connector into the side air bag module. Damage to the connector or component may occur.**

1. With the locking button released, install the side air bag module electrical connector fully into the side air bag module and seat the locking button.

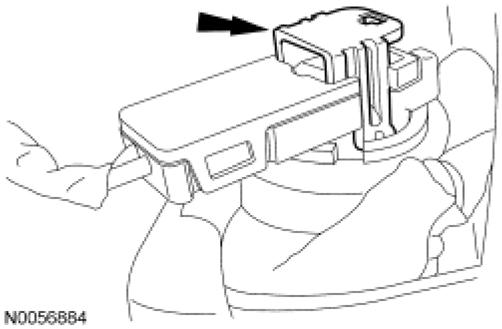


Fig. 255: Identifying Side Air Bag Module Electrical Connector
Courtesy of FORD MOTOR CO.

NOTE: Make sure the side air bag module wiring harness is not pinched between the side air bag module and the backrest frame. Damage to wire harness may occur.

NOTE: When installing the side air bag module, new bolts must be installed.

2. Position the side air bag module in the seat backrest and install the 2 new bolts.
   - Tighten to 23 Nm (17 lb-ft).
3. Position the seat backrest trim cover rear panel and attach the 2 pin-type retainers.
4. Attach the 2 seat backrest cover rear side J-clips. Then attach the seat backrest cover rear lower J-clip.
5. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

SIDE IMPACT SENSOR - B-PILLAR

NOTE: RH shown in illustration, LH similar.

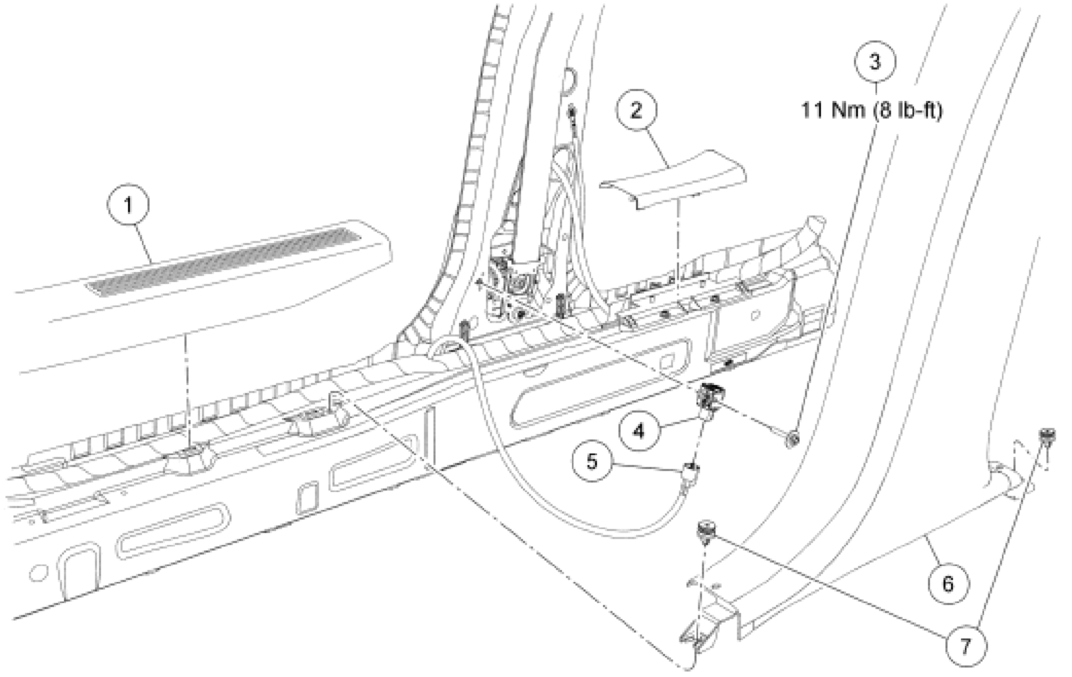


Fig. 256: Exploded View Of Side Impact Sensor - B-Pillar With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	13208	Front door scuff plate
2	13228	Rear door scuff plate
3	W712191	Side impact sensor retaining bolt
4	14B345	Side impact sensor
5	-	Side impact sensor electrical connector (part of 14A005)
6	24356	Lower B-pillar trim panel
7	-	Lower B-pillar trim panel retainers (part of 24356)

WARNING: **If a vehicle has been in a crash, inspect the restraints control module (RCM) and the impact sensor (if equipped) mounting areas for deformation. If damaged, restore the mounting areas to the original production configuration. A new RCM and sensors must be installed whether or not the air bags have deployed. Failure to follow these instructions may result in serious personal injury or death in a crash.**

NOTE: **The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.**

NOTE: The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

1. Position the front seat forward.
2. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
3. Remove the front door scuff plate.
   - Pull up to release the 5 retaining clips.
4. Remove the rear door scuff plate.
   - Pull up to release the 2 retaining clips.
5. Detach the 2 retainers and remove the lower B-pillar trim panel.

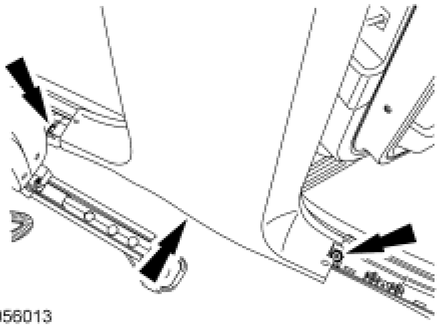


Fig. 257: Identifying Clips & Lower B-Pillar Trim Panel
Courtesy of FORD MOTOR CO.

6. Disconnect the side impact sensor electrical connector.

WARNING: **Always tighten the fasteners of the restraints control module (RCM) and impact sensor (if equipped) to the specified torque. Failure to do so may result in incorrect restraint system operation, which increases the risk of personal injury or death in a crash.**

NOTE: Make sure the B-pillar and side impact sensor mating surfaces are clean and free of foreign material.

NOTE: Note the position of the side impact sensor locating tabs for installation.

7. Remove the bolt and the side impact sensor.
   - To install, tighten to 11 Nm (8 lb-ft).
8. To install, reverse the removal procedure.
9. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.

SIDE IMPACT SENSOR - C-PILLAR

NOTE: RH shown in illustration, LH similar.

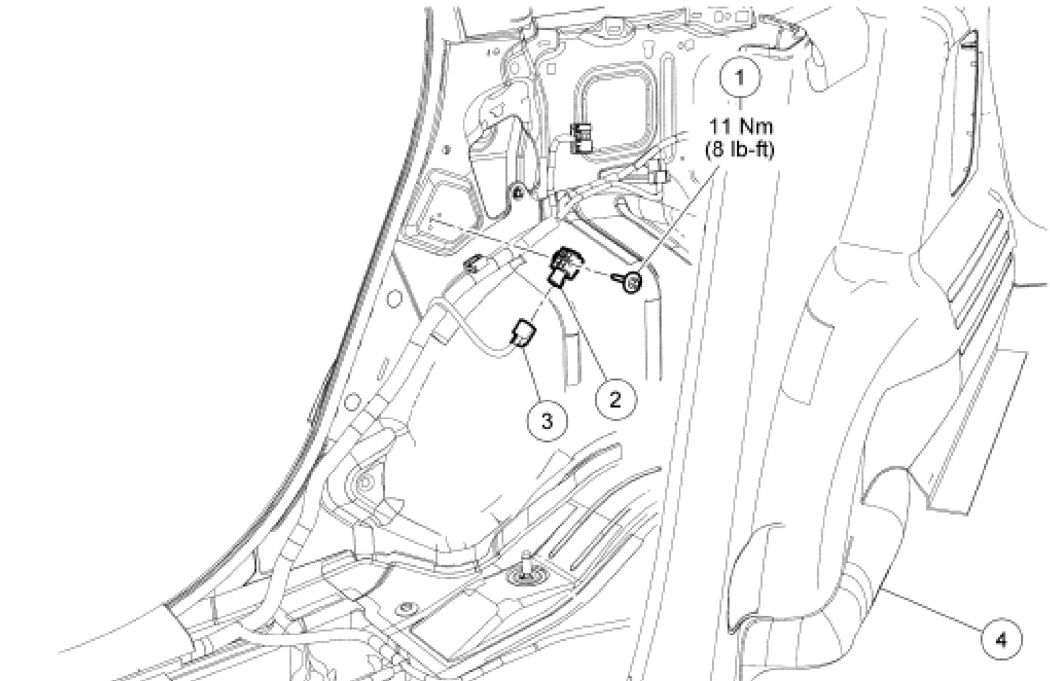


Fig. 258: Exploded View Of Side Impact Sensor - C-Pillar With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W712191	Side impact sensor bolt
2	14B345	Side impact sensor
3	-	Side impact sensor electrical connector (part of 14A005)
4	31012	Quarter trim panel

WARNING: **If a vehicle has been in a crash, inspect the restraints control module (RCM) and the impact sensor (if equipped) mounting areas for deformation. If damaged, restore the mounting areas to the original production configuration. A new RCM and sensors must be installed whether or not the air bags have deployed. Failure to follow these instructions may result in serious personal injury or death in a crash.**

NOTE: The air bag warning indicator illuminates when the correct restraints control module (RCM) fuse is removed and the ignition switch is ON.

NOTE: **The supplemental restraint system (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.**

1. Fold down the rear seat backrest.
2. Depower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.
3. Position aside the quarter trim panel.
   - Pull out on the quarter trim panel at the C-pillar to release the retaining clip to access the C-pillar side impact sensor.

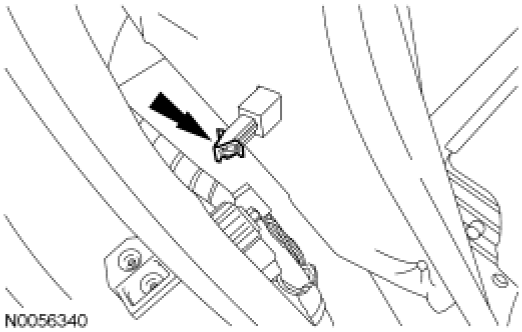


Fig. 259: Pulling Out On Quarter Trim Panel At C-Pillar
Courtesy of FORD MOTOR CO.

4. Disconnect the side impact sensor electrical connector.

NOTE: **Note the position of the 2 side impact sensor locating tabs for installation.**

NOTE: **Make sure the C-pillar and side impact sensor mating surfaces are clean and free of foreign material.**

5. Remove the bolt and the side impact sensor.
   - To install, tighten to 11 Nm (8 lb-ft).

WARNING: **Always tighten the fasteners of the restraints control module (RCM) and impact sensor (if equipped) to the specified torque. Failure to do so may result in incorrect restraint system operation, which increases the risk of personal injury or death in a crash.**

6. To install, reverse the removal procedure.
7. Repower the SRS. For additional information, refer to **Supplemental Restraint System (SRS) Depowering and Repowering**.